1 Appendix A: Barcode Metric, Threshold-Gap Geometry, and Repaired-Zero Transport

1.1 Scope, canonical ownership, and v21 migration status

Appendix A is the expanded-proof and persistence-metric record owner for Chapter 2 of AK-HPDST v21.0.0, *Finite Globalization, Unified Bridge Calculus, and Certificate Complexity.* It belongs to the Foundation Core-P / Verification package.

1.1.0.1 Normative status. The canonical statement owner for the Chapter 2 metric, comparison, and Core-P-to-Core-DG handoff results is the v21 Main. This appendix supplies expanded proofs, hypothesis calculus, counterexamples, record schemas, regression witnesses, and implementation-facing consequences. A theorem reproduced here under the same mathematical content is a proof expansion of the corresponding Main theorem and does not acquire an independent competing theorem identity.

1.1.0.2 Migration status. Migration status is assigned componentwise, not by adding semantic tokens at appendix level. Each MigrationID row has exactly one semantic status. The frozen v20 metric, crop, and repaired-zero statements are `inherited`; the new v21 comparison-mode, discrepancy-level, and P2DG-input statements are distinct `strengthened` successors. The token `relocated` is a disposition modifier for proof ownership and does not change theorem strength. A row is `repaired` only when the Migration Matrix identifies an exact predecessor reading that is unavailable for v21 theorem use and binds it to a corrected successor. Historical v18–v19 repair ancestry, the valid v20 three-mode schema, and a new v21 guard-rail are not by themselves grounds for a blanket repaired classification. The appendix-level phrase "mixed component migration" is summary metadata only; it is not an admissible semantic status or invocation rule.

1.1.0.3 Immutable lower baseline. Every inherited v20 theorem is used with the same hypotheses, endpoint convention, infinite-bar convention, evaluation order, equality boundary, metric constant, and prohibited stronger reading. A v21 strengthening receives its own Main identity and does not overwrite the v20 source statement.

1.1.0.4 Core-P responsibility. The mathematical responsibility of this appendix is restricted to

$$\text{barcode or persistence profile} \longrightarrow \text{typed comparison} \longrightarrow \text{repaired zero transport.}$$

It does not construct a finite diagram object, prove finite Morita reconstruction, perform effective object descent, or establish an external Return. Those responsibilities belong to the persistence-to-diagram Interface, Appendix DG, the unified Bridge / Return layer, and the Problem Interface layer.

The inherited metric repair is:

> For a positive threshold $\tau > 0$, hard barcode threshold deletion T_τ^{bar} is neither a global exact Serre localization nor a globally non-expansive operation under the standard bottleneck metric. Metric transport through hard deletion is licensed only by the sharp two-sided threshold-gap-safe theorem or by a separately proved exact comparison at its declared strength.

The inherited v20 repaired-zero extension is:

A bottleneck bound between repaired profiles does not preserve arbitrary pointwise detector values. It preserves the exact zero/nonzero predicate only inside the τ-reduced image and under the sharp post-threshold separation condition $2\varepsilon \leq \tau$. The v21 operational contract retains the strict reserve $2\varepsilon < \tau$.

The inherited crop repair is:

Window restriction and hard threshold deletion commute exactly when every globally retained bar, including every infinite bar, is either disjoint from the window or remains retained after clipping. A criterion that checks only finite globally retained bars is false on bounded windows.

The v21 architectural extension is:

Every consumed comparison has exactly one evidence mode:

$$\texttt{exact}, \quad \texttt{metric\_gap\_certified}, \\ \texttt{theorem\_controlled\_nonmetric}, \quad \texttt{not\_compared}.$$

No mode implies another without a separately proved conversion theorem. Every discrepancy also declares its mathematical level. A P2DG readiness record prepares a persistence profile for a later finite diagram Interface, but it does not itself construct or certify a Core-DG object.

Accordingly, this appendix establishes and records:

(i) constructible one-parameter persistence and barcode normal forms over a field;

(ii) endpoint, germ, window-cofinal, and finite-grid input discipline;

(iii) the standard finite/infinite bottleneck convention and interleaving isometry;

(iv) failure of the positive-threshold Serre property;

(v) failure of global non-expansiveness for hard threshold deletion;

(vi) sharp two-sided threshold-gap-safe stability with closed band $[\tau - \varepsilon, \tau + \varepsilon]$;

(vii) the identity $\mathrm{gap}_\tau = \mathrm{clear}_\tau$, with no factor $1/2$;

(viii) typed scalarization of a quantale-valued metric ledger, together with evidence that the scalarized value bounds the actual bottleneck discrepancy consumed by the proof;

(ix) the τ-reduced image of hard deletion and the exact distance-to-zero formula;

(x) sharp zero separation for finite τ-reduced profiles at $2\varepsilon \leq \tau$;

(xi) metric-safe repaired-zero transport and its strict scalarized operational form;

(xii) detector-certified zero transport, restricted to the already certified zero predicate;

(xiii) the explicit failure of bottleneck stability for raw pointwise detector ranks;

(xiv) the fixed order window $\to$ pre-threshold profile $\to$ hard deletion;

(xv) the distinction among exact, metric-gap-certified, theorem-controlled nonmetric, and not-compared evidence;

(xvi) the separation of barcode, persistence-module, filtered-chain, finite-diagram, and derived discrepancies;

(xvii) the no-automatic-upward-transport theorem;

(xviii) the infinite-bar obstruction to the former finite-only commutation condition;

(xix) an exact necessary-and-sufficient crop–collapse commutation criterion;

(xx) conditional functoriality and exactness only in declared τ-split or exact-admissible regimes;

(xxi) the P2DG input, readiness, loss, and handoff boundary;

(xxii) actual-morphism, terminal-tameness, and monitored-scope requirements for Type IV diagnostics;

(xxiii) the boundary that object-distance control does not automatically control kernel or cokernel defect profiles;

(xxiv) the distinction between manifest completeness and mathematical certification;

(xxv) the local P1 regression and G21 contribution profile.

Remark 1.1 (Conservative evolution relative to v18–v20). Earlier v18 formulations correctly rejected the overstrong Serre-localization interpretation of positive-threshold deletion, but still retained language suggesting global barcode non-expansiveness. The v19 repair removed that overclaim and replaced it by Theorem 1.37. The inherited v20 additions below are consequences on the image of hard deletion. The new v21 records do not make hard deletion globally non-expansive, exact, functorial, detector-rank stable, diagram-faithful, or derived-faithful.

1.1.0.5 Division of responsibility. Appendix A proves metric transport, exact profile transport, post-threshold zero separation, comparison-mode separation, discrepancy-level guardrails, and the P2DG input boundary. It does not prove detector soundness, completeness, birth or germ exhaustiveness, stencil coverage, family membership, MinCert bounds, finite-linear reconstruction, general finite-category reconstruction, effective descent, or finite-to-infinite reduction. A zero-looking finite observation therefore remains undefined until the governing detector certificate is complete, and a ready P2DG input remains only an Interface input until the relevant Chapter 4 or Appendix DG theorem passes.

Remark 1.2 (What this appendix does not do). This appendix is a Foundation Core-P Appendix. It does not prove a higher Ext-edge, a finite Morita theorem, an Alexandrov realization theorem, an arithmetic, PDE, geometric, categorical, cryptographic, or external Return theorem. Any external-domain conclusion requires a separately proved and registered Interface–Core–Bridge–Return chain.

1.2 Constructible one-parameter persistence

Fix a field k. Let

$$\mathsf{Pers}_k := [(\mathbb{R}, \leq), \mathsf{Vect}_k]$$

be the category of one-parameter persistence modules over k, indexed by the ordered real line.

Definition 1.3 (Constructible persistence module). *A persistence module $M \in \mathsf{Pers}_k$ is called* constructible *if:*

(i) $M(t)$ is finite-dimensional over k for every $t \in \mathbb{R}$;

(ii) *on every bounded interval $J \subset \mathbb{R}$, there is a finite critical set outside which all structure maps are isomorphisms;*

(iii) *there is no accumulation of critical parameters inside a bounded interval.*

The full subcategory is denoted by

$$\mathsf{Pers}_k^{\mathrm{cons}} \subset \mathsf{Pers}_k.$$

Remark 1.4 (Working regime)*.* All Core statements in this appendix are made in the pointwise finite-dimensional constructible one-parameter regime over k, or for finite barcode profiles obtained from declared windowed Core evaluations. Wild, nonconstructible, infinite-rank, or uncontrolled multiparameter data are outside scope unless an Interface theorem reduces them to declared constructible one-parameter profiles.

Definition 1.5 (Finite-type filtered chain complex)**.** *Let $\mathsf{Ch}^b(k)$ be the category of bounded chain complexes of finite-dimensional k-vector spaces. A* finite-type filtered chain complex *is a pair*

$$F = (C_\bullet, \{F^t C_\bullet\}_{t\in\mathbb{R}})$$

such that:

(i) *$F^t C_\bullet \subseteq F^{t'} C_\bullet$ whenever $t \leq t'$;*

(ii) *the filtration is exhaustive and left-bounded;*

(iii) *for every degree i, the persistence module*

$$\mathbf{P}_i(F) : \mathbb{R} \longrightarrow \mathsf{Vect}_k, \qquad t \longmapsto H_i(F^t C_\bullet)$$

belongs to $\mathsf{Pers}_k^{\mathrm{cons}}$.

The category of such objects and filtration-preserving chain maps is denoted by

$$\mathsf{FiltCh}(k)\,.$$

Remark 1.6 (Persistence layer first)*.* The Core reads F through monitored persistence profiles

$$\mathbf{P}_i(F).$$

Any filtered-level or derived-level use requires an additional representative policy. Barcode-level deletion does not by itself supply a global strict filtered lift.

1.3 Barcode normal forms and bar lengths

Definition 1.7 (Interval module)**.** *For an interval $I \subseteq \mathbb{R}$, the interval module k_I is defined by*

$$k_I(t) = \begin{cases} k, & t \in I, \\ 0, & t \notin I, \end{cases}$$

with identity structure maps when both parameters lie in I, and zero otherwise.

Definition 1.8 (Bar and barcode)**.** *An interval summand k_I is a* bar*. A barcode of $M \in \mathsf{Pers}_k^{\mathrm{cons}}$ is an interval decomposition*

$$M \cong \bigoplus_{\lambda\in\Lambda} k_{I_\lambda}$$

with locally finite support on bounded intervals. The multiset of bars is

$$\bar{}(M) = \{I_\lambda\}_{\lambda\in\Lambda}.$$

Theorem 1.9 (Expanded proof record for Main Theorem 2.11: constructible interval decomposition)**.** *Every $M \in \mathsf{Pers}_k^{\mathrm{cons}}$ admits a locally finite interval decomposition*

$$M \cong \bigoplus_{\lambda \in \Lambda} k_{I_\lambda}.$$

The barcode is unique up to permutation of interval summands.

Proof sketch. This is the standard theorem for interval decomposition in the pointwise finite-dimensional one-parameter regime over a field. Constructibility gives local finite type. Uniqueness follows from the Krull–Schmidt property. □

Definition 1.10 (Length of a bar)**.** *If I is finite with endpoint values $a < b$, define*

$$\ell(I) := b - a.$$

If $I = [a, \infty)$, up to harmless endpoint decoration at a, define

$$\ell(I) := +\infty$$

for thresholding purposes.

Remark 1.11 (Endpoint independence)*.* The value $\ell(I)$ depends only on endpoint values, not on finite-endpoint open/closed decoration. Every threshold rule below is therefore independent of that bookkeeping.

Definition 1.12 (Endpoint-discipline record)**.** *An endpoint-discipline record contains:*

(i) the interval-decoration convention used by the source profile;

(ii) the rule by which finite endpoint decorations are forgotten for bar length and bottleneck cost;

(iii) the birth-point or right-germ convention consumed by any downstream detector;

(iv) the policy for a bar meeting the right endpoint of a bounded window;

(v) the finite critical set, endpoint sentinels, or locally finite critical-data source used on the declared audit range;

(vi) the exact artifact identity of the endpoint data.

Length-based thresholding is decoration-independent, but detector completeness, finite-grid reconstruction, and boundary coverage need not be.

Definition 1.13 (Window-cofinal bar)**.** *Let $W = [u, u')$ be bounded. A nonempty clipped bar is* window-cofinal *when its right endpoint is u' in the declared right-open convention. It is finite for threshold-length purposes, even when it is the crop of a globally infinite bar.*

Remark 1.14 (Finite critical data are not automatically a certificate)*.* A finite critical set supplies a possible finite presentation of a constructible profile. It does not by itself prove endpoint completeness, detector completeness, zero reflection, reconstruction faithfulness, grid independence, or a finite proof theorem. Those properties require the corresponding Chapter 3, Chapter 4, or Appendix DG theorem.

1.4 Standard bottleneck metric and infinite-bar convention

Definition 1.15 (Standard bottleneck distance with infinite bars)**.** *Let B, C be finite barcode profiles, or locally finite constructible profiles restricted to a declared bounded audit range. The bottleneck distance $d_B(B, C)$ is computed by partial matchings of bars and diagonal copies. For finite bars with endpoint values*

$$I = [a, b), \qquad J = [a', b'),$$

up to the declared finite-endpoint decoration convention, the finite-to-finite matching cost is

$$\mathrm{cost}(I, J) := \max\{|a - a'|,\ |b - b'|\}.$$

The factor $1/2$ appears only in the finite-bar-to-diagonal cost below; it does not divide the endpoint L^∞-cost between two finite bars.

The infinite-bar convention is:

(i) a finite bar I may be matched to the diagonal with cost $\ell(I)/2$;

(ii) an infinite bar is never matched to the diagonal at finite cost;

(iii) a finite bar and an infinite bar never match at finite cost;

(iv) $[a, \infty)$ and $[a', \infty)$ may match with cost

$$|a - a'|.$$

Lemma 1.16 (Infinite-bar threshold invariance)**.** *For every finite $\tau > 0$,*

$$T_\tau^{\mathrm{bar}}([a, \infty)) = [a, \infty).$$

Moreover, an infinite bar never lies in a finite threshold band

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Proof. An infinite bar has threshold length $+\infty$, whereas hard deletion removes only finite bars of length at most τ. □

Remark 1.17 (Bottleneck–interleaving isometry). For q-tame or constructible one-parameter persistence modules over k, the isometry theorem gives

$$d_{\mathrm{int}}(M, N) = d_B(\bar(M), \bar(N)).$$

The equality is interpreted under Definition 1.15. Consequently, a d_B-statement may be transferred to d_{int} only when the persistence objects and barcode profiles compared are exactly those appearing in the certified record.

1.5 Positive-threshold short bars are not Serre in general

Proposition 1.18 (Positive-threshold short-bar objects are not Serre in general)**.** *Let $\tau > 0$. The class of constructible persistence modules all of whose barcode summands have finite length at most τ is not extension-closed in general. Hence it is not a Serre subcategory of $\mathsf{Pers}_k^{\mathrm{cons}}$ in general.*

Proof. For $\tau = 1$, consider

$$0 \longrightarrow k_{[1,2)} \longrightarrow k_{[0,2)} \longrightarrow k_{[0,1)} \longrightarrow 0.$$

The subobject and quotient have length 1, while the middle interval has length 2. $\square$

Corollary 1.19 (No global exact reflector claim)**.** *Hard positive-threshold deletion is not asserted to arise globally from a Serre localization, an abelian exact reflector, or a global exact endofunctor on all of* $\mathsf{Pers}_k^{\mathrm{cons}}$.

Remark 1.20 (Why this guard-rail matters)*.* Without this restriction, exactness, functoriality, filtered lifts, and tower comparison maps would inherit unsupported categorical strength. Stronger calculus must be proved in a declared τ-split or exact-admissible regime.

1.6 Hard barcode threshold deletion

Definition 1.21 (Hard barcode threshold deletion)**.** *For* $\tau > 0$, *define*

$$T_\tau^{\mathrm{bar}}(B) := \{\, I \in B \mid \ell(I) > \tau \ \text{or} \ \ell(I) = +\infty \,\}.$$

Thus finite bars of length $\leq \tau$ *are deleted, finite bars of length* $> \tau$ *are retained, and infinite bars are retained. For a module* M, *the notation* $T_\tau^{\mathrm{bar}}(M)$ *means that the operation is applied to* $\overline{(}M)$ *at barcode-profile level.*

Remark 1.22 (Status of T_τ^{bar})*.* In the general Core, T_τ^{bar} is an operation on barcode normal forms, hence on isomorphism classes. It is not asserted to be a strict exact endofunctor on all morphisms.

Proposition 1.23 (Basic properties of hard threshold deletion)**.** *For every* $\tau > 0$, T_τ^{bar} *is:*

(i) endpoint-independent;

(ii) idempotent;

(iii) infinite-bar preserving;

(iv) threshold-monotone.

Explicitly,

$$T_\tau^{\mathrm{bar}} T_\tau^{\mathrm{bar}}(B) = T_\tau^{\mathrm{bar}}(B).$$

Proof. Each assertion follows directly from deletion by the predicate $\ell(I) \leq \tau$ on finite bars. $\square$

Proposition 1.24 (Threshold composition)**.** *For* $\tau, \sigma > 0$,

$$T_\sigma^{\mathrm{bar}} T_\tau^{\mathrm{bar}}(B) = T_{\max\{\tau,\sigma\}}^{\mathrm{bar}}(B).$$

Proof. The composite deletes exactly the finite bars whose length is at most one of the two thresholds, equivalently at most their maximum. $\square$

1.7 Counterexample to global non-expansiveness

Example 1.25 (Failure of global non-expansiveness). Fix $\tau > 0$ and $0 < \eta < \tau/4$. Let

$$B = \{[0, \tau + \eta)\}, \qquad C = \{[0, \tau - \eta)\}.$$

The direct finite-to-finite matching has cost

$$\max\{|0 - 0|, |(\tau + \eta) - (\tau - \eta)|\} = 2\eta.$$

Matching both bars to the diagonal has bottleneck cost

$$\max\left\{\frac{\tau + \eta}{2}, \frac{\tau - \eta}{2}\right\} = \frac{\tau + \eta}{2} > 2\eta,$$

where the final inequality follows from $\eta < \tau/4$. Hence

$$d_B(B, C) = 2\eta.$$

After deletion,

$$T_\tau^{\mathrm{bar}}(B) = \{[0, \tau + \eta)\}, \qquad T_\tau^{\mathrm{bar}}(C) = \varnothing,$$

and therefore

$$d_B(T_\tau^{\mathrm{bar}}(B), T_\tau^{\mathrm{bar}}(C)) = \frac{\tau + \eta}{2} > 2\eta.$$

Thus global non-expansiveness fails.

Remark 1.26 (Meaning of the counterexample). A small endpoint perturbation can move a finite bar across the hard classification boundary $\ell = \tau$. The correct stability theorem must therefore exclude both input barcodes from a neighborhood of the threshold.

Corollary 1.27 (Rejected global non-expansiveness claim). *Global non-expansiveness of hard positive-threshold deletion under the standard bottleneck metric was rejected by the v19 repair and remains unavailable for v21 theorem use.*

1.8 Threshold clearance and gap

Definition 1.28 (Threshold clearance). *For a barcode profile B, define*

$$\mathrm{clear}_\tau(B) := \inf_{I \in B_{\mathrm{fin}}} |\ell(I) - \tau|.$$

If B has no finite bars, set $\mathrm{clear}_\tau(B) = +\infty$*. If B contains a finite bar of length exactly τ, then* $\mathrm{clear}_\tau(B) = 0$.

Definition 1.29 (Threshold gap as ledger alias). *For one profile,*

$$\mathrm{gap}_\tau(B) := \mathrm{clear}_\tau(B).$$

For a finite family,

$$\mathrm{gap}_\tau(B_1, \ldots, B_m) := \min_j \mathrm{clear}_\tau(B_j).$$

Thus gap_τ *is a ledger-facing alias, not a second invariant, and there is no factor* $1/2$.

Definition 1.30 (Typed bottleneck scalarization). *Let*

$$(\mathsf{V}, \oplus, 0_{\mathsf{V}}, \preceq)$$

be the declared metric-ledger quantale. A bottleneck scalarization is a monotone, subadditive map

$$\mathrm{val}_{\mathrm{B}} : \mathsf{V} \longrightarrow [0, +\infty]$$

with

$$\mathrm{val}_{\mathrm{B}}(0_{\mathsf{V}}) = 0, \qquad \mathrm{val}_{\mathrm{B}}(q \oplus q') \leq \mathrm{val}_{\mathrm{B}}(q) + \mathrm{val}_{\mathrm{B}}(q').$$

For $\mathfrak{d}_{\mathrm{met}} \in \mathsf{V}$, define

$$\varepsilon_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}) := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}).$$

A certified use additionally contains a theorem, verified computation, or artifact establishing that the relevant actual discrepancy satisfies

$$d_B(B, C) \leq \varepsilon_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}).$$

The shorthand

$$\Sigma\delta < \mathrm{gap}_\tau$$

means the typed real inequality

$$\varepsilon_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}) < \mathrm{gap}_\tau(\mathfrak{B})$$

for the declared protected family $\mathfrak{B}$.

Only metric defects belong to $\mathfrak{d}_{\mathrm{met}}$. Representative existence, actual morphisms, eligibility, detector applicability, endpoint convention, completeness provenance, terminal tameness, threshold reconciliation, higher coherence, claim status, manifest consistency, and artifact identity are non-scalar obligations.

Definition 1.31 (Metric-use identity and no-double-counting rule)**.** *Every consumed scalarized discrepancy is assigned an immutable* MetricUseID*. The record identifies the source defect terms, aggregation path, scalarization, protected profile family, and exact theorem invocation. One physical or mathematical defect contribution may be consumed at most once in the same certified bound unless a proved decomposition theorem shows that the repeated entries are disjoint contributions. A copied ledger token, repeated manifest row, or reused artifact reference does not create additional metric reserve.*

Remark 1.32 (Sharp gap versus scalarization tightness)*.* An ε-matching changes the length of a matched finite bar by at most 2ε. Nevertheless, for a two-sided certificate the sharp forbidden band is

$$[\tau - \varepsilon, \tau + \varepsilon],$$

not

$$[\tau - 2\varepsilon, \tau + 2\varepsilon].$$

The word *sharp* refers to the optimal half-width for a true bottleneck bound; it does not assert that a chosen scalarization is numerically tight.

Definition 1.33 (Windowed pre-threshold profile)**.** *For $F \in \mathsf{FiltCh}(k)$, monitored degree i, and window W, define*

$$B_{i,W}(F) := \mathsf{W}_W \mathbf{P}_i(F).$$

Threshold clearance and threshold gap are measured on this profile. The repaired evaluation is

$$\mathrm{Eval}_{i,W,\tau}(F) = T_\tau^{\mathrm{bar}}(B_{i,W}(F)).$$

Remark 1.34 (Clearance is measured before deletion, after windowing)*.* Clearance is measured on

$$B_{i,W}(F) = \mathsf{W}_W \mathbf{P}_i(F),$$

not on the already deleted profile. Once a bar has been deleted, its threshold proximity can no longer be audited.

Definition 1.35 (Threshold-gap record)**.** *A threshold-gap record for a metric comparison of two repaired evaluations contains:*

(i) immutable identifiers for the two windowed pre-threshold profiles B and C;

(ii) a finiteness witness for the compared barcode profiles, or a declared bounded audit-range restriction producing the finite profiles to which the matching theorem is applied;

(iii) the monitored degree, common window, and threshold;

(iv) the finite-bar length data or certified summaries for both profiles;

(v) $\mathrm{clear}_\tau(B)$*,* $\mathrm{clear}_\tau(C)$*, and* $\mathrm{gap}_\tau(B, C)$*;*

(vi) the metric-ledger element $\mathfrak{d}_{\mathrm{met}}$*, its immutable* MetricUseID*, and the no-double-counting declaration;*

(vii) the scalarization $\mathrm{val}_{\mathrm{B}}$ *and*

$$\varepsilon_{\mathrm{B}} := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}});$$

(viii) evidence that

$$d_B(B, C) \leq \varepsilon_{\mathrm{B}};$$

(ix) confirmation that neither finite-bar family meets

$$[\tau - \varepsilon_{\mathrm{B}}, \tau + \varepsilon_{\mathrm{B}}],$$

equivalently

$$\varepsilon_{\mathrm{B}} < \mathrm{gap}_\tau(B, C);$$

(x) every crop, window-edge, endpoint, or threshold-reconciliation component used to derive the windowed bound;

(xi) artifact references sufficient to reconstruct the comparison;

(xii) the specialized status

`gap_certified`, `gap_failed`, `gap_pending`.

The record is two-sided by definition. The specialized statuses map respectively to the atomic statuses `pass`*,* `reject`*, and* `undefined`*. Here* `gap_failed` *means that a well-typed required gap condition is proved or artifact-evidenced to fail. Missing, incomplete, stale, or pending evidence is* `gap_pending`*, never* `gap_failed`*.*

For a single profile, a clearance record may be stored as a readiness record, but it does not by itself certify a comparison.

Remark 1.36 (Length exactly τ is not gap-safe)*.* A finite bar of length exactly τ is deleted by the rule $\ell(I) \leq \tau$, but it gives $\mathrm{clear}_\tau = 0$. The evaluation is defined, whereas a positive robustness certificate at that threshold fails.

1.9 Threshold-gap-safe stability

Theorem 1.37 (Expanded proof of Main Theorem 2.46: sharp threshold-gap-safe stability)**.** *Let B, C be finite barcodes, possibly with infinite bars, under Definition 1.15. Let $\tau > 0$, $\varepsilon \geq 0$, and suppose*

$$d_B(B, C) \leq \varepsilon.$$

Assume no finite bar of either barcode has length in

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Then

$$d_B(T_\tau^{\mathrm{bar}}(B), T_\tau^{\mathrm{bar}}(C)) \leq \varepsilon.$$

Proof. If $\varepsilon = 0$, the barcodes agree and deterministic deletion gives the result.

Assume $\varepsilon > 0$ and choose a bottleneck matching of cost at most ε, which exists for finite barcodes.

Infinite bars match only infinite bars and are retained on both sides.

Let I be a retained finite bar of B. The closed-band hypothesis gives

$$\ell(I) > \tau + \varepsilon.$$

If I is matched to the diagonal, retain that diagonal match. If it is matched to a finite bar $J \in C$, then

$$|\ell(I) - \ell(J)| \leq 2\varepsilon.$$

If J were deleted, then the closed-band hypothesis on C would force

$$\ell(J) < \tau - \varepsilon,$$

hence

$$\ell(I) - \ell(J) > 2\varepsilon,$$

a contradiction. Thus J is retained.

The symmetric argument applies to retained bars of C. After removing pairs deleted on both sides, the original matching therefore restricts, together with retained-to-diagonal matches, to a matching of the retained barcodes of cost at most ε. ☐

Example 1.38 (Sharpness of the ε-band). Let $0 < c < \varepsilon$, choose $0 < \eta < \varepsilon - c$, and choose τ so large that the diagonal costs below exceed ε. Set

$$L_+ = \tau + c + \eta, \qquad L_- = \tau - c - \eta$$

and take centered bars I, J of lengths L_+, L_-. Both lengths avoid the smaller band $[\tau - c, \tau + c]$, and

$$d_B(\{I\}, \{J\}) = c + \eta < \varepsilon.$$

But I is retained, J is deleted, and the post-deletion distance is $L_+/2 > \varepsilon$. Thus no smaller two-sided half-width $c < \varepsilon$ suffices in general.

Remark 1.39 (Two-sided certificate). It is insufficient for only one barcode to avoid the threshold band. A one-sided 2ε-type margin may be used as a separately declared conservative policy, but it is not the sharp inherited two-sided theorem.

Corollary 1.40 (Scalarized-ledger-safe threshold classification)**.** *Let*

$$\varepsilon_{\mathrm{B}} := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}).$$

Suppose the record proves

$$d_B(B, C) \leq \varepsilon_{\mathrm{B}}$$

and

$$\varepsilon_{\mathrm{B}} < \mathrm{gap}_\tau(B, C).$$

Then

$$d_B(T_\tau^{\mathrm{bar}}(B), T_\tau^{\mathrm{bar}}(C)) \leq \varepsilon_{\mathrm{B}}.$$

Proof. The strict inequality implies two-sided avoidance of the closed band with half-width ε_{B}. Apply Theorem 1.37. □

1.10 Sharp zero separation inside the repaired image

Definition 1.41 (τ-reduced barcode profile)**.** *A barcode profile P is* τ-reduced *if*

$$T_\tau^{\mathrm{bar}}(P) = P.$$

Equivalently, every finite bar of P has length strictly greater than τ, while every infinite bar is retained. Every repaired profile $T_\tau^{\mathrm{bar}}(B)$ is τ-reduced.

Definition 1.42 (Zero radius of a finite barcode profile)**.** *For a finite barcode profile P, define*

$$\mathrm{zrad}(P) := d_B(P, 0),$$

where 0 *denotes the empty barcode. Under the standard convention,*

$$\mathrm{zrad}(P) = \begin{cases} 0, & P = 0, \\ \dfrac{1}{2} \max\limits_{I \in P} \ell(I), & P \neq 0 \textit{ and every bar of } P \textit{ is finite}, \\ +\infty, & P \textit{ contains an infinite bar.} \end{cases}$$

Lemma 1.43 (Distance-to-zero formula)**.** *Definition 1.42 gives the exact bottleneck distance from P to the zero profile.*

Proof. If every bar is finite, every bar must be matched to the diagonal, and the cost of the unique type of admissible matching is the maximum of the diagonal costs $\ell(I)/2$. If an infinite bar is present, it cannot be matched to the diagonal at finite cost. The empty profile has distance zero from itself. □

Theorem 1.44 (Expanded proof of Main Theorem 2.53: sharp zero separation for τ-reduced profiles)**.** *Let P, Q be finite τ-reduced barcode profiles, possibly including infinite bars. Let $\varepsilon \geq 0$, and suppose*

$$d_B(P, Q) \leq \varepsilon, \qquad 2\varepsilon \leq \tau.$$

Then

$$P = 0 \iff Q = 0.$$

The constant is sharp: if $2\varepsilon > \tau$, the conclusion can fail.

Proof. Assume $P = 0$. If Q contained an infinite bar, then $d_B(P, Q) = +\infty$, contradicting the hypothesis. If $Q \neq 0$ and all bars of Q were finite, τ-reducedness would give

$$\ell(I) > \tau \geq 2\varepsilon \qquad (I \in Q).$$

Lemma 1.43 would then imply

$$d_B(0, Q) = \frac{1}{2} \max_{I \in Q} \ell(I) > \varepsilon,$$

again a contradiction. Hence $Q = 0$. The converse is symmetric.

For sharpness, suppose $2\varepsilon > \tau$. Choose L with

$$\tau < L \leq 2\varepsilon,$$

let $P = 0$, and let Q consist of one finite bar of length L. Then Q is τ-reduced and

$$d_B(P, Q) = L/2 \leq \varepsilon,$$

but $P = 0$ and $Q \neq 0$. ☐

Remark 1.45 (Strict operational reserve). Theorem 1.44 is valid at the closed boundary $2\varepsilon = \tau$, because every retained finite bar has length strictly greater than τ. The v21 operational contract uses

$$2\varepsilon < \tau$$

to retain a positive reserve and to avoid equality-sensitive scalarization, rounding, or finite-precision policies. The strict policy is conservative, not mathematically sharp.

Proposition 1.46 (Exact zero transport)**.** *If $P \cong Q$ as persistence profiles, then*

$$P = 0 \iff Q = 0.$$

No metric ledger or threshold-gap certificate is required for this exact claim. The actual isomorphism and its immutable proof artifact are required.

Proof. The zero object is invariant under isomorphism. ☐

Theorem 1.47 (Expanded proof of Main Theorem 2.56: metric-safe repaired-zero transport)**.** *Let B, C be finite barcode profiles, possibly including infinite bars. Let $\tau > 0$, $\varepsilon \geq 0$, and suppose*

$$d_B(B, C) \leq \varepsilon.$$

Assume that no finite bar of either profile has length in

$$[\tau - \varepsilon, \tau + \varepsilon],$$

and assume

$$2\varepsilon \leq \tau.$$

Then

$$T_\tau^{\mathrm{bar}}(B) = 0 \iff T_\tau^{\mathrm{bar}}(C) = 0.$$

Proof. Theorem 1.37 gives

$$d_B(T_\tau^{\mathrm{bar}}(B), T_\tau^{\mathrm{bar}}(C)) \leq \varepsilon.$$

Both repaired profiles are τ-reduced by Definition 1.41. Apply Theorem 1.44. ☐

Corollary 1.48 (Scalarized metric-safe repaired-zero transport)**.** *Let $\mathfrak{d}_{\mathrm{met}} \in \mathsf{V}$, set*

$$\varepsilon_{\mathrm{B}} := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met}}),$$

and suppose the record proves

$$d_B(B, C) \le \varepsilon_{\mathrm{B}}, \qquad \varepsilon_{\mathrm{B}} < \mathrm{gap}_\tau(B, C), \qquad 2\varepsilon_{\mathrm{B}} < \tau.$$

Then

$$T_\tau^{\mathrm{bar}}(B) = 0 \quad \Longleftrightarrow \quad T_\tau^{\mathrm{bar}}(C) = 0.$$

This is the strict operational form consumed by the v21 metric-safe detector chain.

Proof. The strict gap inequality supplies the two-sided forbidden-band hypothesis. Apply Theorem 1.47. □

Definition 1.49 (Post-threshold zero-separation record)**.** *A post-threshold zero-separation record contains:*

(i) immutable identifiers for $P = T_\tau^{\mathrm{bar}}(B)$ and $Q = T_\tau^{\mathrm{bar}}(C)$;

(ii) the common threshold τ, the proved bound $d_B(P, Q) \le \varepsilon$, and its derivation;

(iii) evidence that P and Q are τ-reduced;

(iv) the mathematical inequality $2\varepsilon \le \tau$, or the stricter operational inequality $2\varepsilon < \tau$;

(v) the theorem reference Theorem 1.44;

(vi) exact source, scalarization, computation, proof, and artifact identities.

Missing required evidence yields `undefined`. An evidenced false required inequality yields `reject`. This record creates no independent status algebra.

1.11 Detector-zero transport and the rank-instability boundary

Definition 1.50 (Metric-safe detector-zero transport record)**.** *A metric-safe detector-zero transport record from B to C contains:*

(i) a complete Chapter 3 detector certificate establishing $T_\tau^{\mathrm{bar}}(B) = 0$, including its exact profile stage;

(ii) the two-sided threshold-gap comparison record for B and C;

(iii) the repaired-profile distance bound produced by Theorem 1.37;

(iv) the post-threshold zero-separation record of Definition 1.49;

(v) the conclusion $T_\tau^{\mathrm{bar}}(C) = 0$;

(vi) immutable detector, comparison, theorem, and replay artifacts.

The record transports only the exact zero predicate. It does not transport detector sample values, ranks, birth sets, germ sets, sample counts, coverage witnesses, or certificate sizes.

Corollary 1.51 (Detector-certified metric zero transport)**.** *Assume a complete detector certificate proves*

$$T_\tau^{\mathrm{bar}}(B) = 0.$$

If the hypotheses of Corollary 1.48 hold for (B, C)*, then*

$$T_\tau^{\mathrm{bar}}(C) = 0.$$

The converse holds after exchanging B *and* C*. Detector completeness is required only for a profile whose zero statement is obtained from detector data; the metric theorem transports an already certified zero predicate.*

Proof. The detector certificate supplies the source zero statement. Apply Corollary 1.48. □

Definition 1.52 (Pointwise span-rank observation)**.** *For a persistence module* M*, a point* t*, and a span* $s > 0$*, define*

$$r_{t,s}(M) := \dim_k \operatorname{im}\big(M(t) \longrightarrow M(t+s)\big),$$

when the displayed structure map is defined. This is a finite observation, not by itself a complete detector certificate.

Example 1.53 (Arbitrarily small perturbation can change a pointwise detector rank)*.* Fix $s > 0$ and $L > 0$. For any

$$0 < \delta < \min\{L, (s+L)/2\},$$

let

$$M_\delta := k_{[0,s+L)}, \qquad N_\delta := k_{[\delta,s+L+\delta)}.$$

Their one-bar barcode profiles B_δ, C_δ satisfy

$$d_B(B_\delta, C_\delta) = \delta.$$

Both bars have the same length $s+L$, and therefore have the same clearance from every threshold that depends only on bar length. Nevertheless,

$$r_{0,s}(M_\delta) = 1, \qquad r_{0,s}(N_\delta) = 0.$$

Thus pointwise span-rank observations can change under arbitrarily small bottleneck perturbations, even when no bar crosses a length threshold.

Proposition 1.54 (Expanded proof of Main Proposition 2.85: no bottleneck stability for raw pointwise detector ranks)**.** *There is no general implication*

$$d_B(B, C) \le \varepsilon \quad \Longrightarrow \quad r_{t,s}(B) = r_{t,s}(C)$$

for a fixed sample point t *and span* s*, even for one-bar finite barcodes and arbitrarily small positive* ε*. Here the rank notation is read on the interval modules represented by the barcodes.*

Proof. Apply Example 1.53 with $0 < \delta \le \varepsilon$. □

Remark 1.55 (Zero predicate, not raw detector output). The v21 metric chain preserves the repaired zero/nonzero predicate under the sharp separation hypotheses. It does not preserve the raw output of a fixed-point detector. Birth, germ, endpoint, cofinality, coverage, mesh, family-membership, excess-bound, and profile-stage conditions remain separate non-scalar obligations and cannot be replaced by metric slack.

1.12 Window restriction and window-edge budget

Definition 1.56 (Right-open window)**.** *A domain window is a nonempty right-open interval*

$$W = [u, u'), \qquad u \in \mathbb{R}, \qquad u' \in \mathbb{R} \cup \{+\infty\}, \qquad u < u'.$$

A finite u' gives a bounded audit window; $u' = +\infty$ gives a terminal audit window.

Definition 1.57 (Window restriction)**.** *Define $\mathsf{W}_W(M)$ by restriction to W, with extension by zero outside W whenever a globally indexed module is required. At barcode level, bars are clipped to $I \cap W$, and empty intersections are discarded.*

Remark 1.58 (Infinite bars under window restriction)*.* On a bounded window, an infinite bar $[a, \infty)$ with nonempty intersection is clipped to

$$[\max\{a, u\}, u')$$

and therefore becomes finite; it is then classified by its clipped finite length. On a terminal window it remains infinite and is retained by every finite threshold.

Proposition 1.59 (No unconditional crop–collapse commutation)**.** *In general,*

$$\mathsf{W}_W T_\tau^{\mathrm{bar}}(B) \neq T_\tau^{\mathrm{bar}} \mathsf{W}_W(B).$$

Proof. Take $B = \{[0, 10)\}$, $W = [0, 1)$, and $\tau = 2$. Global deletion retains the length-10 bar, which clips to $[0, 1)$. Windowing first produces a length-1 bar, which is then deleted. □

Definition 1.60 (Window-edge clearance)**.** *A window-edge clearance record supplies a lower bound on the distance of relevant endpoints or critical parameters from the active window boundaries. Its role is to justify that cropping does not silently gain or lose relevant bars under a source perturbation.*

Definition 1.61 (Crop budget)**.** *A crop budget*

$$\delta_{\mathrm{crop}}(i, W)$$

is a metric-ledger component bounding discrepancy introduced by restriction, clipping, endpoint uncertainty, or moving window boundaries. A source-level perturbation may be transported to a windowed pre-threshold bound only through a theorem proving zero crop error or through a ledger that includes the crop contribution.

Remark 1.62 (Gap-safe stability consumes windowed profile distance)*.* Theorem 1.37 consumes

$$d_B(B, C) \leq \varepsilon$$

for the actual profiles to which hard deletion is applied. In AK applications these are the windowed pre-threshold profiles. It does not itself prove source-to-window metric stability.

Example 1.63 (Infinite-bar obstruction to the former finite-only condition)*.* Let

$$B = \{[0, +\infty)\}, \qquad W = [0, 1), \qquad \tau = 2.$$

Global threshold deletion retains the infinite bar, after which cropping gives $[0, 1)$. Windowing first turns the bar into a finite length-1 bar, which is deleted. Thus infinite globally retained bars must be checked.

Definition 1.64 (τ-window-adapted condition)**.** *The pair (B, W) is* τ-window-adapted *if every bar retained globally by T_τ^{bar}, including every infinite bar, is either:*

(i) disjoint from W; or

(ii) clipped to a bar retained by threshold deletion on W.

Proposition 1.65 (Expanded proof of Main Proposition 2.65: exact crop–collapse commutation criterion)**.** *For a barcode B, right-open window W, and $\tau > 0$, the following are equivalent:*

(i) (B, W) is τ-window-adapted;

(ii)

$$\mathsf{W}_W T_\tau^{\mathrm{bar}}(B) = T_\tau^{\mathrm{bar}} \mathsf{W}_W(B)$$

as barcode multisets, including multiplicity.

Proof. A globally deleted finite bar has length at most τ, and clipping cannot increase its length, so it is deleted in both orders. A globally retained bar contributes equally in both orders exactly when it is disjoint from the window or its nonempty crop remains retained. Multiplicity makes this criterion necessary as well as sufficient. ☐

Remark 1.66 (Default policy and audit use). The Core does not assume τ-window-adaptedness. The default order is window first, threshold second. Commutation may be used only when Definition 1.64 is verified for every globally retained bar and recorded as an artifact-backed non-scalar obligation.

1.13 Canonical v21 comparison-mode calculus

Definition 1.67 (Canonical v21 comparison modes)**.** *Every comparison inference consumed by Appendix A records exactly one mode:*

$$\texttt{exact}, \qquad \texttt{metric\_gap\_certified},$$
$$\texttt{theorem\_controlled\_nonmetric}, \qquad \texttt{not\_compared}.$$

Their meanings are:

(i) `exact`: an actual equality or persistence-profile isomorphism is proved for the displayed source and target;

(ii) `metric_gap_certified`: the actual pre-threshold bottleneck bound, the sharp two-sided gap hypotheses, and every zero-separation hypothesis required by the conclusion are supplied;

(iii) `theorem_controlled_nonmetric`: a named theorem transports the declared predicate or relation without asserting a bottleneck estimate or an exact isomorphism;

(iv) `not_compared`: no comparison conclusion is consumed.

The mode is part of the claim type and is not selected retrospectively from whichever interpretation would produce the desired verdict.

Definition 1.68 (Theorem-controlled nonmetric comparison record)**.** *A theorem-controlled nonmetric comparison record contains:*

(i) immutable source and target profile identities;

(ii) the exact predicate or relation transported;

(iii) the canonical theorem identity and proof locator;

(iv) every hypothesis and assumption context;

(v) the direction of transport;

(vi) the naturality or family scope, when any;

(vii) the downward-closed set of supported conclusion or Return strengths;

(viii) , only when a unique greatest supported strength is separately proved, that greatest supported strength;

(ix) immutable proof and execution artifacts;

(x) prohibited stronger readings and failure routes.

Such a record does not create a bottleneck bound, threshold-gap certificate, robustness radius, actual comparison morphism, inverse implication, or exact profile isomorphism unless the cited theorem explicitly proves that item.

Theorem 1.69 (Expanded proof of Main Theorem 2.69: comparison-mode separation)**.** *No canonical comparison mode implies another merely by its status. In particular:*

(i) an exact comparison need not have positive threshold clearance;

(ii) a metric-gap-certified comparison need not provide an actual profile isomorphism;

(iii) a theorem-controlled nonmetric comparison need not supply either a metric bound or an exact comparison;

(iv) `not_compared` *discharges no invoked comparison obligation.*

A conversion between modes is permitted only through a separate theorem whose conclusion supplies exactly the target mode's required data.

Proof. The modes encode different evidence types.

For item (i), let B contain a finite bar of length exactly τ. The identity $B \cong B$ is an exact comparison, but $\mathrm{clear}_\tau(B) = 0$, so no positive-radius threshold-gap certificate exists at that threshold.

For item (ii), choose a finite bar $I = [0, L)$ with $L \neq \tau$, and let

$$J = [\delta, L + \delta), \qquad 0 < \delta < \min\{|L - \tau|, L/2\}.$$

The one-bar profiles satisfy

$$d_B(\{I\}, \{J\}) = \delta$$

and are two-sided gap-safe, but they are not equal as barcode profiles and their interval modules are not isomorphic as $\mathbb{R}$-indexed persistence modules.

For item (iii), the defining data of Definition 1.68 contain neither an exact isomorphism nor a metric estimate unless the cited theorem separately supplies one. Item (iv) follows because `not_compared` is a positive scope exclusion, not comparison evidence. Therefore no mode conversion is licensed without an additional theorem constructing the target record. $\square$

Remark 1.70 (Exactness is not robustness)*.* An exact comparison at one input pair does not provide a positive-radius stability theorem. Conversely, a metric-gap-certified comparison gives a controlled radius but does not choose an actual isomorphism. Theorem-controlled transport returns only the named predicate or relation at its declared direction and strength.

1.14 Typed discrepancy levels and no automatic level change

Definition 1.71 (Typed discrepancy level)**.** *Every discrepancy record declares exactly one mathematical level:*

$$\texttt{barcode},\quad \texttt{persistence\_module},\quad \texttt{filtered\_chain},$$
$$\texttt{finite\_diagram},\quad \texttt{derived}.$$

It additionally records the source and target objects, coefficient and indexing conventions, comparison mode, relation or metric, and the theorem permitting any change of level.

Definition 1.72 (Barcode and persistence-module discrepancy)**.** *Barcode discrepancy is measured by the standard bottleneck distance under the declared endpoint and infinite-bar conventions. Persistence-module discrepancy may be measured by interleaving distance in the constructible or q-tame regime. The bottleneck–interleaving isometry identifies these values only for the exact modules and barcode conventions covered by the isometry theorem.*

Definition 1.73 (Higher-level discrepancy placeholder)**.** *Appendix A assigns no canonical scalar to filtered-chain, finite-diagram, or derived discrepancy. Such a record is meaningful only after a later theorem declares the object category, equivalence relation or metric, actual comparison data, and the Interface relating that level to the persistence profile.*

Theorem 1.74 (Expanded proof of Main Theorem 2.74: no automatic upward discrepancy transport)**.** *Let a barcode or persistence-module comparison be supplied.*

(i) A positive-radius bound, or any bound not accompanied by an exact identification theorem, does not imply chain isomorphism, chain-homotopy equivalence, filtered quasi-isomorphism, equality of finite diagrams, quasi-isomorphism, derived equivalence, or stability of kernels, cokernels, cones, or descent data.

(ii) Even an exact barcode or persistence-module isomorphism does not by itself identify filtered-chain representatives, branching finite diagrams, derived enhancements, cones of independently chosen morphisms, or descent data.

(iii) An exact persistence-module isomorphism induces a natural isomorphism after restriction to any fixed finite linearly ordered subcategory, but only through that declared restriction functor. It supplies neither a canonical artifact-level representative nor a stronger reconstruction theorem.

Every other level change requires a separately proved Interface with its own functoriality, exactness, reflection, information, and artifact profiles.

Proof. A positive bottleneck or interleaving radius permits non-isomorphic persistence modules, so it cannot entail any exact higher-level relation. Even an exact monitored persistence profile forgets filtered-chain extension data, representative choices, branching data, independently chosen morphisms, and derived enhancement data. A barcode matching is not a morphism of any of those objects.

For item (iii), precomposition of a natural isomorphism with the inclusion of a finite linearly ordered subcategory preserves naturality and invertibility. This formal restriction does not prove endpoint completeness, full-profile reconstruction, grid independence, branching reconstruction, or descent. □

Remark 1.75 (Downward level change is also typed). A filtered-chain, finite-diagram, or derived comparison may imply a persistence or barcode comparison only through a declared persistence functor and a theorem proving the required preservation or stability. The absence of automatic upward transport does not license automatic downward transport.

1.15 Core-P to Core-DG input and handoff boundary

Definition 1.76 (Finite critical-grid datum). *For a constructible profile P on a declared window W, a finite critical-grid datum is a finite ordered set*

$$\Gamma = \{t_0 < \cdots < t_m\} \subset W$$

together with endpoint sentinels or germ data sufficient for the declared finite-presentation task. The record states whether Γ is complete, merely sampled, or supplied by an external characterization theorem. Finiteness alone is not completeness.

Definition 1.77 (Core-P to Core-DG input record). *A P2DG input record is*

$$\mathrm{P2DGInput} = (P, \Gamma, W, \tau, \mathrm{endpt}, \mathrm{variance}, \mathrm{comparison}, \mathrm{loss}, \mathrm{artifacts}),$$

where:

(i) the source kind is declared as `actual_module` *or* `barcode_profile`*, and the exact source P and evaluation stage are identified;*

(ii) Γ *is a finite critical-grid datum;*

(iii) W, τ*, and the window-first evaluation stage are fixed;*

(iv) endpt *records endpoint, sentinel, cofinal, and germ conventions;*

(v) variance *records the direction of every proposed diagram arrow;*

(vi) comparison *records one mode from Definition 1.67;*

(vii) loss *lists every feature not certified to survive the proposed passage;*

(viii) artifacts *binds the source profile, critical data, computations, proofs, and immutable object identities.*

Definition 1.78 (P2DG readiness status). *A P2DG input has exactly one workflow-readiness value:*

`ready`, `incomplete`, `inconsistent`, `not_invoked`.

Ready means only that the fields required by the selected downstream Interface are present and internally consistent. It does not mean that a target finite diagram exists, that reconstruction is faithful, or that a Chapter 8 verifier status is `pass`*.*

Theorem 1.79 (Expanded proof of Main Theorem 2.79: P2DG handoff boundary). *Let a ready P2DG input record be given.*

(i) If the source kind is `actual_module`*, restriction to the finite ordered grid Γ canonically gives the preliminary linear diagram $P|_\Gamma$.*

(ii) If the source kind is `barcode_profile`*, the barcode determines the isomorphism class of the one-parameter persistence module in the standing interval-decomposable regime, but it does not canonically choose an actual representative, structure-map artifact, or immutable object identity.*

(iii) In either case, readiness does not prove endpoint completeness, reconstruction of the full windowed profile, detector preservation, zero reflection, grid independence, or extension to a branching Core-DG object.

Canonical finite linear reconstruction and its loss or reflection claims belong to the persistence-to-diagram Interface. General finite-category reconstruction, cone detection, Morita reconstruction, Alexandrov realization, and effective descent belong to Appendix DG.

Proof. For an actual persistence module, evaluation at the grid objects and use of the actual structure maps define the restriction diagram. For barcode input, interval decomposition determines only an isomorphism class; it does not select a representative with artifact-level maps. Neither construction contains the completeness, reflection, branching, cone, reconstruction, or descent theorems excluded in item (iii). □

Remark 1.80 (P2DG supported handoff outputs). The supported handoff outputs of Appendix A are a typed and internally consistent input record and, for actual-module input, the formal finite linear restriction. This is a supported-output boundary, not an automatic claim that a unique greatest Return strength exists. Appendix A does not return a reconstructed full profile, a filtered-chain object, a general finite diagram, a derived equivalence, or a global object.

1.16 Core windowed evaluation

Definition 1.81 (Windowed repaired evaluation)**.** *For $F \in \mathsf{FiltCh}(k)$, monitored degree i, window W, and $\tau > 0$, define*

$$\mathrm{Eval}_{i,W,\tau}(F) := T_\tau^{\mathrm{bar}}(\mathsf{W}_W \mathbf{P}_i(F)).$$

Remark 1.82 (Fixed order). The notation means the ordered pipeline

$$F \longmapsto \mathbf{P}_i(F) \longmapsto B_{i,W}(F) \longmapsto T_\tau^{\mathrm{bar}}(B_{i,W}(F)).$$

Definition 1.83 (Repaired evaluation record)**.** *A repaired evaluation record contains:*

(i) immutable source-object and profile identifiers;

(ii) degree i, window W, and threshold τ;

(iii) the pre-threshold profile $B_{i,W}(F)$;

(iv) the repaired profile $\mathrm{Eval}_{i,W,\tau}(F)$;

(v) the comparison mode

$$\texttt{exact}, \qquad \texttt{metric\_gap\_certified},$$
$$\texttt{theorem\_controlled\_nonmetric}, \qquad \texttt{not\_compared}.$$

(vi) reconstructible artifact references.

Definition 1.84 (Threshold-gap-certified repaired evaluation record)**.** *A repaired evaluation record is threshold-gap-certified only when its comparison mode is `metric_gap_certified` and every metric comparison transporting through hard deletion is supplemented by the two-sided record of Definition 1.35.*

An exact equality or proved profile isomorphism is stored with comparison mode `exact`. A named nonmetric transport theorem is stored with mode `theorem_controlled_nonmetric`. No comparison inference is consumed in mode `not_compared`. None of these modes automatically supplies another mode's evidence.

Remark 1.85 (Evaluation is definable without certification). The repaired profile is definitionally available once the pre-threshold profile is available. Threshold-gap certification is required whenever a metric discrepancy is transported through hard deletion. If the transported conclusion is an exact repaired-zero predicate, the post-threshold record of Definition 1.49 is additionally required; threshold-gap stability alone bounds distance and does not identify zero and nonzero profiles. No metric certificate is fabricated for a purely exact identity.

1.17 Overlap Gate: post-threshold comparison and pre-threshold gap

Definition 1.86 (Overlap pre-threshold profiles)**.** *For local inputs F_α, F_β and*

$$W_{\alpha\beta} = W_\alpha \cap W_\beta,$$

define

$$B^{\alpha}_{i,W_{\alpha\beta}} := \mathsf{W}_{W_{\alpha\beta}}\mathbf{P}_i(F_\alpha),$$

$$B^{\beta}_{i,W_{\alpha\beta}} := \mathsf{W}_{W_{\alpha\beta}}\mathbf{P}_i(F_\beta).$$

The repaired overlap profiles are obtained by applying T^{bar}_τ to these two inputs.

Remark 1.87 (Two-level structure of the Overlap Gate)*.* The threshold-gap theorem requires a discrepancy bound on the *pre-threshold* profiles. The theorem then transports that bound to the *post-threshold* repaired profiles. A post-threshold comparison alone is not a substitute for the pre-threshold hypothesis.

Definition 1.88 (Comparison-typed overlap handoff)**.** *An overlap comparison handoff records one requested overlap conclusion and exactly one canonical comparison mode.*

(i) In mode `exact`, it supplies an actual equality or profile-isomorphism proof on the common overlap.

(ii) In mode `metric_gap_certified`, it supplies the pre-threshold bottleneck bound, two-sided threshold-gap record, and, when zero equivalence is requested, the post-threshold zero-separation record.

(iii) In mode `theorem_controlled_nonmetric`, it supplies the exact named theorem and transports only that theorem's declared overlap predicate or relation.

(iv) In mode `not_compared`, no overlap conclusion is consumed and a scope-exclusion policy is required.

This handoff is record-level. It does not supply a cocycle, natural transformation, descent datum, or global object.

Proposition 1.89 (Expanded proof of Main Proposition 2.89: overlap transfer under a scalarized threshold-gap certificate)**.** *Let*

$$\varepsilon_{\alpha\beta,i} := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\alpha\beta,i}).$$

Suppose the record proves

$$d_B\big(B^{\alpha}_{i,W_{\alpha\beta}}, B^{\beta}_{i,W_{\alpha\beta}}\big) \le \varepsilon_{\alpha\beta,i}$$

and

$$\varepsilon_{\alpha\beta,i} < \mathrm{gap}_\tau\big(B^{\alpha}_{i,W_{\alpha\beta}}, B^{\beta}_{i,W_{\alpha\beta}}\big).$$

Then

$$d_B\big(\mathrm{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha), \mathrm{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta)\big) \le \varepsilon_{\alpha\beta,i}.$$

In the declared constructible regime, the same bound may be read using d_{int}.

Proof. Apply Corollary 1.40 to the two pre-threshold overlap profiles. □

Corollary 1.90 (Overlap repaired-zero equivalence)**.** *Under the hypotheses of Proposition 1.89, assume in addition*

$$2\varepsilon_{\alpha\beta,i} \leq \tau.$$

Then

$$\mathrm{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha) = 0 \quad \Longleftrightarrow \quad \mathrm{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta) = 0.$$

For operational audit consumption, the strict reserve $2\varepsilon_{\alpha\beta,i} < \tau$ is used.

Proof. Proposition 1.89 supplies the repaired-profile bottleneck bound. Both repaired profiles are τ-reduced. Apply Theorem 1.44. □

1.18 Exact-admissible regimes

Definition 1.91 (τ-split object)**.** *An object $M \in \mathsf{Pers}_k^{\mathrm{cons}}$ is τ-split if it is equipped with declared data*

$$M \cong M_{\leq\tau} \oplus M_{>\tau}$$

such that finite bars in $M_{\leq\tau}$ have length $\leq \tau$, finite bars in $M_{>\tau}$ have length $> \tau$, and all infinite bars lie in $M_{>\tau}$.

Definition 1.92 (τ-split morphism)**.** *A morphism $f : M \to N$ between τ-split objects is τ-split if*

$$f(M_{\leq\tau}) \subseteq N_{\leq\tau}, \qquad f(M_{>\tau}) \subseteq N_{>\tau}.$$

Definition 1.93 (τ-split regime)**.** *A τ-split regime is a declared subcategory or workflow domain in which every object and morphism under consideration is τ-split with the chosen decompositions as part of the data.*

Proposition 1.94 (Expanded proof of Main Proposition 2.94: functorial deletion in a τ-split regime)**.** *Inside a declared τ-split regime, the assignment*

$$M \longmapsto M_{>\tau}$$

is functorial and agrees with $T_\tau^{\mathrm{bar}}(M)$ on barcode profiles.

Proof. Every declared morphism restricts to the retained summands. □

Definition 1.95 (Exact-admissible collapse regime)**.** *An exact-admissible collapse regime is a τ-split regime in which the retained-summand assignment is additionally proved to preserve every exact sequence used by the relevant argument.*

Remark 1.96 (Exactness is conditional)*.* Exactness is not a global Core theorem. A proof object invoking it records:

(i) the τ-split decomposition policy;

(ii) the morphism-compatibility policy;

(iii) the exact sequences being tested;

(iv) the proof or immutable artifact establishing exact preservation.

1.19 Filtered representative handoff to Appendix B

This subsection mirrors the Chapter 2 input contract only. Appendix B is the canonical detailed owner of filtered-representative calculus, compatibility, and certificate use. No independent representative theorem authority is created here.

Definition 1.97 (Windowed collapsed persistence profile)**.** *For $F \in \mathsf{FiltCh}(k)$, define*

$$\mathcal{P}_{i,W,\tau}(F) := \mathrm{Eval}_{i,W,\tau}(F).$$

Definition 1.98 (Admissible filtered representative)**.** *A filtered object $C_{\tau,W}F$ is an admissible representative if, for every monitored degree i,*

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathcal{P}_{i,W,\tau}(F)$$

at the declared profile-equivalence strength. Existence is an independent non-scalar obligation.

Remark 1.99 (No global strict lift by default). The Core does not assert a global strict endofunctor

$$C_\tau : \mathsf{FiltCh}(k) \longrightarrow \mathsf{FiltCh}(k)\,.$$

The default notation $C_{\tau,W}F$ denotes a declared window-relative representative.

Definition 1.100 (Representative equivalence)**.** *Two representatives are equivalent for a declared gate if their monitored persistence profiles agree in every monitored degree under the declared barcode or persistence-profile equivalence policy. This does not automatically imply chain isomorphism, filtered quasi-isomorphism, or derived equivalence.*

Remark 1.101 (Filtered quasi-isomorphism). Filtered quasi-isomorphism may be required when the argument uses filtered-chain or derived constructions. It is not required merely to define a barcode-level repaired profile.

1.20 B-Gate input handoff to Appendices B and C

This subsection records only the metric-side handoff. Appendix B owns the representative and detector obligations, and Appendix C owns eligibility and higher-Ext realization. Their downstream theorems are not duplicated here.

Definition 1.102 (Windowed B-side input)**.** *The B-side input is*

$$\mathrm{Eval}_{i,W,\tau}(F),$$

or a verified representative $C_{\tau,W}F$ realizing that profile.

Definition 1.103 (Windowed topological clause)**.** *The topological clause is*

$$\mathrm{Eval}_{1,W,\tau}(F) = 0.$$

If representative compatibility is verified, it may equivalently be recorded as

$$\mathrm{PH}_1(C_{\tau,W}F) = 0.$$

Definition 1.104 (Windowed categorical clause)**.** *The categorical clause may be invoked only when an admissible, realization-compatible representative exists and the eligible regime, including*

$$R(C_{\tau,W}F) \in D^{[-1,0]}(k\text{-}mod),$$

is verified. Only then is

$$\mathrm{Ext}^1(R(C_{\tau,W}F), k) = 0$$

a well-typed Core clause.

Definition 1.105 (Comparison-typed B-Gate^{+} input)**.** *A B-Gate^{+} input records:*

(i) the post-threshold repaired profile $\mathrm{Eval}_{1,W,\tau}(F)$*;*

(ii) the pre-threshold profile $B_{1,W}(F)$*;*

(iii) the comparison mode for every comparison involving that profile;

(iv) in mode `metric_gap_certified`*, the two compared pre-threshold profiles, their two-sided threshold-gap comparison record, and every ledger component needed to justify the actual bottleneck bound, including crop and window-edge contributions;*

(v) in mode `exact`*, the actual equality or profile-isomorphism proof and its immutable artifact;*

(vi) in mode `theorem_controlled_nonmetric`*, the named transport theorem, exact transported predicate, direction, hypotheses, downward-closed supported conclusion-strength set, and immutable proof artifact;*

(vii) in mode `not_compared`*, the positive scope-exclusion policy showing that no comparison conclusion is consumed;*

(viii) every required non-scalar obligation.

A single-profile clearance value is readiness information only and is not substituted for a two-sided metric comparison record. No purely exact comparison is assigned a fabricated positive-radius reserve.

Remark 1.106 (PH before Ext)*.* The repaired persistence profile is primary. The Ext clause is read only through an eligible representative and the one-way bridge. No reverse implication is supplied here.

1.21 Tower and Type IV input handoff to Appendix D

This subsection fixes the Chapter 2 input schema only. Appendix D is the canonical detailed owner of tower, Type IV, transient-defect, derived-limit, and finite-to-infinite mathematics.

Definition 1.107 (Windowed collapsed tower objects)**.** *For a tower*

$$F_0 \to F_1 \to F_2 \to \cdots ,$$

write

$$M_{n;i,W,\tau} := \mathrm{Eval}_{i,W,\tau}(F_n).$$

The sequence of barcode profiles is not by itself a directed system: actual profile morphisms and composition laws must be declared, unless a registered τ-split or exact-admissible theorem supplies them.

Definition 1.108 (Tower comparison artifact)**.** *A tower comparison artifact contains:*

(i) an actual repaired profile system and its transition morphisms;

(ii) a source profile $\mathrm{ColimProfile}_{i,W,\tau}(F_\bullet)$*;*

(iii) an apex profile $\mathrm{ApexProfile}_{i,W,\tau}(F_\infty)$*;*

(iv) an actual morphism

$$\phi_{i,W,\tau} : \mathrm{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \mathrm{ApexProfile}_{i,W,\tau}(F_\infty);$$

(v) *kernel/cokernel typing, terminal-tameness evidence, and immutable artifact references.*

Barcode multisets or bottleneck matchings without actual morphisms are not a tower comparison artifact.

Remark 1.109 (No automatic tower comparison). The comparison morphism is not supplied automatically by hard threshold deletion, a barcode matching, or the mere coexistence of a source tower and an apex.

Definition 1.110 (Typed Type IV scope status)**.** *Every run records exactly one Type IV status:*

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{certified\_zero}, \qquad \texttt{obstructed}.$$

They mean respectively:

(i) *no terminal-generic kernel/cokernel Type IV diagnostic belongs to the claim; a direct or inverse tower may still be used by a finite-to-infinite, Roos, stable-image, or apex-agreement lane;*

(ii) *a terminal-generic Type IV diagnostic is invoked, but its artifact or another required terminal-generic input is absent or ill-typed;*

(iii) *a valid terminal-generic diagnostic is defined and zero;*

(iv) *a valid terminal-generic diagnostic is defined and nonzero.*

A missing required Type IV artifact is `undefined`*, never a numerical zero and never* `not_invoked`*. The latter status is available only when the scope policy excludes the terminal-generic comparison diagnostic itself; it does not exclude every use of tower semantics.*

Definition 1.111 (Collapse diagnostics notation contract)**.** *For a valid, terminal-tame comparison artifact, define*

$$\mu_{i,W,\tau} := \operatorname{tgdim}_W(\ker \phi_{i,W,\tau}), \qquad \nu_{i,W,\tau} := \operatorname{tgdim}_W(\operatorname{coker} \phi_{i,W,\tau}).$$

On a right-unbounded window, tgdim_W *is the eventual generic dimension and counts defect bars cofinal to* $+\infty$*. On a bounded window* $W = [u, u')$*, for a terminal-tame defect profile* D*,*

$$\operatorname{tgdim}_W(D) := \lim_{t \uparrow u'} \dim_k D(t),$$

so it counts defect bars cofinal to the right endpoint and is not replaced by the dimension after zero extension to $+\infty$*.*

For one fixed declared pair (W, τ) *and a declared finite monitored degree set* I_{mon}*, define*

$$\mu_{\text{Collapse}}(W, \tau) := \sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau}, \qquad \nu_{\text{Collapse}}(W, \tau) := \sum_{i \in I_{\text{mon}}} \nu_{i,W,\tau}.$$

The pair (W, τ) *and* I_{mon} *are part of the diagnostic scope. Diagnostics from different windows or thresholds are not silently summed. An infinite monitored degree set requires a separately proved aggregation or summability theorem.*

The pre-v20 Latin-letter cokernel notation is non-normative and may appear only inside historical migration metadata; it is never used as a v21 mathematical symbol.

Remark 1.112 (Type IV remains Core, but zero is typed). For a fixed declared scope, the shorthand equality

$$(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$$

means

$$(\mu_{\mathrm{Collapse}}(W, \tau), \nu_{\mathrm{Collapse}}(W, \tau)) = (0, 0)$$

and may be used only under status `certified_zero`. It means absence of monitored terminal-generic defect relative to the declared artifact. It does not imply full isomorphism on finite interior bars.

Remark 1.113 (Object-distance control does not automatically control defect profiles). A bottleneck bound between the source and target objects of two comparison morphisms does not, by itself, produce bottleneck bounds between their kernels or cokernels. Metric transport of a detector on `kernel_defect` or `cokernel_defect` therefore requires actual comparison morphisms between the defect profiles, or a separately proved kernel/cokernel stability theorem with the required commutative data. Object proximity cannot manufacture Type IV or transient-defect evidence.

1.22 Manifest consequences

Definition 1.114 (Collapse policy manifest entry)**.** *A collapse-policy entry contains:*

(i) *immutable run, object, profile, comparison, theorem, and artifact identifiers;*

(ii) *the threshold τ, monitored degrees, window, and window type;*

(iii) *the evaluation order*

$$\mathbf{P}_i \longrightarrow \mathsf{W}_W \mathbf{P}_i \longrightarrow T_\tau^{\mathrm{bar}}(\mathsf{W}_W \mathbf{P}_i);$$

(iv) *the collapse status*

$$\texttt{barcode\_level}, \qquad \texttt{tau\_split}, \qquad \texttt{exact\_admissible};$$

(v) *exactly one comparison mode for every consumed comparison;*

(vi) *the repaired evaluation records;*

(vii) *for every finite-barcode theorem invocation, a finiteness witness or bounded audit-range record identifying the exact finite profiles consumed by the proof;*

(viii) *for each metric discrepancy transported through hard deletion, the complete two-sided threshold-gap comparison record;*

(ix) *for every exact comparison, the actual equality or profile-isomorphism proof and immutable artifact;*

(x) *for every theorem-controlled nonmetric comparison, the named theorem, transported predicate, direction, assumptions, downward-closed supported conclusion-strength set, artifacts, and prohibited stronger readings;*

(xi) *for every not-compared entry, the positive scope-exclusion record;*

(xii) *for every metric transport of a repaired-zero predicate, the post-threshold zero-separation record, including $2\varepsilon \le \tau$ or the stricter operational reserve;*

(xiii) *for every detector-derived zero statement, the complete detector certificate and, when transported, the metric-safe detector-zero transport record;*

(xiv) *the metric ledger,* MetricUseID*, no-double-counting record, bottleneck scalarization, scalarized budget, and actual discrepancy upper-bound evidence whenever metric mode is invoked;*

(xv) *crop and window-edge records required to reach the actual windowed pre-threshold comparison;*

(xvi) *any τ-window-adapted commutation certificate, including explicit verification of globally retained infinite bars on bounded windows;*

(xvii) *the endpoint-discipline record and the discrepancy level for every consumed comparison;*

(xviii) *the P2DG input, readiness, variance, loss, and artifact records when the Core-P-to-Core-DG handoff is invoked;*

(xix) *representative and eligibility records when invoked;*

(xx) *Type IV status, monitored scope, terminal-generic input, and actual comparison artifact when invoked;*

(xxi) *all non-scalar obligations, including detector applicability, completeness provenance, profile stage, and artifact identity;*

(xxii) *canonical Main statement identities, Appendix proof locators, migration records, repair ancestry, and Claim Register references.*

Definition 1.115 (Metric, zero-transport, and crop repair record)**.** *An Appendix A repair record contains:*

(i) *the deprecated global non-expansiveness claim being avoided;*

(ii) *the replacement theorem Theorem 1.37;*

(iii) *the counterexample Example 1.25;*

(iv) *the protected pre-threshold profiles and their two-sided threshold-gap comparison record;*

(v) *the metric ledger,* MetricUseID*, no-double-counting declaration, scalarization, and actual discrepancy upper-bound evidence;*

(vi) *every crop or window-edge component required to obtain the windowed profile comparison;*

(vii) *if crop–collapse commutation is used, the exact criterion of Proposition 1.65;*

(viii) *explicit confirmation that globally retained infinite bars were included in that criterion;*

(ix) *every post-threshold zero-separation record consumed by the proof;*

(x) *every detector-zero transport record consumed by the proof;*

(xi) *explicit confirmation that no raw pointwise detector rank, birth or germ set, coverage witness, sample count, or certificate-size stability was inferred from bottleneck proximity;*

(xii) *Type IV scope and actual artifact references when tower semantics are invoked;*

(xiii) *confirmation that canonical ν-notation is used and the pre-v20 Latin-letter cokernel notation does not occur in normative mathematics.*

Remark 1.116 (Audit consequence). A proof object that merely says "apply T_τ" is incomplete. It must identify whether T_τ means barcode-level hard threshold deletion, a τ-split functorial model, or an exact-admissible regime.

A metric discrepancy transported through hard deletion must cite the two-sided scalarized threshold-gap record rather than global non-expansiveness. A transported repaired-zero conclusion must additionally cite a post-threshold zero-separation record. Threshold-gap stability alone bounds distance; it does not identify zero and nonzero profiles.

A separately proved exact comparison is stored in exact mode and is not misrepresented as a positive-radius robustness certificate. Crop–collapse commutation must verify all globally retained bars, including infinite bars clipped by bounded windows. Type IV use requires an actual comparison morphism. Detector use requires a complete detector certificate at the exact declared profile stage.

Manifest completeness is necessary for certification but is not sufficient for mathematical truth. A complete manifest may faithfully record a failed comparison, a pending detector, or an obstructed tower.

1.23 Operating summary

For each monitored (F, i, W, τ), the Foundation pipeline is

$$F \longmapsto \mathbf{P}_i(F) \longmapsto B_{i,W}(F) = \mathsf{W}_W \mathbf{P}_i(F) \longmapsto T_\tau^{\mathrm{bar}}(B_{i,W}(F)) = \mathrm{Eval}_{i,W,\tau}(F).$$

Threshold clearance is measured after windowing and before deletion. For a metric comparison, the repaired-zero chain is

$$\mathfrak{d}_{\mathrm{met}} \xrightarrow{\mathrm{val}_{\mathrm{B}}} \varepsilon_{\mathrm{B}}, \qquad d_B(B, C) \le \varepsilon_{\mathrm{B}}, \qquad \varepsilon_{\mathrm{B}} < \mathrm{gap}_\tau(B, C),$$

hence

$$d_B(T_\tau^{\mathrm{bar}}(B), T_\tau^{\mathrm{bar}}(C)) \le \varepsilon_{\mathrm{B}},$$

and, under

$$2\varepsilon_{\mathrm{B}} \le \tau,$$
$$T_\tau^{\mathrm{bar}}(B) = 0 \iff T_\tau^{\mathrm{bar}}(C) = 0.$$

The mathematically sharp zero-separation boundary is $2\varepsilon_{\mathrm{B}} \le \tau$. The v21 operational contract uses the strict reserve $2\varepsilon_{\mathrm{B}} < \tau$.

A complete detector may establish the source zero predicate. Appendix A transports that already certified predicate through the displayed metric chain. It does not transport raw detector ranks, sample values, birth or germ sets, coverage, certificate size, or completeness.

The default order is window first and threshold second. Source-level perturbations require crop and window-edge control before they become bounds on the actual pre-threshold profiles. Filtered or derived arguments require declared representatives. Tower and transient-defect arguments require actual comparison morphisms. Object-level bottleneck proximity does not automatically control kernel or cokernel defect profiles. Exactness requires a declared exact-admissible regime.

1.24 Foundation Core-P package and local closure

Definition 1.117 (Appendix A v21 proof-package closure record)**.** *The Appendix A proof-package closure record binds, without creating a new competing theorem identity, the following canonical statements and local records in the constructible one-parameter regime under the standard infinite-bar convention.*

(i) *constructible modules admit interval decompositions;*

(ii) *hard deletion removes exactly finite bars of length $\leq \tau$, preserves infinite bars, is idempotent, threshold-compositional, and threshold-monotone;*

(iii) *positive-threshold short-bar objects are not extension-closed in general;*

(iv) *hard deletion is not globally non-expansive;*

(v) *the sharp two-sided threshold-gap theorem holds with closed forbidden band $[\tau - \varepsilon, \tau + \varepsilon]$;*

(vi) $\mathrm{gap}_\tau = \mathrm{clear}_\tau$*, with no factor* $1/2$*;*

(vii) *scalarized budgets are usable only with actual discrepancy upper-bound evidence and a unique* MetricUseID*;*

(viii) *every repaired profile is τ-reduced;*

(ix) *the bottleneck distance to zero is given exactly by Lemma 1.43;*

(x) *finite τ-reduced profiles have the sharp zero-separation boundary $2\varepsilon \leq \tau$;*

(xi) *metric-safe repaired-zero transport follows from threshold-gap stability plus zero separation;*

(xii) *a complete detector certificate may be composed with metric zero transport, but only its certified zero predicate is transported;*

(xiii) *raw pointwise detector ranks are not bottleneck-stable in general;*

(xiv) *clearance is measured after windowing and before deletion;*

(xv) *crop and hard deletion do not commute in general;*

(xvi) *they commute exactly under the all-retained-bars criterion, including globally infinite bars clipped by bounded windows;*

(xvii) *every comparison uses exactly one of the four canonical v21 modes;*

(xviii) *the four comparison modes are mutually nonpromotional without a separate conversion theorem;*

(xix) *discrepancy level is part of the record type;*

(xx) *barcode or persistence discrepancy does not automatically lift to filtered-chain, finite-diagram, or derived discrepancy;*

(xxi) *a finite critical grid is not automatically complete;*

(xxii) *a P2DG readiness record prepares, but does not prove, finite-diagram reconstruction;*

(xxiii) *functoriality and exactness are conditional on declared split or exact-admissible regimes;*

(xxiv) *representatives are window-relative declared representatives, not a global collapse functor;*

(xxv) *Type IV requires actual morphisms, terminal tameness, finite or otherwise justified aggregation scope, and typed status;*

(xxvi) *object-distance control does not automatically control kernel or cokernel defect profiles;*

(xxvii) *manifest completeness does not replace mathematical validity;*

(xxviii) Main statement identity, Appendix proof location, migration status, and artifact identity remain separate fields.

Remark 1.118 (Source map for the closure record)*.* Items (i)–(ii) follow from Theorem 1.9, Lemma 1.16, and Propositions 1.23–1.24. Item (iii) is Proposition 1.18. Item (iv) is Example 1.25. Item (v) is Theorem 1.37 and Example 1.38. Items (vi)–(vii) are Definitions 1.29, 1.30, and 1.31. Items (viii)–(ix) are Definition 1.41 and Lemma 1.43. Item (x) is Theorem 1.44. Item (xi) is Theorem 1.47 and Corollary 1.48. Item (xii) is Corollary 1.51. Item (xiii) is Example 1.53 and Proposition 1.54. Item (xiv) follows from Definitions 1.33, 1.60, and 1.61. Items (xv)–(xvi) are Proposition 1.59, Example 1.63, and Proposition 1.65. Items (xvii)–(xviii) are Definition 1.67 and Theorem 1.69. Items (xix)–(xx) are Definition 1.71 and Theorem 1.74. Items (xxi)–(xxii) are Definitions 1.76, 1.77, 1.78, and Theorem 1.79. Item (xxiii) is Proposition 1.94 and Definition 1.95. Item (xxiv) is Definition 1.98. Item (xxv) is Definitions 1.108, 1.110, and 1.111. Item (xxvi) is Remark 1.113. Item (xxvii) is Remark 1.116. Item (xxviii) is bound to the ownership and manifest records of Subsections 1.1 and 1.22.

This source map is operational and provenance-only; it does not register a second mathematical theorem for any Main-owned statement.

1.25 P1 local gates and G21 contribution

Definition 1.119 (Appendix A P1 completion record)**.** *The Appendix A P1 completion record contains:*

(i) regression of every mandatory v20 metric, crop, repaired-zero, and non-claim statement at its unchanged strength;

(ii) a one-to-one canonical Main statement owner and Appendix proof locator map;

(iii) four-mode comparison-schema conformance;

(iv) endpoint-discipline, discrepancy-level, and MetricUseID *conformance;*

(v) a complete P2DG input and readiness schema;

(vi) replayable counterexamples for the Serre, global non-expansiveness, crop–collapse, detector-rank, and level-change boundaries;

(vii) manifest serialization readiness for Appendix G;

(viii) absence of conflicting theorem identities;

(ix) artifact, build, and review status.

This is a construction-stage completion record. It is not the final v21 Claim Register or release gate verdict.

Proposition 1.120 (Appendix A contribution ceiling)**.** *A completed Appendix A may contribute evidence to:*

G21-2, G21-4, G21-8, G21-9.

It cannot by itself pass any of these gates. In particular:

(i) G21-2 additionally requires the complete Core-P regression package;

(ii) G21-4 additionally requires a proved nontrivial P2DG Interface and finite-diagram reconstruction downstream;

(iii) G21-8 additionally requires global namespace and ownership closure;

(iv) G21-9 additionally requires final registers, source bindings, manifests, hashes, replay, render, and package verification.

Proof. Appendix A supplies only the persistence-metric and input-record components of the listed gates. Each gate quantifies over downstream artifacts or the complete release package, none of which is constructed by this appendix alone. ☐

1.26 Explicit non-claims

The following are not asserted.

(i) a P2DG readiness record as a finite-diagram realization, reconstruction, zero-reflection, or descent theorem;

(ii) silent retyping of a barcode or persistence discrepancy as a filtered-chain, finite-diagram, or derived discrepancy;

(iii) representation of theorem-controlled nonmetric evidence as exact or metric-gap-certified without a separate conversion theorem;

(iv) use of `not_compared` to discharge an invoked comparison;

(v) endpoint completeness, reconstruction faithfulness, or zero reflection from finiteness of a critical grid alone;

(vi) construction in this appendix of a general finite diagram, cone, Morita reconstruction, Alexandrov realization, descent datum, or global object;

(vii) a new canonical theorem identity merely because a Main theorem has an expanded proof in this appendix;

(viii) passage of G21-2, G21-4, G21-8, or G21-9 by Appendix A alone;

(ix) global non-expansiveness of hard positive-threshold deletion under the standard bottleneck metric;

(x) a positive-threshold Serre subcategory in general;

(xi) a global exact endofunctor, abelian reflector, or Serre localization;

(xii) unconditional crop–collapse commutation;

(xiii) sufficiency of a finite-only commutation criterion;

(xiv) source-to-window metric transport without a crop theorem, clearance argument, or crop budget;

(xv) direct comparison of an unscalarized quantale value with real clearance;

(xvi) scalarization without evidence that it bounds the actual consumed bottleneck discrepancy;

(xvii) duplicated use of one defect contribution as additional metric reserve without a disjointness theorem;

(xviii) numerical repair of a missing non-scalar obligation;

(xix) one-sided threshold-gap certification;

(xx) replacement of a two-sided comparison record by a single-profile readiness value;

(xxi) gap safety for a finite bar of length exactly τ;

(xxii) a sharp 2ε-band for the inherited two-sided theorem;

(xxiii) $\mathrm{gap}_\tau = \frac{1}{2}\,\mathrm{clear}_\tau$;

(xxiv) a global strict filtered lift;

(xxv) an automatic tower comparison map;

(xxvi) a kernel or cokernel defined merely by a barcode matching;

(xxvii) identifying undefined Type IV input with zero;

(xxviii) fabricating numerical zero for a no-tower claim;

(xxix) full comparison isomorphism from terminal-generic zero alone;

(xxx) an aggregate Type IV value over an undeclared infinite scope;

(xxxi) identifying a bounded-window cofinal bar with a global infinite bar;

(xxxii) a well-typed Ext clause without an eligible representative;

(xxxiii) identifying manifest completeness with proof;

(xxxiv) replacing a Core obligation by a Search artifact, advisory score, or AI ranking;

(xxxv) repaired-zero equivalence from threshold-gap stability alone without post-threshold zero separation;

(xxxvi) metric zero transport when $2\varepsilon > \tau$;

(xxxvii) bottleneck stability of raw pointwise detector ranks, birth sets, germ sets, sample counts, coverage witnesses, or certificate sizes;

(xxxviii) promotion of detector soundness to completeness by metric proximity or repeated zero observations;

(xxxix) silent change of detector profile stage under metric transport;

(xl) automatic metric control of kernel or cokernel defect profiles from object-level bottleneck bounds;

(xli) proof of a Chapter 3 finite detector certificate merely because its already certified zero predicate can be transported metrically;

(xlii) an exact comparison as a positive-radius robustness certificate;

(xliii) a metric-gap certificate as an actual profile isomorphism;

(xliv) any higher Ext, arithmetic, PDE, geometric, cryptographic, or external Return theorem.

1.27 Provenance, theorem identity, and migration status

Interval decomposition, barcode uniqueness, and the bottleneck/interleaving isometry are standard one-parameter persistence results at their displayed hypotheses.

The v19 contribution inherited by the immutable lower baseline is the metric and crop repair:

(i) rejection of positive-threshold Serre localization in general;

(ii) rejection of global non-expansiveness;

(iii) replacement by the sharp two-sided threshold-gap theorem;

(iv) the all-retained-bars crop criterion including infinite bars clipped by bounded windows.

The v20 conservative extension inherited without change is:

(i) the exact distance-to-zero formula;

(ii) sharp zero separation in the τ-reduced image;

(iii) metric-safe repaired-zero transport;

(iv) detector-zero composition;

(v) the explicit pointwise-rank instability boundary;

(vi) actual-morphism and terminal-tameness guard-rails for Type IV input.

The genuinely new v21 material in Appendix A is architectural:

(i) the four-mode comparison calculus, including `theorem_controlled_nonmetric`;

(ii) comparison-mode separation and explicit conversion discipline;

(iii) typed discrepancy-level separation;

(iv) the no-automatic-upward-transport theorem;

(v) the finite critical-grid and P2DG input contract;

(vi) the P2DG readiness and supported-handoff boundary;

(vii) canonical Main statement ownership versus Appendix proof ownership;

(viii) MetricUseID and no-double-counting integration;

(ix) P1 and G21 contribution ceilings.

None of these additions strengthens the inherited bottleneck, threshold-gap, zero-separation, or crop constants.

1.27.0.1 Canonical statement ownership. The v21 Main is the canonical statement owner for Main Theorems 2.11, 2.46, 2.53, 2.56, 2.69, 2.74, and 2.79 and the associated Main propositions and records. The corresponding theorem environments in this appendix are expanded proof records, counterexample owners, or schema expansions under the same canonical mathematical identities. They are not independent competing theorem registrations.

1.27.0.2 Appendix-local closure authority. Definition 1.117 is an operational and provenance-only package-closure record; it creates no independent theorem identity. Theorem 1.69, Theorem 1.74, and Theorem 1.79 are proof expansions of their Main owners. Appendix-local definitions and propositions may be registered separately only where they add genuinely distinct record, regression, implementation, or counterexample content.

1.27.0.3 Migration classification. Migration is registered per component row. A frozen v20 mathematical statement retained at the same strength is `inherited`; a distinct v21 comparison, discrepancy, ownership, or P2DG successor with greater declared architectural strength is `strengthened`; and `repaired` is used only for an exact predecessor reading identified by the Migration Matrix as unavailable for v21 theorem use. The valid v20 three-mode schema remains directly invocable on its frozen v20 route and is not retroactively declared erroneous merely because v21 adds a fourth mode. Historical finite-only crop and notation repairs remain repair ancestry unless a specific v20 row requires a new v21 repair. `relocated` is a disposition modifier and does not change theorem strength. No single row carries more than one semantic migration status.

1.27.0.4 Status index. *Standard persistence input:* Theorem 1.9 and Remark 1.17.

Inherited metric-repair package: Proposition 1.18, Corollary 1.19, Example 1.25, Theorem 1.37, Example 1.38, and Corollary 1.40.

Inherited repaired-zero transport package: Theorem 1.44, Proposition 1.46, Theorem 1.47, Corollaries 1.48, 1.51, and 1.90.

New v21 comparison and Interface guard-rails: Definition 1.67, Definition 1.68, Theorem 1.69, Definition 1.71, Theorem 1.74, Definitions 1.76, 1.77, and 1.78, and Theorem 1.79.

Foundation closure and release contribution: Definition 1.117, Definition 1.119, and Proposition 1.120.

Operational record ownership: Definitions 1.12, 1.35, 1.49, 1.50, 1.60, 1.61, 1.31, 1.83, 1.84, 1.88, 1.105, 1.114, and 1.115.

Explicit exclusions: Subsection 1.26.

1.27.0.5 Final governance boundary. This source may reach

`ready_for_final_claim_extraction`

after mathematical, cross-Appendix, build, render, and artifact review. Canonical ClaimID assignment, final migration closure, source-manifest binding, release-gate evaluation, and package release remain later governance acts.

2 Appendix B: Finite Detector Calculus, Certificate Complexity, and Filtered Representatives

2.1 Scope, ownership, migration, and dependence on the repaired Core

2.1.0.1 Normative status and theorem ownership. Appendix B is the expanded-proof, certificate-model, counterexample, record-schema, and regression owner for the Core-P finite-detector and certificate-complexity statements canonically owned by Main Chapter 3 of AK-HPDST v21.0.0, the *Finite Globalization, Unified Bridge Calculus, and Certificate Complexity*

Release. The Main carries the canonical statement identity whenever a result is a Main theorem. This appendix may reproduce such a statement only as an expanded proof under that identity; it does not create a competing theorem UID. Appendix-local lemmas, model-relative lower bounds, benchmark theorems, counterexamples, and implementation corollaries receive local identities only when they are not duplicates of a Main theorem.

The immutable v20 Appendix B remains directly invocable from its frozen source artifact at its original strength. The v21 migration is componentwise:

(i) the birth, right-germ, finite-stencil, certified-coverage, family-mesh, robust-sandwich, finite-realization, representative, and right-unbounded no-go results are `inherited`;

(ii) the certificate model $\mathfrak{M} = (\mathsf{I}, \mathsf{Q}, \mathsf{E}, \mathsf{V})$, vector-valued cost, Pareto frontier, finite-probe schema, information-model lower bounds, finite composition calculus, and certificate-to-representative handoff are `strengthened` successor material with new v21 identities;

(iii) the scalar v20 MinCert theorem is `inherited` exactly as the query-coordinate projection in the exact nonadaptive stencil model; the vector-valued Pareto object is a distinct `strengthened` successor, and a model-independent scalar reading is prohibited rather than retroactively attributed to v20;

(iv) diagram and descent probes are Interface-stage inputs only. General finite generators, cone detection, Morita reconstruction, and effective descent are `relocated` to Appendix DG and are not proved here.

Here `relocated` is a disposition modifier, not a semantic migration status.

The canonical project acronym is AK-HPDST. The legacy transposed acronym AK-HDPST may occur in historical artifacts, but it has no normative use in this appendix.

The Core does not treat positive-threshold deletion as a global exact Serre reflector, a globally non-expansive endofunctor, or a strict functor on all persistence morphisms. Its primary persistence data are the declared windowed pre-threshold profile

$$B_{i,W}(F) := \mathsf{W}_W \mathbf{P}_i(F)$$

and the repaired barcode-level evaluation

$$\mathrm{Eval}_{i,W,\tau}(F) := T_\tau^{\mathrm{bar}}(B_{i,W}(F)).$$

Appendix B has three coupled but non-interchangeable responsibilities.

(i) It proves when finitely many structure maps of a declared pre-threshold profile give a sound or complete certificate for the exact zero/nonzero predicate of the repaired profile.

(ii) It fixes the information, query, encoding, and verifier models under which finite proof size and lower bounds are mathematically meaningful.

(iii) It specifies when repaired profiles may be handed to a filtered representative construction for filtered, derived, higher-Ext, or tower-level use, without pretending that the detector itself creates that representative.

The central rules are:

> A finite observation is not a finite proof certificate until its source stage, endpoint convention, soundness, completeness or coverage, family membership, query transcript, encoding, verifier semantics, cost vector, and immutable artifacts are certified.

Certificate complexity is model-relative. Query count, encoded bit length, verification cost, and persistent replay footprint are separate coordinates; no scalar weighting is canonical without a declared model-specific utility.

A filtered representative does not define collapse and does not retroactively supply a detector certificate. It realizes a declared repaired persistence profile, at a declared equivalence strength, for a declared monitored scope.

Accordingly, Appendix B separates:

(i) windowed pre-threshold and repaired post-threshold profiles;

(ii) the six detector stages `windowed_prethreshold`, `repaired_postthreshold`, `kernel_defect`, `cokernel_defect`, `diagram_probe`, and `descent_coherence`;

(iii) finite-stencil soundness from detector-zero completeness;

(iv) left-closed birth detection from decoration-invariant right-germ detection;

(v) externally certified coverage from family-mesh completeness;

(vi) object-specific certificate records from family-relative schemes;

(vii) information, query, encoding, and verifier models;

(viii) adaptive from nonadaptive access, deterministic from randomized semantics, and exact from approximate responses;

(ix) the four cost coordinates (q, b, t, a) and their Pareto order;

(x) upper bounds from lower bounds and sharpness;

(xi) model-relative MinCert, the inherited scalar query projection, and the right-unbounded no-go regime;

(xii) local finite certificate composition from global certificate complexity;

(xiii) metric transport of a certified zero predicate from stability of raw detector ranks;

(xiv) object-level realization of a finite barcode family;

(xv) monitored profile equivalence, levelwise filtered quasi-isomorphism, and realization-compatible equivalence;

(xvi) actual representative morphisms and tower-comparison artifacts;

(xvii) detector preservation at an exact source stage from mere representative admissibility.

Remark 2.1 (Conservative-evolution boundary). This appendix does not restore a global exact collapse functor, a Serre localization, global non-expansiveness, automatic tower comparison, or a reverse higher-Ext-to-persistence bridge. It does not assert pointwise detector-rank stability under bottleneck perturbation.

The finite-probe schema is conditional: a diagram or descent probe is complete only after Appendix DG or another named Core-DG theorem supplies the actual object and a zero-reflection or decision theorem. Merely changing the stage token does not transport a barcode completeness theorem.

The object-level barcode realization theorem concerns finite barcode profile families only. It does not provide functoriality on arbitrary morphisms, preserve additional geometric structure, establish higher eligibility, construct a Core-DG object, or prove an external Return Theorem.

Remark 2.2 (Dependency and handoff direction). Appendix B consumes the metric, window, threshold, comparison-mode, discrepancy-level, and P2DG-input boundaries of Appendix A. Main Chapter 3 is the canonical statement owner for the v21 detector and certificate-complexity results whose expanded proofs are recorded here. The representative sublayer of this appendix may turn a candidate into an actual representative by the finite-realization and compatibility theorems below. Appendix C consumes only a certified representative record and separately proves P2DG and eligible higher-edge data. Appendix D may consume representative tower data and defect-stage detector records, but an actual tower comparison morphism and kernel/cokernel profiles are never generated by Appendix B alone. Appendix DG owns arbitrary finite categories, general projective generators, cone detection, and object-level descent.

2.1.0.2 Division of responsibility.

Appendix A proves metric-safe transport of an already certified repaired-zero predicate. Appendix B proves finite detector characterizations and model-relative certificate-complexity statements. Appendix C proves the actual persistence-to-diagram and higher eligible realization interfaces. Appendix D proves terminal/transient defect and finite-to-infinite tower results. Appendix F owns proof-DAG admission semantics, and Appendix G owns canonical serialization and replay. No one layer silently discharges another.

2.2 Filtered chain complexes and monitored persistence

Definition 2.3 (Finite-type filtered chain complex). *A* finite-type filtered chain complex *over a field k is a bounded chain complex $C_\bullet$ of finite-dimensional k-vector spaces, together with an increasing filtration by subcomplexes*

$$F^t C_\bullet \subseteq F^{t'} C_\bullet \qquad (t \le t'),$$

such that:

(i) the filtration is exhaustive and left-bounded;

(ii) only finitely many filtration-critical values occur;

(iii) for every homological degree i, the persistence module

$$\mathbf{P}_i(F) : t \longmapsto H_i(F^t C_\bullet)$$

belongs to $\mathsf{Pers}_k^{\mathrm{cons}}$.

The category of such objects and filtration-preserving chain maps is denoted by

$$\mathsf{FiltCh}(k)\,.$$

Definition 2.4 (Persistence functor). *For every homological degree i, define*

$$\mathbf{P}_i : \mathsf{FiltCh}(k) \longrightarrow \mathsf{Pers}_k^{\mathrm{cons}}$$

by

$$\mathbf{P}_i(F)(t) := H_i(F^t C_\bullet).$$

Definition 2.5 (Monitored scope). *A* monitored scope *is a finite set of homological degrees*

$$I_{\mathrm{mon}} \subset \mathbb{Z}.$$

All representative assertions in this appendix are relative to a declared monitored scope unless a full-degree statement is explicitly proved.

Remark 2.6 (Monitored-degree convention). Agreement in I_{mon} does not imply agreement in unmonitored degrees. A proof object must not silently upgrade a monitored representative to a full filtered-chain equivalence.

2.3 Three levels of representative equivalence

Definition 2.7 (Monitored persistence equivalence)**.** *Two filtered objects $F, G \in \mathsf{FiltCh}(k)$ are* monitored-persistence equivalent*, written*

$$F \simeq_{\mathrm{PH}, I_{\mathrm{mon}}} G,$$

if for every $i \in I_{\mathrm{mon}}$ there is an isomorphism

$$\mathbf{P}_i(F) \cong \mathbf{P}_i(G)$$

in $\mathsf{Pers}_k^{\mathrm{cons}}$.

Definition 2.8 (Levelwise filtered quasi-isomorphism)**.** *A filtration-preserving chain map*

$$f : F \longrightarrow G$$

is a levelwise filtered quasi-isomorphism *if, for every parameter t, the map of chain complexes*

$$F^t C_\bullet \longrightarrow G^t D_\bullet$$

is a quasi-isomorphism.

Such a map induces

$$\mathbf{P}_i(F) \cong \mathbf{P}_i(G)$$

for every degree i, and hence monitored-persistence equivalence for every monitored scope.

Definition 2.9 (Realization-compatible equivalence)**.** *Fix a realization functor or realization procedure*

$$\mathcal{R} : \mathsf{FiltCh}(k) \dashrightarrow D^{\mathrm{b}}(k\text{-}mod)$$

on a declared domain.

Two representatives F, G are $\mathcal{R}$-equivalent *if:*

(i) both $\mathcal{R}(F)$ and $\mathcal{R}(G)$ are defined;

(ii) an actual isomorphism

$$\mathcal{R}(F) \cong \mathcal{R}(G)$$

is supplied in $D^{\mathrm{b}}(k\text{-}mod)$;

(iii) the isomorphism is compatible with every structure consumed by the relevant Ext or bridge argument.

Proposition 2.10. *A levelwise filtered quasi-isomorphism gives monitored-persistence equivalence. The converse fails: monitored persistence alone gives neither chain-level, chain-homotopy, nor $\mathcal{R}$-equivalence.*

Proof. The first statement follows by taking homology at every filtration level and using naturality of the structure maps.

For the second statement, monitored persistence records only the declared homology persistence modules. It forgets chain-level extension data, unmonitored degrees, and any extra structure consumed by $\mathcal{R}$. Hence no stronger equivalence follows without an additional theorem or artifact. □

Remark 2.11 (Why the equivalence hierarchy matters). Levelwise filtered quasi-isomorphism is sufficient for every monitored persistence clause, but it is stronger than mere profile agreement. The weaker profile relation is therefore used only where the proof consumes no additional chain-level or derived structure.

Remark 2.12 (No automatic localization). The v21 Core-P contract does not require a global localization of $\mathsf{FiltCh}(k)$ at monitored-persistence equivalences. If a later construction uses a localized category, it must declare the class of weak equivalences, prove that the localization is suitable for the intended argument, and record the actual morphisms used there.

Definition 2.13 (Localized representative category). *When a declared workflow has fixed a class $\mathcal{W}$ of filtered quasi-isomorphisms for which the required localization exists, write*

$$\mathrm{Ho}_{\mathcal{W}}(\mathsf{FiltCh}(k))$$

for that declared localization.

This notation is optional and has no default global meaning in Appendix B.

2.4 Windowed repaired persistence profiles

Recall the fixed evaluation order from Chapter 1 and Appendix A:

$$F \longmapsto \mathbf{P}_i(F) \longmapsto \mathsf{W}_W\mathbf{P}_i(F) \longmapsto T_\tau^{\mathrm{bar}}(\mathsf{W}_W\mathbf{P}_i(F)).$$

Definition 2.14 (Windowed repaired persistence profile). *For $F \in \mathsf{FiltCh}(k)$, $i \in I_{\mathrm{mon}}$, a right-open window W, and $\tau > 0$, define*

$$\mathcal{P}_{i,W,\tau}(F) := \mathrm{Eval}_{i,W,\tau}(F) = T_\tau^{\mathrm{bar}}(\mathsf{W}_W\mathbf{P}_i(F)).$$

Definition 2.15 (Profile family over the monitored scope). *For a finite monitored scope I_{mon}, define*

$$\mathcal{P}_{W,\tau}^{I_{\mathrm{mon}}}(F) := \{\mathcal{P}_{i,W,\tau}(F)\}_{i\in I_{\mathrm{mon}}}.$$

Remark 2.16 (The profile is primary). The family $\mathcal{P}_{W,\tau}^{I_{\mathrm{mon}}}(F)$ is the primary Core object. A filtered representative is secondary evidence used only when a chain-level, derived, Ext, or tower-level construction requires it.

Definition 2.17 (Canonical four-mode comparison calculus). *Every consumed runtime profile comparison is declared as exactly one of*

`exact`, `metric_gap_certified`,
`theorem_controlled_nonmetric`, `not_compared`.

(i) `exact` *requires an actual equality or isomorphism of the exact compared source-stage profiles.*

(ii) `metric_gap_certified` *requires an artifact-backed discrepancy bound on the actual windowed pre-threshold profiles, the two-sided threshold-gap record, and every post-threshold zero-separation hypothesis consumed by the conclusion.*

(iii) `theorem_controlled_nonmetric` *names a theorem, its complete hypotheses, exact source and target, and the nonmetric conclusion licensed by that theorem. It supplies neither a bottleneck radius nor an actual isomorphism unless the named theorem explicitly does so.*

(iv) `not_compared` *is permitted only when the claim consumes no comparison inference and an explicit scope policy records that fact.*

The modes are mutually exclusive and no favorable mode may be substituted for another. In particular, an exact comparison is not automatically a positive-radius robustness certificate, and a theorem-controlled nonmetric transport cannot replace the actual profile isomorphisms required by an admissible filtered representative.

Definition 2.18 (Comparison evidence record). *A comparison evidence record stores the selected mode, exact source and target stage, theorem or metric witness, discrepancy level, complete hypotheses, conclusion strength, immutable artifacts, and prohibited stronger readings. A metric-zero transport branch references one parent metric comparison through one* `MetricUseID`*; it is not a fifth comparison mode and does not spend the metric budget twice.*

Proposition 2.19 (Comparison modes do not self-promote). *No one of the four comparison modes implies another without a separately proved conversion theorem whose hypotheses and conclusion are recorded.*

Proof. The modes carry different evidence types: an actual isomorphism, a metric bound with threshold reserve, a named nonmetric theorem, or absence of a comparison inference. None of these data is definitionally identical to the others. A conversion therefore requires additional mathematical evidence. □

2.5 Certificate models and vector-valued cost

Definition 2.20 (Certificate information model). *An information model* I *specifies all input data available before queries are issued. It records the source family, exactness or precision, promise conditions, provenance, and permitted primitive views. Primitive views may include*

`full_barcode`, `critical_value_list`, `point_evaluations`, `rank_access`,
`enriched_projective_probes`, `local_object_restrictions`, `overlap_morphisms`.

Each view is marked as explicit input, derived data, charged oracle access, or mere promise. A proof may not use object-dependent information outside I*.*

Definition 2.21 (Certificate query model). *A query model* Q *fixes the query alphabet and response type, adaptive or nonadaptive access, deterministic or randomized strategy, exact or approximate responses, stopping and aggregation rules, random seeds and error parameters when used, producer/verifier charging, cache policy, and the theorem converting approximate responses into the claimed verdict. For an object* X*, the ordered charged query–answer transcript is* $\mathrm{Tr}_{\mathsf{Q}}(X)$*.*

Definition 2.22 (Certificate encoding model). *An encoding model* E *fixes canonical representations for queries, answers, witnesses, theorem and dependency references, verifier traces, and artifacts. It declares when a content-addressed object may be referenced rather than duplicated and how the reference is charged. For a certificate* C*, define*

$$\mathrm{cost}_{\mathfrak{M}}(C) := (q(C), b(C), t(C), a(C)) \in \overline{\mathbb{N}}^4, \qquad \overline{\mathbb{N}} := \mathbb{N} \cup \{+\infty\},$$

where q *counts logical source probes,* b *encoded bits,* t *verifier cost in the declared computational model, and* a *persistent artifact footprint required for independent replay. An arbitrary exact real value without a finite code has* $b = +\infty$*.*

Definition 2.23 (Coordinatewise certificate-cost order). *For* $c, c' \in \overline{\mathbb{N}}^4$*, write*

$$c \preceq_{\mathrm{cost}} c'$$

when every coordinate of c *is at most the corresponding coordinate of* c'*. This is a partial order. No scalar weighting of the coordinates is canonical unless the certificate model explicitly supplies one.*

Definition 2.24 (Certificate verifier model). *A verifier model* V *declares the algorithms, arithmetic, proof rules, trusted theorem registry, artifact access, resource bounds, and acceptance semantics available to the checker. It separates producer work from verifier work, oracle answers from recomputation, trusted dependencies from unverified claims, deterministic from probabilistic acceptance, and mathematical rejection from undefined verification caused by missing or stale evidence.*

Definition 2.25 (Certificate model). *A certificate model is the typed quadruple*

$$\mathfrak{M} = (\mathsf{I}, \mathsf{Q}, \mathsf{E}, \mathsf{V}).$$

Certificate costs are compared only inside one common model or through a proved model-translation theorem that charges every added information item, query, encoding, verification step, and artifact.

Definition 2.26 (Family-relative certificate scheme). *Fix a family* $\mathcal{F}$*, a decision problem* Θ*, and a certificate model* $\mathfrak{M}$*. A family-relative certificate scheme* Π *consists of a producer or acquisition strategy, an object-specific certificate* C_X *for every* $X \in \mathcal{F}$*, one declared verifier, and a proof that acceptance is sound and complete for* Θ *throughout the family. In the default deterministic exact regime, its family cost is the coordinatewise worst-case vector*

$$\mathrm{cost}^{\mathcal{F}}_{\mathfrak{M}}(\Pi) := \left(\sup_{X\in\mathcal{F}} q(C_X), \sup_{X\in\mathcal{F}} b(C_X), \sup_{X\in\mathcal{F}} t(C_X), \sup_{X\in\mathcal{F}} a(C_X) \right).$$

A randomized or average-case aggregation defines a different model and is not compared directly with this default rule.

Definition 2.27 (Oracle-closed model). *An exact certificate model is oracle-closed for* $\mathcal{F}$ *when every object-dependent fact available to the producer or verifier is either fixed in* I *or obtained through a charged response in* $\mathrm{Tr}_{\mathsf{Q}}(X)$*. No hidden full-object access, omniscient prover message, or uncharged object-dependent advice is permitted.*

Policy 2.28 (Exact-zero use of randomized or approximate access). *A randomized or approximate procedure is not consumed as an exact zero certificate unless its verifier semantics and a proved theorem establish the exact claim at the declared error strength. A confidence value, finite-precision zero, or empirical absence of a witness is not silently retyped as deterministic exact vanishing.*

2.6 Detector profile stages, Interface probes, and effective domains

Definition 2.29 (Detector profile stage). *Every detector record declares exactly one source stage:*

$$\texttt{windowed\_prethreshold},$$
$$\texttt{repaired\_postthreshold},$$
$$\texttt{kernel\_defect}, \quad \texttt{cokernel\_defect},$$
$$\texttt{diagram\_probe}, \quad \texttt{descent\_coherence}.$$

The standard stencil, birth, germ, and mesh theorems below use only the stage

$$\texttt{windowed\_prethreshold}.$$

A result at one stage is not reused at another without an explicit stage-transport theorem or an actual source-stage isomorphism.

The kernel and cokernel stages are available only after Appendix D supplies an actual comparison morphism and well-typed defect profiles. The `diagram_probe` *stage is available only after*

Appendix C or Appendix DG supplies an actual finite diagram or bounded derived object. The `descent_coherence` *stage is available only after Appendix DG supplies actual local objects, overlap morphisms, and coherence cells. The last two are Interface stages and inherit no barcode completeness theorem from their names.*

Definition 2.30 (Typed finite probe family)**.** *For a declared family $\mathcal{F}$ and predicate Θ, a typed finite probe family is a finite list*

$$\mathfrak{P} = (p_1, \ldots, p_m)$$

with one common source stage, response types, actual evaluation procedures, and a theorem candidate relating the response tuple to $\Theta(X)$. At stage `diagram_probe`*, examples include declared* $\mathrm{Hom}(G_j, X)$ *or* $\mathrm{RHom}(G_j, X)$ *probes. At stage* `descent_coherence`*, examples include prescribed cocycle squares, triple-overlap cells, and effectivity comparison maps. Merely listing the probes proves neither completeness nor effectivity.*

Theorem 2.31 (Expanded proof of Main Theorem 3.52: certified finite-probe upper-bound schema)**.** *Let $\mathfrak{P} = (p_1, \ldots, p_m)$ be a typed finite probe family for $(\mathcal{F}, \Theta)$. Suppose a proved decision theorem establishes*

$$\Theta(X) = 0 \iff p_j(X) = 0 \text{ for all } j = 1, \ldots, m$$

for every $X \in \mathcal{F}$, with all source-stage and family hypotheses verified. Querying every probe defines a nonadaptive exact family-relative certificate scheme with worst-case query coordinate at most m. Its (b, t, a)-coordinates are determined separately by E and V.

Proof. The fixed producer queries all m probes and the verifier applies the assumed decision theorem. Its two implications give soundness and completeness uniformly over the family. Finiteness controls only the query coordinate; the remaining costs depend on their declared encodings and verification rules. □

Remark 2.32 (No automatic generator or descent completeness)*.* Theorem 2.31 is conditional. It does not assert that a proposed projective object is a generator, that $\mathrm{RHom}(G, -)$ reflects zero, that pairwise overlap tests imply a cocycle, or that cocycle verification implies effective descent. Those are Appendix DG theorems and hypotheses.

Definition 2.33 (Effective detector domain)**.** *Let*

$$W = [a, b), \qquad b \in \mathbb{R} \cup \{+\infty\}, \qquad \tau > 0.$$

Define

$$W_\tau := \{t \in W \mid t + \tau \in W\}.$$

Thus

$$W_\tau = \begin{cases} [a, b - \tau), & b < +\infty \text{ and } b - a > \tau, \\ \varnothing, & b < +\infty \text{ and } b - a \le \tau, \\ [a, +\infty), & b = +\infty. \end{cases}$$

Only points of W_τ may be sampled by the exact τ-stencil detector.

Remark 2.34 (Window-relative endpoint and cofinal policy)*.* Every bar in the persistence detector calculus is a bar of the already windowed profile. A bar reaching the right endpoint of a bounded window is window-cofinal but finite for threshold-length purposes. Its length is the clipped window-relative length and it is included in every coverage, excess, and boundary calculation. A bar is genuinely infinite only on a right-unbounded window.

2.7 Exact finite stencil and active-start regions

Definition 2.35 (Exact finite stencil detector)**.** *Let B be a constructible persistence module on W, let $T \subseteq W_\tau$ be finite, and fix the left-closed right-open interval registration convention. Define*

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) := \bigoplus_{t\in T} \operatorname{im}\big(B(t) \longrightarrow B(t+\tau)\big).$$

Its value is zero exactly when every displayed image is zero. The direct sum is finite and retains vector-space multiplicity.

Definition 2.36 (Retained bars and active-start region)**.** *Let $\bar{\ }(B)$ be the barcode of a declared windowed profile and define, with multiplicity,*

$$\operatorname{Ret}_{\tau,W}(B) := \{I \in \bar{\ }(B) \mid \ell_W(I) > \tau\}.$$

For an interval summand I, define its τ-active-start region by

$$\operatorname{Act}_\tau(I) := \{t \in W_\tau \mid t, t+\tau \in I\}.$$

If $I = [p, q)$ in the left-closed right-open convention, then

$$\operatorname{Act}_\tau(I) = \begin{cases} [p, q-\tau), & q < +\infty \text{ and } q - p > \tau, \\ [p, +\infty), & q = +\infty, \\ \varnothing, & q - p \le \tau. \end{cases}$$

Thus $\operatorname{Act}_\tau(I) \neq \varnothing$ exactly when I survives T^{bar}_τ.

Proposition 2.37 (Expanded proof of Main Proposition 3.58: stencil-image decomposition)**.** *Under Definition 2.35,*

$$\dim_k \operatorname{im}\big(B(t) \to B(t+\tau)\big)$$

is the number, counted with multiplicity, of interval summands I for which $t \in \operatorname{Act}_\tau(I)$. Consequently,

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) \neq 0 \quad \Longleftrightarrow \quad \exists I \in \operatorname{Ret}_{\tau,W}(B) \text{ with } T \cap \operatorname{Act}_\tau(I) \neq \varnothing.$$

Proof. Decompose B into interval modules. On an interval summand k_I, the structure map from t to $t+\tau$ is the identity exactly when both parameters lie in I, and is zero otherwise. Images and finite direct sums decompose componentwise in the pointwise finite-dimensional constructible regime. □

Theorem 2.38 (Expanded proof of Main Theorem 3.59: finite-stencil soundness)**.** *For every finite $T \subseteq W_\tau$,*

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) \neq 0 \quad \Longrightarrow \quad T^{\mathrm{bar}}_\tau(B) \neq 0.$$

Equivalently,

$$T^{\mathrm{bar}}_\tau(B) = 0 \quad \Longrightarrow \quad \mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) = 0.$$

No coverage or mesh hypothesis is required for this direction.

Proof. A nonzero detector component supplies a retained interval whose active-start region contains a sample point, by Proposition 2.37. □

Remark 2.39 (Soundness is not completeness)*.* A zero stencil may miss a retained bar whose active-start region contains no sample point. The converse of Theorem 2.38 therefore requires a birth, germ, externally certified coverage, or family-mesh theorem. Missing completeness evidence makes a detector-based zero verdict `undefined`; it does not manufacture profile vanishing.

2.8 Birth and right-germ characterizations

Definition 2.40 (Exhaustive birth set). *For a finite left-closed right-open windowed barcode*

$$\bar{(}B) = \{[p_\lambda, q_\lambda)\}_{\lambda\in\Lambda},$$

define

$$T_{\text{birth}}(B) := \{p_\lambda \mid \lambda \in \Lambda\}, \qquad T_{\text{birth},\tau}(B) := T_{\text{birth}}(B) \cap W_\tau.$$

A birth-set certificate is exhaustive when it proves that every interval summand has its left endpoint in the recorded finite set. Only the effective subset is sampled. Multiplicity remains part of the barcode artifact even when repeated birth values are sampled once.

Theorem 2.41 (Expanded proof of Main Theorem 3.62: birth-detector characterization). *Let B be a finite windowed barcode in the left-closed right-open convention and let its exhaustive birth set be certified. Then*

$$\mathcal{D}^{\text{stc}}_{T_{\text{birth},\tau}(B),\tau}(B) \neq 0 \iff T^{\text{bar}}_\tau(B) \neq 0.$$

Equivalently,

$$\mathcal{D}^{\text{stc}}_{T_{\text{birth},\tau}(B),\tau}(B) = 0 \iff T^{\text{bar}}_\tau(B) = 0.$$

Proof. Soundness is Theorem 2.38. Conversely, a retained interval $I = [p, q)$ satisfies $p + \tau < q$, so its structure map at its certified birth point is nonzero. □

Remark 2.42 (Endpoint dependence of the point-birth theorem). Theorem 2.41 evaluates $B(p)$ and therefore consumes the left-closed registration convention. It is not decoration-invariant. Decorated intervals are handled by the right-germ theorem below or by another separately proved endpoint-aware theorem.

Definition 2.43 (Stable right germ and shifted germ map). *Let B be constructible and let $p \in \mathbb{R}$. Choose $\delta_p > 0$ such that every structure map*

$$B(p+\epsilon) \longrightarrow B(p+\epsilon') \qquad (0 < \epsilon \le \epsilon' < \delta_p)$$

is an isomorphism. The stable right germ *$B(p^+)$ is the resulting canonical isomorphism class of these stable values. Equivalently, after fixing any $0 < \epsilon_p < \delta_p$, one may take*

$$B(p^+) := B(p+\epsilon_p),$$

with independence of the choice supplied by the displayed transition isomorphisms.

The traditional notation

$$B(p^+) = \varinjlim_{\epsilon\downarrow 0} B(p+\epsilon)$$

is used only as shorthand for this constructible stable-right-value construction. It is not an unrestricted colimit over all $\epsilon > 0$.

For $\tau > 0$, choose $\delta > 0$ small enough that the stable-right conditions hold simultaneously at p and $p + \tau$. For any $0 < \epsilon < \delta$, the structure map

$$B(p+\epsilon) \longrightarrow B(p+\tau+\epsilon)$$

induces a choice-independent map

$$\gamma_{p,\tau} : B(p^+) \longrightarrow B((p+\tau)^+).$$

Naturality and the stable transition isomorphisms make $\gamma_{p,\tau}$ independent of the auxiliary ϵ.

Definition 2.44 (Germ detector)**.** *Let $T_{\text{gbirth}}(B)$ be the finite set of geometric lower endpoint values of the decorated interval summands, and put*

$$T_{\text{gbirth},\tau}(B) := T_{\text{gbirth}}(B) \cap W_\tau.$$

Define

$$\mathcal{D}^{\text{germ}}_\tau(B) := \bigoplus_{p \in T_{\text{gbirth},\tau}(B)} \operatorname{im}(\gamma_{p,\tau}).$$

An exhaustive geometric-birth certificate is part of the detector record.

Theorem 2.45 (Expanded proof of Main Theorem 3.66: decoration-invariant germ characterization)**.** *Let B be a finite decorated windowed barcode with an exhaustive geometric-birth set. Then, independently of finite-endpoint open or closed decoration,*

$$\mathcal{D}^{\text{germ}}_\tau(B) \neq 0 \iff T^{\text{bar}}_\tau(B) \neq 0.$$

Equivalently,

$$\mathcal{D}^{\text{germ}}_\tau(B) = 0 \iff T^{\text{bar}}_\tau(B) = 0.$$

Proof. For a decorated interval with geometric endpoints $p < q$, the shifted right germ is nonzero exactly when points immediately to the right of p and $p + \tau$ remain in the interval. This is equivalent to $p+\tau < q$, hence to length strictly greater than τ. At equality $q-p = \tau$, the target right germ lies strictly to the right of q and is zero, including when q is closed. Summing over the exhaustive geometric-birth set proves the result. □

2.9 Externally certified stencil completeness

Definition 2.46 (Stencil coverage certificate)**.** *A stencil coverage certificate for (B, W, τ, T) is an artifact-backed proof of*

$$\forall I \in \operatorname{Ret}_{\tau,W}(B), \qquad T \cap \operatorname{Act}_\tau(I) \neq \varnothing.$$

The record declares the source of the coverage evidence and includes bounded-window cofinal bars. Coverage is a non-scalar obligation.

Theorem 2.47 (Expanded proof of Main Theorem 3.68: certified stencil completeness)**.** *Let $T \subseteq W_\tau$ be finite and suppose a valid stencil coverage certificate is supplied. Then*

$$\mathcal{D}^{\text{stc}}_{T,\tau}(B) = 0 \iff T^{\text{bar}}_\tau(B) = 0.$$

Proof. The implication from right to left is Theorem 2.38. If the repaired profile is nonzero, choose a retained interval. Coverage supplies a sample point in its active-start region, and Proposition 2.37 makes the stencil nonzero. □

Remark 2.48 (Logical validity versus proof compression)*.* A genuine coverage proof establishes mathematical completeness even if it was obtained by reconstructing the complete target barcode. A claim of proof compression additionally requires a non-circular source, such as a family-generation theorem, finite critical-data theorem, endpoint lattice, separation theorem, filtration constraint, or external theorem. A posteriori full-profile enumeration does not by itself establish a compression gain.

2.10 Family threshold excess, augmented mesh, and sharpness

Definition 2.49 (Window-relative threshold excess)**.** *For a windowed barcode B, define*

$$\operatorname{exc}_{\tau,W}(B) := \inf_{I \in \operatorname{Ret}_{\tau,W}(B)} (\ell_W(I) - \tau),$$

with $\inf \varnothing := +\infty$*. A genuinely infinite retained interval contributes* $+\infty$*; a bounded-window cofinal interval contributes its finite clipped excess.*

For a nonempty barcode family $\mathcal{F}$, a uniform excess lower-bound certificate is a number

$$e_{\tau,W}(\mathcal{F}) > 0$$

with artifact-backed proof that

$$e_{\tau,W}(\mathcal{F}) \leq \operatorname{exc}_{\tau,W}(B) \qquad (B \in \mathcal{F}).$$

Remark 2.50 (Independent source of family excess)**.** When family-mesh completeness is advertised as proof compression, the lower bound must be obtained independently of complete retrospective reading of the target barcode. Permitted sources include a model-class theorem, endpoint lattice, Lipschitz or separation theorem, filtration constraint, external-domain theorem, or certified family generator. An absent or unverifiable source makes family-mesh completeness `undefined`.

Definition 2.51 (Augmented mesh)**.** *Assume $W = [a, b)$ is bounded and $b - a > \tau$. Put*

$$c := b - \tau, \qquad W_\tau = [a, c), \qquad L_{\text{eff}} := c - a = b - a - \tau.$$

For

$$T = \{t_1 < \cdots < t_m\} \subseteq [a, c),$$

define

$$\operatorname{Mesh}_{W_\tau}(T) := \max\{t_1 - a, t_2 - t_1, \ldots, t_m - t_{m-1}, c - t_m\}.$$

For $T = \varnothing$, set

$$\operatorname{Mesh}_{W_\tau}(\varnothing) := L_{\text{eff}}.$$

The endpoints a, c are virtual boundary points for measuring gaps; they are not automatically detector samples.

Lemma 2.52 (Augmented-mesh hitting lemma)**.** *Let $J = [p, r) \subseteq [a, c)$. If*

$$r - p > \operatorname{Mesh}_{W_\tau}(T),$$

then $J \cap T \neq \varnothing$.

Proof. If $J \cap T = \varnothing$, then J lies in one component between two successive augmented boundary or sample points. Its length is at most that augmented gap, hence at most the mesh. □

Theorem 2.53 (Expanded proof of Main Theorem 3.74: family-mesh completeness)**.** *Let $W = [a, b)$ be bounded, let $\mathcal{F}$ be a nonempty family of left-closed right-open windowed barcodes, and suppose a valid uniform excess lower bound $e_{\tau,W}(\mathcal{F}) > 0$ is supplied. If a finite stencil $T \subseteq W_\tau$ satisfies*

$$\operatorname{Mesh}_{W_\tau}(T) < e_{\tau,W}(\mathcal{F}),$$

then T is complete for every $B \in \mathcal{F}$:

$$\forall B \in \mathcal{F}, \qquad \mathcal{D}^{\text{stc}}_{T,\tau}(B) = 0 \iff T^{\text{bar}}_\tau(B) = 0.$$

If $b - a \leq \tau$, then $W_\tau = \varnothing$ and every finite windowed bar has length at most τ, so the finite-bar zero conclusion is trivial.

Proof. Let $I = [p, q)$ be retained. Its active-start region has length

$$q - p - \tau \geq e_{\tau,W}(\mathcal{F}) > \mathrm{Mesh}_{W_\tau}(T).$$

The hitting lemma gives a sample in every retained active-start region, so Theorem 2.47 applies. □

Theorem 2.54 (Expanded proof of Main Theorem 3.75: strict mesh sharpness)**.** *Fix a bounded effective domain $W_\tau = [a, c)$ and a finite stencil T.*

(i) If $\mathrm{Mesh}_{W_\tau}(T) > e > 0$, there exists a one-bar barcode B satisfying

$$\mathrm{exc}_{\tau,W}(B) > e, \qquad T_\tau^{\mathrm{bar}}(B) \neq 0, \qquad \mathcal{D}_{T,\tau}^{\mathrm{stc}}(B) = 0.$$

(ii) The strict hypothesis in Theorem 2.53 cannot uniformly be replaced by $\mathrm{Mesh}_{W_\tau}(T) \leq e$. For example, when $t_1 - a = e$, the retained interval

$$I = [a, a + \tau + e)$$

has active region $[a, t_1)$, containing no detector sample.

(iii) At equality $\mathrm{Mesh}_{W_\tau}(T) = e$, completeness is sensitive to boundary convention and the location of the maximal gap. No universal pass or fail conclusion follows without additional placement data.

Proof. For (i), choose an augmented gap of length $\mathrm{Mesh}_{W_\tau}(T)$, choose $e < e' < \mathrm{Mesh}_{W_\tau}(T)$, and place a half-open active-start interval $[p, p+e')$ inside the sample-free gap. The bar $[p, p+\tau+e')$ lies in W, survives thresholding, and is missed by T. Item (ii) is the displayed left-boundary construction. Item (iii) follows because internal and boundary equality gaps have different half-open endpoint behavior. □

2.11 Detector certificates and two-axis status

Definition 2.55 (Finite detector certificate record)**.** *A finite detector certificate record contains at least:*

(i) detector, theorem, run, and source-profile identifiers;

(ii) the exact source object and profile stage;

(iii) monitored degree, window, threshold, endpoint convention, and window-cofinal policy;

(iv) detector mode

$$\texttt{birth}, \quad \texttt{germ}, \quad \texttt{certified\_stencil}, \quad \texttt{family\_mesh};$$

(v) the finite stencil, birth set, or geometric-birth set;

(vi) soundness evidence;

(vii) completeness, characterization, or coverage evidence;

(viii) family membership and independently sourced excess evidence when family mesh is invoked;

(ix) exact or metric transport records when the consumed predicate belongs to a compared profile;

(x) *certificate size, family-relative* MinCert *data, and immutable replay artifacts;*

(xi) *separate certificate and mathematical-value statuses.*

Definition 2.56 (Two-axis detector status)**.** *Detector certification and mathematical value are independent fields. The certificate axis is*

$$\texttt{detector\_certified}, \qquad \texttt{detector\_failed}, \qquad \texttt{detector\_pending},$$

mapping respectively to the Chapter 8 atomic statuses

$$\texttt{pass}, \qquad \texttt{reject}, \qquad \texttt{undefined}.$$

Here `detector_failed` *means that a well-typed required detector condition is evidenced false. Missing, incomplete, stale, or pending evidence is* `detector_pending`*, not failure.*

The value axis is

$$\texttt{zero}, \qquad \texttt{nonzero}, \qquad \texttt{undefined}.$$

A zero value discharges a zero clause only with certified completeness. A certified nonzero value rejects a required zero clause. A raw zero-looking computation with missing completeness remains undefined. When detector semantics are outside the declared scope, the atomic status is `not_invoked`*; no certificate or value token is fabricated.*

Proposition 2.57 (Expanded proof of Main Proposition 3.78: detector invariance)**.** *Let*

$$\eta : B \xrightarrow{\cong} C$$

be an isomorphism of persistence modules on the declared window. For every finite $T \subseteq W_\tau$,

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B) \cong \mathcal{D}^{\mathrm{stc}}_{T,\tau}(C).$$

The corresponding right-germ detectors are isomorphic when the same geometric-birth convention is used. Hence detector zero/nonzero is invariant under an actual isomorphism of the exact source-stage profiles.

Proof. Naturality gives a commutative square for every structure map from t to $t+\tau$. Vertical isomorphisms induce isomorphisms of image spaces, and finite direct sums preserve them. Passing through the canonical stable right-neighborhood identifications gives the germ assertion. □

2.12 Metric-safe detector composition and robust support

Theorem 2.58 (Expanded proof of Main Theorem 3.79: metric zero-verdict composition)**.** *Let B, C be windowed pre-threshold profiles. Suppose:*

(i) *complete detector certificates establish*

$$\mathcal{D}^{\mathrm{stc}}_{T_B,\tau}(B) = 0 \iff T^{\mathrm{bar}}_\tau(B) = 0$$

and

$$\mathcal{D}^{\mathrm{stc}}_{T_C,\tau}(C) = 0 \iff T^{\mathrm{bar}}_\tau(C) = 0;$$

(ii) $d_B(B,C) \le \varepsilon$*;*

(iii) *the two-sided threshold-gap hypotheses of Appendix A hold at the same ε;*

(iv) $2\varepsilon \le \tau$.

Then
$$\mathcal{D}^{\mathrm{stc}}_{T_B,\tau}(B)=0 \quad\Longleftrightarrow\quad \mathcal{D}^{\mathrm{stc}}_{T_C,\tau}(C)=0.$$
The operational v21 certificate uses the strict reserve $2\varepsilon<\tau$.

Proof. Theorem 1.47 gives
$$T^{\mathrm{bar}}_{\tau}(B)=0 \Longleftrightarrow T^{\mathrm{bar}}_{\tau}(C)=0.$$
Compose this with the two complete detector characterizations. □

Corollary 2.59 (One-way transport of a detector-certified zero)**.** *If a complete detector certificate proves $T^{\mathrm{bar}}_{\tau}(B)=0$ and the metric hypotheses of Theorem 2.58 hold, then*
$$T^{\mathrm{bar}}_{\tau}(C)=0.$$
A second detector certificate is required only when the target conclusion is to be restated as a target detector value.

Remark 2.60 (Zero predicate, not raw detector-rank stability)*.* The metric theorem transports a certified repaired-zero predicate. It does not compare raw ranks at common sample points, preserve birth or germ sets, transport coverage witnesses, or preserve certificate size. Appendix A supplies explicit counterexamples to pointwise rank stability.

Definition 2.61 (Boundary-safe robust stencil)**.** *Let B,C be equipped with actual ε-interleaving morphisms. Define*
$$W^{\mathrm{rob}}_{\tau,\varepsilon}:=\{t\mid t-\varepsilon,\ t+\tau+\varepsilon\in W\}.$$
For finite $T\subseteq W^{\mathrm{rob}}_{\tau,\varepsilon}$, define
$$\mathcal{D}^{\mathrm{rob}}_{T,\tau,\varepsilon}(B):=\bigoplus_{t\in T}\mathrm{im}\big(B(t-\varepsilon)\to B(t+\tau+\varepsilon)\big).$$
For $W=[a,b)$, the boundary-safe domain is $[a+\varepsilon,b-\tau-\varepsilon)$ when nonempty.

Proposition 2.62 (Expanded proof of Main Proposition 3.82: robust-to-exact sandwich)**.** *Let B,C carry specified ε-interleaving morphisms on the required boundary-safe domain. Then*
$$\mathcal{D}^{\mathrm{rob}}_{T,\tau,\varepsilon}(B)\neq 0 \quad\Longrightarrow\quad \mathcal{D}^{\mathrm{stc}}_{T,\tau}(C)\neq 0,$$
and symmetrically with B,C exchanged.

Proof. For each $t\in T$, the structure map
$$B(t-\varepsilon)\longrightarrow B(t+\tau+\varepsilon)$$
factors, by naturality and the interleaving identities, through
$$B(t-\varepsilon)\longrightarrow C(t)\longrightarrow C(t+\tau)\longrightarrow B(t+\tau+\varepsilon).$$
A nonzero composite forces the middle exact structure map of C to be nonzero. □

Remark 2.63 (Supporting status of the robust sandwich)*.* The sandwich is one-way and requires actual interleaving morphisms, not merely a scalar distance. It is a supporting detector theorem, not a substitute for completeness or for the repaired-zero metric chain.

2.13 Family-relative minimal certificates and Pareto complexity

Definition 2.64 (Detector decision problem)**.** *A detector decision problem is*

$$\Theta = (\mathcal{F}, W, \tau, \mathrm{stage}, \mathrm{endpt}, \mathrm{predicate}, \mathrm{promises}),$$

where $\mathcal{F}$ is a nonempty family and the remaining fields fix the source stage, endpoint convention, target predicate, and all promises. A certificate is family-relative when one fixed verifier rule is sound and complete for every member, rather than being designed after unrestricted inspection of one target object.

Definition 2.65 (Achievable cost set and model-relative minimal certificate)**.** *Fix $(\mathcal{F}, \Theta, \mathfrak{M})$. Let*

$$\mathcal{C}_{\mathcal{F}}^{\mathfrak{M},\mathrm{fin}}(\Theta) \subseteq \mathbb{N}^4$$

be the finite family-cost vectors of all valid sound-and-complete family-relative schemes. Define

$$\mathrm{MinCert}_{\mathcal{F}}^{\mathfrak{M}}(\Theta) := \mathrm{ParetoMin}\Big(\mathcal{C}_{\mathcal{F}}^{\mathfrak{M},\mathrm{fin}}(\Theta)\Big).$$

If no finite-cost scheme exists, the achievable set and Pareto frontier are empty. For $x \in \{q, b, t, a\}$, define the coordinate lower envelope

$$\mathrm{MinCert}_{x,\mathcal{F}}^{\mathfrak{M}}(\Theta) := \inf_{\Pi} x\Big(\mathrm{cost}_{\mathfrak{M}}^{\mathcal{F}}(\Pi)\Big),$$

with value $+\infty$ when that coordinate is never finite.

The legacy notation $\mathrm{MinCert}(\Theta)$ *is retained only for the worst-case query projection in the exact nonadaptive point-stencil model. It is not the canonical v21 certificate complexity.*

Remark 2.66 (Legacy minimum and infimum projection)*.* In the exact nonadaptive stencil model, every finite query count lies in $\mathbb{N}$. Whenever a finite complete stencil exists, the attainable query counts therefore have a least element, so the legacy minimum equals its infimum. This observation concerns only the scalar query projection and does not collapse the v21 Pareto frontier.

Theorem 2.67 (Expanded proof of Main Theorem 3.86: finite Pareto-frontier well-posedness)**.** *Suppose*

$$\mathcal{C}_{\mathcal{F}}^{\mathfrak{M},\mathrm{fin}}(\Theta) \neq \varnothing.$$

Then $\mathrm{MinCert}_{\mathcal{F}}^{\mathfrak{M}}(\Theta)$ *is a nonempty finite set, and every finite achievable cost vector dominates at least one frontier element.*

Proof. The coordinatewise order on $\mathbb{N}^4$ is a well-quasi-order by Dickson's lemma. Hence every nonempty subset has finitely many minimal elements. Descending inside the finite coordinate box below any fixed vector reaches a minimal element, proving domination by the frontier. □

Remark 2.68 (Why the frontier is set-valued)*.* A scheme with fewer queries may require larger answers, more verifier work, or a larger replay artifact. Without a declared scalar utility, incomparable cost vectors remain simultaneously minimal.

Proposition 2.69 (Expanded proof of Main Proposition 3.88: birth-set query upper bound)**.** *Let $\mathfrak{M}_{\mathrm{birth}}$ be the exact nonadaptive model in which one query evaluates one birth stencil map and suppose*

$$T_{\mathrm{birth},\tau}(\mathcal{F}) := \bigcup_{B \in \mathcal{F}} T_{\mathrm{birth},\tau}(B)$$

is finite and uniformly exhaustive, with its provenance independent of the target verdict. Then a valid family-relative scheme exists whose query coordinate satisfies

$$q \le |T_{\mathrm{birth},\tau}(\mathcal{F})|,$$

and therefore

$$\mathrm{MinCert}_{q,\mathcal{F}}^{\mathfrak{M}_{\mathrm{birth}}}(\Theta) \le |T_{\mathrm{birth},\tau}(\mathcal{F})|.$$

The remaining coordinates are determined by the declared encoding and verifier models.

Proof. Query the finite union of effective birth locations. The birth characterization detects every retained bar in every family member. □

Definition 2.70 (Minimal augmented-mesh stencil number)**.** *For $L_{\mathrm{eff}} \ge 0$ and $\gamma > 0$, define*

$$m_{\mathrm{mesh}}(L_{\mathrm{eff}}, \gamma) := \max\left\{0, \left\lceil \frac{L_{\mathrm{eff}}}{\gamma} \right\rceil - 1\right\}.$$

Theorem 2.71 (Expanded proof of Main Theorem 3.90: exact augmented-mesh query count)**.** *Fix the exact nonadaptive point-stencil model and let $W_\tau = [a, c)$ have length $L_{\mathrm{eff}} = c - a$. Then:*

(i) there exists $T \subseteq [a, c)$ with

$$|T| = m_{\mathrm{mesh}}(L_{\mathrm{eff}}, \gamma), \qquad \mathrm{Mesh}_{W_\tau}(T) \le \gamma;$$

(ii) every stencil satisfying $\mathrm{Mesh}_{W_\tau}(T) \le \gamma$ *has*

$$|T| \ge m_{\mathrm{mesh}}(L_{\mathrm{eff}}, \gamma).$$

Thus the displayed number is the exact query coordinate for the augmented-mesh design problem. It determines neither bit length, verifier time, nor artifact footprint.

Proof. Put $N = \lceil L_{\mathrm{eff}}/\gamma \rceil$. If $N = 0$, the domain is empty. If $N \ge 1$, divide it into N equal gaps and place $N - 1$ internal samples. Conversely, m samples create $m + 1$ augmented gaps whose lengths sum to L_{eff}; if each is at most γ, then $L_{\mathrm{eff}} \le (m + 1)\gamma$. □

Corollary 2.72 (Expanded proof of Main Corollary 3.91: family-mesh query bound)**.** *Under Theorem 2.53, choose $0 < \gamma < e_{\tau,W}(\mathcal{F})$. In the exact nonadaptive stencil model, a valid family-relative scheme exists with a finite cost vector and*

$$q \le m_{\mathrm{mesh}}(L_{\mathrm{eff}}, \gamma).$$

Hence the same quantity bounds the query-coordinate lower envelope. It need not describe the entire Pareto frontier of a more structured family.

2.14 Information-model-relative lower bounds

Definition 2.73 (Private witness cells)**.** *Fix a deterministic exact oracle-closed model with query universe $\mathcal{U}$, response function $\mathrm{Ans}_X(u)$, predicate Θ, baseline X_0, and witnesses $X_1, \dots, X_m$. Pairwise disjoint sets $U_1, \dots, U_m \subseteq \mathcal{U}$ are private witness cells when*

$$\Theta(X_j) \ne \Theta(X_0)$$

and

$$\mathrm{Ans}_{X_j}(u) = \mathrm{Ans}_{X_0}(u) \qquad (u \notin U_j).$$

Thus witness X_j is indistinguishable from the baseline until a query in its private cell is issued.

Theorem 2.74 (Expanded proof of Main Theorem 3.93: indistinguishability lower bound)**.** *Under Definition 2.73, every deterministic exact oracle-closed family-relative scheme that is sound and complete on $\{X_0, X_1, \ldots, X_m\}$ must query at least one element of each private cell on the baseline transcript. Consequently,*

$$q\Big(\mathrm{cost}_{\mathfrak{M}}^{\mathcal{F}}(\Pi)\Big) \geq m.$$

The conclusion applies to adaptive and nonadaptive deterministic schemes.

Proof. Run the scheme on X_0. If no query enters some U_j, every charged answer agrees on X_0 and X_j. Oracle-closedness implies inductively that an adaptive producer asks the same subsequent queries and creates the same verifier-visible record on both objects. The verifier must return the same verdict, contradicting soundness and completeness because the predicate values differ. Disjointness forces at least m queried cells. □

Corollary 2.75 (Expanded proof of Main Corollary 3.94: private-cell lower bounds)**.** *Theorem 2.74 specializes to:*

(i) disjoint private active-start regions for point-stencil persistence queries;

(ii) disjoint private generator-probe cells at stage `diagram_probe`*, provided Appendix DG supplies the actual probe semantics;*

(iii) disjoint private coherence cells at stage `descent_coherence`*, provided Appendix DG supplies the actual local and overlap data.*

These are information-model-relative statements. A stronger information model may destroy the indistinguishability hypotheses.

Remark 2.76 (Randomized lower bounds require a separate theorem)*.* The deterministic adversary theorem does not by itself give an optimal randomized lower bound. A randomized claim requires an explicit error model and a distributional or minimax argument.

Theorem 2.77 (Common-model exact query complexity benchmark)**.** *Fix rational $\tau > 0$, a bounded rational effective domain $W_\tau = [a, c)$, and pairwise disjoint nonempty rational half-open cells*

$$U_j = [p_j, p_j + e_j) \subseteq W_\tau, \qquad j = 1, \ldots, m.$$

Let

$$B_0 := 0, \qquad B_j := \{[p_j, p_j + \tau + e_j)\},$$

and let $\mathcal{F}_{\mathrm{priv}} = \{B_0, B_1, \ldots, B_m\}$. Consider the deterministic exact oracle-closed point-stencil model in which I contains the rational family specification and the promise of membership in $\mathcal{F}_{\mathrm{priv}}$, but not the identity of the unknown member; a charged query at $t \in W_\tau$ returns

$$\dim_k \mathrm{im}(B(t) \to B(t+\tau)).$$

For the predicate

$$\Theta(B) = 0 \iff T_\tau^{\mathrm{bar}}(B) = 0,$$

the exact worst-case query complexity is

$$\mathrm{MinCert}_{q,\mathcal{F}_{\mathrm{priv}}}^{\mathfrak{M}_{\mathrm{priv}}}(\Theta) = m.$$

Moreover, under any finite encoding of the rational cells and finite field or finite-coded coefficient data, a valid scheme has a finite cost vector whose query coordinate is m.

Proof. Choose one rational point $s_j \in U_j$ for every cell and query all of them. The zero profile returns zero everywhere, while B_j returns a nonzero image exactly on its active-start cell U_j; hence the fixed stencil is a sound-and-complete nonadaptive scheme with $q = m$. For the lower bound, take B_0 as baseline and B_j as the private witness for cell U_j. Outside U_j, its query answers coincide with those of B_0. The cells are pairwise disjoint, so Theorem 2.74 gives $q \geq m$ even for adaptive deterministic schemes. The rational and finite-coded assumptions make the remaining certificate coordinates finite for the exhibited scheme. □

Remark 2.78 (G21-6 local witness). Theorem 2.77 supplies a nontrivial upper bound and an information-model lower bound on the same family, source stage, predicate, and certificate model. It witnesses the local certificate-complexity content required for G21-6; it does not determine a global local-to-global complexity theorem.

2.15 Right-unbounded-window no-go theorem

Theorem 2.79 (Uniform finite-detector no-go on an unbounded birth family)**.** *Let*

$$W = [a, +\infty), \qquad \tau > 0, \qquad e > 0,$$

and consider

$$\mathcal{U}_{a,\tau,e} := \{B_p = \{[p, p + \tau + e)\} \mid p \geq a\}.$$

For every finite stencil $T \subseteq W_\tau$, there exists $B_p \in \mathcal{U}_{a,\tau,e}$ such that

$$\mathcal{D}^{\mathrm{stc}}_{T,\tau}(B_p) = 0, \qquad T^{\mathrm{bar}}_{\tau}(B_p) \neq 0.$$

For the corresponding information class $\Theta_{\mathcal{U}}$,

$$\mathrm{MinCert}(\Theta_{\mathcal{U}}) = +\infty.$$

The same conclusion holds for every family admitting retained active-start regions arbitrarily far to the right of every finite sample set.

Proof. A finite T is bounded above. Choose $p > \max T$, with the empty-stencil case arbitrary. The retained bar has active-start region $[p, p+e)$, disjoint from T, while its length $\tau + e$ exceeds the threshold. □

Theorem 2.80 (Expanded proof of Main Theorem 3.97: adaptive right-unbounded no-go)**.** *In the family and exact point-stencil setting of Theorem 2.79, no deterministic oracle-closed scheme that halts after finitely many queries on every input is sound and complete for the repaired-zero predicate. Consequently the finite achievable scheme-cost set is empty and*

$$\mathrm{MinCert}^{\mathfrak{M}_{\mathrm{stencil}}}_{q,\mathcal{U}_{a,\tau,e}}(\Theta) = +\infty$$

in that model. The conclusion holds for adaptive as well as nonadaptive strategies.

Proof. Run the scheme on the zero profile. Its finite transcript queries a bounded set of parameters. Choose a one-bar witness whose active-start interval lies strictly to the right of every queried parameter. Every response on the zero transcript is unchanged, so oracle-closedness forces the same adaptive query path, certificate, and verdict. The witness survives thresholding, contradicting soundness and completeness. □

Remark 2.81 (Admissible escape conditions). The theorem does not exclude finite detectors when additional structure is proved, such as a finite known birth set, global birth bound, eventual zero, eventual stabilization, finite-to-infinite reduction theorem, or separately certified countable cofinal detector. Finite observed stability alone establishes none of these conditions.

2.16 Finite certificate composition and evidence reuse

Definition 2.82 (Composable certificate schemes)**.** *Two family-relative schemes Π_1, Π_2 are composable when they use one compatible family and certificate model, their source stages and predicates are typed, their dependencies form an acyclic verifier subgraph, and* V *contains a rule deriving the requested conjunction or typed composite predicate from the accepted component verdicts. Shared queries, answers, theorem references, and artifacts are reusable only through identical content addresses and a declared replay rule.*

Definition 2.83 (Composition overhead and reuse discipline)**.** *Let*

$$h_{12} = (h_q, h_b, h_t, h_a) \in \overline{\mathbb{N}}^4$$

record the additional query, encoding, verification, and artifact cost of the composition node. Pure logical conjunction may have $h_q = 0$. Any new compatibility probe or dependent source query contributes to h_q. Duplicate artifacts are counted once only when the same immutable object is actually referenced by both records. A textual assertion that evidence is shared creates no reduction.

Theorem 2.84 (Expanded proof of Main Theorem 3.101: certificate-scheme composition)**.** *Let Π_1, Π_2 be composable finite-cost schemes with family costs $c_1, c_2 \in \mathbb{N}^4$. Then a scheme $\Pi_1 \otimes \Pi_2$ exists for the verifier-approved composite predicate and satisfies*

$$\mathrm{cost}^{\mathcal{F}}_{\mathfrak{M}}(\Pi_1 \otimes \Pi_2) \preceq_{\mathrm{cost}} c_1 + c_2 + h_{12}.$$

For an input X, if $Q_i(X)$ are the content-addressed query sets and $H_{12}(X)$ the additional composition queries, the object-specific query count may be taken as

$$|Q_1(X) \cup Q_2(X) \cup H_{12}(X)|.$$

Exact savings in (b, t, a) require the declared reuse rules; no claim of Pareto optimality is made.

Proof. Run or replay the two certificates, store each distinct content-addressed query once, and append the verifier's composition node. Concatenation gives the coordinatewise sum plus overhead, while the query union gives the refined object-specific count. Taking the model's family aggregation proves the bound. □

Proposition 2.85 (Replayable shared-query composition benchmark)**.** *Use the family and model of Theorem 2.77, with $m \geq 2$, and choose $1 < r < m$. Let $\Theta_L(B) = 0$ mean that no witness occurs in cells $U_1, \dots, U_r$, and let $\Theta_R(B) = 0$ mean that no witness occurs in cells $U_r, \dots, U_m$. The canonical schemes query one point in each listed cell. Their conjunction is the repaired-zero predicate, and the content-addressed query in U_r is shared. Consequently the composite query record has*

$$q = r + (m - r + 1) - 1 = m,$$

not $m + 1$. The saving is valid because the identical query, answer, and artifact identifier are replayed by both component verifiers.

Proof. Each component predicate is decided exactly by its listed private cells. On the promised family, both predicates vanish exactly when no nonzero witness cell is present, hence exactly when the barcode is zero. The two query sets intersect only in the content-addressed probe from U_r, so their union has size m. The declared reuse rule permits one stored answer and artifact to discharge both component nodes. □

Remark 2.86 (Local composition is not global certificate complexity)*.* Theorem 2.84 is finite and model-relative. It does not prove a global inequality obtained by summing local MinCert-values. A global theorem must account for overlap coherence, dependency cycles, shared encodings, object descent, and artifact reuse under one model; that theorem belongs to Appendix DG.

2.17 Certificate-to-representative handoff

Definition 2.87 (Certificate-to-representative handoff)**.** *A v21 handoff record is*

$$\begin{aligned}\mathrm{Cert2RepInput} := (&\mathrm{source}, \mathrm{stage}, \Theta, \mathrm{verdict}, \mathfrak{M},\\ &\mathrm{certificate}, \mathrm{representative\_candidate}, \mathrm{strength}, \mathrm{artifacts}).\end{aligned}$$

It records exactly which source-stage predicate the object-specific certificate proves, the family and model for which the governing scheme is complete, the object-specific and family-level costs, and any candidate representative identifier proposed to the representative-construction sublayer and, after certification, to Appendix C. The representative field may be absent. When present, it is only a candidate until actual profile isomorphisms and every required realization-compatibility artifact are proved.

Theorem 2.88 (Expanded proof of Main Theorem 3.104: representative handoff boundary)**.** *A valid detector certificate establishes only its declared source-stage predicate and may populate* Cert2RepInput*. Without a separate construction theorem it produces no filtered representative, finite diagram, chain equivalence, derived object, Ext edge, descent object, or external Return.*

Proof. The certificate relation contains only the declared source object, predicate, transcript, verifier, and artifacts. The promoted mathematical objects and equivalences are not data of that relation and therefore require their own construction and proof artifacts. □

2.18 Legacy whole-profile degree-one detector

Definition 2.89 (Residual degree-one bar mass)**.** *Let $\mathcal{B}^{\mathrm{res}}_{1,W,\tau}(F)$ denote the residual barcode of* $\mathrm{Eval}_{1,W,\tau}(F)$*, counted with multiplicity. Define*

$$E_1(F; W, \tau) := \sum_{I \in (\mathcal{B}^{\mathrm{res}}_{1,W,\tau}(F))_{\mathrm{fin}}} \ell(I).$$

If an infinite residual bar occurs, set $E_1 = +\infty$, unless a separately declared truncated functional is used.

Proposition 2.90 (Expanded proof of Main Proposition 3.106: whole-profile E_1 characterization)**.** *For a finite residual barcode under the stated convention,*

$$E_1(F; W, \tau) = 0 \quad \Longleftrightarrow \quad \mathrm{Eval}_{1,W,\tau}(F) = 0.$$

Proof. Every nonempty finite interval has positive length and every infinite interval forces $+\infty$. The zero barcode has empty sum. □

Remark 2.91 (Why E_1 is not finite proof compression)*.* The functional reads the already reconstructed residual barcode. It is a whole-profile vanishing functional, not a finite-map stencil certificate, not a bottleneck-Lipschitz quantity, and not a threshold-gap budget. It cannot be counted as a proof-compression result.

2.19 Explicit realization of finite barcode profile families

The following construction turns the object-level existence of a bare filtered representative into a theorem in the finite barcode regime. It does not assert functoriality or preservation of additional structures.

Definition 2.92 (Elementary interval complex)**.** *Fix a degree i.*

For a finite right-open interval $I = [b, d)$ with $b < d$, define the filtered chain complex $E_i(I)$ by generators

$$x_I \in C_i, \qquad y_I \in C_{i+1},$$

with

$$\partial y_I = x_I, \qquad \partial x_I = 0,$$

and filtration values

$$\mathrm{fil}(x_I) = b, \qquad \mathrm{fil}(y_I) = d.$$

For an infinite interval $I = [b, \infty)$, define $E_i(I)$ to have one generator $x_I \in C_i$, with $\partial x_I = 0$ and $\mathrm{fil}(x_I) = b$.

Lemma 2.93 (Persistence of an elementary interval complex)**.** *For every interval I in Definition 2.92,*

$$\mathbf{P}_i(E_i(I)) \cong k_I,$$

and

$$\mathbf{P}_j(E_i(I)) = 0 \qquad (j \neq i).$$

Proof. For $I = [b, d)$, no generator is present before b. For $b \leq t < d$, only x_I is present, so it represents one nonzero class in degree i. At $t \geq d$, the generator y_I appears and $\partial y_I = x_I$, so the pair is acyclic. This gives the interval module $k_{[b,d)}$ in degree i and no other homology.

For $I = [b, \infty)$, the class x_I appears at b and is never killed, producing $k_{[b,\infty)}$. ☐

Definition 2.94 (Bare barcode realization)**.** *Let I_{mon} be finite, and let*

$$\mathcal{B} = \{B_i\}_{i \in I_{\mathrm{mon}}}$$

be a family of finite barcodes written in right-open interval normal form. Define

$$\mathrm{Rep}_{\mathrm{bar}}(\mathcal{B}) := \bigoplus_{i \in I_{\mathrm{mon}}} \bigoplus_{I \in B_i} E_i(I).$$

Theorem 2.95 (Finite barcode family realization)**.** *Let I_{mon} be finite and let each B_i be a finite barcode. Then* $\mathrm{Rep}_{\mathrm{bar}}(\{B_i\})$ *is a finite-type filtered chain complex and, for every $i \in I_{\mathrm{mon}}$,*

$$\mathbf{P}_i(\mathrm{Rep}_{\mathrm{bar}}(\{B_j\})) \cong \bigoplus_{I \in B_i} k_I$$

as persistence modules. Equivalently,

$$\bar{}(\mathbf{P}_i(\mathrm{Rep}_{\mathrm{bar}}(\{B_j\}))) = B_i$$

as barcode multisets, including multiplicity.

Consequently, every finite windowed repaired profile family

$$\mathcal{P}^{I_{\mathrm{mon}}}_{W,\tau}(F)$$

admits at least one bare filtered representative.

Proof. The direct sum is finite because both the monitored scope and every barcode are finite. It is bounded and finite-dimensional. The persistence homology of a finite direct sum is the direct sum of the persistence homologies of its summands. The result follows from Lemma 2.93. ☐

Remark 2.96 (Strength and limitation of the realization theorem). Theorem 2.95 proves object-level existence of a filtered complex realizing the declared finite profiles. It does not supply:

- a canonical choice independent of barcode decomposition data;
- a functor on arbitrary persistence morphisms;
- compatibility with the original filtered object F;
- geometric, arithmetic, sheaf-theoretic, or categorical structure;
- $\mathcal{R}$-equivalence between different representatives;
- the amplitude or edge-realization hypotheses needed for Appendix C.

Those are independent obligations.

2.20 Admissible filtered representatives and invocation status

Definition 2.97 (Admissible filtered representative). *Let $F \in \mathsf{FiltCh}(k)$, let W be a right-open window, let $\tau > 0$, and let I_{mon} be finite.*

A filtered object

$$C_{\tau,W}F \in \mathsf{FiltCh}(k)$$

is an admissible filtered representative *if it is equipped with actual profile isomorphisms*

$$\rho_i : \mathbf{P}_i(C_{\tau,W}F) \xrightarrow{\cong} \mathcal{P}_{i,W,\tau}(F)$$

for every $i \in I_{\mathrm{mon}}$.

The tuple

$$\left(C_{\tau,W}F, I_{\mathrm{mon}}, \{\rho_i\}_{i \in I_{\mathrm{mon}}}\right)$$

is part of the representative artifact.

Definition 2.98 (Representative compatibility). *Call a representative* profile-compatible *when its maps ρ_i identify the monitored persistence modules with the repaired profile family.*

Proposition 2.99 (Compatibility is exactly the representative condition). *A filtered object is an admissible representative in the sense of Definition 2.97 if and only if it is supplied with profile-compatibility isomorphisms for every monitored degree.*

Proof. This is the defining condition, with the requirement that the isomorphisms and monitored scope are explicit artifact data. □

Definition 2.100 (Representative lifecycle and canonical record status). *The optional lifecycle axis is*

$$\texttt{not\_required}, \quad \texttt{constructed}, \quad \texttt{verified}, \quad \texttt{failed}, \quad \texttt{missing}.$$

It records workflow progress only.

The canonical representative-record status consumed by the v21 verifier is exactly one of

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{certified}.$$

A well-typed supplied representative that is evidenced to violate a required compatibility condition produces the Chapter 8 atomic status `reject`; it is not relabeled as missing evidence.

When a representative is required:

$$\texttt{verified} \mapsto \texttt{certified} \mapsto \texttt{pass},$$

$$\texttt{constructed} \text{ or } \texttt{missing} \mapsto \texttt{undefined},$$

while `failed` *maps to* `reject`*. The lifecycle value* `not_required` *maps to* `not_invoked` *only under an explicit scope-exclusion policy.*

Remark 2.101 (Lifecycle status versus canonical audit status)*.* Lifecycle labels do not replace the canonical atomic status calculus. Missing, incomplete, stale, or pending evidence remains `undefined`; an evidenced false compatibility condition is `reject`; and a clause is `not_invoked` only when it is explicitly outside scope.

Remark 2.102 (Representative, not functor)*.* The notation $C_{\tau,W}F$ denotes a chosen representative artifact. It does not assert that $C_{\tau,W}$ is globally defined, canonical, or functorial.

2.21 Implementable range, non-uniqueness, and invariance

Definition 2.103 (Implementable representative range)**.** *The* implementable range *consists of the declared tuples*

$$(F, W, \tau, I_{\text{mon}})$$

for which the proof object supplies:

(i) the finite repaired profile family $\mathcal{P}_{W,\tau}^{I_{\text{mon}}}(F)$*;*

(ii) a constructed representative;

(iii) verified profile-compatibility isomorphisms;

(iv) immutable artifact references;

(v) every stronger compatibility required by the invoked downstream clause.

Remark 2.104 (Why an implementable range is needed)*.* The bare realization theorem proves that finite barcode profiles can be realized by filtered complexes. It does not prove that a representative preserves the extra structure required by a particular derived, geometric, or tower-level argument. The implementable range records exactly where those stronger obligations have also been discharged.

Axiom 2.105 (Structure-compatible representative availability)**.** *A run may claim membership in the implementable representative range only after an actual representative artifact has been constructed and verified.*

The finite barcode realization theorem discharges bare object-level existence. Any stronger requirement—for example realization compatibility, geometric compatibility, functoriality, or tower coherence—must be discharged separately.

Remark 2.106 (Status of the availability axiom)*.* Axiom 2.105 is an audit admissibility rule, not an assertion that every repaired profile has a unique canonical or structure-compatible representative.

Proposition 2.107 (Non-uniqueness of filtered representatives)**.** *An admissible filtered representative is generally not unique as a strict filtered chain complex.*

Proof. A monitored persistence profile does not determine chain-level contractible summands, unmonitored degrees, bases, or additional structure. Different filtered complexes can therefore realize the same monitored profile family. □

Definition 2.108 (Admissible representative choice)**.** *An admissible representative choice consists of:*

(i) the representative $C_{\tau,W}F$;

(ii) the finite monitored scope;

(iii) the profile isomorphisms ρ_i;

(iv) the comparison mode;

(v) the declared equivalence strength;

(vi) any realization-compatibility data;

(vii) artifact identifiers and hashes.

Proposition 2.109 (Representative invariance of persistence-level clauses)**.** *If $C_{\tau,W}F$ and $C'_{\tau,W}F$ are admissible representatives of the same profile family, then every Core clause depending only on the monitored persistence modules has the same value for both representatives.*

Proof. For each monitored degree,

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathcal{P}_{i,W,\tau}(F) \cong \mathbf{P}_i(C'_{\tau,W}F).$$

Hence every predicate invariant under persistence-module isomorphism has the same value. □

Remark 2.110 (No derived invariance from profile invariance)*.* Proposition 2.109 applies only to persistence-level clauses. It does not imply invariance of $\mathcal{R}$, Ext, products, higher operations, or tower comparison maps. Those require stronger compatibility artifacts.

Definition 2.111 (Stage-compatible detector transport)**.** *A stage-compatible detector transport from a profile B to a profile C consists of:*

(i) the same declared detector stage on source and target;

(ii) an actual isomorphism between the exact source-stage persistence modules;

(iii) identical window, threshold, endpoint, and sample conventions, or a separately proved transport of those conventions;

(iv) artifact-backed compatibility with every birth, germ, coverage, family-membership, and excess record consumed by the target claim.

Proposition 2.112 (Detector preservation by an exact stage-compatible isomorphism)**.** *A stage-compatible detector transport preserves the exact stencil and germ detector zero/nonzero predicates. It preserves a complete detector certificate only when the certificate's completeness and artifact fields are also transported or independently reverified.*

Proof. The zero/nonzero assertion follows from Proposition 2.57. Completeness is an additional theorem-and-artifact obligation and therefore does not follow from value invariance alone. □

Remark 2.113 (A repaired representative does not preserve the pre-threshold detector)*.* The standard finite stencil reads $B_{i,W}(F) = \mathsf{W}_W \mathbf{P}_i(F)$ at stage `windowed_prethreshold`. An admissible representative supplies only

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathrm{Eval}_{i,W,\tau}(F),$$

which lies at stage `repaired_postthreshold`. Therefore representative admissibility alone does not transport the standard pre-threshold stencil, birth set, active-start regions, coverage certificate, or MinCert data. A separate stage-transport theorem is mandatory.

2.22 Global representative notation and its restricted use

Definition 2.114 (Global representative construction)**.** *A* global representative construction *at threshold τ is a declared assignment*

$$F \longmapsto C_\tau F$$

with actual profile isomorphisms

$$\mathbf{P}_i(C_\tau F) \cong T_\tau^{\mathrm{bar}}(\mathbf{P}_i(F))$$

for every declared monitored degree.

Remark 2.115 (Global representative construction is not default)*.* Definition 2.114 is optional. It does not assert functoriality on morphisms or compatibility with window restriction.

Definition 2.116 (Window compatibility of a global representative)**.** *A global representative construction is* window-compatible on W *if for every monitored degree there is an actual isomorphism*

$$\mathsf{W}_W \mathbf{P}_i(C_\tau F) \cong T_\tau^{\mathrm{bar}}(\mathsf{W}_W \mathbf{P}_i(F)).$$

Proposition 2.117 (Restriction under window compatibility)**.** *If a global representative construction is window-compatible on W, then the restricted object $(C_\tau F)|_W$ is an admissible representative of the windowed repaired profile.*

Proof. By the definition of restriction to the declared window,

$$\mathbf{P}_i\big((C_\tau F)|_W\big) \cong \mathsf{W}_W \mathbf{P}_i(C_\tau F).$$

The defining window-compatibility isomorphism therefore gives

$$\mathbf{P}_i\big((C_\tau F)|_W\big) \cong T_\tau^{\mathrm{bar}}(\mathsf{W}_W \mathbf{P}_i(F)),$$

which is precisely the monitored profile isomorphism required by Definition 2.97. □

Remark 2.118 (Why restriction is not the default)*.* Appendix A, Proposition 1.59, proves that

$$\mathsf{W}_W T_\tau^{\mathrm{bar}} \not\cong T_\tau^{\mathrm{bar}} \mathsf{W}_W$$

in general. The infinite-bar obstruction is exhibited in Example 1.63, and the exact window-adapted criterion is Definition 1.64 together with Proposition 1.65. Therefore

$$C_{\tau,W}F = (C_\tau F)|_W$$

may be used only under a verified window-compatibility or τ-window-adapted criterion.

2.23 Profile algebra at the representative level

Definition 2.119 (Window-normalized representative)**.** *An admissible representative $C_{\tau,W}F$ is* window-normalized *if its monitored persistence profiles are supported on W with the same extension-by-zero convention used in Appendix A.*

Proposition 2.120 (Windowed representative idempotence)**.** *Let $C_{\tau,W}F$ be a verified window-normalized representative. Suppose a second representative is constructed using the same window, monitored scope, and profile-equivalence policy. Then*

$$C_{\tau,W}(C_{\tau,W}F) \simeq_{\mathrm{PH},I_{\mathrm{mon}}} C_{\tau,W}F.$$

Proof. Window normalization ensures that reapplying W_W does not clip the monitored profile further. Every finite bar in the first repaired profile has length strictly greater than τ, and every infinite bar is retained. Barcode-level idempotence therefore gives identical monitored repaired profiles. □

Remark 2.121 (Idempotence is profile-level). The proposition does not assert strict equality of filtered complexes or the existence of a globally idempotent endofunctor.

Proposition 2.122 (Windowed threshold composition). *Let $\tau, \sigma > 0$. Suppose all relevant representatives exist, are window-normalized, and use the same monitored scope and window. Then*

$$C_{\sigma,W}(C_{\tau,W}F) \simeq_{\mathrm{PH},I_{\mathrm{mon}}} C_{\max\{\tau,\sigma\},W}F.$$

Proof. On every monitored profile,

$$T_\sigma^{\mathrm{bar}} T_\tau^{\mathrm{bar}} = T_{\max\{\tau,\sigma\}}^{\mathrm{bar}}.$$

Window normalization prevents a second crop from creating new short bars. Thus both representatives realize the same monitored profile family. □

Remark 2.123 (Threshold sweeps). Threshold composition is a statement about repaired profiles and verified representatives. It does not create a strict nested family of filtered complexes or actual transition morphisms between threshold levels.

2.24 Representative morphisms and conditional functoriality

Definition 2.124 (Repaired profile morphism artifact). *Let $f : F \to G$ be filtration-preserving. A* repaired profile morphism artifact *for (i, W, τ) is an actual persistence-module morphism*

$$\overline{f}_{i,W,\tau} : \mathcal{P}_{i,W,\tau}(F) \longrightarrow \mathcal{P}_{i,W,\tau}(G)$$

with declared provenance and compatibility with the intended source map.

A bottleneck matching or a small bottleneck distance is not such a morphism.

Definition 2.125 (Representative morphism datum). *Let $C_{\tau,W}F$ and $C_{\tau,W}G$ be admissible representatives. A* representative morphism datum *consists of an actual filtration-preserving chain map, or a declared morphism in a justified localized category,*

$$C_{\tau,W}f : C_{\tau,W}F \longrightarrow C_{\tau,W}G,$$

such that the induced monitored persistence morphisms commute with the profile isomorphisms and the repaired profile morphisms $\overline{f}_{i,W,\tau}$.

Definition 2.126 (Weak representative functoriality). *A representative construction is* weakly functorial *on a declared subcategory if:*

(i) every object has a verified representative;

(ii) every morphism has a repaired profile morphism artifact and a representative morphism datum;

(iii) identity morphisms are represented by identity data up to the declared equivalence;

(iv) composition is compatible up to a specified coherent representative equivalence.

Proposition 2.127 (Weak functoriality is conditional)**.** *Weak representative functoriality is available only relative to the declared subcategory, actual morphism artifacts, and composition-coherence data. It is not automatic in the general Core.*

Proof. Barcode-level hard deletion is not a globally defined exact functor on all persistence morphisms, and arbitrary representative choices do not determine chain maps. The listed artifacts are therefore additional data, not consequences of object-level profile realization. □

Remark 2.128 (No strict global functor). Appendix B does not assert a strict global endofunctor

$$C_\tau : \mathsf{FiltCh}(k) \longrightarrow \mathsf{FiltCh}(k)$$

or a strict bifunctorial construction $(F, W) \mapsto C_{\tau,W}F$.

2.25 Compatibility with realization and Ext-level use

Definition 2.129 (Realization-compatible representative)**.** *An admissible representative $C_{\tau,W}F$ is* realization-compatible *for a declared realization $\mathcal{R}$ if:*

(i) $\mathcal{R}(C_{\tau,W}F)$ is defined in $D^{\mathrm{b}}(k\text{-mod})$;

(ii) the representative artifact records the exact realization used;

(iii) every alternative representative used interchangeably is supplied with an actual $\mathcal{R}$-equivalence;

(iv) the degree-n amplitude, edge-extractor, edge-realization, naturality, and eligibility predicates required by Appendix C can be evaluated;

(v) all additional structure consumed by the Ext argument is preserved.

Proposition 2.130 (Realization invariance under stronger equivalence)**.** *If two admissible representatives are $\mathcal{R}$-equivalent, then their realized objects are isomorphic in $D^{\mathrm{b}}(k\text{-mod})$, and every isomorphism-invariant derived predicate has the same value on both.*

Proof. This is immediate from the supplied isomorphism in the derived category. □

Remark 2.131 (Profile equivalence is insufficient for Ext). Monitored-persistence equivalence alone does not justify replacing one representative by another inside Ext. The degree-n persistence-to-Ext discharge consumes a realization-compatible representative and the corresponding n-eligible regime of Appendix C.

2.26 Relation to B-Gate$^+$

Definition 2.132 (Detector and representative input to B-Gate$^+$)**.** *A degreewise input contains:*

(i) F, the monitored degree n, W, τ, and the finite monitored scope;

(ii) the exact pre-threshold profile $B_{n,W}(F)$;

(iii) the repaired profile $\mathcal{P}_{n,W,\tau}(F)$;

(iv) the detector stage, mode, certificate status, and mathematical value;

(v) every soundness, completeness, endpoint, coverage, family, excess-bound, and artifact obligation;

(vi) the comparison mode and every exact or metric-safe transport record;

(vii) a verified admissible representative when a filtered, derived, higher-Ext, or tower clause is invoked;

(viii) realization-compatible and n-eligible data when the degree-n Ext clause is invoked;

(ix) canonical atomic statuses and immutable artifact references.

Definition 2.133 (Degreewise topological zero clause)**.** *The primary degree-n topological clause is*

$$\mathrm{Eval}_{n,W,\tau}(F) = 0.$$

It may be discharged directly from the repaired profile or through a complete finite detector certificate at the exact permitted stage. If a verified admissible representative is supplied, the same repaired profile may be read as

$$\mathbf{P}_n(C_{\tau,W}F) = 0$$

through the declared profile isomorphism. The degree-one clause is the specialization $n = 1$.

Definition 2.134 (Degreewise categorical clause via representative)**.** *The categorical clause*

$$\mathrm{Ext}^n(\mathcal{R}_n(C_{\tau,W}F), k) = 0$$

may be invoked only when the representative is certified, realization-compatible, and n-eligible under Appendix C. Appendix B does not prove this implication; it supplies the required representative and detector-side input records.

Remark 2.135 (No representative, no derived clause). Barcode-level evaluation and detector certification can proceed without a filtered representative. A missing or unverified in-scope representative makes a required filtered, derived, or higher-Ext clause `undefined`; it is not a pass and cannot be repaired by metric slack or a favorable detector value.

2.27 Representative towers and Type IV input

Definition 2.136 (Representative tower)**.** *Let*

$$F_0 \longrightarrow F_1 \longrightarrow F_2 \longrightarrow \cdots$$

be a directed tower.

A representative tower *for (i, W, τ) consists of:*

(i) verified representatives $C_{\tau,W}F_n$;

(ii) actual representative morphism data between consecutive levels;

(iii) identity and composition coherence;

(iv) compatibility with the repaired profile system.

Definition 2.137 (Representative tower comparison artifact)**.** *A representative tower comparison artifact is declared data specifying:*

(i) a repaired directed profile system;

(ii) a source profile $\mathrm{ColimProfile}_{i,W,\tau}(F_\bullet)$*;*

(iii) an apex profile $\mathrm{ApexProfile}_{i,W,\tau}(F_\infty)$*;*

(iv) an actual comparison morphism

$$\phi_{i,W,\tau} : \mathrm{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \mathrm{ApexProfile}_{i,W,\tau}(F_\infty);$$

(v) kernel, cokernel, terminal-tameness, monitored-scope, and provenance data required by Chapter 5 and Appendix D.

Remark 2.138 (Type IV input)*.* Representative existence, representative tower existence, and comparison morphism existence are three distinct obligations. The first two do not supply the third.

A barcode matching, small bottleneck distance, expected colimit map, or sequence of barcode multisets is not an actual Type IV comparison artifact.

2.28 Certificate, obligation, and manifest requirements

Definition 2.139 (Appendix B certificate and manifest extension)**.** *A proof object invoking Appendix B records, as applicable:*

(i) source object, monitored degrees, windows, thresholds, and exact pre-/post-threshold profile identities;

(ii) detector stage, endpoint and cofinal policies, detector mode, finite sample or critical set, and exact computation artifacts;

(iii) the full model $\mathfrak{M}$*, including its information, query, encoding, and verifier fields, plus adaptive, randomized, and approximation policies;*

(iv) object-specific query transcripts, finite encodings, verifier traces, cost vectors (q, b, t, a)*, family aggregation, and any model-translation theorem;*

(v) soundness theorem identity and separate completeness, characterization, or coverage theorem identity;

(vi) exhaustive birth/germ records, coverage witnesses, family membership, augmented mesh, and every independently sourced excess lower bound;

(vii) detector certificate and mathematical-value statuses on separate axes;

(viii) the finite achievable cost set, Pareto frontier, coordinate lower envelopes, legacy query projection, upper bounds, lower bounds, sharpness records, no-go records, and any proved escape condition;

(ix) private witness cells, oracle-closedness, advice boundaries, and the deterministic/randomized scope of every lower bound;

(x) composition nodes, content-addressed reuse records, overhead vectors, and Cert2RepInput *handoffs when invoked;*

(xi) one comparison mode with its exact, metric-gap, or named theorem-controlled evidence, discrepancy level, and prohibited readings; metric-zero transport additionally records one parent `MetricUseID` *and the Appendix A separation evidence;*

(xii) actual interleaving morphisms and boundary-safe domain when the robust sandwich is used;

(xiii) *representative identifier, canonical status, lifecycle metadata, monitored profile isomorphisms, and equivalence strength;*

(xiv) *exact source-stage isomorphisms and preservation evidence whenever a detector record is transported through a representative;*

(xv) *window-normalization, realization compatibility, and degree-n eligibility when invoked;*

(xvi) *actual representative morphisms and coherence when functoriality or a tower is invoked;*

(xvii) *actual tower comparison artifacts when Type IV is invoked;*

(xviii) *claim status, dependency closure, repair ancestry, schema versions, artifact hashes, and canonical atomic status for every obligation.*

Definition 2.140 (Appendix B non-scalar obligation bundle)**.** *The non-scalar bundle includes:*

(i) *detector applicability and exact source-stage typing;*

(ii) *endpoint and window-cofinal conventions;*

(iii) *soundness and completeness theorem applicability;*

(iv) *birth/germ exhaustiveness, coverage, family membership, and excess-bound provenance;*

(v) *representative existence and monitored profile compatibility;*

(vi) *equivalence strength and detector-preservation scope;*

(vii) *realization compatibility and degree-n eligibility when used;*

(viii) *actual morphisms and composition coherence when used;*

(ix) *tower comparison existence when Type IV is used;*

(x) *claim status, manifest consistency, and artifact identity.*

None of these obligations is a numerical metric-ledger element.

Remark 2.141 (Appendices F and G handoff)*.* Appendix F elaborates status, dependency closure, failure routing, and non-compensation. Appendix G serializes the fields and artifact references. A complete record does not prove its mathematical assertions; theorem applicability and artifact validity remain separate checks.

2.29 Appendix B v21 proof-package closure

Definition 2.142 (Appendix B v21 proof-package closure record)**.** *The Appendix B proof-package closure record confirms, without creating a second theorem authority, that the expanded proofs and Appendix-local supporting results jointly establish the following at their registered strengths:*

(i) *the inherited stencil, birth, germ, coverage, mesh, metric, realization, representative, and right-unbounded results;*

(ii) *the six-stage detector taxonomy and its Interface boundaries;*

(iii) *the certificate model $\mathfrak{M} = (\mathsf{I}, \mathsf{Q}, \mathsf{E}, \mathsf{V})$;*

(iv) vector-valued object and family cost in (q, b, t, a)*;*

(v) a finite Pareto frontier whenever a finite-cost scheme exists;

(vi) model-relative upper bounds, private-cell lower bounds, and the common-model exact query benchmark;

(vii) the adaptive oracle-closed right-unbounded no-go theorem;

(viii) finite composition with content-addressed evidence reuse and explicit overhead;

(ix) the Cert2RepInput *handoff and its nonpromotion boundary;*

(x) the filtered-representative equivalence hierarchy, morphism, realization, and tower boundaries;

(xi) complete non-scalar obligation and manifest fragments for Appendices F and G.

The canonical Main statements remain owned by Chapter 3. Appendix-local Theorem 2.77 is a supporting model-specific theorem and does not duplicate a Main theorem.

Remark 2.143 (Historical v20 package theorem). The v20 summary theorem formerly labeled as the Appendix B package remains available only through the frozen v20 source artifact. It is not reproduced as an independent v21 theorem because v21 separates canonical Main statement ownership from Appendix proof-package closure.

2.30 Explicit exclusions

The following are not asserted in Appendix B.

(i) No finite observation is treated as a complete zero certificate without an applicable completeness theorem.

(ii) No detector theorem is silently moved between profile stages.

(iii) No point-birth theorem is treated as decoration-invariant.

(iv) No zero stencil implies repaired-profile zero without birth, germ, coverage, family-mesh, or another proved completeness route.

(v) No family-mesh theorem is invoked without a strictly smaller mesh than a positive, independently sourced family excess.

(vi) No equality-boundary mesh case is assigned a universal pass.

(vii) No bounded-window cofinal bar is omitted from coverage or excess.

(viii) No retrospective full-barcode enumeration is counted automatically as proof compression.

(ix) No object-relative empty stencil defines a nontrivial family-relative MinCert.

(x) No uniform finite complete detector exists for the unrestricted right-unbounded translation family.

(xi) No finite observed stabilization defeats the right-unbounded no-go theorem without a proved escape condition.

(xii) No bottleneck bound implies pointwise detector-rank, birth-set, germ-set, coverage, family-membership, or certificate-size stability.

(xiii) No scalar bottleneck distance replaces actual interleaving morphisms in the robust sandwich.

(xiv) No whole-profile E_1 functional is counted as finite proof compression.

(xv) No strict global endofunctor $C_\tau : \mathsf{FiltCh}(k) \to \mathsf{FiltCh}(k)$ is asserted.

(xvi) No strict global functor $(F, W) \mapsto C_{\tau,W}F$ is asserted.

(xvii) No uniqueness or hidden canonicality of a representative is asserted.

(xviii) No monitored-profile equivalence is automatically promoted to a filtered quasi-isomorphism, derived equivalence, detector preservation, or realization equivalence.

(xix) No admissible repaired representative automatically transports the standard pre-threshold detector.

(xx) No bare barcode realization is automatically compatible with source geometry, arithmetic, sheaves, categories, or higher realization.

(xxi) No unconditional identity $C_{\tau,W}F = (C_\tau F)|_W$ is asserted.

(xxii) No global exactness, Serre localization, reflector, or global non-expansiveness claim is restored.

(xxiii) No profile idempotence becomes strict filtered-complex idempotence.

(xxiv) No threshold-composition identity creates transition morphisms.

(xxv) No bottleneck matching is treated as a persistence morphism.

(xxvi) No representative morphism follows from object-level representative existence.

(xxvii) No higher-Ext clause is invoked without realization compatibility and degree-n eligibility.

(xxviii) No detector output is identified with a natural higher Ext-edge.

(xxix) No representative tower automatically supplies a colimit, apex, or comparison morphism.

(xxx) No missing detector, representative, morphism, completeness theorem, or artifact is repaired by numerical slack.

(xxxi) No lifecycle vocabulary replaces the canonical audit statuses.

(xxxii) No complete manifest replaces mathematical verification.

(xxxiii) No external-domain theorem or Return Theorem is created here.

(xxxiv) No query count is identified with bit length, verifier time, or artifact footprint.

(xxxv) No scalar MinCert is canonical for v21; the canonical object is a Pareto frontier in one declared certificate model.

(xxxvi) No costs from different information, query, encoding, or verifier models are compared without a proved model translation.

(xxxvii) No per-object cost is substituted for family-level scheme cost.

(xxxviii) No deterministic private-cell lower bound is promoted to a randomized lower bound without a distributional or minimax theorem.

(xxxix) No approximate or randomized zero-looking transcript is an exact zero certificate without explicit error and separation semantics.

(xl) No private-witness or unbounded lower bound is asserted in a model with hidden full-object access or uncharged object-dependent advice.

(xli) No finite query bound implies finite bit or artifact complexity when exact values lack finite encodings.

(xlii) No finite certificate-composition inequality implies global certificate complexity or object-level descent.

(xliii) No Cert2RepInput handoff constructs a representative, diagram, derived object, Ext edge, or Return.

(xliv) No Interface-stage probe is consumed without the actual Appendix DG objects and the zero-reflection theorem required by its declared stage.

2.31 P1 completion criteria and G21 contribution

Appendix B is mathematically complete for the P1 construction stage only when all of the following are artifact-backed:

(i) regression of every inherited v20 detector and representative result at unchanged strength;

(ii) one fully declared certificate model exhibiting a nontrivial finite family-cost upper bound;

(iii) one information-model-relative lower bound with oracle/advice boundary;

(iv) explicit adaptive/nonadaptive and deterministic/randomized policies;

(v) separate query and bit-complexity accounting;

(vi) compatibility with the right-unbounded no-go theorem;

(vii) the replayable shared-query composition benchmark with overhead and reuse accounting;

(viii) one typed Cert2RepInput handoff whose strength does not exceed the certified source-stage predicate;

(ix) unique Main statement ownership and no duplicate theorem authority;

(x) a complete Appendix G-compatible manifest fragment.

Theorem 2.77 closes the common-model upper/lower requirement locally. This appendix contributes principally to G21-2, G21-6, G21-8, and G21-9. It does not by itself pass Core-DG closure, object-level globalization, or unified external Return gates.

2.32 Provenance and status

The barcode threshold operation, fixed window-first order, four comparison modes, discrepancy-level separation, sharp threshold-gap repair, repaired-zero separation, and metric-safe zero transport are supplied by Main Chapter 2 and Appendix A. The v20 Appendix B source supplies the inherited finite-detector and filtered-representative theorem pack at its frozen strength. The v21 additions in this appendix are the certificate-model and complexity architecture, typed finite-probe schema, information-model lower bounds, composition calculus, and certificate-to-representative handoff.

2.32.0.1 Canonical Main-owned statements with expanded proofs here.

The inherited detector statements correspond to Main Chapter 3 Proposition 3.58; Theorems 3.59, 3.62, 3.66, 3.68, 3.74, 3.75, 3.79, 3.90, and 3.97; and their displayed lemmas, propositions, and corollaries. The new v21 certificate-model statements correspond to Main Definitions 3.41–3.50, 3.84–3.85, 3.92, 3.99–3.100, and 3.103, together with Main Theorems 3.52, 3.86, 3.93, 3.101, and 3.104. These Main identities are not re-registered here.

2.32.0.2 Appendix-local supporting theorem.

Theorem 2.77 proves an exact query complexity equality for one rational private-cell barcode family under one fixed oracle-closed model. It is narrower than the Main general lower-bound schema and supplies a concrete G21-6 witness.

2.32.0.3 Migration status.

The v20 mathematical results are inherited. The four-coordinate certificate model and Pareto complexity are strengthened successor material. The scalar v20 MinCert statement is inherited as the exact nonadaptive stencil query projection. The Pareto formulation is a distinct strengthened successor; a model-independent scalar interpretation is a prohibited stronger reading, not a mutation of the v20 theorem. Diagram and descent mathematics are relocated to Appendix DG. The v20 source remains immutable and directly invocable at its original scope.

2.32.0.4 Construction-stage status.

This artifact is a P1 mathematical draft and expanded-proof package. Final canonical UIDs, MigrationIDs, hashes, release verdicts, and Claim Register promotion are deferred until all Appendix artifacts are frozen and the final v21 governance layer is generated.

3 Appendix C: Eligible Realization, Persistence-to-Diagram Interfaces, and Higher Ext Edges

3.1 Scope, canonical ownership, migration, and v21 dual-Core boundary

3.1.0.1 Normative status and theorem ownership.

Appendix C is the expanded-proof, finite-linear-interface, eligibility-calculus, counterexample, record-schema, and regression owner for Main Chapter 4 of AK-HPDST v21.0.0, the *Finite Globalization, Unified Bridge Calculus, and Certificate Complexity Release*. Whenever a result is canonically stated in the Main, the Main owns its theorem identity. This appendix reproduces it only as an expanded proof, a typed specialization, or an implementation record under that identity; it creates no competing theorem UID. Appendix-local identities are reserved for lemmas, counterexamples, finite-linear refinements, and migration statements not duplicated in the Main.

The immutable v20 Appendix C remains directly invocable from its frozen source artifact at its original strength. The v21 migration is componentwise:

(i) the field-coefficient higher Ext edge, the inherited degree-one bridge, standard degree-n eligibility, one-way discharge, zero-reflection boundary, natural nonfaithful-edge counterexample, and representative-invariance discipline are `inherited`;

(ii) the two-axis realization classification, endpoint-complete atom grid, finite linear-quiver equivalence, decorated-barcode reconstruction, enriched projective/Ext signature, and finite-linear P2DG handoff are `strengthened` successors with v21 Main identities;

(iii) the v20 Appendix-local package-theorem form is not repeated as an independent authority. Its mathematical components remain inherited, while v21 records their closure through an operational proof-package record. This is a governance repair, not a rejection of the v20 mathematics;

(iv) arbitrary finite categories, branching posets, general generators, general cone reconstruction, Morita–Alexandrov realization, effective descent, and object-level globalization are assigned to Appendix DG. The disposition modifier is `relocated`; it is not a semantic migration status.

The canonical project acronym is AK-HPDST. The legacy transposed acronym AK-HDPST may occur in historical artifacts, but it has no normative use in this appendix.

Appendix C consumes:

(i) the barcode, threshold-gap, repaired-zero, comparison-mode, and discrepancy-level contracts of Appendix A, including its P2DG input;

(ii) the finite detector, certificate-model, proof-complexity, and filtered-representative contracts of Appendix B, including Cert2RepInput;

(iii) the canonical Main Chapter 4 statements for finite linear reconstruction and eligible Ext edges.

Its four non-interchangeable responsibilities are:

(i) classify persistence-to-diagram and persistence-to-derived constructions by provenance and semantic fidelity;

(ii) prove the exact finite-linear $\mathsf{Pers}_k^\Gamma(W) \simeq \mathrm{Rep}_k(A[m])$ interface on an endpoint-complete enriched atom grid;

(iii) prove and audit the conditional degree-n persistence-to-Ext discharge at exactly its eligible strength;

(iv) hand actual finite-linear diagram objects and morphisms to Appendix DG without importing the general finite-category, Morita–Alexandrov, cone-probe, or descent theorems owned there.

The central v21 rules are:

> A Cert2RepInput record may nominate a source or representative candidate, but it constructs neither a persistence module, a finite quiver object, nor a derived realization. Each promoted object requires its own construction and compatibility proof.
>
> Restriction of an actual Γ-constructible persistence module is a functorial exact finite-linear realization. Encoding only a barcode isomorphism class may be lossless on objects while remaining `objectwise_only` on morphisms.
>
> The field-coefficient isomorphism
>
> $$\mathrm{Ext}^n(A, k) \cong \mathrm{Hom}_k(H^{-n}(A), k)$$
>
> is unconditional derived algebra. The persistence-to-derived identification
>
> $$H^{-n}(A_{n;F,W,\tau}) \cong \mathrm{Edge}_n(\mathrm{Eval}_{n,W,\tau}(F))$$
>
> is a separately certified realization interface. Neither statement alone is an object-level identification $\mathrm{PH}_n \cong \mathrm{Ext}^n$.

Every consumed comparison records exactly one of

$$\texttt{exact}, \quad \texttt{metric\_gap\_certified},$$
$$\texttt{theorem\_controlled\_nonmetric}, \quad \texttt{not\_compared}.$$

No mode silently promotes to another, and no barcode- or persistence-level comparison is retyped as a finite-diagram or derived comparison without a named theorem and actual data at the target level.

Remark 3.1 (Conservative-evolution boundary)*.* For $n = 1$, the degree-one eligible bridge is inherited unchanged. For every $n \geq 1$, v21 retains the standard amplitude policy

$$A_{n;F,W,\tau} \in D^{[-n,0]}(k\text{-mod})$$

while recording that the canonical field-coefficient Ext edge itself needs no such amplitude restriction.

The new finite-linear P2DG theorem is an exact equivalence on the fixed endpoint-complete Γ-constructible ambient category. On the full subcategory of repaired profiles it remains full, faithful, conservative, zero-reflecting, and lossless on objects; it preserves and reflects every ambient exact sequence whose terms and relevant kernels and cokernels remain inside that subcategory. No abelian, Serre, or extension-closed structure of the repaired subcategory is inferred. It is not a full faithful embedding of all constructible persistence modules on all grids into general Core-DG.

Remark 3.2 (Division of responsibility)*.* Appendix A owns metric transport and P2DG input readiness. Appendix B owns detector completeness, certificate complexity, and candidate-to-representative handoff. Appendix C owns the actual finite-linear persistence-to-quiver interface and eligible Ext edge. Appendix DG owns arbitrary finite categories, general generators and cones, Morita–Alexandrov reconstruction, effective descent, and global zero/isomorphism certification. Appendix F owns proof-DAG semantics, and Appendix G owns canonical serialization and replay.

Remark 3.3 (No external-domain upgrade)*.* Nothing in this appendix proves an arithmetic, PDE, geometric, motivic, Fukaya-categorical, cryptographic, or other external-domain theorem. Every output here is an internal persistence, finite-linear-diagram, finite-signature, or typed Ext-vanishing Return. External semantics require the separately proved Problem-Interface and Return chain.

3.2 Two-axis realization classification and verdict independence

Definition 3.4 (Two-axis realization classification)**.** *Every construction from a persistence source to a finite diagram or derived target records two independent fields.*

The construction-provenance axis *is exactly one of*

$$\mathtt{natural\_realization}, \quad \mathtt{synthetic\_encoding}, \quad \mathtt{target\_shaped}.$$

A natural realization is fixed at scope level from source semantics and has a declared action on morphisms or an explicit declaration that only objects are handled. A synthetic encoding is a mathematically valid encoding of selected source data without an independent theorem identifying the target with the intended Core or external object. A target-shaped construction is chosen or modified so that the desired verdict is built into its definition.

The semantic-fidelity axis *is exactly one of*

$$\mathtt{purpose\_faithful}, \quad \mathtt{purpose\_preserving\_only},$$
$$\mathtt{observation\_only}, \quad \mathtt{not\_established}.$$

The label `purpose_faithful` *requires a theorem proving both preservation and reflection of the declared predicate or invariant on the stated source class. It does not imply full faithfulness or essential surjectivity of the underlying functor.*

Definition 3.5 (Action mode and audit-use class)**.** *Every realization record separately declares one action mode*

$$\texttt{full\_faithful}, \quad \texttt{functorial}, \quad \texttt{partially\_functorial}, \quad \texttt{objectwise\_only},$$

and one audit-use class

$$\texttt{principal}, \qquad \texttt{validation}, \qquad \texttt{toy}.$$

The audit-use class changes neither the mathematical theorem nor the provenance axis. In particular, a validation object may be natural or synthetic, and a toy object does not become principal by passing a local computation.

Policy 3.6 (No verdict-conditioned realization)**.** *The grid, representative, realization functor, edge extractor, target category, and comparison policy are fixed from source semantics and declared scope before the target zero/nonzero verdict is evaluated. A construction selected after inspecting the desired conclusion is* `target_shaped` *and is inadmissible as independent Bridge or Return evidence unless a separate theorem removes the verdict dependence.*

Remark 3.7 (Natural restriction versus objectwise barcode encoding). Evaluation of an actual persistence module on a fixed enriched grid is a functor on actual persistence morphisms. Choosing an interval decomposition of a barcode isomorphism class and writing down a direct sum of interval representations is generally objectwise unless a morphism action is supplied. The two routes must not share one action-mode label.

3.3 Endpoint-complete atom grids

Definition 3.8 (Finite endpoint set)**.** *Let P be a finite persistence profile on a right-open window $W = [u, u')$, where $u' \in \mathbb{R} \cup \{+\infty\}$. A finite set*

$$K = \{c_0 < \cdots < c_r\} \subseteq W$$

is endpoint-complete for P when it contains u, every finite geometric birth and death value lying in W, and every additional critical value at which the declared module or morphism data change. For a family or a morphism, endpoint completeness is simultaneous for every source, target, and comparison object in scope.

Definition 3.9 (Atom decomposition and enriched grid)**.** *Given W and endpoint-complete K, decompose W into the nonempty ordered atoms formed by singleton critical values, open intervals between successive critical values, the possible initial atom, and the terminal atom. Write*

$$\mathcal{A}_\Gamma = (A_0 < \cdots < A_m).$$

Choose $\gamma_j \in A_j$ when A_j is not a singleton and put $\gamma_j = c_i$ on a singleton atom. The ordered probe set

$$\Gamma = \{\gamma_0 < \cdots < \gamma_m\}$$

together with the atom dictionary is the enriched finite grid. The monotone atom projection is

$$q_\Gamma : W \longrightarrow [m] = \{0 < \cdots < m\}, \qquad q_\Gamma(A_j) = j.$$

Remark 3.10 (Endpoint decoration is retained). A list of geometric critical values alone can lose open/closed decoration. The enriched grid separates singleton endpoint atoms from adjacent open atoms, so left- and right-germ behavior remains recoverable.

Definition 3.11 (Terminal sentinel policy). *On a bounded window, the final nonempty atom below the excluded endpoint is a bounded terminal atom; no value at the excluded endpoint is invented. On a terminal window $W = [u, +\infty)$, the last unbounded atom is represented by one sentinel probe and the record declares* `cofinal_tail`*. The module must be constant on that tail. A finite sentinel without the cofinal-tail proof does not certify an infinite bar.*

Definition 3.12 (Γ-constructible profile). *A persistence module P on W is Γ-constructible when every $P(x)$ is finite-dimensional and every structure map between two points of the same atom is an isomorphism. On a declared cofinal-tail atom, this is the required eventual-constancy condition.*

Definition 3.13 (Grid completeness record). *A grid completeness record contains:*

(i) source profile, window, coefficient field, and profile stage;

(ii) K, $\mathcal{A}_\Gamma$, Γ, and the atom dictionary;

(iii) proof that every endpoint and module/morphism critical value in scope is present;

(iv) endpoint-decoration and singleton/open-atom conventions;

(v) bounded-terminal or cofinal-tail policy;

(vi) proof of Γ-constructibility for every source and target;

(vii) the action mode, comparison mode, and discrepancy level;

(viii) immutable artifacts and prohibited stronger readings.

An incomplete endpoint list or unproved tail policy yields `undefined`*; it is not reported as a complete but merely lossy reconstruction.*

Remark 3.14 (Non-circular endpoint acquisition). Endpoint completeness may come from a previously reconstructed barcode or from an independent critical-data theorem, filtration rule, endpoint lattice, or source artifact. In the first case, the finite-grid theorem is a valid serialization and consistency theorem but not by itself a discovery or proof-compression result.

3.4 Finite linear-quiver equivalence

Definition 3.15 (Finite linear-quiver category). *Let $A[m]$ be the linearly oriented quiver*

$$0 \longrightarrow 1 \longrightarrow \cdots \longrightarrow m,$$

and define

$$\mathrm{Rep}_k(A[m]) := \mathrm{Fun}([m], \mathsf{Vect}_k^{\mathrm{fd}}).$$

Kernels, cokernels, and exactness are computed pointwise.

Definition 3.16 (Grid restriction and step realization). *Define*

$$\mathrm{Res}_\Gamma(P) := \big(P(\gamma_0) \to \cdots \to P(\gamma_m)\big) \in \mathrm{Rep}_k(A[m]),$$

and, conversely,

$$\mathrm{Step}_\Gamma(V) := q_\Gamma^* V.$$

Both constructions act on actual morphisms by evaluation and pullback.

Lemma 3.17 (Expanded proof of Main Lemma 4.14: canonical atom transport)**.** *Let P be Γ-constructible. For $x \in A_j$, use the structure map from γ_j to x, or its inverse when the order is reversed, to define*

$$\theta_{P,x} : P(\gamma_j) \xrightarrow{\sim} P(x).$$

Then these maps form a natural isomorphism

$$\theta_P : \mathrm{Step}_\Gamma \mathrm{Res}_\Gamma(P) \xrightarrow{\sim} P,$$

natural in actual persistence morphisms.

Proof. Every map used lies inside one atom and is therefore invertible. For $x \leq y$, functoriality of the persistence structure maps, with cancellation of the within-atom isomorphisms, gives the parameter-naturality square. Applying the same square to an actual persistence morphism gives naturality in P. □

Theorem 3.18 (Expanded proof of Main Theorem 4.15: finite-grid encoding equivalence)**.** *Let $\mathsf{Pers}_k^\Gamma(W)$ be the full category of Γ-constructible persistence modules on W. Then*

$$\mathrm{Res}_\Gamma : \mathsf{Pers}_k^\Gamma(W) \rightleftarrows \mathrm{Rep}_k(A[m]) : \mathrm{Step}_\Gamma$$

are quasi-inverse exact equivalences. Hence Res_Γ is full, faithful, conservative, and zero-reflecting on this declared category.

Proof. By construction $\mathrm{Res}_\Gamma \mathrm{Step}_\Gamma(V) = V$ strictly. Lemma 3.17 gives $\mathrm{Step}_\Gamma \mathrm{Res}_\Gamma(P) \cong P$ naturally. A natural transformation between step modules is uniquely determined by its components at the probes, and every natural transformation of finite quiver representations pulls back to one of the step modules. Thus restriction is full and faithful. Pointwise kernels and cokernels of maps between Γ-constructible modules remain constant up to isomorphism on every atom, so the source category is closed under them and both functors preserve them. The equivalence is exact; conservativity and zero-reflection follow. □

Proposition 3.19 (Expanded proof of Main Proposition 4.16: probe-choice invariance)**.** *If Γ and Γ' choose different probes in the same atom decomposition, then within-atom transport gives a natural isomorphism*

$$\mathrm{Res}_\Gamma(P) \xrightarrow{\sim} \mathrm{Res}_{\Gamma'}(P)$$

for every atom-constructible P. Consequently the quiver isomorphism class, rank table, interval multiplicities, and decorated-barcode Return are independent of probe choice inside fixed atoms.

Proof. At node j, use the structure map between $P(\gamma_j)$ and $P(\gamma'_j)$, inverted when necessary. Since both points lie in one atom, the map is an isomorphism. Functoriality gives the commuting squares with adjacent arrows and with every actual persistence morphism. □

Remark 3.20 (Grid refinement is comparison, not literal equality)*.* A refinement changes the indexing quiver. The resulting diagrams are compared through their common step realization or through an explicitly proved subdivision/contraction functor whose inserted arrows are invertible. No literal equality is inferred merely because both grids are complete.

Remark 3.21 (Exactness is not a theorem about hard deletion)*.* The exactness above concerns restriction of actual Γ-constructible persistence modules. It does not make T_τ^{bar} exact. A repaired source supplied only as a barcode isomorphism class has an objectwise finite encoding until an actual module and morphism action are supplied.

Definition 3.22 (Objectwise barcode encoding). *For a finite decorated barcode B whose intervals are unions of atoms, define*

$$\mathrm{GridRep}_\Gamma(B) := \bigoplus_{I \in B} k_{J_\Gamma(I)},$$

where $J_\Gamma(I) \subseteq [m]$ is the consecutive node interval formed by the atoms contained in I. Multiplicity and endpoint decoration are preserved through the atom dictionary. Without a declared action on barcode morphisms, the action mode is `objectwise_only`.

Definition 3.23 (Grid encoding strength record). *The record independently declares:*

(i) object status: `reconstructive` *or* `observational`*;*

(ii) morphism status, exactly one of

$$\texttt{full\_faithful},\quad \texttt{functorial},$$
$$\texttt{partially\_functorial},\quad \texttt{objectwise\_only};$$

(iii) exactness status: `exact`, `theorem_controlled`, *or* `not_asserted`*;*

(iv) barcode status: `endpoint_complete`, `undecorated_only`, *or* `incomplete`*;*

(v) terminal status: `bounded_terminal`, `cofinal_tail`, *or* `unresolved`*;*

(vi) information profile: `lossless`, `feature_faithful`, `observation_only`, *or* `lossy`*;*

(vii) exact retained, reflected, and discarded data.

These fields are independent. In particular, an objectwise encoding may be lossless on objects without having any functorial morphism claim.

Theorem 3.24 (Expanded proof of Main Theorem 4.22: endpoint-complete reconstruction). *Let P be a finite Γ-constructible persistence profile and assume that the grid completeness record contains every finite geometric endpoint and the terminal policy. Then:*

(i) P is reconstructed up to persistence-module isomorphism from $\mathrm{Res}_\Gamma(P)$;

(ii) each decorated barcode interval corresponds bijectively, with multiplicity, to an interval representation on a consecutive node set of $A[m]$;

(iii) the atom dictionary reconstructs geometric endpoints, endpoint decorations, and finite/cofinal terminal type.

Proof. The first assertion is Theorem 3.18. Every interval summand is constant on each atom and has consecutive atom support, so restriction gives an indecomposable interval representation of $A[m]$. Conversely, the union of atoms in a consecutive node interval is one decorated real interval; singleton atoms determine endpoint inclusion. The terminal atom and sentinel policy distinguish bounded cofinal support from a genuine infinite tail. Krull–Schmidt uniqueness in the finite type-A category preserves multiplicities. □

Remark 3.25 (Loss modes). Omitting a critical value, merging a singleton endpoint atom with an adjacent open atom, or omitting the terminal policy can identify distinct persistence profiles. Exact computation on an incomplete grid is therefore not a reconstructive Return.

3.5 Rank and Möbius reconstruction

Definition 3.26 (Finite rank table). *For $V \in \mathrm{Rep}_k(A[m])$ and $0 \le i \le j \le m$, define*

$$r_V(i,j) := \mathrm{rk}(V_i \longrightarrow V_j),$$

where the map is the composite along the quiver. Set

$$r_V(-1,j) = r_V(i,m+1) = r_V(-1,m+1) = 0.$$

Theorem 3.27 (Expanded proof of Main Theorem 4.25: finite Möbius multiplicity formula). *The multiplicity $\mu_V(i,j)$ of the interval representation supported on $[i,j]$ is*

$$\begin{aligned} \mu_V(i,j) = r_V(i,j) &- r_V(i-1,j) \\ &- r_V(i,j+1) + r_V(i-1,j+1). \end{aligned}$$

All $\mu_V(i,j)$ are nonnegative integers, and the collection determines V up to isomorphism.

Proof. If

$$V \cong \bigoplus_{a \le b} k_{[a,b]}^{\oplus \mu_V(a,b)},$$

then

$$r_V(i,j) = \sum_{a \le i,\, b \ge j} \mu_V(a,b).$$

Möbius inversion on the product of two finite chains gives the displayed second-difference formula. Nonnegativity follows because the values are interval multiplicities, and uniqueness of the type-A interval decomposition proves completeness. ☐

Corollary 3.28 (Expanded proof of Main Corollary 4.26: decorated-barcode Return). *Under Theorem 3.24, the finite rank table of $\mathrm{Res}_\Gamma(P)$, the atom dictionary, and the terminal policy reconstruct the complete decorated barcode of P.*

Proof. Apply Theorem 3.27 and translate each consecutive node interval back to the union of its atoms. ☐

Remark 3.29 (Rank Return is not chain-level Return). The rank table returns the persistence-module isomorphism class and decorated barcode. It does not return a filtered chain complex, chosen chain map, derived realization, or external-domain object.

3.6 Finite projective and quiver-Ext signature

Definition 3.30 (Projective and simple probes). *For $0 \le i \le m$, let P_i be the indecomposable projective supported on $[i,m]$, and let S_i be the simple representation at node i. For $i < m$,*

$$0 \longrightarrow P_{i+1} \longrightarrow P_i \longrightarrow S_i \longrightarrow 0$$

is a projective resolution, while $S_m = P_m$ is projective.

Definition 3.31 (Enriched finite projective/Ext signature). *For $V \in \mathrm{Rep}_k(A[m])$, the signature $\Sigma_\Gamma^{\mathrm{proj/Ext}}(V)$ contains:*

(i) the spaces $\mathrm{Hom}(P_i, V)$*;*

(ii) the precomposition maps induced by $P_{i+1} \to P_i$;

(iii) the local spaces $\operatorname{Hom}(S_i, V)$*;*

(iv) the groups $\operatorname{Ext}^1_{\operatorname{Rep}_k(A[m])}(S_i, V)$*;*

(v) identification and artifact data relating them to the grid diagram.

The word enriched *includes the actual linear maps, not only their dimensions.*

Proposition 3.32 (Expanded proof of Main Proposition 4.30: local projective/Ext exact sequence)**.** *For $i < m$, there is a natural exact sequence*

$$0 \to \operatorname{Hom}(S_i, V) \to V_i \to V_{i+1} \to \operatorname{Ext}^1(S_i, V) \to 0.$$

Moreover,

$$\operatorname{Hom}(P_i, V) \cong V_i, \qquad \operatorname{Ext}^p_{\operatorname{Rep}_k(A[m])}(-,-) = 0 \quad (p \geq 2).$$

Proof. Apply $\operatorname{Hom}_{\operatorname{Rep}_k(A[m])}(-, V)$ to the projective resolution of S_i. Projectivity gives the exact sequence and vanishing beyond degree one. Yoneda evaluation identifies $\operatorname{Hom}(P_i, V)$ with V_i, and the induced map is the quiver arrow $V_i \to V_{i+1}$. The linearly oriented type-A path algebra is hereditary. $\square$

Theorem 3.33 (Expanded proof of Main Theorem 4.31: enriched-signature reconstruction)**.** *The enriched projective part of* $\Sigma^{\mathrm{proj/Ext}}_{\Gamma}(V)$ *reconstructs V functorially:*

$$V_i \cong \operatorname{Hom}(P_i, V),$$

and the arrow $V_i \to V_{i+1}$ is the precomposition map induced by $P_{i+1} \to P_i$. Consequently the full enriched signature reconstructs the finite diagram and, on an endpoint-complete grid, the decorated barcode. The dimensions of the simple Hom and Ext groups alone are not complete.

Proof. The Yoneda identifications recover every vertex space and every arrow, hence the complete quiver representation. The barcode conclusion follows from Corollary 3.28. Forgetting the actual projective-probe maps can identify nonisomorphic diagrams having the same local dimensions, so dimension-only data are not reconstructive. $\square$

Remark 3.34 (Two distinct Ext theories). The quiver groups $\operatorname{Ext}^1_{\operatorname{Rep}_k(A[m])}(S_i, V)$ measure local arrow cokernels. The field-valued groups $\operatorname{Ext}^n_{D^b(k\text{-mod})}(A_n, k)$ use the monitored persistence degree and a separate derived realization. They are not identified. Hereditary vanishing in the quiver category does not trivialize the higher eligible Ext-edge.

3.7 Derived category setting and canonical higher Ext algebra

Fix a field k. Let

$$D^{\mathrm{b}}(k\text{-mod})$$

denote the bounded derived category of finite-dimensional k-vector spaces, equipped with the standard t-structure.

Definition 3.35 (Bounded derived object)**.** *An object*

$$A \in D^{\mathrm{b}}(k\text{-}mod)$$

is represented by a bounded cochain complex of finite-dimensional k-vector spaces. Its cohomology objects are denoted by $H^j(A)$.

Definition 3.36 (Amplitude)**.** *An object A has amplitude contained in $[a, b]$ if*

$$H^j(A) = 0 \qquad (j \notin [a, b]).$$

We write

$$A \in D^{[a,b]}(k\text{-}mod).$$

Definition 3.37 (Inherited two-term eligible amplitude)**.** *An object has the inherited degree-one eligible amplitude when*

$$A \in D^{[-1,0]}(k\text{-}mod).$$

This is the specialization of the standard degree-n policy to $n = 1$.

Definition 3.38 (Ext^1 in the derived category)**.** *For $A \in D^{\mathrm{b}}(k\text{-}mod)$, define*

$$\mathrm{Ext}^1(A, k) := \mathrm{Hom}_{D^{\mathrm{b}}(k\text{-}mod)}(A, k[1]).$$

Definition 3.39 (Higher derived Ext)**.** *For $n \geq 0$ and $A \in D^{\mathrm{b}}(k\text{-}mod)$, define*

$$\mathrm{Ext}^n(A, k) := \mathrm{Hom}_{D^{\mathrm{b}}(k\text{-}mod)}(A, k[n]).$$

Lemma 3.40 (Semisimplicity over a field)**.** *The abelian category of finite-dimensional k-vector spaces is semisimple. Consequently,*

$$\mathrm{Ext}^p_k(V, k) = 0 \qquad (p > 0)$$

for every finite-dimensional vector space V.

Proof. Every vector space over a field is projective and injective, so every short exact sequence splits and all positive derived Ext groups vanish. □

Lemma 3.41 (Expanded proof of Main Lemma 4.52: formality over a field)**.** *Every $A \in D^{\mathrm{b}}(k\text{-}mod)$ admits an isomorphism*

$$A \cong \bigoplus_{j \in \mathbb{Z}} H^j(A)[-j].$$

Such a direct-sum isomorphism generally depends on choices and is not used as the source of the canonical naturality below.

Proof. Choose splittings of cycles, boundaries, and cohomology in a bounded complex representing A. The resulting cohomology complex with zero differential is quasi-isomorphic to A. The chosen splittings need not be functorial. □

Theorem 3.42 (Expanded proof of Main Theorem 4.50: canonical higher field-coefficient Ext edge)**.** *Let $A \in D^{\mathrm{b}}(k\text{-}mod)$ and $n \geq 0$. There is a canonical isomorphism, natural in A,*

$$\mathrm{Ext}^n(A, k) \xrightarrow{\sim} \mathrm{Hom}_k(H^{-n}(A), k).$$

No amplitude restriction is required for this algebraic theorem.

Proof. The standard t-structure supplies the natural hyper-Ext spectral sequence

$$E_2^{p,q} = \mathrm{Ext}^p_k(H^{-q}(A), k) \quad \Longrightarrow \quad \mathrm{Ext}^{p+q}(A, k).$$

By Lemma 3.40, all terms with $p > 0$ vanish. For total degree n, the unique possibly nonzero term is

$$E_2^{0,n} = \mathrm{Hom}_k(H^{-n}(A), k).$$

There are no nontrivial incoming or outgoing differentials and the induced filtration on $\mathrm{Ext}^n(A, k)$ has one nonzero graded piece. The resulting edge map is therefore a canonical isomorphism. Naturality follows from the naturality of the spectral sequence. □

Corollary 3.43 (Expanded proof of Main Corollary 4.51: higher edge vanishing and reflection)**.** *For every morphism $f : A \to A'$, the canonical square*

$$\begin{array}{ccc} \operatorname{Ext}^n(A', k) & \longrightarrow & \operatorname{Ext}^n(A, k) \\ \big\downarrow{\sim} & & \big\downarrow{\sim} \\ \operatorname{Hom}_k(H^{-n}(A'), k) & \longrightarrow & \operatorname{Hom}_k(H^{-n}(A), k) \end{array}$$

commutes. Both horizontal maps are contravariantly induced by f.

Proof. This is the naturality assertion of Theorem 3.42. □

Corollary 3.44 (Higher edge vanishing criterion)**.** *For $A \in D^{\mathrm{b}}(k\text{-mod})$,*

$$H^{-n}(A) = 0 \quad \Longrightarrow \quad \operatorname{Ext}^n(A, k) = 0.$$

Conversely, because $H^{-n}(A)$ is finite-dimensional,

$$\operatorname{Ext}^n(A, k) = 0 \quad \Longrightarrow \quad H^{-n}(A) = 0.$$

Proof. The first direction follows directly from Theorem 3.42. For the converse, a nonzero finite-dimensional vector space has a nonzero linear functional, so $\operatorname{Hom}_k(H^{-n}(A), k) = 0$ implies $H^{-n}(A) = 0$. □

Corollary 3.45 (Two-term splitting)**.** *If $A \in D^{[-1,0]}(k\text{-mod})$, then there exists an isomorphism*

$$A \cong H^{-1}(A)[1] \oplus H^0(A).$$

The displayed splitting is explanatory and need not be natural.

Proof. Apply Lemma 3.41 and the amplitude restriction. □

Theorem 3.46 (Expanded proof of Main Theorem 4.53: canonical two-term Ext^1-edge identification)**.** *Let $A \in D^{[-1,0]}(k\text{-mod})$. Then there is a canonical isomorphism, natural in A,*

$$\operatorname{Ext}^1(A, k) \xrightarrow{\sim} \operatorname{Hom}_k(H^{-1}(A), k).$$

Proof. The standard t-structure gives the canonical truncation triangle

$$H^{-1}(A)[1] \longrightarrow A \longrightarrow H^0(A) \xrightarrow{+1} H^{-1}(A)[2].$$

Apply the contravariant cohomological functor

$$\operatorname{Hom}_{D^{\mathrm{b}}(k\text{-mod})}(-, k[1]).$$

The resulting long exact sequence contains

$$\begin{aligned} \operatorname{Hom}_D(H^0(A), k[1]) &\longrightarrow \operatorname{Hom}_D(A, k[1]) \\ &\longrightarrow \operatorname{Hom}_D(H^{-1}(A)[1], k[1]) \\ &\longrightarrow \operatorname{Hom}_D(H^0(A), k[2]), \end{aligned}$$

where $D = D^{\mathrm{b}}(k\text{-mod})$. Semisimplicity gives

$$\operatorname{Hom}_D(H^0(A), k[1]) = 0, \qquad \operatorname{Hom}_D(H^0(A), k[2]) = 0.$$

Hence the middle arrow is an isomorphism, and

$$\operatorname{Hom}_D(H^{-1}(A)[1], k[1]) \cong \operatorname{Hom}_k(H^{-1}(A), k).$$

Naturality follows from the canonical truncation triangle. □

Corollary 3.47 (Expanded proof of Main Corollary 4.54: naturality square)**.** *For every morphism $f : A \to A'$ in $D^{[-1,0]}(k\text{-mod})$, the square*

$$\begin{array}{ccc} \mathrm{Ext}^1(A',k) & \longrightarrow & \mathrm{Ext}^1(A,k) \\ \downarrow\sim & & \downarrow\sim \\ \mathrm{Hom}_k(H^{-1}(A'),k) & \longrightarrow & \mathrm{Hom}_k(H^{-1}(A),k) \end{array}$$

commutes.

Proof. This is the naturality assertion of Theorem 3.46. ☐

Corollary 3.48 (Expanded proof of Main Corollary 4.55: edge vanishing criterion)**.** *If $A \in D^{[-1,0]}(k\text{-mod})$ and $H^{-1}(A) = 0$, then*

$$\mathrm{Ext}^1(A,k) = 0.$$

Proof. Apply Theorem 3.46. ☐

Remark 3.49 (Amplitude is an eligibility policy, not an algebraic necessity)*.* The standard v21 n-eligible regime uses

$$A_{n;F,W,\tau} \in D^{[-n,0]}(k\text{-mod})$$

as a conservative typing and audit policy. Theorem 3.42 itself is valid for every bounded derived object and does not require this amplitude condition.

Remark 3.50 (No reverse information from the algebraic edge)*.* The canonical Ext edge reflects the vector space $H^{-n}(A)$, not the full persistence profile. Reverse recovery of a repaired profile requires a separate zero-reflection theorem for the declared persistence edge.

Remark 3.51 (Why the field hypothesis matters)*.* Over a general coefficient ring, the terms

$$\mathrm{Ext}^p(H^{-q}(A),k) \qquad (p > 0)$$

need not vanish. The spectral sequence need not collapse to one row, so the displayed higher-edge isomorphism requires separate homological hypotheses. Non-field extensions belong to downstream Technical Extension material.

Remark 3.52 (Splitting versus canonical naturality)*.* Lemma 3.41 explains the semisimple shape of the answer, but a chosen splitting is generally noncanonical. The canonical identification used by the Core is the natural edge map of Theorem 3.42.

3.8 Repaired higher-degree input and admissible representatives

Definition 3.53 (Repaired windowed profile)**.** *For $F \in \mathsf{FiltCh}(k)$, degree i, right-open window W, and $\tau > 0$, define*

$$\mathrm{Eval}_{i,W,\tau}(F) := T_\tau^{\mathrm{bar}}(\mathsf{W}_W \mathbf{P}_i(F)).$$

This is the repaired post-threshold profile. A Chapter 3 Cert2RepInput *record may point to this source and to a representative candidate, but it is not itself an existence or compatibility proof. Threshold clearance, whenever metric transport is invoked, is measured on the windowed pre-threshold profile*

$$\mathsf{W}_W \mathbf{P}_i(F).$$

Definition 3.54 (Admissible representative)**.** *An object*

$$C_{\tau,W} F \in \mathsf{FiltCh}(k)$$

is an admissible representative on a declared finite monitored degree set when it carries the explicit profile isomorphisms required by Appendix B, Definition 2.97.

Definition 3.55 (Degree-one compatible representative)**.** *An admissible representative is degree-one compatible if it carries an actual isomorphism*

$$\rho_{1;F,W,\tau} : \mathbf{P}_1(C_{\tau,W}F) \xrightarrow{\sim} \mathrm{Eval}_{1,W,\tau}(F).$$

When this condition is part of the record, the notation

$$\mathrm{PH}_1(C_{\tau,W}F) = 0$$

is used only as shorthand for the vanishing of the repaired degree-one profile.

Definition 3.56 (Degree-n compatible representative)**.** *Fix $n \geq 1$. An admissible representative is* degree-n compatible *if it carries an actual isomorphism*

$$\rho_{n;F,W,\tau} : \mathbf{P}_n(C_{\tau,W}F) \xrightarrow{\sim} \mathrm{Eval}_{n,W,\tau}(F).$$

The notation

$$\mathrm{PH}_n(C_{\tau,W}F) = 0$$

is licensed only as shorthand for the vanishing of this declared repaired profile.

Remark 3.57 (Representative, not collapse functor). The notation $C_{\tau,W}F$ denotes a chosen representative artifact. It does not define a global strict collapse functor and does not imply

$$C_{\tau,W}F = (C_\tau F)|_W.$$

Remark 3.58 (Detector stage and representative stage). An admissible representative normally realizes the stage

`repaired_postthreshold`.

It does not automatically preserve detectors at any of

`windowed_prethreshold`, `kernel_defect`,
`cokernel_defect`.

Any detector-to-representative transport requires the exact stage-compatible data and preservation evidence of Appendix B; representative admissibility alone is insufficient.

3.9 Higher realization data and edge extractors

3.9.1 Inherited degree-one realization interface

Definition 3.59 (Realization functor)**.** *A realization functor or realization procedure is a declared functor*

$$\mathcal{R} : \mathcal{C}_{\mathrm{rep}} \longrightarrow D^{\mathrm{b}}(k\text{-}mod),$$

where $\mathcal{C}_{\mathrm{rep}}$ is a declared representative subcategory. Its domain, version, morphism policy, and representative-equivalence policy are part of the record.

Definition 3.60 (Realization-compatible representative)**.** *A degree-one compatible representative $C_{\tau,W}F$ is realization-compatible when:*

(i) it belongs to the domain of $\mathcal{R}$;

(ii) $\mathcal{R}(C_{\tau,W}F)$ is well-defined;

(iii) every alternative representative used interchangeably is related by an actual realization equivalence;

(iv) the inherited amplitude and edge-realization conditions can be audited.

Definition 3.61 (Eligible realized object)**.** *For a realization-compatible degree-one representative define*

$$A_{F,W,\tau} := \mathcal{R}(C_{\tau,W}F).$$

Remark 3.62 (Why realization compatibility is required)*.* Persistence profiles and realized derived objects live in different categories. Profile equivalence alone does not identify realizations or their Ext groups. A fixed representative commitment or an actual realization equivalence is therefore mandatory.

Definition 3.63 (Residual degree-one persistence edge)**.** *A residual degree-one edge extractor is a declared vector-space-valued assignment*

$$\mathrm{Edge}_1 : \mathfrak{P}_1 \longrightarrow \mathsf{Vect}_k$$

on a declared class containing the zero profile, with

$$\mathrm{Edge}_1(0) = 0.$$

No faithfulness, conservativity, or zero-reflection is inferred.

Definition 3.64 (Edge-realization condition)**.** *An inherited degree-one edge-realization certificate is an actual isomorphism*

$$\epsilon_{F,W,\tau} : H^{-1}(A_{F,W,\tau}) \xrightarrow{\sim} \mathrm{Edge}_1(\mathrm{Eval}_{1,W,\tau}(F)).$$

The record declares whether this is objectwise or belongs to a natural family.

Remark 3.65 (Persistence-to-vector-space type correction)*.* The repaired profile is a persistence module, while $H^{-1}(A_{F,W,\tau})$ is a finite-dimensional vector space. They are not directly identifiable. The declared edge extractor is the required type-changing interface.

Definition 3.66 (Eligible regime)**.** *The inherited degree-one eligible regime consists of:*

(i) the repaired degree-one evaluation record and any governing Cert2RepInput *handoff;*

(ii) a degree-one compatible admissible representative;

(iii) a realization-compatible $\mathcal{R}$;

(iv) the object $A_{F,W,\tau}$, together with its two-axis realization classification, action mode, and audit-use class;

(v) an amplitude certificate

$$A_{F,W,\tau} \in D^{[-1,0]}(k\text{-}mod);$$

(vi) a fixed residual edge extractor with $\mathrm{Edge}_1(0) = 0$*;*

(vii) the edge-realization certificate;

(viii) the field coefficient declaration;

(ix) representative- and realization-change policies;

(x) complete claim, manifest, proof, and artifact records.

Eligibility is independent of whether the repaired profile is zero or nonzero. Missing required data make an invoked categorical clause `undefined`*.*

Remark 3.67 (Meaning of the edge-realization condition). Amplitude alone does not connect persistence to derived cohomology. The edge-realization isomorphism is an independent, non-scalar obligation that records exactly which vector-space edge is realized.

Remark 3.68 (No raw persistence-to-derived shortcut). The edge extractor consumes the repaired post-threshold profile, not the raw unwindowed persistence module. No raw barcode, detector number, or metric budget supplies the edge-realization certificate.

3.9.2 Higher realization interface

Remark 3.69 (Eligibility is a gate condition). Eligibility licenses use of the inherited categorical edge. Missing eligibility is neither categorical vanishing nor rejection; an invoked incomplete clause is `undefined`.

Definition 3.70 (Declared repaired-profile class)**.** *For each invoked degree $n \geq 1$, a repaired-profile class*

$$\mathfrak{P}_n$$

is a declared category or class of constructible repaired degree-n profiles, together with:

(i) its object and morphism scopes, including the zero profile;

(ii) its profile stage;

(iii) endpoint, window-cofinal, and threshold conventions;

(iv) a membership test or proof obligation;

(v) immutable scope and policy identifiers.

The class is fixed independently of the desired categorical verdict.

Definition 3.71 (Higher realization functor)**.** *A degree-n realization is a declared functor or procedure*

$$\mathcal{R}_n : \mathcal{C}^{(n)}_{\mathrm{rep}} \longrightarrow D^{\mathrm{b}}(k\text{-}mod)$$

on a declared representative subcategory. Its domain, morphism policy, version, representative-change policy, and artifact identity are part of the realization record.

Definition 3.72 (Realization-compatible representative)**.** *A degree-n compatible representative is* realization-compatible *when:*

(i) it lies in the domain of $\mathcal{R}_n$;

(ii) the realized object is well-defined;

(iii) every alternative representative used interchangeably is related by an actual realization equivalence in the sense of Appendix B, Definition 2.9;

(iv) the amplitude, edge-realization, and naturality records required below can be audited;

(v) all additional structures consumed by the categorical clause are preserved at the declared strength.

Definition 3.73 (Eligible realized object)**.** *For a realization-compatible representative define*

$$A_{n;F,W,\tau} := \mathcal{R}_n(C_{\tau,W}F).$$

This object must be independently meaningful for the declared realization purpose. It may not be defined solely by encoding the desired detector output unless the record explicitly classifies the construction as synthetic.

Definition 3.74 (Higher edge extractor)**.** *A degree-n edge extractor is a fixed vector-space-valued functor or objectwise assignment*

$$\mathrm{Edge}_n : \mathfrak{P}_n \longrightarrow \mathsf{Vect}_k$$

with declared object and morphism semantics. For the one-way bridge it must satisfy zero compatibility:

$$P = 0 \quad \Longrightarrow \quad \mathrm{Edge}_n(P) = 0.$$

Functoriality, faithfulness, conservativity, and zero-reflection are not inferred from zero compatibility.

Definition 3.75 (Edge-realization certificate)**.** *An edge-realization certificate is an actual isomorphism*

$$\epsilon_{n;F,W,\tau} : H^{-n}(A_{n;F,W,\tau}) \xrightarrow{\sim} \mathrm{Edge}_n(\mathrm{Eval}_{n,W,\tau}(F)).$$

The record states whether this is:

`objectwise` *or* `natural_family`*.*

A natural-family assertion additionally contains the compatible morphism class and the commutative naturality squares.

Definition 3.76 (Non-adaptive policy)**.** *The realization, edge extractor, profile class, detector policy, and edge-realization theorem are* non-adaptive *when their identifiers and scope are fixed before the desired pass/reject value is inspected. An adaptively chosen zero extractor, realization, sample point, or profile class cannot support theorem evidence.*

Definition 3.77 (Standard n-eligible regime)**.** *Fix $n \geq 1$. A standard n-eligible record for $(F; W, \tau)$ contains:*

(i) the repaired degree-n profile, its exact profile identity, and any governing Cert2RepInput *handoff;*

(ii) a degree-n compatible admissible representative;

(iii) a realization-compatible $\mathcal{R}_n$;

(iv) the independently meaningful realized object $A_{n;F,W,\tau}$, together with its two-axis realization classification, action mode, and audit-use class;

(v) the conservative amplitude certificate

$$A_{n;F,W,\tau} \in D^{[-n,0]}(k\text{-}mod),$$

or a separately proved and registered admissible replacement;

(vi) the declared profile class $\mathfrak{P}_n$ and membership evidence;

(vii) a fixed edge extractor Edge_n*;*

(viii) zero compatibility;

(ix) the edge-realization certificate;

(x) the non-adaptive policy record;

(xi) the field coefficient declaration;

(xii) representative-change and realization-change policies;

(xiii) claim-status, manifest, proof, and immutable artifact data.

If an invoked item is missing or ill-typed, the degree-n categorical clause is `undefined`.

Remark 3.78 (Eligibility is verdict-independent). A standard n-eligible record types the categorical object and its edge. It does not require the repaired profile to vanish. Evidence establishing a zero statement belongs to the discharge record, not to eligibility. It is typed by exactly one canonical comparison mode when a comparison is consumed. This separation is necessary for nonzero direct computations, reject certificates, and the reverse-failure examples below.

Remark 3.79 (Standard amplitude is conservative). The amplitude

$$D^{[-n,0]}$$

is the standard release typing policy. It makes the intended obstruction edge and other cohomological degrees explicit. It is not a necessary hypothesis for Theorem 3.42.

Remark 3.80 (Non-scalar eligibility obligations). Representative existence, profile-class membership, realization compatibility, amplitude, edge naturality, edge realization, detector completeness, non-adaptivity, claim status, and artifact identity are non-scalar obligations. No metric reserve or ledger slack can manufacture them.

Theorem 3.81 (Expanded proof of Main Theorem 4.58: eligible higher Ext-edge identification)**.** *Let $n \geq 1$ and suppose $(F; W, \tau)$ has a complete standard n-eligible realization record. Then there is an artifact-backed isomorphism*

$$\operatorname{Ext}^n(A_{n;F,W,\tau}, k) \xrightarrow{\sim} \operatorname{Hom}_k(\operatorname{Edge}_n(\operatorname{Eval}_{n,W,\tau}(F)), k)\,.$$

It is natural exactly on the morphism scope on which both the realization family and the edge-realization certificate are natural.

Proof. Theorem 3.42 gives the canonical natural isomorphism

$$\operatorname{Ext}^n(A_{n;F,W,\tau}, k) \cong \operatorname{Hom}_k(H^{-n}(A_{n;F,W,\tau}), k).$$

Precomposition with the inverse of the recorded edge-realization isomorphism

$$H^{-n}(A_{n;F,W,\tau}) \xrightarrow{\sim} \operatorname{Edge}_n(\operatorname{Eval}_{n,W,\tau}(F))$$

gives the displayed map. Its naturality scope is the intersection of the naturality scopes of those two ingredients. □

3.10 The higher one-way repaired bridge

Proposition 3.82 (Repaired degree-n vanishing implies edge vanishing)**.** *Let $(F; W, \tau)$ have a complete standard n-eligible record. If*

$$\operatorname{Eval}_{n,W,\tau}(F) = 0,$$

then

$$H^{-n}(A_{n;F,W,\tau}) = 0.$$

Proof. The edge-realization certificate gives

$$H^{-n}(A_{n;F,W,\tau}) \cong \operatorname{Edge}_n(\operatorname{Eval}_{n,W,\tau}(F)).$$

Zero compatibility sends the zero profile to the zero vector space. □

Theorem 3.83 (Expanded proof of Main Theorem 4.59: higher one-way repaired discharge)**.** *Let $n \geq 1$, and let $(F; W, \tau)$ have a complete standard n-eligible record. Then*

$$\mathrm{Eval}_{n,W,\tau}(F) = 0 \quad \Longrightarrow \quad \mathrm{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

Proof. By Proposition 3.82,

$$H^{-n}(A_{n;F,W,\tau}) = 0.$$

Apply Corollary 3.44. ☐

Proposition 3.84 (Repaired degree-one vanishing implies edge vanishing)**.** *Let $(F; W, \tau)$ have a complete inherited eligible record. If*

$$\mathrm{Eval}_{1,W,\tau}(F) = 0,$$

then

$$H^{-1}(A_{F,W,\tau}) = 0.$$

Proof. The inherited edge-realization certificate gives

$$H^{-1}(A_{F,W,\tau}) \cong \mathrm{Edge}_1(\mathrm{Eval}_{1,W,\tau}(F)).$$

Zero compatibility gives $\mathrm{Edge}_1(0) = 0$. ☐

Theorem 3.85 (Expanded proof of Main Theorem 4.66: conditional one-way repaired bridge)**.** *Let $(F; W, \tau)$ have a complete inherited eligible record. If*

$$\mathrm{Eval}_{1,W,\tau}(F) = 0,$$

then

$$\mathrm{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Equivalently, when the degree-one representative record licenses the notation,

$$\mathrm{PH}_1(C_{\tau,W}F) = 0 \quad \Longrightarrow \quad \mathrm{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Proof. Set $A_{F,W,\tau} = \mathcal{R}(C_{\tau,W}F)$. Eligibility gives $A_{F,W,\tau} \in D^{[-1,0]}$. Proposition 3.84 gives $H^{-1}(A_{F,W,\tau}) = 0$. Apply Corollary 3.48. ☐

Remark 3.86 (Where the topological input enters). The sole topological input is exact vanishing of the repaired degree-one profile. The edge extractor and its realization certificate translate that statement into $H^{-1} = 0$.

Remark 3.87 (Metric transport is independent). Appendix C does not reprove threshold-gap or repaired-zero transport. If the topological zero statement is transported from another profile, the record must cite the Appendix A two-sided gap theorem, the actual discrepancy bound, post-threshold zero separation, and the applicable strict operational reserve.

Remark 3.88 (Discharge theorem). The inherited one-way theorem discharges one categorical gate clause. It does not reconstruct persistence from Ext vanishing and does not supply any missing metric, representative, or realization artifact.

Remark 3.89 (Independence from the metric repair proof). Metric repair controls transport of an exact repaired-zero statement between profiles. Eligibility and the Ext edge control categorical discharge after that exact zero statement has been established. Neither proof replaces the other.

3.11 Natural zero-reflection and vanishing equivalence

Definition 3.90 (Zero-reflecting edge extractor)**.** *An edge extractor* Edge_n *is* zero-reflecting *on* $\mathfrak{P}_n$ *when a theorem proves*

$$\mathrm{Edge}_n(P) = 0 \quad \Longrightarrow \quad P = 0 \qquad (P \in \mathfrak{P}_n).$$

Together with zero compatibility, this gives

$$\mathrm{Edge}_n(P) = 0 \quad \Longleftrightarrow \quad P = 0$$

on the declared class. Zero-reflection is weaker than full faithfulness and does not identify objects with their edge values.

Definition 3.91 (Natural zero-reflection package)**.** *A natural zero-reflection package contains:*

(i) a profile class $\mathfrak{P}_n$ *fixed independently of the verdict;*

(ii) a fixed functorial edge extractor Edge_n*;*

(iii) a natural-family edge-realization isomorphism;

(iv) a zero-reflection theorem on $\mathfrak{P}_n$*;*

(v) representative-change compatibility;

(vi) realization-change compatibility;

(vii) profile-class membership evidence;

(viii) non-adaptivity, proof, claim, and immutable artifact records.

An objectwise edge certificate alone is not a natural zero-reflection package.

Theorem 3.92 (Expanded proof of Main Theorem 4.61: zero-reflecting higher vanishing equivalence)**.** *Let* $n \geq 1$*. Suppose* $(F; W, \tau)$ *has a complete standard* n*-eligible record and a complete natural zero-reflection package. Then*

$$\mathrm{Eval}_{n,W,\tau}(F) = 0 \quad \Longleftrightarrow \quad \mathrm{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

This is an equivalence of zero predicates only.

Proof. The forward direction is Theorem 3.83.

Conversely, suppose the Ext group vanishes. By Corollary 3.44,

$$H^{-n}(A_{n;F,W,\tau}) = 0.$$

The edge-realization isomorphism gives

$$\mathrm{Edge}_n(\mathrm{Eval}_{n,W,\tau}(F)) = 0.$$

Zero-reflection on $\mathfrak{P}_n$ yields

$$\mathrm{Eval}_{n,W,\tau}(F) = 0.$$

□

Remark 3.93 (No object-level PH_n-to-Ext^n isomorphism)*.* Theorem 3.92 does not assert

$$\mathrm{PH}_n \cong \mathrm{Ext}^n$$

or an equivalence of persistence and derived categories. The two zero predicates are connected through a typed extractor and realization interface.

3.12 Detector-induced zero-reflection at the exact profile stage

Definition 3.94 (Fixed-threshold stencil edge functor). *Fix a right-open window W, a threshold $\tau > 0$, and a finite stencil*

$$T \subseteq W_\tau$$

in the valid Appendix B detector domain. For a repaired post-threshold profile P supported on W, define

$$\mathrm{Edge}_n^{T,\tau}(P) := \bigoplus_{t \in T} \mathrm{im}\big(P(t) \longrightarrow P(t+\tau)\big).$$

Proposition 3.95 (Expanded proof of Main Proposition 4.62: fixed-stencil functoriality). *On every declared category of repaired profiles and actual persistence morphisms on the fixed domain (W, τ, T), the construction $\mathrm{Edge}_n^{T,\tau}$ is a functor to finite-dimensional vector spaces.*

Proof. For each $t \in T$, naturality of a persistence morphism gives a commutative square between the structure maps at t and $t + \tau$. The target component sends the source image into the target image. These induced maps preserve identities and composition, and finite direct sum preserves functoriality. □

Definition 3.96 (Family-mesh repaired profile class). *Let $W = [a, b)$ be bounded. Let $\mathcal{F}$ be a nonempty family of left-closed right-open windowed barcodes with a valid Appendix B uniform excess certificate*

$$e_{\tau,W}(\mathcal{F}) > 0.$$

Let $T \subseteq W_\tau$ be finite and satisfy

$$\mathrm{Mesh}_{W_\tau}(T) < e_{\tau,W}(\mathcal{F}).$$

Define

$$\mathfrak{P}_n^{\mathcal{F},T,\tau} := \left\{ T_\tau^{\mathrm{bar}}(B) \,\middle|\, B \in \mathcal{F} \right\} \cup \{0\},$$

with actual persistence morphisms on the fixed window. When nontriviality is advertised, the record also proves that some $B \in \mathcal{F}$ has $T_\tau^{\mathrm{bar}}(B) \neq 0$.

Theorem 3.97 (Non-synthetic family-mesh zero-reflection). *On the class $\mathfrak{P}_n^{\mathcal{F},T,\tau}$, the fixed-threshold stencil edge is natural and zero-reflecting:*

$$\mathrm{Edge}_n^{T,\tau}(P) = 0 \iff P = 0.$$

If the nontriviality clause in Definition 3.96 is discharged, this is a non-synthetic natural zero-reflection theorem on a declared class containing a nonzero profile.

Proof. Naturality is Proposition 3.95. The implication $P = 0 \Rightarrow \mathrm{Edge}_n^{T,\tau}(P) = 0$ is immediate. Conversely, let

$$P = T_\tau^{\mathrm{bar}}(B), \qquad B \in \mathcal{F},$$

and suppose $P \neq 0$. Choose a retained interval I of B. Theorem 2.53 gives a point

$$t \in T \cap \mathrm{Act}_\tau(I).$$

The interval I is a summand of P, and its structure map

$$P(t) \longrightarrow P(t+\tau)$$

is nonzero. Hence $\mathrm{Edge}_n^{T,\tau}(P) \neq 0$, proving zero-reflection. □

Corollary 3.98 (Expanded proof of Main Corollary 4.63: fixed-stencil higher vanishing equivalence)**.** *Assume:*

(i) standard n-eligibility;

(ii) membership of the repaired profile in $\mathfrak{P}_n^{\mathcal{F},T,\tau}$;

(iii) a natural-family, independently meaningful edge-realization isomorphism

$$H^{-n}(A_{n;F,W,\tau}) \xrightarrow{\sim} \mathrm{Edge}_n^{T,\tau}(\mathrm{Eval}_{n,W,\tau}(F));$$

(iv) representative-change and realization-change compatibility.

Then

$$\mathrm{Eval}_{n,W,\tau}(F) = 0 \quad \Longleftrightarrow \quad \mathrm{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

Proof. Theorem 3.97 supplies a proved natural zero-reflection package. Apply Theorem 3.92. □

Proposition 3.99 (Expanded proof of Main Proposition 4.82: complete-detector higher Ext discharge)**.** *Let a complete Appendix B detector certificate, on the exact source profile stage and certified profile class, establish*

$$\mathcal{D}(\mathrm{Eval}_{n,W,\tau}(F)) = 0 \quad \Longrightarrow \quad \mathrm{Eval}_{n,W,\tau}(F) = 0.$$

If its detector value is certified zero and the object is standard n-eligible, then

$$\mathrm{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

Proof. Detector completeness gives repaired-profile vanishing. Apply Theorem 3.83. □

Remark 3.100 (Birth and germ certificates)*.* The Appendix B birth and germ characterizations may supply objectwise zero-reflection on their certified endpoint and profile classes. They become part of a *natural* zero-reflection package only after a functorial family construction and natural edge realization have been proved. Object-dependent birth sets do not become a natural transformation merely because each objectwise certificate is complete.

Remark 3.101 (Stage discipline)*.* A detector proved complete on `windowed_prethreshold` data does not automatically become a zero-reflecting edge on the repaired profile. Likewise, kernel- and cokernel-defect detectors have different source objects. Every detector edge must cite its exact Appendix B stage-transport theorem.

Remark 3.102 (Detector output is not a metric ledger)*.* A stencil, birth, germ, bar-mass, or other finite-dimensional detector is not a bottleneck bound, a threshold-gap certificate, or a metric-ledger component merely because its value is numerical or finite-dimensional.

3.13 Inherited interaction with the E_1 detector

Definition 3.103 (Imported residual degree-one bar-mass detector)**.** *The symbol*

$$E_1(F; W, \tau)$$

denotes the canonical non-truncated residual degree-one bar-mass detector of the inherited Core: it is the sum of the lengths of all finite residual bars in $\mathrm{Eval}_{1,W,\tau}(F)$, counted with multiplicity, and has value $+\infty$ when a residual infinite bar is present. It is an operational exact-zero detector, not a complete persistence invariant or metric certificate.

Proposition 3.104 (Expanded proof of Main Proposition 4.83: operational E_1-detector discharge)**.** *If*

$$E_1(F; W, \tau) = 0$$

and the degree-one eligible record is complete, then

$$\mathrm{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Proof. The canonical non-truncated definition excludes both finite and infinite residual bars when its value is zero, hence

$$\mathrm{Eval}_{1,W,\tau}(F) = 0.$$

Apply Theorem 3.85. $\square$

Remark 3.105 (No metric, stability, or converse role for E_1)*.* The preceding proposition does not make E_1 bottleneck-stable, a threshold-gap record, a metric-ledger component, a complete profile invariant, or a reverse Ext-to-persistence theorem. A truncated variant requires its own theorem and artifact policy.

3.14 Natural nonfaithful edge and reverse-failure boundary

Proposition 3.106 (Expanded proof of Main Proposition 4.85: reverse-inference boundary)**.** *Under standard n-eligibility alone, the equality*

$$\mathrm{Ext}^n(A_{n;F,W,\tau}, k) = 0$$

implies only

$$\mathrm{Edge}_n(\mathrm{Eval}_{n,W,\tau}(F)) = 0.$$

It does not imply repaired-profile vanishing without zero-reflection.

Proof. Ext vanishing implies $H^{-n}(A_{n;F,W,\tau}) = 0$ by Corollary 3.44. The edge-realization certificate then gives edge-value zero. No conclusion about the full profile follows without Definition 3.90. $\square$

Proposition 3.107 (Expanded proof of Main Proposition 4.90: natural nonfaithful-edge counterexample)**.** *For every $n \geq 1$, there is a fixed non-adaptive functorial realization, a fixed functorial edge extractor, and a natural edge-realization family on a declared category for which standard n-eligibility holds, but*

$$\mathrm{Ext}^n(A_{n;F,W,\tau}, k) = 0 \qquad \textit{and} \qquad \mathrm{Eval}_{n,W,\tau}(F) \neq 0.$$

Proof. Choose $0 < \tau < L$ and $\delta > 0$, and set

$$W = [0, L + \delta), \qquad t_* = L + \frac{\delta}{2}.$$

Let

$$C_I := E_n([0, L))$$

be the elementary interval representative from Appendix B, and set

$$F := C_I, \qquad C_{\tau,W}F := C_I.$$

Because W contains $[0, L)$ and $L > \tau$, this is an admissible degree-n representative with repaired profile

$$P_I := \mathrm{Eval}_{n,W,\tau}(F) = k_{[0,L)}.$$

Let $\mathcal{C}_*$ be the declared representative category generated by the zero representative and C_I, with their actual filtration-preserving chain maps. Let $\mathfrak{P}_*$ be the corresponding repaired-profile category. For a filtered complex C, write $C_{\le t_*}$ for its filtration subcomplex at t_*. Fix, independently of the verdict,

$$\mathrm{Edge}_n(P) := P(t_*)$$

and

$$\mathcal{R}_n(C) := H_n(C_{\le t_*})[n].$$

Both are functorial. The canonical identity

$$H^{-n}(\mathcal{R}_n(C)) = H_n(C_{\le t_*}) = \mathbf{P}_n(C)(t_*),$$

combined with the degree-n representative isomorphism, forms a natural edge-realization family. For $C = C_I$,

$$\mathcal{R}_n(C_I) \in D^{[-n,-n]}(k\text{-mod}) \subseteq D^{[-n,0]}(k\text{-mod}),$$

but $t_* > L$, so

$$\mathrm{Edge}_n(P_I) = P_I(t_*) = 0.$$

Therefore

$$\mathrm{Ext}^n(\mathcal{R}_n(C_I), k) = 0,$$

while

$$P_I = \mathrm{Eval}_{n,W,\tau}(F) \neq 0.$$

Thus naturality, non-adaptivity, and standard eligibility do not imply zero-reflection. □

Proposition 3.108 (Reverse implication fails in the inherited degree-one regime)**.** *Under the inherited degree-one eligibility axioms,*

$$\mathrm{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0$$

does not imply

$$\mathrm{Eval}_{1,W,\tau}(F) = 0.$$

Proof. Apply Proposition 3.107 with $n = 1$, using $\mathcal{R}_1 = \mathcal{R}$ and $\mathrm{Edge}_1 = \mathrm{Edge}_1$. □

Remark 3.109 (Why the counterexample is stronger than a constant-zero edge). The realization and edge above are fixed functorially at scope level and have ordinary mathematical meaning as point evaluation. The failure is therefore not caused by selecting a constant-zero policy after seeing the verdict. It is caused by genuine information loss: a point evaluation can miss a transient retained bar.

Remark 3.110 (What a reverse theorem requires). A reverse theorem requires a separately proved zero-reflection or faithfulness condition on the declared profile class, together with exact compatibility between that edge and the realized H^{-n}. Neither standard eligibility nor Ext vanishing supplies this condition.

3.15 Synthetic shifted detector objects and non-promotion

Definition 3.111 (Expanded record of Main Definition 4.86: synthetic shifted detector object)**.** *Let $\mathcal{D}(P)$ be a finite-dimensional detector value. Define the shifted detector encoding*

$$\mathrm{Syn}_{n,\mathcal{D}}(P) := \mathcal{D}(P)[n].$$

Then

$$H^{-n}(\mathrm{Syn}_{n,\mathcal{D}}(P)) = \mathcal{D}(P).$$

Unless an independent realization theorem identifies this object with an externally or internally meaningful $A_{n;F,W,\tau}$, its classification is

$$\texttt{synthetic} \quad \text{or} \quad \texttt{toy}.$$

Proposition 3.112 (Expanded proof of Main Proposition 4.87: encoded detector duality)**.** *For every finite-dimensional detector value,*

$$\mathrm{Ext}^n(\mathrm{Syn}_{n,\mathcal{D}}(P), k) \cong \mathrm{Hom}_k(\mathcal{D}(P), k)$$

canonically.

Proof. Apply Theorem 3.42 to $\mathrm{Syn}_{n,\mathcal{D}}(P) = \mathcal{D}(P)[n]$. □

Theorem 3.113 (No synthetic promotion)**.** *Proposition 3.112 does not by itself establish:*

(i) a natural persistence-to-derived realization theorem;

(ii) representative-independent categorical meaning;

(iii) an object-level PH_n-to-Ext^n bridge;

(iv) a problem-specific external obstruction theorem.

Promotion to the principal natural higher bridge requires an independently defined realization, natural edge certificate, representative compatibility, and all applicable Interface and Return Theorems.

Proof. The shifted object is defined from the detector output itself. The algebraic identity therefore expresses the tautological duality of that encoding. It supplies none of the independent semantic, naturality, representative, or external-transport obligations listed above. □

3.16 Degreewise categorical status and gate integration

Definition 3.114 (Ext-edge discharge record)**.** *An inherited degree-one discharge record contains:*

(i) the exact repaired evaluation artifact establishing $\mathrm{Eval}_{1,W,\tau}(F) = 0$;

(ii) the canonical comparison mode and its exact, metric-gap, or named theorem-controlled non-metric evidence;

(iii) the complete inherited eligible record;

(iv) the theorem identifier for Theorem 3.85;

(v) the canonical two-term Ext edge and its naturality data;

(vi) the resulting Ext-vanishing conclusion;

(vii) immutable claim, proof, manifest, and artifact references.

Definition 3.115 (Typed Ext-edge clause status)**.** *The inherited degree-one categorical clause has one of*

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{pass}, \qquad \texttt{reject}.$$

A passing clause requires a well-typed Ext group evidenced to vanish by the bridge theorem or a separately verified direct computation. A rejecting clause requires evidenced nonvanishing. Missing data are `undefined`, never zero.

Definition 3.116 (Typed degree-n Ext-clause status)**.** *For each invoked degree n, the categorical clause has exactly one canonical atomic meaning:*

(i) `not_invoked`*: an explicit scope policy excludes the degree-n clause;*

(ii) `undefined`*: the clause is invoked but eligibility, detector completeness, naturality, comparison, claim, or artifact evidence is missing or ill-typed;*

(iii) `pass`*: the well-typed Ext group is evidenced to vanish;*

(iv) `reject`*: the well-typed Ext group is evidenced to be nonzero.*

Missing data are never encoded as a zero group.

Definition 3.117 (Finite monitored degree set and aggregate categorical status)**.** *Let*

$$N_{\mathrm{mon}} \subseteq \mathbb{Z}_{\geq 1}$$

be finite. The aggregate higher categorical clause:

(i) is `not_invoked` *when every degree is explicitly out of scope;*

(ii) is `reject` *when at least one invoked, well-typed degree is evidenced nonzero;*

(iii) is `undefined` *when no reject is forced but at least one invoked required degree is undefined;*

(iv) is `pass` *when every invoked degree passes and all out-of-scope degrees are explicitly recorded.*

No numerical sum of Ext dimensions replaces this conjunctive status.

Definition 3.118 (Higher Ext discharge record)**.** *A degree-n discharge record contains:*

(i) the exact repaired profile and zero-evidence artifact;

(ii) the standard n-eligible record;

(iii) the theorem identifier Theorem 3.83;

(iv) the higher Ext edge and naturality data;

(v) the resulting categorical conclusion;

(vi) the degreewise and aggregate statuses;

(vii) the canonical comparison mode and the exact, metric-gap, or named theorem-controlled nonmetric evidence used to obtain the topological zero statement;

(viii) detector identities and completeness evidence when detector-derived;

(ix) immutable Claim, CM, TB, proof, hash, and provenance references.

Proposition 3.119 (Expanded proof of Main Proposition 4.75: degree-one B-Gate discharge)**.** *Under the inherited degree-one eligible regime,*

$$\mathrm{Eval}_{1,W,\tau}(F) = 0$$

discharges the degree-one categorical clause:

$$\mathrm{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Proof. Apply Theorem 3.85. □

Remark 3.120 (Non-compensation)*.* A passing categorical clause does not cancel a failed or undefined detector, metric, representative, Type IV, transient-defect, finite-to-infinite, manifest, claim-status, or external Return-Theorem obligation.

3.17 Representative-change and realization-change compatibility

Definition 3.121 (Realization-equivalent eligible representatives)**.** *Two inherited degree-one eligible representatives are realization-equivalent when an actual isomorphism*

$$\alpha : \mathcal{R}(C_{\tau,W}F) \xrightarrow{\sim} \mathcal{R}(C'_{\tau,W}F)$$

is supplied and the edge-realization certificates commute with $H^{-1}(\alpha)$*.*

Proposition 3.122 (Eligibility is invariant under admissible representative equivalence)**.** *The inherited amplitude, edge-realization, and categorical typing transport across a declared degree-one realization equivalence.*

Proof. Derived isomorphisms preserve cohomology and amplitude, and compatibility of the edge certificates transports the inherited edge condition. ☐

Proposition 3.123 (Expanded proof of Main Proposition 4.80: representative invariance of Ext discharge)**.** *For realization-equivalent inherited eligible representatives, the Ext groups are identified contravariantly and the truth value of degree-one categorical vanishing is unchanged.*

Proof. Apply $\mathrm{Hom}_{D^{\mathrm{b}}(k\text{-mod})}(-, k[1])$ to the recorded realization isomorphism and use Corollary 3.47. ☐

Definition 3.124 (Higher realization-equivalent representatives)**.** *Two standard n-eligible representatives are* realization-equivalent *when an actual isomorphism*

$$\alpha : A_{n;F,W,\tau} \xrightarrow{\sim} A'_{n;F,W,\tau}$$

is supplied in $D^{\mathrm{b}}(k\text{-mod})$, and their edge-realization certificates are compatible with $H^{-n}(\alpha)$. For a natural zero-reflection package, the declared profile class and edge functor must also be preserved.

Proposition 3.125 (Expanded proof of Main Proposition 4.78: higher representative invariance)**.** *Under a declared realization equivalence, amplitude, cohomology, edge-realization, degreewise categorical typing, and the applicable zero-reflection package transport to the second representative.*

Proof. Derived isomorphisms preserve cohomology and amplitude. Compatibility with $H^{-n}(\alpha)$ transports the edge isomorphism. The explicitly required profile-class and edge compatibility transport the zero-reflection data. ☐

Proposition 3.126 (Representative independence of the higher discharge)**.** *The realization isomorphism induces an identification*

$$\mathrm{Ext}^n(A'_{n;F,W,\tau}, k) \xrightarrow{\sim} \mathrm{Ext}^n(A_{n;F,W,\tau}, k),$$

and the truth value of Ext vanishing is independent of the representative choice relative to the recorded equivalence.

Proof. Apply the contravariant functor

$$\mathrm{Hom}_{D^{\mathrm{b}}(k\text{-mod})}(-, k[n])$$

to α. Corollary 3.43 identifies this with dualizing $H^{-n}(\alpha)$. ☐

Remark 3.127 (Relative independence, not hidden canonicality). The proposition gives independence only relative to the recorded realization equivalence. It does not make the filtered representative, realization functor, or edge extractor unique.

Remark 3.128 (Profile equivalence remains weaker). Monitored persistence-profile equivalence from Appendix B does not by itself imply realization equivalence or detector-edge preservation.

3.18 Actual finite-diagram morphisms and the linear cone handoff

Definition 3.129 (Actual finite-diagram morphism record)**.** *An actual finite-diagram morphism record contains $V, V' \in \mathrm{Rep}_k(A[m])$, an actual natural transformation*

$$f : V \longrightarrow V',$$

all component maps, every commuting arrow square, common or explicitly compared grid identities, variance and coefficient conventions, and immutable artifacts. A bottleneck matching, equality of rank tables, or an objectwise barcode correspondence is not an actual diagram morphism.

Definition 3.130 (Linear cone handoff record)**.** *For an actual $f : V \to V'$, regard V, V' as complexes concentrated in cohomological degree zero and form*

$$\mathrm{Cone}(f) \in D^b(\mathrm{Rep}_k(A[m])).$$

The handoff is

$$\begin{aligned} \mathrm{ConeHandoff}_\Gamma(f) := (&\Gamma, V, V', f, \mathrm{Cone}(f), \\ &\mathrm{variance}, \mathrm{coefficients}, \mathrm{supported\_strengths}, \mathrm{artifacts}). \end{aligned}$$

It records the actual chain-level representative used to form the cone.

Proposition 3.131 (Expanded proof of Main Proposition 4.94: cone-zero criterion in the finite linear category)**.** *For an actual morphism $f : V \to V'$ in $\mathrm{Rep}_k(A[m])$,*

$$f \text{ is an isomorphism} \quad \Longleftrightarrow \quad \mathrm{Cone}(f) \simeq 0 \text{ in } D^b(\mathrm{Rep}_k(A[m])).$$

Proof. In every derived category, a morphism is an isomorphism exactly when its cone is a zero object. Since V, V' are concentrated in degree zero, derived isomorphism agrees with isomorphism in the abelian category. □

Remark 3.132 (Local cone result versus Appendix DG). This proposition is a local standard fact for the finite linear category. Appendix DG owns arbitrary finite-category generators, complete cone-probe systems, branching diagrams, and object/morphism descent. No cone is formed from a distance value or matching without an actual morphism.

3.19 Finite-linear Core-P to Core-DG interface

Definition 3.133 (Finite-linear P2DG interface)**.** *For an actual grid-complete persistence module P, define*

$$\mathrm{P2DG}_\Gamma(P) := \mathrm{Res}_\Gamma(P) \in \mathrm{Rep}_k(A[m]).$$

For a finite barcode isomorphism class, define the objectwise branch

$$\mathrm{P2DG}_\Gamma^{\mathrm{obj}}(B) := \mathrm{GridRep}_\Gamma(B).$$

The interface record independently declares action mode, exactness profile, reflection profile, information profile, preserved and discarded data, comparison mode, discrepancy level, proof artifacts, and prohibited stronger readings.

Definition 3.134 (Finite-linear P2DG record)**.** *A complete record contains:*

(i) the Appendix A P2DGInput *and any Appendix B* Cert2RepInput *record;*

(ii) *route kind* `object_route` *or* `morphism_route`*;*

(iii) *the actual source persistence module, or an explicit declaration that only a barcode isomorphism class is available;*

(iv) *for a morphism route, an actual persistence morphism, one common complete grid for source and target, and the induced natural transformation after restriction;*

(v) *the endpoint set, atom dictionary, orientation, variance, terminal policy, grid completeness, and encoding-strength records;*

(vi) *the finite quiver object and, when applicable, its actual morphism action;*

(vii) *rank, Möbius, projective/Ext-signature, and decorated-barcode reconstruction artifacts when invoked;*

(viii) *the two-axis realization classification, action mode, and audit-use class;*

(ix) *any standard n-eligible derived realization and Ext-edge record;*

(x) *a downward-closed route-supported set of Return strengths and the exact requested strength, chosen from observation, profile isomorphism class, decorated barcode, quiver object, quiver morphism, or Ext-vanishing predicate;*

(xi) *immutable dependencies, failure leaves, and verifier status.*

A quiver-morphism Return is admissible only on a morphism route with an actual source morphism. An Ext-vanishing Return requires the separate eligible realization route; it is not a consequence of finite-grid equivalence.

Theorem 3.135 (Expanded proof of Main Theorem 4.98: P-to-DG finite-linear handoff)**.** *On the ambient endpoint-complete category,*

$$\mathrm{P2DG}_\Gamma : \mathsf{Pers}_k^\Gamma(W) \xrightarrow{\sim} \mathrm{Rep}_k(A[m])$$

is an exact equivalence and therefore is full, faithful, conservative, zero-reflecting, and lossless at persistence-module strength. Restricted to the full subcategory of repaired profiles at fixed positive threshold, it is full, faithful, conservative, zero-reflecting, and lossless onto the repaired essential image, which may be a proper full subcategory. It also preserves and reflects each ambient exact sequence whose terms and relevant kernels and cokernels remain in the repaired subcategory. This relative statement does not assert that repaired profiles form an abelian, Serre, or extension-closed subcategory. For a barcode-only source, the objectwise branch is lossless on the declared finite decorated barcode but has no asserted morphism fullness or functoriality.

Proof. The ambient actual-module assertion is Theorem 3.18 together with Theorem 3.24. Restricting a full and faithful equivalence to a full subcategory preserves fullness and faithfulness onto its essential image; conservativity and zero-reflection are inherited as well. Exactness is used only by testing a sequence in the ambient abelian categories. If every term and the relevant kernel and cokernel stay in the repaired subcategory, the ambient exact equivalence preserves and reflects that sequence. No closure of repaired profiles under extensions, kernels, or cokernels is used. The barcode-only assertion follows from the atom-by-atom interval dictionary and preservation of multiplicity, while the absence of a morphism action prevents a stronger functorial conclusion. The fixed window, grid, linear target, and declared source subcategory prohibit promotion to the general Core-DG setting. □

Remark 3.136 (Exactness qualifier for the repaired essential image)*.* The phrase "the same functor has those properties" in the canonical Main statement is consumed here in the preceding relative ambient sense. It does not restore exactness of hard threshold deletion and does not assert that the class of τ-reduced profiles is Serre or closed under arbitrary kernels, cokernels, or extensions.

Theorem 3.137 (Expanded proof of Main Theorem 4.99: finite profile-to-quiver internal Return closure)**.** *Let $P = \mathrm{Eval}_{n,W,\tau}(F)$ be a finite repaired profile with complete grid and P2DG records. Then the internal finite route*

$$P \longrightarrow \mathrm{P2DG}_{\Gamma}(P) \longrightarrow (r(i,j), \mu(i,j), \Sigma_{\Gamma}^{\mathrm{proj/Ext}}) \longrightarrow \bar{}(P)$$

closes at decorated persistence-profile strength. If a complete standard n-eligible record is also supplied, the same source-zero predicate closes the conditional branch

$$P = 0 \Longrightarrow \mathrm{Ext}^{n}(A_{n;F,W,\tau}, k) = 0.$$

Under a separately proved natural zero-reflection package, the latter branch is an equivalence of vanishing predicates.

Proof. The diagram construction is Theorem 3.135; rank and barcode reconstruction are Theorem 3.27 and Corollary 3.28; the finite signature is Theorem 3.33; and the conditional Ext branch is Theorem 3.83, upgraded by Theorem 3.92 under its additional hypotheses. ☐

Remark 3.138 (Internal Return only)*.* The closure returns a persistence profile, decorated barcode, finite linear diagram, finite signature, or typed Ext-vanishing predicate. It is not an external-domain Return theorem. The supported Return strengths form a downward-closed set; no unique scalar "ceiling" is asserted unless a greatest element is separately proved.

3.20 Appendix C and Appendix DG ownership boundary

Appendix C owns one-parameter finite atom chains, the linearly oriented $A[m]$-quiver, finite barcode/rank reconstruction, persistence-specific projective/Ext signatures, and the local finite-linear cone handoff. Appendix DG owns arbitrary finite categories and posets, branching diagrams, general projective generators, complete cone probes, Morita reconstruction, Alexandrov realization with fixed variance, effective descent, and global zero/isomorphism certificates. A linear P2DG theorem is not promoted to the general finite-category theorem by relabeling its indexing category.

3.21 Higher eligibility manifest and obligation handoff

Definition 3.139 (Eligibility manifest extension)**.** *An inherited degree-one eligibility entry records at least:*

(i) repaired and pre-threshold profile identities;

(ii) the representative and actual degree-one profile isomorphism;

(iii) the realization functor, realized object, two-axis realization classification, action mode, and audit-use class;

(iv) the two-term amplitude certificate;

(v) the fixed degree-one edge extractor;

(vi) *the edge-realization certificate and its objectwise/natural scope;*

(vii) *coefficient field, equivalence policy, status, theorem, claim, manifest, hash, and provenance records;*

(viii) *the canonical comparison mode and its evidence only when used to establish the separate topological zero statement.*

Definition 3.140 (Eligible-realization obligation bundle). *The inherited non-scalar obligation bundle contains representative availability, degree-one profile compatibility, realization-domain membership, amplitude, fixed edge extraction, edge realization, field declaration, equivalence policy, claim admissibility, manifest consistency, and immutable artifact identity.*

Definition 3.141 (Higher eligibility manifest extension). *A proof object invoking a degree-n Ext clause records at least:*

(i) *the repaired evaluation and pre-threshold profile identifiers;*

(ii) *the detector source stage and zero-evidence route;*

(iii) *the representative identifier, construction artifact, and actual degree-n profile isomorphism;*

(iv) n, W, τ, *endpoint convention, and monitored scope;*

(v) *the realization domain, functor, version, and artifact identity;*

(vi) *the independently meaningful realized object;*

(vii) *the amplitude certificate or registered replacement;*

(viii) *the profile class and membership proof;*

(ix) *the fixed edge extractor, object/morphism scope, and version;*

(x) *zero compatibility;*

(xi) *the edge-realization certificate and its* `objectwise` *or* `natural_family` *classification;*

(xii) *when a converse is used, the complete natural zero-reflection package;*

(xiii) *representative-change and realization-change compatibility;*

(xiv) *the non-adaptive policy record;*

(xv) *the degreewise and aggregate categorical statuses;*

(xvi) *the theorem or direct-computation identities supporting pass or reject;*

(xvii) *the canonical comparison mode and its exact, metric-gap, or named theorem-controlled non-metric records independently supporting the topological zero statement;*

(xviii) *detector completeness, profile stage, class, and artifact records when a detector is consumed;*

(xix) *the two-axis realization classification, action mode, and audit-use class for every shifted detector or encoded Ext object;*

(xx) *all Claim, CM, TB, proof, hash, and provenance links required for replay;*

(xxi) *the endpoint set, atom grid, terminal policy, and encoding-strength records whenever a finite-linear route is invoked;*

(xxii) the quiver object, rank/Möbius, enriched signature, actual morphism, cone handoff, and complete P2DG records whenever those outputs are consumed;

(xxiii) the downward-closed supported Return set and the exact requested Return strength.

For $n = 1$, this extension contains the inherited v19 eligibility fields.

Definition 3.142 (Higher eligible-realization obligation bundle)**.** *The non-scalar obligation bundle contains:*

(i) representative availability and degree-n profile compatibility;

(ii) realization-domain membership and independent semantic meaning;

(iii) amplitude or admissible replacement;

(iv) profile-class declaration and membership;

(v) fixed edge extractor and zero compatibility;

(vi) edge-realization isomorphism;

(vii) naturality when claimed;

(viii) zero-reflection when a converse is consumed;

(ix) detector completeness at the exact profile stage when detector-based;

(x) representative- and realization-change compatibility;

(xi) non-adaptivity;

(xii) field coefficient declaration;

(xiii) claim-status admissibility;

(xiv) manifest completeness and consistency;

(xv) immutable artifact identity.

Remark 3.143 (Appendix F/G handoff). Appendix F may integrate the eligible-realization obligations into status, dependency, and non-compensation logic. Appendix G may serialize their fields and immutable references. Neither appendix creates a missing mathematical proof by recording it.

Remark 3.144 (Failure routing). An invoked clause with missing required data is `undefined`, not a mathematical counterexample and not `not_invoked` unless an explicit scope exclusion applies. An internally inconsistent manifest is routed to Type V rejection under the canonical audit architecture.

3.22 Appendix C v21 proof-package closure

Definition 3.145 (Appendix C v21 proof-package closure record)**.** *The closure record confirms, without creating a second Main theorem identity, that:*

(i) every Main Chapter 4 statement reproduced here is marked as an expanded proof under Main ownership;

(ii) the complete v20 degree-one and higher eligible Ext package regresses unchanged at its proved field-coefficient strength;

(iii) realization provenance, semantic fidelity, action mode, and audit-use class are independently recorded;

(iv) the endpoint-complete atom grid and terminal policy preserve decorated endpoint and cofinal information;

(v) the actual-module P2DG branch is an exact equivalence on the declared Γ*-constructible category, while the barcode branch is objectwise unless a morphism action is supplied;*

(vi) rank/Möbius and enriched projective/Ext signatures reconstruct only at their declared persistence or finite-linear strength;

(vii) the eligible higher Ext edge is one-way unless a separately proved natural zero-reflection package is present;

(viii) synthetic and target-shaped encodings are not promoted to principal natural realizations;

(ix) the cone output is a typed finite-linear handoff, not the general Appendix DG cone/descent theorem;

(x) every comparison mode, discrepancy level, supported Return set, artifact, and failure leaf is present in the manifest.

The v20 labels `thm:C-package` *and* `thm:C-higher-package` *remain historical source locators only; this record is not a replacement mathematical theorem.*

Definition 3.146 (Appendix C local completion gates)**.** *Appendix C is locally complete only when:*

(i) the v20 Ext-edge and counterexample regression suite passes;

(ii) theorem ownership and namespace checks pass;

(iii) one nontrivial endpoint-complete actual-module P2DG witness and one barcode-only objectwise witness are replayable;

(iv) one actual morphism route and its cone handoff are replayable;

(v) eligible, zero-reflecting, nonfaithful, and synthetic routes are distinguished by records and counterexamples;

(vi) the Appendix DG non-overlap audit passes;

(vii) the Appendix F/G manifest handoff is complete.

Remark 3.147 (Contribution to G21 gates)*.* Appendix C contributes to G21-2 through Core-P regression, to G21-4 through a nontrivial P2DG closure, to G21-7 by supplying typed finite-linear and eligible Ext inputs, and to G21-8 through ownership and namespace separation. It does not by itself pass G21-3 general Core-DG closure, G21-5 object-level globalization, G21-7 unified external Return, or G21-9 final release closure.

3.23 Explicit exclusions

The following are not asserted.

(i) No unconditional object-level equivalence $\mathrm{PH}_n \cong \mathrm{Ext}^n$ is asserted in any degree.

(ii) No reverse inference from Ext vanishing to repaired-profile vanishing is used without a separately proved natural zero-reflection package.

(iii) No functorial or zero-compatible edge extractor is automatically faithful, conservative, or zero-reflecting.

(iv) No objectwise edge certificate is a natural transformation without a natural-family theorem.

(v) No realization, grid, edge extractor, stencil, or target category is selected after inspecting the desired verdict.

(vi) No Cert2RepInput record constructs a representative, persistence module, quiver object, derived object, or Ext edge.

(vii) No incomplete endpoint set, unresolved terminal policy, missing morphism action, or absent artifact is interpreted as reconstruction.

(viii) No objectwise barcode encoding returns a quiver morphism.

(ix) No finite rank table returns a filtered chain complex or derived realization.

(x) No dimensions-only local Hom/Ext signature is treated as complete.

(xi) No quiver Ext^1 is identified with the higher field-valued $\mathrm{Ext}^n(A_n, k)$ edge.

(xii) No actual finite-diagram morphism is manufactured from a bottleneck matching, rank-table equality, or barcode correspondence.

(xiii) No local linear cone theorem is promoted to a complete general finite-category cone-probe theorem.

(xiv) No linear P2DG theorem is promoted to Morita, Alexandrov, effective descent, or global-object reconstruction.

(xv) No barcode or persistence discrepancy is silently lifted to finite diagram or derived discrepancy.

(xvi) No standard amplitude certificate is claimed to be algebraically necessary for the field-coefficient Ext isomorphism.

(xvii) No synthetic shifted detector object is promoted to a principal natural realization merely because its Ext group is dual to the detector output.

(xviii) No missing eligibility, naturality, detector completeness, or artifact is interpreted as zero; the invoked clause is `undefined`.

(xix) No categorical pass compensates for failed metric, detector, Type IV, finite-to-infinite, manifest, or external Return obligations.

(xx) No route-supported Return set is forced into a total order or assigned a unique greatest strength without proof.

(xxi) No external-domain theorem follows without a separately proved Interface–Core–Reduction–Return chain.

3.24 Provenance, status, and final local verdict

The semisimplicity of finite-dimensional vector spaces, hyper-Ext spectral sequence, type-A interval decomposition, Möbius inversion, Yoneda projective probes, and cone-zero criterion are standard inputs. The inherited AK contribution is the typed eligible realization and one-way persistence-to-Ext discharge with explicit nonconverse. The v21 strengthened contribution is the endpoint-complete finite-linear P2DG interface, decorated barcode and enriched-signature reconstruction, two-axis realization taxonomy, and typed cone/Return handoff.

3.24.0.1 Canonical Main statement ownership. The canonical statements are Main Definitions 4.3, 4.5–4.13, 4.20–4.21, 4.24, 4.28–4.29, 4.36–4.47, 4.60, 4.69–4.74, 4.77, 4.79, 4.86, 4.92–4.93, 4.96–4.97, and 4.101; Main Lemmas 4.14 and 4.52; Main Theorems 4.15, 4.22, 4.25, 4.31, 4.50, 4.53, 4.58, 4.59, 4.61, 4.66, 4.98, 4.99, and 4.103; Main Corollaries 4.26, 4.51, 4.54–4.55, and 4.63; and Main Propositions 4.16, 4.30, 4.62, 4.71, 4.75, 4.78, 4.80, 4.82–4.83, 4.85, 4.87, 4.90, and 4.94. This appendix owns expanded proofs, local lemmas, counterexamples, record detail, and regression artifacts only.

3.24.0.2 Appendix-local supporting results. The objectwise barcode route, loss-mode counterexamples, stage-specific zero-reflection specializations, natural nonfaithful-edge construction, and synthetic nonpromotion proof are Appendix-local only to the extent that they do not duplicate a Main identity. The same restriction applies to the local completion gates and migration statement.

3.24.0.3 Final local verdict. Subject to cross-Appendix locator freeze and final manifest extraction, the Appendix C mathematical draft is locally closed at finite-linear P2DG and eligible Ext-edge strength. General Core-DG and external Return closure remain downstream obligations.

4 Appendix D: Towers, Type IV, Derived Limits, and Finite-to-Infinite Reduction

4.1 Scope, canonical ownership, migration, and standing regime

4.1.0.1 Normative status and theorem ownership. Appendix D is the expanded-proof, counterexample, tail-certificate, derived-limit, defect-profile, record-schema, and regression owner for the Core-P tower and finite-to-infinite statements canonically owned by Main Chapter 5 of AK-HPDST v21.0.0, the *Finite Globalization, Unified Bridge Calculus, and Certificate Complexity Release.* A result canonically stated in the Main retains the Main theorem identity. This appendix reproduces it only as an expanded proof, typed specialization, explicit counterexample, or implementation record; it creates no competing theorem UID. Appendix-local identities are reserved for strict refinements not duplicated in the Main.

The immutable v20 Appendix D remains directly invocable from its frozen source artifact at its original strength. The v21 migration is componentwise:

(i) the bounded-window terminal repair, actual-comparison requirement, terminal-generic μ/ν diagnostics, cofinal/interior separation, transient-defect calculus, short-defect exclusion, stable bands, and eventual-constancy reduction are `inherited`;

(ii) explicit direct/inverse variance, the countable Roos operator, $\varprojlim^1$, Mittag–Leffler control, inverse apex separation, finite-length and Noetherian lanes, finite-generation windows, deterministic finite-state certification, periodic stable-image inverse limits, and the partially ordered internal Return profile are `strengthened` successor material with v21 Main identities;

(iii) the previous package-theorem wrappers are not repeated as independent v21 authorities. Their mathematical components remain inherited or strengthened, while closure is recorded operationally. This is a governance repair, not a rejection of the v20 mathematics;

(iv) source-specific Iwasawa, p-adic L-function, characteristic ideal, Selmer, PDE, and other external tower Returns remain in the Problem-Interface, HT/UB, or successor layers. They are not promoted here.

Appendix D consumes:

(i) Appendix A's repaired evaluation, four comparison modes, discrepancy levels, threshold-gap, and repaired-zero contracts;

(ii) Appendix B's actual-representative, actual-morphism, certificate-model, and freshly registered `kernel_defect`/`cokernel_defect` detector records;

(iii) Appendix C's eligible degreewise Ext discharge, without treating it as tower or limit evidence;

(iv) Main Chapter 5's canonical tower, Type IV, derived-limit, and finite-to-infinite statements.

All persistence-level tower diagnostics are formed after the repaired windowed evaluation

$$\mathrm{Eval}_{i,W,\tau}(F) := T_\tau^{\mathrm{bar}}\big(\mathsf{W}_W \mathbf{P}_i(F)\big).$$

Hard deletion does not automatically lift the transition maps of a filtered or persistence tower. A sequence of repaired barcode isomorphism classes, pairwise bottleneck matchings, equality of dimensions, or a finite numerical plateau is therefore not a repaired tower-profile system.

The central v21 rules are:

> Variance is theorem identity. A direct colimit, an inverse limit, a mixed diagram, and a completed tower are not interchangeable notations.
>
> The terminal Type IV pair measures cofinal kernel/cokernel defect of an actual typed comparison morphism. The Roos defect $\varprojlim^1$, an inverse apex defect, a long transient defect, and a short residual defect are separate fields; no one of them is renamed as another.
>
> A finite-to-infinite Return requires a named theorem, a finite witness, a quantified or finite-state tail certificate, source authority, all required limit or derived-limit fields, and an actual apex comparison at the strength claimed.

Every consumed comparison records exactly one of

`exact`, `metric_gap_certified`,
`theorem_controlled_nonmetric`, `not_compared`.

A terminal Type IV diagnostic requires an actual morphism and therefore cannot be created by `not_compared`. Metric comparison of source and target objects does not create a morphism or a kernel/cokernel comparison.

Remark 4.1 (Appendix D does not enlarge the external Core). This appendix proves internal tower, limit, derived-limit, stable-image, comparison-isomorphism, and apex-agreement statements only. It does not identify an undeclared apex with a categorical limit, restore a global exact collapse functor, or prove an arithmetic, analytic, geometric, PDE, cryptographic, or other external-domain Return.

Remark 4.2 (The bounded-window repair). The shorthand

$$\mathrm{gdim}(M) = \lim_{t \to +\infty} \dim_k M(t)$$

is valid only on a right-unbounded window. On a bounded window, zero extension would force this value to vanish. The canonical invariant is the window-relative terminal-generic dimension tgdim_W.

Remark 4.3 (Terminal, transient, Roos, and apex separation). A nonzero terminal pair is Type IV. A certified nonzero interior detector is a distinct transient obstruction. A nonzero $\varprojlim^1$ is a derived-limit defect in the countable inverse-module lane. Failure of an actual apex map to be an isomorphism is an apex defect. These verdicts are nonexclusive but are never conflated.

Remark 4.4 (Finite observation is not infinite proof). Agreement through finitely many levels proves no tail statement unless a quantified theorem, certified generation bound, finite-state transition rule, or another registered reduction makes the remaining infinite claim a logical consequence of that finite witness.

4.2 Tower variance, semantic authority, and evaluation lift

Definition 4.5 (Expanded form of Main Definition 5.3: typed tower variance). *A tower record declares exactly one primary variance type*

$$\texttt{direct\_system}, \qquad \texttt{inverse\_system}, \qquad \texttt{mixed\_diagram},$$

and may additionally carry the modifiers

$$\texttt{filtered\_tower}, \qquad \texttt{completed\_tower}.$$

A direct system is a functor $(\mathbb{N}, \leq) \to \mathcal{C}$; an inverse system is a functor $(\mathbb{N}, \leq)^{\mathrm{op}} \to \mathcal{C}$. A mixed diagram becomes direct or inverse only after a theorem selects a declared cofinal subdiagram and orientation. Reversing arrows or comparison direction creates a different claim.

Definition 4.6 (Expanded form of Main Definition 5.4: tower semantic role). *Every source, level, and apex records its mathematical role and exactly one authority level*

$$\texttt{categorical}, \qquad \texttt{theorem\_identified}, \qquad \texttt{implementation\_only}.$$

A categorical source is an actual limit or colimit in the declared category. A theorem-identified source is equipped with an actual isomorphism to that limit or colimit under a named theorem. An implementation-only source may support only the implementation-scoped claim stated in its record.

Definition 4.7 (Expanded form of Main Definition 5.5: actual-transition requirement). *Every comparable pair of indices has an actual morphism $d_{n,m}$ in the declared category, with*

$$d_{n,n} = \mathrm{id}, \qquad d_{m,r} d_{n,m} = d_{n,r}$$

in the direction determined by the variance. A barcode matching, equality of isomorphism classes, or predicted transition is not an actual morphism.

Definition 4.8 (Expanded form of Main Definition 5.6: evaluation-lift record)**.** *For a filtered or persistence tower $F_\bullet$, an evaluation-lift record at (i, W, τ) contains the repaired stage objects $M_{n;i,W,\tau} = \mathrm{Eval}_{i,W,\tau}(F_n)$, actual transition morphisms among those repaired profiles, composition proofs, and the theorem or explicit construction lifting the source transitions through windowing and hard deletion. Missing lift data leave the repaired tower undefined.*

4.3 Failure taxonomy and audited structural admissibility

Definition 4.9 (Expanded form of Main Definition 5.7: nonexclusive failure types)**.** *The inherited failure types are nonexclusive.*

- Type I. *a required repaired topological profile is certified nonzero;*
- Type II. *a well-typed invoked eligible Ext group is certified nonzero;*
- Type III. *a detector, metric, representative, transition, comparison, source-authority, tail, derived-limit, or apex obligation is missing, ill-typed, or evidenced false;*
- Type IV. *a valid terminal comparison artifact has $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) \neq (0, 0)$;*
- Type V. *a claim-status, provenance, manifest, stage, or promotion rule is violated, including undefined-as-zero and finite-observation-as-tail substitutions.*

A certified long-transient defect is a mathematical failure of the stronger transient-clean clause while the terminal Type IV field may remain zero.

Definition 4.10 (Expanded form of Main Definition 5.9: failure-check order)**.** *The audit order is: claim/manifest well-formedness; operational, metric, representative, and artifact typing; topological and eligible categorical verdicts; invoked Type IV and transient diagnostics; then the final gate conjunction and budget decision. A later zero never repairs an earlier undefined or rejected obligation.*

Definition 4.11 (Expanded form of Main Definition 5.12: Type IV scope status)**.** *Every run records exactly one terminal Type IV scope status*

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{certified\_zero}, \qquad \texttt{obstructed}.$$

The first means only that no terminal-generic kernel/cokernel diagnostic is consumed. A direct or inverse finite-to-infinite, Roos, stable-image, or apex-agreement lane may remain active. The second means that the diagnostic is in scope but lacks a valid comparison, ambient abelian profile category, source authority, terminal-tameness record, or defect object. The last two require a valid artifact and numerical pair equal or unequal to $(0, 0)$, respectively.

Definition 4.12 (Expanded form of Main Definition 5.13: structural collapse admissibility)**.** *Fix a finite monitored degree set. A run is structurally collapse-admissible when every required repaired topological clause passes, every invoked eligible Ext clause passes, the Type IV status is correctly `not_invoked` or `certified_zero`, every required long-transient or full-isomorphism clause passes, and no required internal finite-to-infinite clause is undefined or rejected.*

Definition 4.13 (Expanded form of Main Definition 5.14: audited collapse admissibility)**.** *A structurally admissible run is audited collapse-admissible when every comparison mode, threshold-gap, metric budget, representative, transition, source-authority, limit, derived-limit, apex, claim-status, dependency, manifest, and artifact-identity obligation required by its declared route also passes.*

Proposition 4.14 (Expanded proof of Main Proposition 5.15: eligible monitored vanishing discharges Type II)**.** *Let* I_{mon} *be finite. If* $\mathrm{Eval}_{n,W,\tau}(F) = 0$ *for every invoked* $n \in I_{\mathrm{mon}}$ *and each degree has a complete eligible Appendix C record, then every invoked Type II clause passes. No Type III, Type IV, transient, finite-to-infinite, apex, or Type V clause is discharged by this proposition.*

Proof. Apply Appendix C's degree-n one-way bridge in each invoked degree and take the finite conjunction of the resulting Ext-vanishing statements. The other clauses concern independent objects or evidence and are untouched. □

4.4 Direct, inverse, and repaired profile systems

Definition 4.15 (Directed tower)**.** *A directed tower is a sequence*

$$F_0 \longrightarrow F_1 \longrightarrow F_2 \longrightarrow \cdots$$

in the declared filtered regime with actual filtration-preserving maps and composition compatibility.

Definition 4.16 (Inverse tower)**.** *An inverse tower is a sequence*

$$F_0 \longleftarrow F_1 \longleftarrow F_2 \longleftarrow \cdots$$

with actual maps toward lower indices and the inverse-system composition law. It is not obtained by typographical reversal of a direct artifact.

Definition 4.17 (Expanded form of Main Definition 5.18: declared apex and comparison direction)**.** *A declared apex* F_∞ *is the object against which a direct source or inverse source is compared. Its record includes construction, semantic role, authority, comparison direction, and every theorem required to identify a proxy or completion with the source strength consumed by the claim.*

Definition 4.18 (Apex identity and authority record)**.** *The apex record classifies the object as a categorical limit/colimit, theorem-identified object, completion, implementation apex, finite proxy, or advisory placeholder. Only the first four may support a Core comparison at their registered authority; a finite proxy requires a named reduction theorem, and an advisory placeholder is never a proof object.*

Remark 4.19 (Apex is declared, not automatic)*.* The notation F_∞ asserts neither existence nor universal property. A direct colimit comparison and an inverse limit comparison generally have opposite directions and different theorem obligations.

Definition 4.20 (Monitored degrees and parameter-indexed scope)**.** *For one fixed* (W, τ)*, let* $I_{\mathrm{mon}} \subset \mathbb{Z}$ *be finite. Numerical totals are formed only over this set. A finite family of window–threshold scopes remains indexed; no cross-scope scalar is defined without a separate aggregation law and theorem.*

Remark 4.21 (Constructibility caveat)*.* A raw limit or colimit may leave the constructible persistence regime. The terminal Type IV calculus is invoked only in a declared abelian constructible profile category containing the compared source, apex, kernel, and cokernel. The countable Roos lane below is separately typed in R-Mod.

Definition 4.22 (Expanded form of Main Definition 5.19: windowed repaired tower objects)**.** *For every stage define*

$$M_{n;i,W,\tau} := \mathcal{P}_{i,W,\tau}(F_n) := \mathrm{Eval}_{i,W,\tau}(F_n).$$

Definition 4.23 (Expanded form of Main Definition 5.20: repaired tower-profile system)**.** *A repaired tower-profile system consists of the repaired stage profiles, actual transition morphisms in the declared variance, identity and composition proofs, and immutable provenance. It is supplied by an evaluation-lift record, not by the objectwise existence of repaired barcodes.*

Definition 4.24 (Expanded form of Main Definitions 5.16 and 5.21: direct-system source profile)**.** *For a valid direct repaired system, the source*

$$\mathrm{ColimProfile}_{i,W,\tau}(F_\bullet)$$

has authority `categorical`*,* `theorem_identified`*, or* `implementation_only`*. The legacy v20 source labels* `genuine_colimit`*,* `verified_stabilization`*, and* `implementation_artifact` *map respectively to these levels, provided the stabilization label includes the theorem identifying it with the colimit.*

Definition 4.25 (Expanded form of Main Definition 5.17: inverse-system source object)**.** *For a valid inverse system $X_0 \leftarrow X_1 \leftarrow \cdots$, the source is $\varprojlim_n X_n$ when that limit exists in the declared category, or a theorem-identified object equipped with an actual isomorphism to it. The first derived inverse-limit field is separate and is not part of the ordinary source object.*

Definition 4.26 (Source-semantic authority record)**.** *The record states the source authority, the semantic strength required by the claim, the theorem connecting them, and one verdict*

`sufficient`, `insufficient`, `undefined`.

A Type IV, derived-limit, or apex verdict is claim-readable only under `sufficient` *authority for that exact semantic claim.*

Definition 4.27 (Expanded form of Main Definition 5.22: collapsed apex profile)**.** *For a declared filtered apex, define*

$$\mathrm{ApexProfile}_{i,W,\tau}(F_\infty) := \mathrm{Eval}_{i,W,\tau}(F_\infty)$$

only when its repaired evaluation and stage identity are supplied.

Remark 4.28 (No automatic limit–collapse interchange). Neither

$$\varinjlim_n \mathrm{Eval}(F_n) \cong \mathrm{Eval}(F_\infty) \qquad \text{nor} \qquad \varprojlim_n \mathrm{Eval}(F_n) \cong \mathrm{Eval}(F_\infty)$$

is a general Core identity. Hard deletion is not globally functorial or exact; the appropriate repaired system and comparison must be constructed.

4.5 Direct Type IV and inverse apex comparison artifacts

Definition 4.29 (Expanded form of Main Definition 5.24: direct Type IV tower comparison artifact)**.** *A direct Type IV artifact at (i, W, τ) contains a valid repaired direct system, a direct source with sufficient authority, a repaired apex, and an actual morphism*

$$\phi_{i,W,\tau} : \mathrm{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \mathrm{ApexProfile}_{i,W,\tau}(F_\infty).$$

It also contains the ambient abelian constructible profile category, comparison direction, construction proof, terminal mode, terminal-tameness data, full kernel/cokernel artifacts, every consumed comparison-mode and metric record, and immutable claim/provenance identifiers.

Definition 4.30 (Expanded form of Main Definition 5.25: inverse apex comparison record)**.** *An inverse apex record contains an actual inverse system and the ordinary source* $\varprojlim X_n$*. It also records the Roos/*$\varprojlim^1$ *field required by the selected lane, a declared apex* X_∞*, and exactly one actual comparison direction*

$$X_\infty \longrightarrow \varprojlim X_n \qquad \textit{or} \qquad \varprojlim X_n \longrightarrow X_\infty.$$

If terminal Type IV is additionally applied to this map, its ambient profile category, direction, kernel/cokernel, and terminal-tameness data are declared anew. The inverse record is not formed by replacing $\varinjlim$ *with* $\varprojlim$ *in the direct artifact.*

Remark 4.31 (Comparison artifact, not automatic map)*.* No filtered tower, barcode decomposition, bottleneck matching, or notation F_∞ creates either comparison map. It must be explicitly constructed or supplied by a theorem whose hypotheses and source authority are recorded.

Remark 4.32 (A bottleneck matching is not a persistence morphism)*.* A matching controls a metric discrepancy between barcode isomorphism classes. It does not define a natural transformation and therefore does not define the kernel or cokernel used by Type IV.

Definition 4.33 (Expanded form of Main Definition 5.27: exact-admissible tower artifact)**.** *An artifact is exact-admissible if its profile transitions, source construction, comparison map, kernels, and cokernels are supplied by a declared regime in which the retained-long-bar assignment is functorial and preserves every exact sequence consumed by the diagnostic.*

Remark 4.34 (Explicit artifacts remain allowed)*.* Exact admissibility is sufficient, not necessary. An explicitly constructed map is admissible when all typing, authority, ambient-category, defect, and terminal-tameness obligations pass.

Remark 4.35 (Order of operations)*.* The order is persistence, windowing, hard deletion, repaired transition lift, source construction, apex construction, actual comparison, then defect and terminal/transient diagnostics. Windowing and deletion are not silently interchanged.

Remark 4.36 (Why repaired collapse precedes diagnostics)*.* The diagnostic belongs to the repaired Core profile selected by the run. This ordering does not assert global exactness or functoriality of deletion; those properties are replaced by the explicit artifact obligations above.

4.6 Countable inverse systems, Roos defects, and apex agreement

Definition 4.37 (Expanded form of Main Definition 5.29: countable inverse-system record)**.** *Fix a ring* R *and an actual countable inverse system in* R*-Mod,*

$$X_0 \xleftarrow{d_0} X_1 \xleftarrow{d_1} X_2 \leftarrow \cdots .$$

The record contains all objects, maps, compositions, coefficient conventions, level semantics, products used below, and the exact claim. No result in this lane is automatically transferred to an arbitrary abelian, derived, Grothendieck, or higher category.

Definition 4.38 (Expanded form of Main Definition 5.30: Roos operator and first derived-limit defect)**.** *Define*

$$\Delta_X : \prod_{n\geq 0} X_n \longrightarrow \prod_{n\geq 0} X_n, \qquad \Delta_X((x_n)_n) = (x_n - d_n x_{n+1})_n.$$

Then

$$\varprojlim_n X_n = \ker \Delta_X, \qquad \varprojlim_n{}^1 X_n := \operatorname{coker} \Delta_X.$$

For countable systems in R*-Mod, the second object canonically computes* $R^1 \varprojlim X_n$*. It is not a definition of derived inverse limits in a different ambient category.*

Definition 4.39 (Expanded form of Main Definition 5.31: Mittag–Leffler condition)**.** *The inverse system is Mittag–Leffler when, for every fixed n, the chain*

$$\mathrm{im}(X_{n+1} \to X_n) \supseteq \mathrm{im}(X_{n+2} \to X_n) \supseteq \cdots$$

stabilizes.

Theorem 4.40 (Expanded proof of Main Theorem 5.32: countable Mittag–Leffler vanishing)**.** *For a countable inverse system of modules satisfying the Mittag–Leffler condition,*

$$\varprojlim_n{}^1 X_n = 0.$$

This clears the first Roos defect and proves neither existence nor agreement of a declared apex.

Proof. We prove surjectivity of Δ_X. Fix $y = (y_n)_{n\geq 0} \in \prod_{n\geq 0} X_n$. For $m \geq 0$, let $P_m(y)$ be the nonempty set of finite solutions

$$(x_0, \ldots, x_m) \quad \text{such that} \quad x_j - d_j x_{j+1} = y_j \quad (0 \leq j < m).$$

It is nonempty because one may choose x_m arbitrarily and solve recursively backwards.

Fix $r \geq 0$, and for $m \geq r$ let $Q_r^{(m)}(y)$ be the image of $P_m(y)$ under truncation to the first $r+1$ coordinates. These sets form a descending chain. Moreover, $Q_r^{(m)}(y)$ is an affine coset whose translation subgroup is the image of $X_m \to X_r$, embedded in the homogeneous solution space by

$$z_r \longmapsto (d_{0,r} z_r, \ldots, d_{r-1,r} z_r, z_r),$$

where $d_{j,r} : X_r \to X_j$ is the composite transition map. The Mittag–Leffler condition at r makes the subgroups $\mathrm{im}(X_m \to X_r)$ stationary for large m. Once the translation subgroup is stationary, two successive nonempty affine cosets in the descending chain are equal. Hence $Q_r^{(m)}(y)$ stabilizes; denote its stable value by $Q_r^{(\infty)}(y)$.

Truncation induces a surjection

$$Q_{r+1}^{(\infty)}(y) \longrightarrow Q_r^{(\infty)}(y).$$

Indeed, an element of the stable set on the right extends to finite solutions of arbitrarily large length; taking the first $r+2$ coordinates after both chains have stabilized gives an extension in the stable set on the left. Starting with any element of $Q_0^{(\infty)}(y)$ and lifting successively along these surjections produces a compatible sequence of finite prefixes, hence an infinite sequence $(x_n)_{n\geq 0}$ with

$$x_n - d_n x_{n+1} = y_n \qquad (n \geq 0).$$

Thus Δ_X is surjective and $\operatorname{coker} \Delta_X = 0$. All products, images, and affine cosets used in the argument lie in the declared countable module lane. ☐

Corollary 4.41 (Expanded proof of Main Corollary 5.33: finite-dimensional and finite-length ML branches)**.** *An inverse system of finite-dimensional k-vector spaces is Mittag–Leffler. More generally, it is Mittag–Leffler when every fixed X_n has finite composition length. Hence $\varprojlim^1 X_n = 0$ in both countable module regimes.*

Proof. For fixed n, the images form a descending chain of subobjects of X_n. Finite dimension or finite composition length admits only finitely many strict decreases. Apply Theorem 4.40. ☐

Remark 4.42 (No uniform finite cutoff from pointwise ML)**.** The stabilization index may depend on n. Pointwise ML therefore does not supply a single finite verifier for the entire tower. A uniform tail theorem, eventual surjectivity bound, finite-state rule, or another effective reduction is separately required.

Definition 4.43 (Expanded form of Main Definition 5.35: inverse apex defect)**.** *For an actual comparison $f : X_\infty \to \varprojlim X_n$ or $f : \varprojlim X_n \to X_\infty$, define*

$$\mathrm{ApexDef}(f) := \ker f \oplus \operatorname{coker} f.$$

The verdict `apex_agreement` *means* $\mathrm{ApexDef}(f) = 0$*. It is independent of* $\varprojlim^1 X_n = 0$*.*

Definition 4.44 (Expanded form of Main Definition 5.36: inverse-limit audit profile)**.** *An inverse run records*

$$\mathrm{InvAudit} = (\mathrm{Sys}, \mathrm{R1Lim}, \mathrm{Apex}).$$

The three codomains are

Sys	`undefined`, `pass`, `reject`
R1Lim	`not_invoked`, `undefined`, `cleared`, `obstructed`
Apex	`not_invoked`, `undefined`, `agreement`, `disagreement`

No component repairs another.

Theorem 4.45 (Expanded proof of Main Theorem 5.37: derived inverse-limit and apex separation)**.** *Without additional hypotheses, neither*

$$\varprojlim_n{}^1 X_n = 0 \Longrightarrow \mathrm{ApexDef}(f) = 0 \qquad \textit{nor} \qquad \mathrm{ApexDef}(f) = 0 \Longrightarrow \varprojlim_n{}^1 X_n = 0$$

holds.

Proof. For the first implication, use the constant zero inverse system. It is ML, so $\varprojlim^1 = 0$, but the actual comparison $0 \to A$ to any nonzero declared apex has nonzero cokernel.

For the converse, consider

$$\mathbb{Z} \xleftarrow{\times 2} \mathbb{Z} \xleftarrow{\times 2} \mathbb{Z} \xleftarrow{\times 2} \cdots .$$

Its ordinary inverse limit is zero. Its Roos cokernel is nonzero: let $y_n = 1$ for even n and $y_n = 0$ for odd n. If $y = \Delta_X(x)$, then for every m

$$x_0 \equiv \sum_{j=0}^{m-1} 2^j y_j \pmod{2^m}.$$

The right side converges in $\mathbb{Z}_2$ to $1 + 4 + 16 + \cdots = -1/3$, which is not an integer; equivalently an integer satisfying all congruences would obey $3x_0 = -1$. Hence the class of y is nonzero in $\operatorname{coker} \Delta_X$. Declare the categorical inverse limit zero itself as apex and use the identity comparison. Apex agreement holds while $\varprojlim^1 \neq 0$. $\square$

Remark 4.46 (Direct Type IV versus inverse derived-limit audit)*.* The canonical inherited μ, ν diagnostic is attached to an actual direct source-to-apex profile morphism. An inverse system carries its own system, Roos, and apex fields. The terminal kernel/cokernel calculus may be applied to an inverse apex map only after its direction and ambient profile hypotheses are explicitly supplied; it is not inferred from inverse-limit notation.

4.7 Terminal-tame profiles and terminal-generic dimension

Definition 4.47 (Expanded form of Main Definition 5.39: terminal-tame profile)**.** *Let $W = [u, v)$, where $v \in \mathbb{R} \cup \{+\infty\}$, and let M be a constructible persistence profile on W. We call M* terminal-tame *if:*

(i) its fibers near the terminal end are finite-dimensional;

(ii) if $v < +\infty$, there exists $s < v$ such that every structure map $M(t) \to M(t')$ with $s \le t \le t' < v$ is an isomorphism;

(iii) if $v = +\infty$, there exists $s \in \mathbb{R}$ such that every structure map $M(t) \to M(t')$ with $s \le t \le t'$ is an isomorphism.

Definition 4.48 (Expanded form of Main Definition 5.40: terminal-generic dimension)**.** *For a terminal-tame profile M on $W = [u, v)$, define*

$$\operatorname{tgdim}_W(M) := \begin{cases} \lim_{t \uparrow v} \dim_k M(t), & v < +\infty, \\ \lim_{t \to +\infty} \dim_k M(t), & v = +\infty. \end{cases}$$

When $v = +\infty$, the same quantity may be written gdim(M)*. On a bounded window,* gdim *after zero extension is not the canonical diagnostic.*

Proposition 4.49 (Expanded proof of Main Proposition 5.41: well-definedness of terminal–generic dimension)**.** *For every terminal-tame profile M on W, the quantity* $\operatorname{tgdim}_W(M)$ *exists and is a finite nonnegative integer.*

Proof. Terminal tameness supplies a terminal subinterval on which all structure maps are isomorphisms. The finite-dimensional fibers on that subinterval therefore have one constant dimension. This value is exactly the displayed limit. □

Definition 4.50 (Expanded form of Main Definition 5.42: cofinal bar)**.** *A bar I in a barcode profile on $W = [u, v)$ is* cofinal in W *if*

$$\sup I = v \quad \textit{when } v < +\infty,$$

or if it is unbounded above when $v = +\infty$. A cofinal bar on a bounded window is a boundary-reaching bar relative to that window; it is not a global infinite bar.

Proposition 4.51 (Expanded proof of Main Proposition 5.44: terminal dimension counts cofinal bars)**.** *Let M be terminal-tame and interval-decomposable on W. Then* $\operatorname{tgdim}_W(M)$ *equals the multiplicity of bars cofinal in W. In particular, on a right-unbounded window it equals the multiplicity of infinite bars.*

Proof. Choose a terminal tail on which every structure map is an isomorphism. An interval summand contributes one dimension to every fiber on that tail exactly when it is cofinal in W. Every noncofinal interval dies before the tail. Summing the surviving summands proves the claim. □

Remark 4.52 (Endpoint conventions). Endpoint bookkeeping at finite endpoints does not alter terminal-generic dimension. What matters is whether the bar reaches the declared terminal end cofinally.

Proposition 4.53 (Expanded proof of Main Proposition 5.46: terminal tameness passes to defects)**.** *Let $\phi : M \to N$ be a morphism of constructible profiles on W. If M and N are terminal-tame, then* $\ker \phi$ *and* $\operatorname{coker} \phi$ *are terminal-tame.*

Proof. Choose a common terminal tail on which the structure maps of M and N are isomorphisms. Naturality of ϕ gives commuting squares on that tail. The induced maps on pointwise kernels and cokernels are therefore isomorphisms, and finite-dimensionality is inherited pointwise. □

4.8 Defect profiles and canonical diagnostics

Definition 4.54 (Expanded form of Main Definition 5.47: kernel and cokernel defect profiles)**.** *For a valid tower comparison artifact, define*

$$\mathsf{Defect}^{\mathrm{ker}}_{i,W,\tau} := \ker(\phi_{i,W,\tau}), \qquad \mathsf{Defect}^{\mathrm{coker}}_{i,W,\tau} := \mathrm{coker}(\phi_{i,W,\tau}).$$

These are objects of the declared constructible profile category on W.

Remark 4.55 (Kernel and cokernel require a profile regime). The defect profiles are not inferred from a tower of filtered objects or a barcode matching. They require the actual comparison morphism and an abelian profile regime in which pointwise kernels and cokernels are defined.

Remark 4.56 (Interpretation of kernel and cokernel defects). The kernel records repaired tower-side classes killed by the comparison. The cokernel records repaired apex classes not supplied by the tower-side source. Both are artifact-relative, after-window, after-threshold defects.

Definition 4.57 (Full defect profile)**.** *The full defect profile is*

$$\mathsf{Defect}_{i,W,\tau} := \big(\mathsf{Defect}^{\mathrm{ker}}_{i,W,\tau}, \mathsf{Defect}^{\mathrm{coker}}_{i,W,\tau}\big).$$

Definition 4.58 (Expanded form of Main Definition 5.48: degreewise Type IV diagnostics)**.** *Assume that both defect profiles are terminal-tame. Define*

$$\mu_{i,W,\tau} := \mathrm{tgdim}_W\big(\mathsf{Defect}^{\mathrm{ker}}_{i,W,\tau}\big), \qquad \nu_{i,W,\tau} := \mathrm{tgdim}_W\big(\mathsf{Defect}^{\mathrm{coker}}_{i,W,\tau}\big).$$

Thus μ measures terminal-generic kernel defect and ν measures terminal-generic cokernel defect.

Definition 4.59 (Expanded form of Main Definition 5.49: total collapse diagnostics)**.** *For fixed (W,τ) and finite I_{mon}, define*

$$\mu_{\mathrm{Collapse}}(W,\tau) := \sum_{i\in I_{\mathrm{mon}}} \mu_{i,W,\tau}, \qquad \nu_{\mathrm{Collapse}}(W,\tau) := \sum_{i\in I_{\mathrm{mon}}} \nu_{i,W,\tau}.$$

When the fixed pair (W,τ) is clear, the arguments may be suppressed. A run containing several windows or thresholds stores the corresponding pairs $(\mu_{\mathrm{Collapse}}(W_a,\tau_a), \nu_{\mathrm{Collapse}}(W_a,\tau_a))$ separately. No cross-pair scalar total is defined by default, and no infinite aggregate is admitted without a separate aggregation theorem.

The symbols $u_{i,W,\tau}$ and u_{Collapse} are deprecated. The canonical cokernel notation is $\nu_{i,W,\tau}$ and ν_{Collapse}.

Remark 4.60 (Meaning of μ). On a bounded window, $\mu_{i,W,\tau}$ counts kernel defect bars cofinal to the right endpoint. On a right-unbounded window it counts infinite kernel bars.

Remark 4.61 (Meaning of ν). On a bounded window, $\nu_{i,W,\tau}$ counts cokernel defect bars cofinal to the right endpoint. On a right-unbounded window it counts infinite cokernel bars.

Proposition 4.62 (Diagnostics count terminal-cofinal defects)**.** *For every valid terminal-tame artifact, $\mu_{i,W,\tau}$ and $\nu_{i,W,\tau}$ equal the multiplicities of cofinal interval summands in the kernel and cokernel defect profiles, respectively. On a right-unbounded window these are precisely infinite-bar defects.*

Proof. Apply Proposition 4.51 to the two defect profiles. □

Definition 4.63 (Expanded record schema for Main Definition 5.50: Type IV diagnostic record)**.** *A Type IV diagnostic record contains the fixed pair (W,τ), the finite monitored degree set, the declared tower scope, and exactly one terminal status*

`not_invoked`, `undefined`, `certified_zero`, `obstructed`.

Its remaining fields are status-dependent.

(i) For `not_invoked`, the record contains an explicit scope rationale showing that no terminal-generic kernel/cokernel diagnostic is consumed by the claim. A direct or inverse finite-to-infinite, Roos, stable-image, or apex-agreement lane may remain active. No numerical μ, ν, full-interior, or transient value is asserted.

(ii) For `undefined`, the record contains the partial or invalid artifact identifiers that exist and a missing/invalid-obligation marker for every absent, insufficient-authority, or ill-typed input. No Core numerical diagnostic, full-interior absence, or transient absence is asserted.

(iii) For `certified_zero` or `obstructed`, the record contains a valid artifact identifier, source provenance, claim-required source semantics, a `sufficient` semantic-authority verdict, apex identity, ambient abelian category, terminal mode, terminal-tameness certificates, full kernel/cokernel defect-profile artifacts, and all degreewise and fixed-(W, τ) total μ, ν values.

(iv) In a valid invoked record, the full-interior-defect field is recorded separately with exactly one value

`not_checked`, `certified_absent`, `certified_present`.

The value `certified_absent` requires proof that both entire noncofinal defect profiles vanish; `certified_present` requires an actual nonzero noncofinal witness.

(v) When transient detection is invoked, the record additionally contains $\tau_{\mathrm{def}} > 0$, detector component, exact detector stage, information class, certificate status, value status, and one aggregate long-transient verdict

`not_invoked`, `undefined`, `certified_absent`, `certified_present`.

Long-transient `certified_absent` excludes only noncofinal bars of length strictly greater than τ_{def}; it is not the full-interior verdict.

The terminal status is `certified_zero` exactly when all invoked terminal inputs are valid for the claim's source semantics and $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$. It is `obstructed` exactly when those inputs are valid and $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) \neq (0, 0)$. The transient field does not alter the terminal Type IV status. Missing, wrong-stage, or insufficient-authority input yields `undefined`, never absence or numerical zero.

4.9 Typed Type IV status

Definition 4.64 (Expanded form of Main Definition 5.51: monitored terminal-generic Type IV obstruction)**.** *Relative to a valid tower comparison artifact, a run has a* Type IV obstruction *if*

$$(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) \neq (0, 0).$$

On a bounded window this is a defect cofinal to the declared right endpoint; on a right-unbounded window it is an infinite-bar defect.

Definition 4.65 (Expanded form of Main Definition 5.51: Type IV-free in the monitored terminal-generic sense)**.** *Relative to a valid terminal-tame artifact, a run is* Type IV-free in the monitored terminal-generic sense *if*

$$(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0).$$

Proposition 4.66 (Expanded proof of Main Proposition 5.52: characterization of certified Type IV-free status)**.** *A tower diagnostic record has status `certified_zero` if and only if:*

(i) the tower comparison artifact is valid;

(ii) its kernel and cokernel defect profiles are terminal-tame;

(iii) $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$.

It has status **`obstructed`** *when the first two clauses hold and* $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) \neq (0, 0)$. *Missing or ill-typed input yields* **`undefined`**, *never zero.*

Proof. These are exactly the validity, terminal-tameness, and numerical clauses in Definition 4.63. The first two clauses are necessary to ensure that the displayed numbers are defined Core diagnostics rather than missing values. □

Remark 4.67 (No full-isomorphism conclusion). The equality $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$ says that no defect bar reaches the terminal end of the declared window. It does not imply

$$\ker \phi_{i,W,\tau} = 0 \quad \text{or} \quad \operatorname{coker} \phi_{i,W,\tau} = 0.$$

Interior defect bars may remain.

Remark 4.68 (Undefined is not zero). Absence of an actual morphism, invalid source provenance, failure of terminal tameness, or unavailable kernel/cokernel artifacts yields `undefined`. Reporting that state as $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$ is a Type V manifest failure.

Remark 4.69 (No scalar repair of Type IV typing). No metric budget or threshold clearance can manufacture a missing profile system, comparison morphism, terminal-tameness certificate, or defect artifact. Those are non-scalar proof obligations.

4.10 Interior defects and the full-isomorphism upgrade

Definition 4.70 (Interior or noncofinal defect). *An* interior defect *is a noncofinal interval summand of either* $\ker \phi_{i,W,\tau}$ *or* $\operatorname{coker} \phi_{i,W,\tau}$. *On a right-unbounded window these are exactly the finite-bar defects. On a bounded window, a globally finite bar reaching the right endpoint is cofinal and is therefore counted by* tgdim_W, *not classified as an interior defect.*

Definition 4.71 (No-interior-defect condition). *A comparison artifact satisfies the* no-interior-defect condition *if its kernel and cokernel have no noncofinal interval summands. The legacy phrase "no-finite-defect" is retained only as a label alias; the window-relative notion is no-interior-defect.*

Proposition 4.72 (Zero diagnostics plus no interior defects imply isomorphism). *Assume the defect profiles are terminal-tame and interval-decomposable. If*

$$(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0)$$

and the no-interior-defect condition holds, then $\phi_{i,W,\tau}$ *is an isomorphism in the declared abelian constructible profile category.*

Proof. Zero terminal diagnostics exclude every cofinal summand of the kernel and cokernel. The no-interior-defect condition excludes every noncofinal summand. Thus both defect profiles have empty interval decompositions and are zero. A morphism with zero kernel and cokernel is an isomorphism in the declared abelian category. □

Example 4.73 (Expanded witness for Main Example 5.54: zero terminal diagnostics with a nonisomorphism). Let $W = [0, 10)$ and consider

$$0 \longrightarrow k_{[2,3)}.$$

Its cokernel is $k_{[2,3)}$, so the map is not an isomorphism. The bar is not cofinal in W, hence

$$\operatorname{tgdim}_W(k_{[2,3)}) = 0.$$

Thus zero terminal diagnostics do not exclude an interior defect.

Remark 4.74 (Why the interior condition is separated)*.* Type IV intentionally monitors terminal-generic failure. Full comparison isomorphism is stronger and requires a separately recorded interior-defect verdict. This prevents a zero terminal count from being overread as complete agreement.

4.11 Cofinal/interior decomposition and transient-defect certification

Definition 4.75 (Expanded form of Main Definition 5.57: cofinal and interior barcode profiles)**.** *Let M be an interval-decomposable constructible profile on $W = [u, v)$. Define $\operatorname{Cof}_W(M)$ to be the barcode subprofile consisting of all bars cofinal in W, and define $\operatorname{Int}_W(M)$ to consist of all noncofinal bars, with multiplicity. At barcode-profile isomorphism strength,*

$$M \cong \operatorname{Cof}_W(M) \oplus \operatorname{Int}_W(M).$$

The decomposition is canonical on barcode isomorphism classes. It is not asserted to be functorial on arbitrary persistence morphisms.

Proposition 4.76 (Expanded proof of Main Proposition 5.58: terminal/interior separation)**.** *If M is terminal-tame and interval-decomposable, then*

$$\operatorname{tgdim}_W(M) = \#\,\mathcal{B}(\operatorname{Cof}_W(M))$$

with multiplicity, and

$$\operatorname{tgdim}_W(M) = 0 \iff \operatorname{Cof}_W(M) = 0 \iff M \cong \operatorname{Int}_W(M).$$

Proof. The first assertion is Proposition 4.51. A direct sum of cofinal intervals has terminal-generic dimension zero exactly when it is empty. The barcode decomposition gives the final equivalence. $\square$

Definition 4.77 (Kernel/cokernel component notation)**.** *Write*

$$K_{i,W,\tau} := \ker \phi_{i,W,\tau}, \qquad Q_{i,W,\tau} := \operatorname{coker} \phi_{i,W,\tau},$$

and define the interior components

$$K^{\circ}_{i,W,\tau} := \operatorname{Int}_W(K_{i,W,\tau}), \qquad Q^{\circ}_{i,W,\tau} := \operatorname{Int}_W(Q_{i,W,\tau}).$$

These are barcode-profile constructions attached to the actual comparison artifact. Their immutable identifiers are distinct from those of the full defect profiles.

Definition 4.78 (Expanded form of Main Definition 5.59: defect threshold and detector component)**.** *A transient-defect run declares a threshold*

$$\tau_{\mathrm{def}} > 0$$

independently of the collapse threshold τ, and declares exactly one component for each detector application:

$$\texttt{full\_profile} \qquad \text{or} \qquad \texttt{interior\_noncofinal}.$$

The full component reads $K_{i,W,\tau}$ or $Q_{i,W,\tau}$; the interior component reads $K^{\circ}_{i,W,\tau}$ or $Q^{\circ}_{i,W,\tau}$. Changing τ_{def}, component, endpoint convention, information class, or detector stage creates a new detector claim and new artifact.

Definition 4.79 (Expanded form of Main Definition 5.60: kernel and cokernel defect detectors). *Let $\mathcal{D}$ be an Appendix B finite detector with all applicability, soundness, and completeness data registered anew at source stage `kernel_defect` or `cokernel_defect`. For the declared component set*

$$K^{\mathrm{det}} \in \{K_{i,W,\tau}, K^{\circ}_{i,W,\tau}\}, \qquad Q^{\mathrm{det}} \in \{Q_{i,W,\tau}, Q^{\circ}_{i,W,\tau}\},$$

and write

$$\mathcal{D}^{\mathrm{ker}}_{\tau_{\mathrm{def}}} := \mathcal{D}_{\tau_{\mathrm{def}}}(K^{\mathrm{det}}), \qquad \mathcal{D}^{\mathrm{coker}}_{\tau_{\mathrm{def}}} := \mathcal{D}_{\tau_{\mathrm{def}}}(Q^{\mathrm{det}}).$$

Stencil, birth, germ, and certified family-mesh detectors may be used only at their exact Appendix B strength. A topological-profile detector artifact is not relabeled as a defect detector.

Theorem 4.80 (Expanded proof of Main Theorem 5.61: complete transient-defect characterization). *Assume a valid tower comparison artifact, full constructible defect profiles, and complete detector certificates at the declared defect stages and components. Then*

$$\mathcal{D}^{\mathrm{ker}}_{\tau_{\mathrm{def}}} = 0 \iff T^{\mathrm{bar}}_{\tau_{\mathrm{def}}}(K^{\mathrm{det}}) = 0,$$

and

$$\mathcal{D}^{\mathrm{coker}}_{\tau_{\mathrm{def}}} = 0 \iff T^{\mathrm{bar}}_{\tau_{\mathrm{def}}}(Q^{\mathrm{det}}) = 0.$$

A certified nonzero detector value is equivalent to the existence of a bar of length strictly greater than τ_{def} in the corresponding declared component.

Proof. Apply the Appendix B detector soundness/completeness theorem at threshold τ_{def} to the supplied constructible defect profile. The source stage, component, endpoint convention, coverage or birth/germ exhaustiveness, family membership, excess provenance, and immutable artifact remain explicit hypotheses. This is a fresh defect-stage certificate, not transport of a detector record from another stage. □

Remark 4.81 (Defect-stage registration). A detector registered at `windowed_prethreshold` or `repaired_postthreshold` cannot be reused by changing its stage label. Kernel and cokernel profiles, information classes, coverage data, and certificate hashes are reconstructed and registered anew.

Corollary 4.82 (Expanded proof of Main Corollary 5.63: full and interior detectors coincide under terminal zero). *Assume the kernel and cokernel defect profiles are terminal-tame and interval-decomposable. If*

$$(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0,0),$$

then

$$K_{i,W,\tau} \cong K^{\circ}_{i,W,\tau}, \qquad Q_{i,W,\tau} \cong Q^{\circ}_{i,W,\tau}$$

at barcode-profile strength. Consequently complete full-profile and corresponding interior detectors have identical zero/nonzero verdicts, after the detector invariance under exact profile isomorphism is recorded.

Proof. Apply Proposition 4.76 to the kernel and cokernel profiles and then Appendix B, Proposition 2.57. □

Definition 4.83 (Expanded form of Main Definition 5.64: aggregate long-transient verdict). *For fixed $(W, \tau, \tau_{\mathrm{def}}, I_{\mathrm{mon}})$, the aggregate long-transient verdict is exactly one of*

`not_invoked`, `undefined`, `certified_absent`, `certified_present`.

It is `certified_absent` exactly when every required kernel and cokernel detector certificate is complete and every detector value is zero. It is `certified_present` when all artifacts needed to interpret the witness are well-typed and at least one required detector value is nonzero. Missing, incomplete, wrong-component, or wrong-stage evidence yields `undefined`, never absence.

Theorem 4.84 (Expanded proof of Main Theorem 5.65: joint terminal/transient defect classification)**.** *Assume a valid tower comparison artifact, terminal-tame interval-decomposable defect profiles, and complete interior defect detectors at τ_{def}. Exactly the following mathematical cases occur degreewise and after finite aggregation over the fixed monitored degrees:*

(i) if $(\mu, \nu) \neq (0, 0)$, a terminal/cofinal Type IV defect is present; the interior detector independently records whether a long noncofinal defect is also present;

(ii) if $(\mu, \nu) = (0, 0)$ and the long-transient verdict is `certified_present`, there is no terminal defect but there is a noncofinal kernel or cokernel bar of length greater than τ_{def};

(iii) if $(\mu, \nu) = (0, 0)$ and the long-transient verdict is `certified_absent`, there is no terminal defect and no noncofinal defect bar longer than τ_{def}. Shorter nonzero defects may remain.

No undefined detector certificate is assigned a mathematical zero verdict.

Proof. Terminal/cofinal presence is characterized by Proposition 4.76; long noncofinal presence or absence is characterized by Theorem 4.80. The conclusions are independent conjunctions. ☐

Definition 4.85 (Expanded form of Main Definition 5.66: short-defect exclusion certificate)**.** *A short-defect exclusion certificate at τ_{def} proves that every nonzero bar in the declared interior kernel and cokernel profiles has length strictly greater than τ_{def}. It is a structural theorem or artifact; it is not inferred from detector zero, terminal-generic zero, metric slack, or finite sampling.*

Theorem 4.86 (Expanded proof of Main Theorem 5.67: conditional full-isomorphism upgrade)**.** *Let $\phi_{i,W,\tau}$ be a valid morphism in the declared abelian profile category, and assume its kernel and cokernel are terminal-tame and interval-decomposable. Suppose:*

(i) $(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0)$;

(ii) complete interior kernel and cokernel detectors at τ_{def} are certified zero;

(iii) a short-defect exclusion certificate holds.

Then

$$\ker \phi_{i,W,\tau} = 0, \qquad \operatorname{coker} \phi_{i,W,\tau} = 0,$$

and $\phi_{i,W,\tau}$ is an isomorphism.

Proof. Terminal zero excludes all cofinal kernel and cokernel bars. Complete interior detector zero excludes every interior bar longer than τ_{def}. The short-defect exclusion certificate says that any remaining nonzero interior bar would have length strictly greater than that threshold, a contradiction. Hence both full defect profiles vanish, and a morphism with zero kernel and cokernel is an isomorphism in an abelian category. ☐

Example 4.87 (Expanded witness for Main Example 5.68: zero terminal and transient diagnostics with a short residual defect)*.* Let $W = [0, 10)$, choose $\tau_{\mathrm{def}} > 0$, and consider

$$0 \longrightarrow k_{[2,\, 2+\tau_{\mathrm{def}}/2)}.$$

The terminal pair is zero, and every complete defect detector at threshold τ_{def} is zero because the sole defect bar is deleted by that threshold. The morphism is not an isomorphism. Thus the short-defect exclusion in Theorem 4.86 is essential.

Remark 4.88 (No automatic metric transport of defect profiles). A bottleneck or interleaving bound between source profiles and a separate bound between apex profiles do not by themselves induce a bound between their kernels or cokernels. Metric transport of a defect-detector verdict requires actual comparison data between the defect profiles and an independently proved kernel/cokernel stability theorem. Only then may Appendix A's threshold-gap and zero-separation machinery be applied at stage `kernel_defect` or `cokernel_defect`.

4.12 Pure kernel, pure cokernel, and mixed modes

Definition 4.89 (Pure kernel Type IV). *A valid artifact has* pure kernel Type IV *if*

$$\mu_{\mathrm{Collapse}} > 0, \qquad \nu_{\mathrm{Collapse}} = 0.$$

Definition 4.90 (Pure cokernel Type IV). *A valid artifact has* pure cokernel Type IV *if*

$$\mu_{\mathrm{Collapse}} = 0, \qquad \nu_{\mathrm{Collapse}} > 0.$$

Definition 4.91 (Mixed Type IV). *A valid artifact has* mixed Type IV *if*

$$\mu_{\mathrm{Collapse}} > 0, \qquad \nu_{\mathrm{Collapse}} > 0.$$

Remark 4.92 (Expanded witness for Main Example 5.55: interpretation of the three modes). Let K_W denote one cofinal interval bar: $k_{[u,v)}$ for bounded $W = [u, v)$, and $k_{[u,+\infty)}$ for a right-unbounded window. Then the zero maps

$$K_W \to 0, \qquad 0 \to K_W, \qquad K_W \oplus 0 \to 0 \oplus K_W$$

have diagnostics $(1, 0)$, $(0, 1)$, and $(1, 1)$, respectively. These are artifact-level kernel loss, apex-only birth, and mixed failure. No external geometric or arithmetic interpretation follows without an interface theorem.

4.13 Finite-stage admissibility does not determine Type IV

Definition 4.93 (Finite-level collapse admissibility). *A finite stage F_n is* finite-level collapse-admissible *on (W, τ) if its repaired topological clause passes and, when the Ext clause is invoked, a valid eligible realization record discharges that clause. This stagewise notion contains no undeclared tower-comparison verdict.*

Proposition 4.94 (Finite admissibility does not imply monitored tower admissibility). *There exists a valid artifact-level witness in which every finite stage is finite-level collapse-admissible while the declared source-to-apex comparison has pure cokernel Type IV.*

Proof. Fix $W = [u, v)$ and let every repaired finite-stage profile be zero, with zero transition morphisms. Take the source profile to be the genuine colimit zero. Declare an apex profile K_W consisting of one cofinal bar and take the actual comparison morphism

$$0 \longrightarrow K_W.$$

Every finite stage has zero repaired degree-one profile and may be represented by the zero filtered complex. The comparison cokernel is K_W, so

$$(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 1).$$

The finite profiles and the cofinal apex barcode admit filtered realizations by the finite-barcode realization theorem of Appendix B, Theorem 2.95. This is a declared apex-mismatch witness; it does not claim that an arbitrary externally chosen apex is the categorical colimit of the tower. ☐

Remark 4.95 (Why Type IV is invisible to finite stages). Finite-stage evaluations do not determine the kernel or cokernel of an undeclared source-to-apex morphism. Type IV becomes meaningful only after the comparison artifact has been supplied.

4.14 Eventual constancy and exact apex Return

Definition 4.96 (Expanded form of Main Definition 5.70: eventual-isomorphism certificate)**.** *An eventual-isomorphism certificate for a direct or inverse tower contains an index N and proof that every transition on the tail $n \geq N$ is an isomorphism, with all tower compatibility. Equality of objects, dimensions, barcodes, or invariants on an inspected finite prefix is not such a certificate.*

Theorem 4.97 (Expanded proof of Main Theorem 5.71: eventually constant direct-system Return)**.** *Let $X_0 \to X_1 \to \cdots$ be a direct system in a category admitting its sequential colimit. If every transition after N is an isomorphism, then*

$$X_N \xrightarrow{\sim} \varinjlim_n X_n.$$

If a declared apex X_∞ has a compatible isomorphism $a_N : X_N \xrightarrow{\sim} X_\infty$, then the induced direct source-to-apex comparison is an isomorphism.

Proof. For $n \leq N$, let $c_n : X_n \to X_N$ be the transition to N; for $n \geq N$, let $c_n : X_n \to X_N$ be the inverse of the tail transition from N to n. These maps form a cocone. Any cocone (Y, y_n) is uniquely determined by y_N, because $y_n = y_N c_n$. Thus X_N has the universal property of the colimit, proving the first isomorphism. The apex cocone $a_N c_n$ induces the comparison $a_N \circ (X_N \to \varinjlim X_n)^{-1}$, hence an isomorphism. □

Theorem 4.98 (Expanded proof of Main Theorem 5.72: eventually constant inverse-system Return)**.** *Let $X_0 \leftarrow X_1 \leftarrow \cdots$ be a countable inverse system of modules. If every transition after N is an isomorphism, then*

$$\varprojlim_n X_n \cong X_N, \qquad \varprojlim_n{}^1 X_n = 0.$$

A declared apex compatibly isomorphic to X_N is therefore Returned at both internal object-isomorphism strength and apex-agreement strength.

Proof. A compatible sequence is uniquely determined by its coordinate in X_N: the invertible tail determines all higher coordinates, and the finite prefix is obtained by the transition maps. Conversely every element of X_N produces a compatible sequence. The tower is ML, indeed eventually constant, so Theorem 4.40 gives $\varprojlim^1 = 0$. Compose the evaluation isomorphism with the declared apex isomorphism. □

Corollary 4.99 (Inherited v20 exact comparison consequence)**.** *In the direct repaired-profile lane, the hypotheses of Theorem 4.97 and compatible apex agreement make the comparison $\phi_{i,W,\tau}$ an isomorphism. Its full kernel and cokernel vanish; hence the invoked Type IV field is* `certified_zero`*, full interior defect is certified absent, and every well-typed complete transient detector is certified absent.*

Proof. An isomorphism in the declared abelian profile category has zero kernel and cokernel. All terminal, cofinal, interior, and thresholded defect predicates therefore vanish at their registered strength. □

Remark 4.100 (Observed stabilization is not eventual constancy). A finite plateau does not prove the quantified tail. The certificate must come from induction, a control theorem, a recurrence, a finite-state rule, a termination argument, or another registered proof.

Remark 4.101 (Main ownership of finite-to-infinite reductions). Main Chapter 5 owns the internal tower reduction theorems, while Main Chapter 8 owns their verifier-DAG and final Return consumption. Appendix D supplies expanded proofs and artifacts; it creates neither a stronger generic reduction principle nor an external-domain theorem.

Remark 4.102 (Finite reduction preserves source semantics only by theorem). An eventual-constancy, finite-state, generation-bound, or stable-image theorem may promote a finite witness to a theorem-identified source at exactly its proved strength. Without that theorem, a finite proxy remains `implementation_only`, even if all inspected levels agree.

4.15 Finite length, Noetherian images, and finite generation

Theorem 4.103 (Expanded proof of Main Theorem 5.73: finite-length chain stabilization). *Let M have finite composition length $\ell(M)$. Every strictly increasing or strictly decreasing chain of subobjects has at most $\ell(M)$ strict inclusions. Consequently every monotone subobject chain stabilizes.*

Proof. For a strict inclusion $A \subsetneq B$, the quotient B/A is nonzero and has positive composition length, so $\ell(B) \geq \ell(A) + 1$. All subobject lengths lie between 0 and $\ell(M)$. The descending assertion follows by applying the same argument to successive quotient lengths. □

Corollary 4.104 (Expanded proof of Main Corollary 5.74: stable image of a finite-length endomorphism). *For an endomorphism $T : M \to M$ of a finite-length object, the image and kernel chains stabilize. At any common stabilization exponent e,*

$$T|_{\operatorname{im} T^e} : \operatorname{im} T^e \xrightarrow{\sim} \operatorname{im} T^e$$

is an automorphism.

Proof. Apply Theorem 4.103. On the stable image the restriction is surjective. A surjective endomorphism of a finite-length object has zero-length cokernel and equal source/target length, hence zero-length kernel; it is an isomorphism. □

Theorem 4.105 (Expanded proof of Main Theorem 5.75: Noetherian ascending-image reduction). *Let M be Noetherian and suppose a direct tower has compatible maps $X_n \to M$ whose images form an ascending chain. Then there is N with*

$$\operatorname{im}(X_N \to M) = \operatorname{im}(X_n \to M) \qquad (n \geq N).$$

Every conclusion depending only on this stable image reduces to a finite stage once the embedding and monotonicity data are certified.

Proof. The ascending-chain condition for the Noetherian object M gives stabilization. The reduction strength is exactly the stable-image predicate; it does not identify the entire tower or its colimit unless another theorem does so. □

Remark 4.106 (Noetherian is not Artinian). Noetherianity alone does not stabilize descending image chains in an inverse system. That direction requires Artinian, finite-length, finite-dimensional, eventual-surjectivity, finite-state, or another separately proved hypothesis.

Theorem 4.107 (Expanded proof of Main Theorem 5.77: finite-generation finite-window zero reduction). *Let $A = \bigoplus_{d \geq 0} A_d$ be nonnegatively graded and $M = \bigoplus_{d \geq 0} M_d$ a graded A-module with certified generation bound D, meaning that M is generated by $\bigoplus_{0 \leq d \leq D} M_d$. Then*

$$M = 0 \iff M_d = 0 \text{ for } 0 \leq d \leq D.$$

If $f : M \to N$ is graded and both $\ker f$ *and* $\operatorname{coker} f$ *have certified generation bound D, then f is an isomorphism exactly when f_d is an isomorphism for every $0 \le d \le D$.*

Proof. Every homogeneous element of M is an A-linear combination of the displayed generating window; if that window is zero, every element is zero. The converse is immediate. Apply the zero criterion to the kernel and cokernel of f for the second assertion. □

Remark 4.108 (Unknown generation bound). Finite generation without a certified degree bound does not license a predetermined finite verification window. The bound is a non-scalar theorem input and belongs to the reduction record.

4.16 Deterministic finite-state tail certification

Definition 4.109 (Expanded form of Main Definition 5.79: deterministic finite-state tail system)**.** *A deterministic tail system is a finite set S, a transition $T : S \to S$, an initial state s_0, and a forward-invariant terminal set $Z \subseteq S$. State equality includes every typed datum consumed by the theorem; equality of dimensions or summary invariants alone is insufficient.*

Theorem 4.110 (Expanded proof of Main Theorem 5.80: finite-state terminality)**.** *Let $|S| = N$. Among the states $s_0, \dots, s_N$, exactly one of the following is certified:*

(i) a state lies in Z, after which all states remain terminal;

(ii) two nonterminal states repeat, after which the orbit is periodic outside Z and never reaches Z.

Thus eventual terminality is decided by at most $N + 1$ states.

Proof. If a terminal state occurs, forward invariance gives permanent terminality. Otherwise the $N + 1$ states all lie in the at most N-element set $S \setminus Z$, so two repeat. Determinism repeats the entire subsequent orbit. If it later entered Z, the corresponding earlier repeated orbit would also enter, contradicting the assumed nonterminal prefix. □

Corollary 4.111 (Expanded proof of Main Corollary 5.81: finite-state eventual-constancy lane)**.** *If terminal states are exactly those after which every tower transition is an isomorphism, the finite-state certificate either activates Theorem 4.97 or Theorem 4.98, or exhibits a nonterminal cycle and rejects that eventual-constancy lane.*

Definition 4.112 (Expanded form of Main Definition 5.82: typed finite-state linear inverse tower)**.** *A typed finite-state linear inverse tower over k consists of a finite state set S, a finite-dimensional based vector space V_s for each state, a deterministic transition $\sigma : S \to S$, a matrix $D_s : V_{\sigma(s)} \to V_s$, and an initial state. It generates an actual inverse tower $V_0 \leftarrow V_1 \leftarrow \cdots$. The first repeated state supplies a certified entry N and period r, with the based identification $V_{N+r} = V_N$.*

Definition 4.113 (Expanded form of Main Definition 5.83: periodic monodromy and stable image)**.** *For the entry-period record define*

$$T = d_N d_{N+1} \cdots d_{N+r-1} : V_N \to V_N.$$

With $e = \dim_k V_N$, set

$$W_{\mathrm{st}} := \operatorname{im} T^e = \bigcap_{q \ge 0} \operatorname{im} T^q.$$

Finite-dimensional image stabilization implies that $T|_{W_{\mathrm{st}}}$ is an automorphism.

Theorem 4.114 (Expanded proof of Main Theorem 5.84: exact finite-state inverse-limit Return)**.** *For a typed finite-state linear inverse tower, evaluation at the periodic entry induces*

$$\mathrm{ev}_N : \varprojlim_n V_n \xrightarrow{\sim} W_{\mathrm{st}}, \qquad \varprojlim_n{}^1 V_n = 0.$$

The isomorphism is canonical relative to the entry-period and based state identifications and is natural only for morphisms commuting with all those data. Moreover

$$\varprojlim_n V_n = 0 \quad\Longleftrightarrow\quad T^e = 0 \quad\Longleftrightarrow\quad T \text{ is nilpotent.}$$

Proof. For a compatible sequence (x_n), periodic compatibility gives $x_N = T^q x_{N+qr}$, hence $x_N \in \bigcap_q \operatorname{im} T^q = W_{\mathrm{st}}$. Conversely, for $w \in W_{\mathrm{st}}$, define the period-boundary values by $x_{N+qr} = T^{-q}w$. Applying the actual transition matrices down each period determines every intermediate coordinate, and the finite prefix is determined by composition to lower levels. These constructions are mutually inverse to evaluation at N. The tower is one of finite-dimensional vector spaces, so Corollary 4.41 clears $\varprojlim^1$. Finally $W_{\mathrm{st}} = 0$ iff $\operatorname{im} T^e = 0$, which is equivalent to nilpotence in finite dimension. ☐

Theorem 4.115 (Expanded proof of Main Theorem 5.85: finite-state apex-agreement criterion)**.** *Let A be a finite-dimensional declared apex with a compatible family $\alpha_n : A \to V_n$. Under Theorem 4.114, the induced map $\alpha : A \to \varprojlim V_n$ is an isomorphism if and only if the entry component*

$$\alpha_N : A \longrightarrow W_{\mathrm{st}}$$

is an isomorphism. Compatibility already forces its image into W_{st}.

Proof. Under the evaluation isomorphism, the composite $A \to \varprojlim V_n \to W_{\mathrm{st}}$ is exactly α_N. Therefore one map is an isomorphism exactly when the other is. ☐

Example 4.116 (Replayable finite-state stable-image and apex witness)*.* Let the state set have one state, let $V = k^3$, and let every inverse transition be

$$T = \operatorname{diag}(1, 1, 0) : V \longrightarrow V.$$

The certified entry is $N = 0$, the period is $r = 1$, and

$$W_{\mathrm{st}} = \operatorname{im} T = \operatorname{span}\{e_1, e_2\}.$$

Hence

$$\varprojlim_n V \cong k^2, \qquad \varprojlim_n{}^1 V = 0.$$

Take the declared apex $A = k^2$ and let every α_n be the standard inclusion into the first two coordinates. Compatibility holds because T is the identity on W_{st}, and $\alpha_0 : A \to W_{\mathrm{st}}$ is an isomorphism. Thus this finite table simultaneously certifies a nontrivial inverse-limit object, a cleared Roos field, and apex agreement, while the tower is not eventually constant by isomorphisms on the whole three-dimensional space.

Remark 4.117 (Finite-state lane exceeds eventual constancy)*.* The individual periodic transition maps need not be isomorphisms on their whole spaces. The inverse limit retains precisely the recurrent stable-image sector, so this lane is strictly stronger than eventual constancy while still being finite and exact.

4.17 Finite-to-infinite records, soundness, and supported Return strengths

Definition 4.118 (Expanded form of Main Definition 5.87: finite-to-infinite reduction record)**.** *A reduction record contains tower variance and actual maps; finite witness and artifact identity; exact theorem and hypotheses; exceptional finite range and tail coverage; every eventual-isomorphism, generation-bound, stable-image, ML, surjective-tail, finite-state, or recurrence certificate used; source and semantic authority; required derived-limit fields; apex and actual comparison; the route-admissible downward-closed internal Return set and its maximal supported strengths; explicit failure boundaries; and Appendix F/G verifier, dependency, provenance, and replay fields. A unique "Return ceiling" is recorded only when the supported set has a proved greatest element.*

Theorem 4.119 (Expanded proof of Main Theorem 5.88: finite-to-infinite soundness)**.** *If a reduction record is complete and consistent, its finite witness passes, every hypothesis and tail obligation of the named theorem passes, and every required ordinary limit, derived-limit, and apex comparison clause passes, then every strength in the registered route-admissible supported set holds. No strength outside that set follows. Missing required evidence makes the requested conclusion undefined; an evidenced false hypothesis rejects the selected lane.*

Proof. Apply the named mathematical theorem to the verified hypotheses and actual maps. The record wrapper contributes only typing, dependency, and status semantics, so it cannot enlarge the theorem's conclusion. The undefined and reject clauses follow from the canonical four-value audit discipline. □

Definition 4.120 (Expanded form of Main Definition 5.89: internal apex Return capability profile)**.** *Let $\mathfrak{S}_{\mathrm{int}}$ be the partially ordered family of typed internal strengths, including*

$$\texttt{tail\_predicate},\quad \texttt{internal\_object},\quad \texttt{internal\_isomorphism},$$
$$\texttt{inverse\_limit\_object},\quad \texttt{derived\_limit\_clear},\quad \texttt{apex\_agreement}.$$

Each required route component e supplies a downward-closed supported set $\mathfrak{S}_e$. The route profile is

$$\mathfrak{S}_{\mathrm{route}} = \bigcap_{e\in E_{\mathrm{required}}} \mathfrak{S}_e.$$

It may have several incomparable maximal elements. In particular, `derived_limit_clear` *and* `apex_agreement` *are independent.*

4.18 Artifact-coherent Type IV and joint defect-clean bands

Definition 4.121 (Degreewise artifact-coherent stable band)**.** *Fix i, a window W, a tower-profile construction rule, source provenance, and a terminal mode. An interval $J \subset (0, +\infty)$ is a* degreewise stable band *if for every $\tau \in J$:*

(i) the same declared construction rule supplies a valid artifact $\phi_{i,W,\tau}$;

(ii) source provenance and terminal mode do not change silently;

(iii) the defect profiles are terminal-tame;

(iv) every repaired comparison used by the artifact declares one v21 comparison mode: metric mode carries its two-sided threshold-gap, crop-budget, ledger, scalarization, and evidenced discrepancy records; exact mode carries its exact proof; theorem-controlled nonmetric mode carries the named theorem and typed conclusion;

(v) the Type IV diagnostic status is `certified_zero`*.*

Definition 4.122 (Expanded form of Main Definition 5.90: artifact-coherent Type IV stable band)**.** *Fix a window, finite monitored degree set, repaired tower-profile construction, source semantics, and terminal mode. An interval $J \subset (0, +\infty)$ is an* artifact-coherent Type IV stable band *when, for every $\tau \in J$:*

(i) one declared construction rule supplies all valid comparison artifacts;

(ii) source semantics, provenance, authority, comparison direction, and terminal mode do not change silently;

(iii) all defect profiles are terminal-tame;

(iv) every repaired comparison records exactly one comparison mode: metric mode carries its ledger, scalarization, evidenced discrepancy, and strict threshold gap; exact mode carries the actual equality or profile-isomorphism proof; theorem-controlled nonmetric mode carries its named theorem, hypotheses, and exact conclusion;

(v) every associated Type IV record is `certified_zero`*.*

The degreewise record of Definition 4.121 is the single-degree specialization.

Proposition 4.123 (Degreewise stable bands exclude degreewise Type IV)**.** *If J is a degreewise stable band for i, then no terminal-generic Type IV obstruction occurs in degree i for any $\tau \in J$, relative to the declared artifact family.*

Proof. Every threshold in the band has a valid record with status `certified_zero`. Theorem 4.66 applies pointwise. □

Proposition 4.124 (Expanded proof of Main Proposition 5.92: no terminal-generic Type IV obstruction on a stable band)**.** *If J is a global stable band, then every $\tau \in J$ is Type IV-free in the monitored terminal-generic sense relative to its declared artifact.*

Proof. Apply Proposition 4.123 in every monitored degree and sum over the finite monitored scope. □

Remark 4.125 (Uniform reserve is a separate claim)*.* If one metric budget is claimed uniformly over J, the manifest must provide a common ledger $\mathfrak{d}_{\mathrm{met},J}$ and prove

$$\mathrm{val}_{\mathrm{B}}(\mathfrak{d}_{\mathrm{met},J}) < \inf_{\tau \in J} \mathrm{gap}_{\tau}(\mathcal{B}_{\tau}).$$

Pointwise positive reserves do not imply that this infimum is positive.

Remark 4.126 (Stable band does not mean full-isomorphism band)*.* A stable band controls terminal-generic defects. Interior defects may remain at some or all thresholds.

Definition 4.127 (Full-isomorphism band)**.** *A threshold interval J is a* full-isomorphism band *if every artifact $\phi_{i,W,\tau}$, for every monitored degree and every $\tau \in J$, is an isomorphism in the declared constructible profile category.*

Proposition 4.128 (Full-isomorphism band implies stable band)**.** *Every artifact-coherent full-isomorphism band is a global stable band.*

Proof. An isomorphism has zero kernel and cokernel, hence zero terminal diagnostics and a `certified_zero` record, provided the remaining artifact fields are present. □

Remark 4.129 (Converse requires interior-defect data). A global stable band becomes a full-isomorphism band only after the no-interior-defect condition is verified throughout the band.

Definition 4.130 (Expanded form of Main Definition 5.93: artifact-coherent joint defect-clean band)**.** *An interval $J \subset (0, +\infty)$ is a joint defect-clean band at fixed $\tau_{\mathrm{def}} > 0$ if it is an artifact-coherent Type IV stable band and, for every $\tau \in J$:*

(i) the same defect threshold, detector stages, endpoint convention, component policy, information class, and completeness rule are used;

(ii) complete kernel and cokernel interior detectors are `certified_zero`*;*

(iii) detector artifacts, source semantics, and semantic authority remain auditable across the band.

Proposition 4.131 (Expanded proof of Main Proposition 5.94: uniform terminal and long–transient cleanliness)**.** *On an artifact-coherent joint defect-clean band, every threshold is free of monitored terminal defects and monitored noncofinal defects longer than τ_{def}. Full isomorphism does not follow without a uniform short-defect exclusion theorem or direct full-defect vanishing.*

Proof. Apply Proposition 4.124 and Theorem 4.84 pointwise. ☐

4.19 Integration with B-Gate^{+}

Definition 4.132 (Expanded form of Main Definition 5.95: Type IV and transient-defect gate discharge)**.** *The terminal and optional transient clauses are discharged as follows:*

(i) if no terminal-generic comparison diagnostic is consumed, the Type IV field is `not_invoked`*; independent direct/inverse reduction, Roos, stable-image, or apex fields may remain active;*

(ii) an invoked direct Type IV record must be valid, have sufficient source authority, and be `certified_zero`*;*

(iii) a required long-transient clause must be `certified_absent` *at the declared τ_{def};*

(iv) a required full-isomorphism clause needs direct full-defect vanishing or all hypotheses of Theorem 4.86;

(v) inverse-system Sys, R1Lim, *and* Apex *fields are evaluated independently at the strength required by the selected route.*

Remark 4.133 (Meaning of terminal zero in the gate). The shorthand $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0, 0)$ is serialized only after the terminal diagnostic has been proved defined. An out-of-scope diagnostic uses `not_invoked`, never a fabricated numerical pair.

Proposition 4.134 (Inherited degree-one eligible reduction of tower-clean admissibility)**.** *In the inherited degree-one eligible regime, the categorical clause is discharged by* $\mathrm{Eval}_{1,W,\tau}(F) = 0$*. The remaining independent structural conditions are the repaired topological zero, a correctly typed Type IV field that is* `not_invoked` *or* `certified_zero`*, and the absence of required transient, full-isomorphism, and manifest failures.*

Proof. Apply Appendix C's degree-one bridge. The tower and manifest clauses concern separate artifacts and remain conjuncts. ☐

Proposition 4.135 (Inherited v20 monitored eligible reduction)**.** *For a finite monitored degree set in the degreewise eligible regime, repaired zero in every invoked degree discharges every invoked Ext clause. Type IV, required long-transient absence, requested full comparison isomorphism, finite-to-infinite route validity, and Type V cleanliness remain independent.*

Proof. Apply the degreewise Appendix C bridge and take the finite conjunction. None of the remaining fields is a logical consequence of Ext vanishing. □

Remark 4.136 (Relation to B-Gate$^+$). The final B-Gate$^+$ pass additionally consumes every applicable metric, comparison, detector, representative, limit, derived-limit, apex, claim-status, and manifest clause. Terminal or categorical zero does not compensate for a failure in another field.

Proposition 4.137 (Expanded proof of Main Proposition 5.97: audited admissibility implies B-Gate$^+$ structural and safety clauses)**.** *An audited collapse-admissible run has passing required topological and eligible categorical clauses; a correctly excluded or certified-zero Type IV field; every required transient/full-isomorphism field; every required metric, exact, limit, derived-limit, and apex field; and no remaining Type III or Type V defect in those records. Hence the corresponding required clauses of B-Gate$^+$ pass. No external-domain conclusion follows without its separate Interface and Return theorem.*

Proof. Each assertion is a conjunct in Definitions 4.12 and 4.13, together with the discharge rule of Definition 4.132. The external limitation is the project-wide Return separation. □

Definition 4.138 (Expanded form of Main Definition 5.98: failure routing record)**.** *A failure routing record contains every applicable Type I–V label, the first failed audit stage, the exact failed clause and artifact or explicit missing marker, the mathematical/operational/diagnostic/manifest class, the terminal and transient fields when applicable, and the permitted repair action without changing claim status. No scalar score replaces this typed record.*

Definition 4.139 (Expanded form of Main Definition 5.99: v21 tower, failure, and finite-to-infinite contract)**.** *The v21 contract requires nonexclusive typed failure routing; degreewise eligibility for Type II; actual comparison and terminal defects for Type IV; separate transient, Roos, and apex fields; bounded-window cofinal semantics; canonical ν-notation; exact direct/inverse variance; a named reduction and finite tail witness for every infinite conclusion; and strict separation of internal apex Return from external mathematical Return. It is a typing and audit contract, not a mixed mathematical theorem bundle.*

4.20 Diagnostic equivalence and invariance

Definition 4.140 (Diagnostically equivalent tower artifacts)**.** *Two valid terminal comparison artifacts are diagnostically equivalent at (i, W, τ) when their kernel and cokernel defect profiles are related by actual isomorphisms compatible with terminal mode, terminal-tameness, cofinal/interior decomposition, and every transient detector stage consumed by the claim.*

Proposition 4.141 (Invariance of terminal and transient diagnostics)**.** *Diagnostically equivalent artifacts have equal degreewise and finite-scope aggregate μ, ν diagnostics. Corresponding complete defect detectors have the same zero/nonzero values when their certificate and stage data are transported or independently replayed.*

Proof. Terminal-generic dimension is invariant under profile isomorphism, giving the terminal statement and its finite sums. Appendix B detector invariance under an exact source-stage isomorphism gives the value statement; completeness and certificate metadata remain separate obligations. □

Remark 4.142 (Relative representative independence). This is artifact-relative invariance, not hidden canonicality of arbitrary filtered representatives. A representative change must preserve or reconstruct the actual source, apex, comparison map, and defect profiles at the strength consumed by the diagnostic.

4.21 Manifest extension and obligation handoff

Definition 4.143 (Expanded form of Main Definition 5.102: Type IV and tower manifest entry)**.** *Every run records the Type IV scope. Depending on the selected lanes, the manifest additionally records tower variance, semantic role, actual transitions, evaluation lift, source and authority, ordinary limit/colimit, Roos and* $\varprojlim^1$*, apex identity and comparison direction, ambient category, full/cofinal/interior defects, terminal mode and tameness, degreewise and total* μ, ν*, transient threshold and detector model, short-defect exclusion, finite-to-infinite witness and theorem, supported internal Return set, comparison modes and metric records, claim/dependency/proof locators, and immutable artifact identities. Undefined fields are explicitly serialized; they are not encoded as zero.*

Definition 4.144 (Expanded form of Main Definition 5.103: failure manifest entry)**.** *Every failed or conditionally accepted run records the applicable failure types, first failed audit stage, exact failed theorem/gate/artifact, canonical status, repairability or out-of-scope decision, and version boundary. Any successor-only theorem used in analysis but excluded from proof is identified explicitly.*

Definition 4.145 (Type IV, derived-limit, and reduction obligation bundle)**.** *Appendix D forwards at least*

`tower_variance`,	`actual_transitions`,
`evaluation_lift`,	`source_authority`,
`ordinary_limit_or_colimit`,	`roos_r1lim`,
`actual_apex_comparison`,	`ambient_category`,
`terminal_tameness`,	`full_defect_profiles`,
`typeIV_scope`,	`transient_detector`,
`short_defect_exclusion`,	`tail_certificate`,
`finite_to_infinite_theorem`,	`return_supported_set`,
`artifact_identity`,	`claim_status`

to Appendix F's proof-DAG semantics and Appendix G's serialization/replay schema. None is purchased by scalar metric slack.

4.22 Appendix D v21 proof-package closure

Definition 4.146 (Appendix D v21 proof-package closure record)**.** *Without creating a second Main theorem authority, the closure record confirms that the expanded proofs and Appendix-local support establish:*

(i) v20 terminal-generic, transient, full-isomorphism, and stable-band regression at unchanged strength;

(ii) explicit direct/inverse variance and source/apex direction;

(iii) the countable Roos operator, ML vanishing, and separation of $\varprojlim^1$ *from apex agreement;*

(iv) eventual-constancy Returns in both variances;

(v) finite-length, Noetherian stable-image, and finite-generation lanes;

(vi) deterministic finite-state terminality and exact periodic stable-image inverse-limit Return;

(vii) finite-to-infinite soundness with a downward-closed supported Return set and no assumed unique ceiling;

(viii) canonical μ/ν Type IV notation with no normative Latin u-cokernel symbol;

(ix) complete gate, proof-DAG, manifest, provenance, and replay handoffs.

The frozen v20 labels `thm:D-package` and `thm:D-v20-package` remain historical source locators only.

Definition 4.147 (Appendix D local completion gates)**.** *Appendix D is locally complete only when:*

(i) the v20 terminal/transient regression suite passes;

(ii) direct and inverse records cannot be interchanged by notation;

(iii) one nonzero Roos-defect witness and one ML-cleared witness replay;

(iv) eventual-constancy, finite-length, and finite-state lanes each have a finite witness and a distinct theorem locator;

(v) one finite-state inverse-limit and apex-agreement witness replays;

(vi) terminal zero, transient absence, derived-limit clear, and apex agreement remain independently serializable;

(vii) theorem ownership, canonical ν-notation, Appendix F/G handoff, and nonpromotion checks pass.

Remark 4.148 (Contribution to G21 gates)*.* Appendix D contributes to G21-2 by Core-P regression, to G21-7 by supplying typed finite-to-infinite and apex inputs, to G21-8 through theorem ownership and namespace separation, and to G21-9 through replayable tower/limit artifacts. It does not by itself pass Core-DG closure, object-level descent, unified external Return, or final release closure.

4.23 Explicit exclusions

The following are not asserted in Appendix D.

(i) No barcode matching is a transition or comparison morphism.

(ii) No direct colimit, inverse limit, completion, or apex is inferred from notation or a finite prefix.

(iii) No colimit–collapse or limit–collapse interchange is automatic.

(iv) No direct and inverse tower are merged by reversing arrows in a formula.

(v) No bounded-window terminal value is computed after zero extension to $+\infty$.

(vi) No missing Type IV artifact is encoded as $(0, 0)$.

(vii) No $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ implies full isomorphism without full-defect or short-defect control.

(viii) No terminal zero implies transient absence, $\varprojlim^1 = 0$, or apex agreement.

(ix) No ML or $\varprojlim^1 = 0$ implies apex agreement.

(x) No apex agreement implies $\varprojlim^1 = 0$.

(xi) No Noetherian hypothesis stabilizes an arbitrary descending inverse image chain.

(xii) No finite generation without a certified degree bound yields a fixed finite verification window.

(xiii) No equality of state summaries is typed state equality.

(xiv) No pointwise ML condition supplies a uniform finite cutoff by itself.

(xv) No source/apex metric closeness automatically controls kernels or cokernels.

(xvi) No detector record is relabeled from a topological stage to a defect stage.

(xvii) No long-transient absence excludes bars of length at most τ_{def}.

(xviii) No finite-state periodic lane is promoted beyond the declared state table and compatible morphism class.

(xix) No supported Return set is assigned a unique ceiling unless a greatest element is proved.

(xx) No legacy Latin u Type IV cokernel notation has normative force.

(xxi) No Search, AI, advisory, or numerical proxy replaces the actual tower, theorem, tail, or apex artifact.

(xxii) No internal tower theorem proves an arithmetic, analytic, PDE, geometric, cryptographic, or other external Return.

4.24 Provenance and status

Appendix A supplies repaired evaluation and metric transport boundaries. Appendix B supplies actual representatives, morphisms, detector models, and fresh defect-stage certificates. Appendix C supplies only eligible Ext discharge. Main Chapter 5 owns the canonical v21 tower and finite-to-infinite statements; this appendix supplies their detailed proofs, counterexamples, and records.

4.24.0.1 Canonical Main statements expanded here. Main Definitions 5.3–5.7, 5.9, 5.12–5.14, 5.16–5.22, 5.24–5.25, 5.27, 5.29–5.31, 5.35–5.36, 5.39–5.40, 5.42, 5.47–5.51, 5.57, 5.59–5.60, 5.64, 5.66, 5.70, 5.79, 5.82–5.83, 5.87, 5.89–5.90, 5.93, 5.95, and 5.98–5.99; Main Proposition 5.15, 5.41, 5.44, 5.46, 5.52, 5.58, 5.92, 5.94, and 5.97; Main Theorems 5.32, 5.37, 5.61, 5.65, 5.67, 5.71–5.73, 5.75, 5.77, 5.80, 5.84–5.85, and 5.88; and Main Corollaries 5.33, 5.63, 5.74, and 5.81.

4.24.0.2 Inherited v20 mathematical baseline. The terminal-generic dimension, cofinal/interior decomposition, Type IV criterion, transient-defect characterization, short-defect exclusion, full-isomorphism upgrade, finite-stage nonimplication, and artifact-coherent stable-band calculus remain available at their frozen and v21-expanded strengths.

4.24.0.3 Explicit exclusions. Subsection 4.23.

5 Appendix E: Windows, Overlap, Certificate Atlases, Restart, and Descent Handoff

5.1 Scope, canonical ownership, migration, and conclusion strength

5.1.0.1 Normative status and theorem ownership. Appendix E is the expanded-proof, counterexample, certificate-atlas, Restart, countable-tail, descent-handoff, record-schema, and regression owner for the window and overlap statements canonically owned by Main Chapter 6 of AK-HPDST v21.0.0, the *Finite Globalization, Unified Bridge Calculus, and Certificate Complexity Release*. A theorem canonically stated in the Main retains its Main identity. This appendix supplies its detailed proof, model calculus, sharp failure boundary, and replay record; it creates no competing Main theorem UID.

The immutable v20 Appendix E remains directly invocable from its frozen artifact at its original strength. The v21 migration is componentwise:

(i) right-open partition and overlap-cover logic, sharp metric overlap, finite Čech depth, conditional κ-Restart, normalized summability, detector atlases, retained-bar coverage, boundary bridges, and record/object separation are `inherited`;

(ii) the canonical four-mode comparison calculus, requested-conclusion typing, theorem-controlled nonmetric overlap and simplex records, model-certified homotopy-coherent cells, atlas bundles, hidden-tail prevention, Restart evidence monotonicity, inverse-audit participation, and the typed DescentInput handoff are `strengthened`;

(iii) the former package-theorem wrappers are not repeated as independent v21 theorem authorities. Their components remain inherited or strengthened and are summarized only by an operational closure record;

(iv) effective descent, existence or uniqueness of a descended object, global object zero/isomorphism, Morita–Alexandrov reconstruction, and external Return strength are relocated to Appendix DG, Main Chapter 7, or the Problem-Interface layer.

Appendix E consumes:

(i) Appendix A's window-first evaluation, crop–collapse criterion, threshold-gap theorem, zero separation, and four comparison modes;

(ii) Appendix B's typed detector models, complete zero certificates, certificate composition, and `Cert2RepInput`;

(iii) Appendix C's finite-diagram and eligible-realization interfaces, without treating record coherence as object descent;

(iv) Appendix D's direct Type IV field, independent inverse audit triple, transient-defect field, and finite-to-infinite tail certificates;

(v) Main Chapter 6's canonical atlas and Restart statements.

Every windowed profile is evaluated in the fixed order

$$\mathbf{P}_i(F) \longmapsto \mathsf{W}_W\mathbf{P}_i(F) \longmapsto T_\tau^{\mathrm{bar}}(\mathsf{W}_W\mathbf{P}_i(F)) = \mathrm{Eval}_{i,W,\tau}(F).$$

No crop-after-collapse substitution is made without the exact all-retained-bars criterion of Appendix A.

Every consumed comparison declares exactly one canonical mode

$$\texttt{exact}, \quad \texttt{metric\_gap\_certified},$$
$$\texttt{theorem\_controlled\_nonmetric}, \quad \texttt{not\_compared}.$$

Exact mode supplies only the registered exact relation. Metric mode supplies only the sharp metric or repaired-zero conclusion whose hypotheses are present. A theorem-controlled nonmetric record supplies only the named theorem's printed conclusion and registered weakenings. The last mode supplies no comparison inference.

The central v21 separations are:

(i) local zero, global record zero, and global object zero are distinct;

(ii) partition covers support ordered Restart, while overlap covers support compatibility and Čech-type coherence;

(iii) pairwise overlap never implies triple, higher-simplex, or homotopy coherence;

(iv) exact cocycles, model-certified homotopy-coherent cells, metric-reference records, and theorem-controlled nonmetric records are different data types;

(v) retained-feature coverage is a universal global obligation and is not implied by local zero, overlap, positive reserve, or summable exposure;

(vi) Restart transports only explicitly inherited evidence and requires a separate target-zero route whenever zero is continued;

(vii) countable claims require complete enumeration and hidden-tail prevention; a finite prefix is not an infinite theorem;

(viii) direct Type IV, inverse-system validity, $\varprojlim^1$ clearance, inverse apex agreement, and transient cleanliness are independent optional fields;

(ix) the strongest unconditional conclusion of this appendix is a typed record-level atlas verdict or a well-typed DescentInput, never an effectively descended object.

Remark 5.1 (Appendix E does not enlarge the object-level Core). This appendix does not create a universal Restart constant, an automatic continuation map, a global profile from pairwise data, a homotopy-coherent object from metric bounds, or an effective descent theorem. Search-side maps, AI proposals, advisory scores, and unverified theorem suggestions remain non-verdict until they re-enter through independently verified Core records.

5.2 Conclusion levels and nonpromotion discipline

Definition 5.2 (Expanded record for Main Definition 6.3: local zero family)**.** *For a declared atlas scope A_0, a* local zero family *is the typed statement*

$$\forall \alpha \in A_0, \qquad Z_\alpha = 0,$$

where every Z_α is the exact local predicate named by its certificate record. The record fixes the source stage, window, threshold, detector or direct proof, comparison route, and verifier status. It contains no global quantifier over retained features unless a separate coverage theorem supplies one.

Definition 5.3 (Expanded record for Main Definition 6.4: global record zero)**.** *A* global record verdict *is a verifier output with status*

$$\texttt{pass}, \qquad \texttt{reject}, \qquad \texttt{undefined}.$$

It is a global record-zero verdict *only when a theorem-backed covered- range predicate cites a declared global target, complete local zero certificates, route-valid interfaces, retained-feature coverage including boundary and tail coverage, and every required overlap and higher-coherence record. It is not a construction of a global object from local objects.*

Definition 5.4 (Expanded record for Main Definition 6.5: global object zero)**.** *A* global object zero *statement is an assertion*

$$E_{\mathrm{glob}} \simeq 0$$

in a declared target category after an effective descent theorem has constructed E_{glob} from actual local objects and descent data. This conclusion is outside Appendix E.

Theorem 5.5 (Expanded proof of Main Theorem 6.6: no object descent from record coherence)**.** *Neither a local zero family nor a global record-zero verdict implies global object zero without a separately proved effective descent theorem and an actual descended object in the declared target category.*

Proof. The first two notions are predicates on registered evidence. They need not contain actual local objects, transition equivalences, cocycles, higher cells, or an effectivity theorem. Those data are required even to type the object E_{glob}. Hence the asserted implication is not well-typed without the additional descent hypotheses. $\square$

5.3 Right-open windows, partitions, and overlap covers

Definition 5.6 (Right-open window)**.** *A* right-open Core window *is an interval*

$$W = [u, v), \qquad -\infty < u < v \leq +\infty.$$

It is bounded when $v < +\infty$ and right-unbounded when $v = +\infty$.

Definition 5.7 (Declared window range)**.** *A* declared window range *is an interval $\Omega \subseteq \mathbb{R}$ together with the cover type, index set, endpoint convention, and audit scope on which local certification is performed.*

Remark 5.8 (Why right-open windows). If

$$W_k = [u_k, u_{k+1}), \qquad W_{k+1} = [u_{k+1}, u_{k+2}),$$

then $W_k \cap W_{k+1} = \varnothing$. Hence a partition does not double-count boundary parameters. This convention does not itself provide a continuation law across the boundary.

Definition 5.9 (Window restriction)**.** *For $M \in \mathsf{Pers}_k^{\mathrm{cons}}$, the window restriction $\mathsf{W}_W M$ is the restriction of M to W, extended by zero outside W only when a globally indexed profile is required. At barcode level, each interval is clipped to W and empty intersections are discarded.*

Definition 5.10 (Core window-evaluation order)**.** *For a filtered object F, monitored degree i, window W, and threshold $\tau > 0$, define*

$$B_{i,W}(F) := \mathsf{W}_W \mathbf{P}_i(F), \qquad \mathrm{Eval}_{i,W,\tau}(F) := T_\tau^{\mathrm{bar}}(B_{i,W}(F)).$$

The pre-threshold profile $B_{i,W}(F)$ is the object on which clearance is measured.

Remark 5.11 (No unconditional crop–collapse commutation). In general,

$$\mathsf{W}_W T_\tau^{\mathrm{bar}} \ncong T_\tau^{\mathrm{bar}} \mathsf{W}_W.$$

A crop-after-collapse substitution is permitted only under the exact all-retained-bars criterion of Appendix A, namely Definition 1.64 and Proposition 1.65, including globally retained infinite bars clipped by bounded windows.

Definition 5.12 (MECE windowing / right-open partition cover). *A family*

$$\mathcal{W}_{\mathrm{MECE}} = \{W_k = [u_k, u_{k+1})\}_{k\in K}$$

is a right-open partition cover *of* Ω *if the windows are pairwise disjoint, their union is* Ω*, and the index order is part of the data whenever sequential continuation is invoked. The term* MECE windowing *is retained as an operational alias.*

Definition 5.13 (MECE certificate sequence). *A* MECE certificate sequence *is an ordered family* $\{\mathcal{C}_k\}_{k\in K}$ *of complete local certificate records indexed by a right-open partition cover. Ordering alone does not certify continuation.*

Remark 5.14 (MECE windowing is not overlap gluing). A disjoint partition has no nonempty overlap on which an Overlap Gate can be evaluated. Conversely, an overlap cover does not by itself supply an ordered continuation path. The two cover types have different proof obligations.

Definition 5.15 (Overlap-admitting cover). *A family* $\mathcal{U} = \{U_\alpha\}_{\alpha\in A}$ *of right-open windows is an* overlap-admitting cover *of* Ω *if*

$$\Omega = \bigcup_{\alpha\in A} U_\alpha$$

and every nonempty finite intersection used by the certificate logic is explicitly indexed and recorded.

Definition 5.16 (Overlap stratum). *For a finite index set* $\sigma = \{\alpha_0, \dots, \alpha_p\}$*, define*

$$U_\sigma := U_{\alpha_0} \cap \cdots \cap U_{\alpha_p}.$$

A stratum participates only when it is nonempty and included in the declared nerve record.

Definition 5.17 (Declared overlap structure). *A* declared overlap structure *records the cover, every nonempty nerve simplex used by the run, its face relations, common thresholds, monitored degrees, metric ledgers and scalarizations, representative and realization provenance, and typed Type IV scope information.*

Remark 5.18 (Why overlap data must be declared). Without an indexed nerve and immutable overlap artifacts, an auditor cannot determine which local records, thresholds, face restrictions, budgets, or Type IV artifacts were compared. Mere set-theoretic overlap is not a certificate.

5.4 Scalarized reserves and non-scalar obligations

Definition 5.19 (Bottleneck scalarization for Appendix E). *Let* $(Q, \oplus, \preceq, 0_Q)$ *be the declared commutative ledger quantale. A bottleneck scalarization is a monotone subadditive map*

$$\mathrm{val_B} : Q \longrightarrow [0, +\infty], \qquad \mathrm{val_B}(0_Q) = 0, \qquad \mathrm{val_B}(q \oplus q') \le \mathrm{val_B}(q) + \mathrm{val_B}(q').$$

Only after this scalarization may a ledger value be compared with a real threshold clearance.

Definition 5.20 (Gate-scoped threshold reserve). *Let $\mathfrak{B} = \{B_j\}_{j\in J}$ be the finite pre-threshold profile family consumed by one local, overlap, simplex, or Restart gate, and let ε be its certified scalarized bottleneck budget. Set*

$$\mathrm{gap}_\tau(\mathfrak{B}) := \min_{j\in J} \mathrm{clear}_\tau(B_j), \qquad \mathrm{res}_\tau(\mathfrak{B};\varepsilon) := \mathrm{gap}_\tau(\mathfrak{B}) - \varepsilon.$$

This reserve belongs only to the declared gate and profile family.

Definition 5.21 (Non-scalar continuation and coherence obligations). *Representative existence, actual comparison or continuation morphisms, threshold reconciliation, κ-admissibility, higher-face coherence, terminal tameness, Type IV typing, claim status, manifest completeness, and artifact readability are non-scalar obligations. No increased budget, larger gap, or summability estimate can replace them.*

Remark 5.22 (Obligation handoff to Appendices F and G). The statuses and dependency closure of the non-scalar obligations in Definition 5.21 are forwarded to the typed status calculus of Appendix F. Their immutable identifiers, evidence references, scope fields, and failure routes are forwarded to the Manifest schema of Appendix G. This handoff records obligations; it does not convert missing evidence into a pass.

5.5 Certificate records

Definition 5.23 (Window-restricted certificate record). *A* window-restricted certificate record *on $(F; W, \tau)$ records at least:*

(i) immutable object, window, threshold, and monitored-degree identifiers;

(ii) the pre-threshold profiles $B_{i,W}(F)$ and post-threshold profiles $\mathrm{Eval}_{i,W,\tau}(F)$;

(iii) clearance, gap, metric-ledger provenance, scalarization, and positive-reserve verdicts;

(iv) the local B-Gate$^+$, representative, eligibility, and typed Type IV statuses;

(v) every invoked representative, realization, tower, comparison, crop, or continuation artifact;

(vi) a local status `pass`*,* `reject`*, or* `undefined`*;*

(vii) artifact references sufficient for independent reconstruction.

Definition 5.24 (Globally coherent certificate record). *A family of local records forms a* globally coherent certificate record system *only if every required local, pairwise, higher-simplex, Restart, budget, Type IV, and manifest clause is present and typed. The phrase does not denote a single global B-Gate$^+$ pass unless an independent global object and global gate record are supplied.*

Remark 5.25 (Certificate record versus proof object). A certificate record is the audit-readable representation of declared proof data. Schema completeness does not by itself prove the mathematical clauses stored in the record, and record gluing does not construct a new mathematical object.

5.6 Direct common-window comparison and the sharp Overlap Gate

Definition 5.26 (Overlap budget). *For a nonempty recorded simplex σ, let $\mathfrak{d}_\sigma \in Q$ be the gate-scoped metric ledger and define its scalarized budget*

$$\varepsilon_\sigma := \mathrm{val}_{\mathrm{B}}(\mathfrak{d}_\sigma) \in [0, +\infty].$$

For a pair $\sigma = \{\alpha, \beta\}$, this is the overlap budget formerly denoted $\Sigma\delta_{\alpha\beta}$.

Definition 5.27 (Preferred repaired overlap profile)**.** *For $U_{\alpha\beta} := U_\alpha \cap U_\beta \neq \varnothing$, define the direct common-window profiles*

$$B^{\alpha}_{i,\alpha\beta} := \mathsf{W}_{U_{\alpha\beta}} \mathbf{P}_i(F_\alpha), \qquad B^{\beta}_{i,\alpha\beta} := \mathsf{W}_{U_{\alpha\beta}} \mathbf{P}_i(F_\beta).$$

After declaring a common overlap threshold $\tau_{\alpha\beta}$, set

$$\mathcal{O}_{\alpha\beta,i}(F_\alpha) := T^{\mathrm{bar}}_{\tau_{\alpha\beta}}(B^{\alpha}_{i,\alpha\beta}), \quad \mathcal{O}_{\beta\alpha,i}(F_\beta) := T^{\mathrm{bar}}_{\tau_{\alpha\beta}}(B^{\beta}_{i,\alpha\beta}).$$

Definition 5.28 (Overlap comparison data)**.** *The overlap data consist of the two direct pre-threshold profiles, the common threshold and threshold- reconciliation record, a pre-threshold bottleneck certificate, two-sided clearance certificates, post-threshold comparison data, representative provenance when invoked, typed Type IV status, and all artifact references.*

Definition 5.29 (Threshold reconciliation record)**.** *If local records use thresholds τ_α and τ_β, an Overlap Gate is well typed only after a common threshold $\tau_{\alpha\beta}$ is declared and both local objects are directly re-evaluated on $U_{\alpha\beta}$ at that threshold. The data justifying this choice form the threshold-reconciliation record.*

Remark 5.30 (Preferred overlap convention)*.* The default comparison is performed directly on $U_{\alpha\beta}$. Cropping profiles already thresholded on U_α or U_β is forbidden unless the exact criterion of Definition 1.64 and Proposition 1.65 is certified.

Definition 5.31 (Two-sided overlap metric certificate)**.** *For each monitored degree i, a two-sided overlap metric certificate at budget $\varepsilon_{\alpha\beta,i}$ consists of*

$$d_B\left(B^{\alpha}_{i,\alpha\beta}, B^{\beta}_{i,\alpha\beta}\right) \leq \varepsilon_{\alpha\beta,i},$$

$$\mathrm{clear}_{\tau_{\alpha\beta}}\left(B^{\alpha}_{i,\alpha\beta}\right) > \varepsilon_{\alpha\beta,i}, \qquad \mathrm{clear}_{\tau_{\alpha\beta}}\left(B^{\beta}_{i,\alpha\beta}\right) > \varepsilon_{\alpha\beta,i},$$

together with the ledger provenance and direct common-window records.

Proposition 5.32 (Expanded proof of Main Proposition 6.28: sharp overlap transfer)**.** *Under a two-sided overlap metric certificate,*

$$d_B(\mathcal{O}_{\alpha\beta,i}(F_\alpha), \mathcal{O}_{\beta\alpha,i}(F_\beta)) \leq \varepsilon_{\alpha\beta,i}.$$

In the declared one-parameter constructible regime, the same bound may be read using interleaving distance.

Proof. The two clearance inequalities exclude finite bars from the closed band $[\tau_{\alpha\beta} - \varepsilon_{\alpha\beta,i}, \tau_{\alpha\beta} + \varepsilon_{\alpha\beta,i}]$ on both sides. Apply Appendix A, Theorem 1.37, to the certified pre-threshold bottleneck bound, then use the bottleneck–interleaving isometry in the declared regime. □

Definition 5.33 (Overlap Gate)**.** *The pairwise Overlap Gate on $U_{\alpha\beta}$ passes exactly when, for every monitored degree:*

(i) a two-sided overlap metric certificate exists;

(ii) the post-threshold discrepancy bound of Proposition 5.32 holds;

(iii) representative and eligible-realization provenance used on the overlap is compatible;

(iv) if Type IV is in scope, its status is `certified_zero`;

(v) `undefined` is never interpreted as zero and `obstructed` is a failure;

(vi) all required crop, threshold-reconciliation, comparison, and artifact records are present.

Remark 5.34 (Overlap Gate is a compatibility test)*.* The discrepancy verdict is post-threshold, but the safety certificate is pre-threshold. Small post-threshold distance alone does not prove stable threshold classification. Passing a pairwise gate also does not create a local pass or higher coherence.

Proposition 5.35 (Overlap Gate preserves pairwise audit readability)**.** *If the pairwise Overlap Gate passes and all of its records are immutable and referenced, then the pairwise compatibility claim is independently reconstructible. No higher-simplex conclusion follows from this proposition.*

Proof. The gate records both pre-threshold inputs, the scalarized metric bound, the two clearance inequalities, the post-threshold verdict, typed statuses, and all required provenance. These data reconstruct precisely the pairwise claim and nothing stronger. □

5.7 Finite Čech depth and simplex coherence

Definition 5.36 (Čech nerve)**.** *The Čech nerve $N(\mathcal{U})$ has a p-simplex for every nonempty intersection of $p+1$ distinct cover windows.*

Definition 5.37 (Finite Čech depth)**.** *The cover has* Čech depth at most K *if every intersection of $K+2$ distinct windows is empty. Equivalently,* $\dim N(\mathcal{U}) \le K$*. Thus nonempty records occur on at most $(K+1)$-fold intersections.*

Definition 5.38 (Simplex compatibility record)**.** *For a nonempty simplex σ, a simplex compatibility record contains direct pre-threshold profiles on U_σ, a common threshold, all required face comparison data, sharp two-sided gap certificates, face-compatibility statements, typed Type IV statuses, scalarized budgets, and artifact references.*

Proposition 5.39 (Expanded proof of Main Proposition 6.46: finite-depth verification bound)**.** *If the cover has Čech depth at most K, no compatibility record is required above simplicial degree K.*

Proof. There are no nonempty intersections indexed by higher-dimensional simplices. This bounds the order of coherence conditions, but not the number of simplices when the cover is infinite. □

Remark 5.40 (Why finite depth is a Core condition)*.* Finite depth bounds overlap order only. It does not imply local finiteness, summability, face compatibility, or finiteness of the record set. Those are separate obligations.

Definition 5.41 (Compatible local certificate family)**.** *A family $\{\mathcal{C}_\alpha\}$ is* Čech-compatible *if every local record passes, every nonempty nerve simplex carries a complete simplex compatibility record, all face restrictions agree under the declared equivalence policy, every scalarized simplex reserve is positive, and no required status is undefined.*

Definition 5.42 (Čech-coherent certificate atlas)**.** *A* Čech-coherent certificate atlas *is a Čech-compatible family indexed by a declared nerve, with immutable local and simplex records and mutually compatible face restrictions.*

Theorem 5.43 (Expanded proof of Main Theorem 6.47: finite-cover record-level gluing)**.** *Let $\mathcal{U}$ be a finite overlap-admitting cover of finite Čech depth. Assume:*

(i) every window carries a passing local certificate record;

(ii) every nonempty nerve simplex carries a simplex compatibility record;

(iii) every Type IV scope has status `not_invoked` or `certified_zero`, never `undefined`;

(iv) every scalarized simplex budget has positive sharp threshold reserve;

(v) all face restrictions and artifact references are audit-readable.

Then the family is a Čech-coherent certificate atlas on the covered range.

Proof. The local records provide the vertices. The simplex records supply every nonempty overlap through the maximal nerve degree, and the face conditions establish coherence under restriction. Finiteness of the cover and finite depth make the required registry finite. Positive reserves and typed Type IV statuses prevent unstable or undefined comparisons from being reported as coherent. □

Remark 5.44 (No unrestricted object-level gluing). Theorem 5.43 glues records into an atlas. It does not glue filtered complexes, persistence modules, derived objects, sheaves, or other mathematical objects.

5.8 κ-admissible Restart

Definition 5.45 (Expanded record for Main Definition 6.58: Restart datum). *A Restart datum from a predecessor state S_k to a target state S_{k+1} contains:*

(i) predecessor and target local-record identities;

(ii) an explicit field-by-field inheritance map;

(iii) an invalidation list for evidence whose hypotheses, stages, windows, thresholds, objects, models, or artifact versions changed;

(iv) the continuation artifact, finite working gaps, scalarized budgets, κ_k, η_k, and reserve fields;

(v) a target-zero route whenever zero is consumed;

(vi) direct Type IV, inverse-audit, transient, and manifest fields at the target whenever required.

Proposition 5.46 (Expanded proof of Main Proposition 6.59: Restart evidence monotonicity). *A certified Restart may inherit only evidence whose complete hypothesis set is preserved by the transition theorem. Every invalidated or fresh field must be reproved at the target; a stronger conclusion cannot be inherited from weaker source evidence.*

Proof. A proof artifact is valid only under its registered hypotheses. If any hypothesis changes, the artifact no longer proves the target clause unless the continuation theorem explicitly transports it. Logical weakening is allowed; strengthening is not. □

Definition 5.47 (Sequential continuation step). *For adjacent partition windows W_k and W_{k+1}, a continuation step is a declared artifact linking the source and target local records. Adjacency itself supplies no inequality and no morphism.*

Definition 5.48 (Per-window budget). *Let $\mathfrak{B}_k$ be the finite pre-threshold profile family used by the local gate. Choose a finite working gap*

$$0 < g_k \le \operatorname{gap}_{\tau_k}(\mathfrak{B}_k),$$

using a finite audit cap when the exact clearance is $+\infty$. Let $\varepsilon_k < +\infty$ be the gate-scoped scalarized local budget. These quantities must have immutable provenance.

Definition 5.49 (Restart margin). *The finite local reserve is*

$$r_k := g_k - \varepsilon_k \in \mathbb{R}.$$

The local record is reserve-positive exactly when $r_k > 0$. A finite working gap avoids undefined extended-real arithmetic in the Restart recursion.

Definition 5.50 (Continuation artifact). *A continuation artifact from W_k to W_{k+1} records source and target identifiers and thresholds, all endpoint-state, initialization, representative, and comparison data used by the transition, a finite continuation loss $\eta_k \geq 0$ not already counted in the target local budget, a degradation factor $\kappa_k \in (0,1]$, and independently checkable evidence for the reserve inequality below. Missing non-scalar obligations may not be encoded numerically inside η_k.*

Definition 5.51 (κ_k-admissible continuation). *The artifact is* κ_k-admissible *if it certifies*

$$r_{k+1} \geq \kappa_k r_k - \eta_k.$$

Neither κ_k nor the inequality is inferred from adjacency, threshold deletion, or a generic "admissible continuation class."

Definition 5.52 (Restart transition status). *Each transition has exactly one status:*

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{certified}, \qquad \texttt{failed}.$$

The status is `certified` *only when the artifact is complete and the declared κ_k-inequality is verified. A certified transition is necessary but not sufficient for a complete Restart pass: the target local record must also pass and have positive reserve.*

Lemma 5.53 (Expanded proof of Main Theorem 6.65: finite-horizon Restart bound). *Assume the transitions $0 \to 1, \dots, n-1 \to n$ are respectively $\kappa_0, \dots, \kappa_{n-1}$-admissible. Then*

$$r_n \geq \left(\prod_{j=0}^{n-1} \kappa_j\right) r_0 - \sum_{m=0}^{n-1} \left(\prod_{j=m+1}^{n-1} \kappa_j\right) \eta_m,$$

where an empty product is 1.

Proof. For $n = 1$ this is the certified continuation inequality. If the formula holds at n, substitute it into $r_{n+1} \geq \kappa_n r_n - \eta_n$, distribute κ_n, and collect the product-weighted loss terms. ☐

Remark 5.54 (Restart is conditional). There is no universal $\kappa > 0$, no default $\kappa = 1$, and no unconditional Restart lemma. Window restriction, threshold changes, and unrelated target profiles can destroy inherited reserve. A missing continuation artifact gives status `undefined`, not harmless continuation.

Definition 5.55 (Normalized Restart loss). *Set*

$$K_0 := 1, \qquad K_n := \prod_{j=0}^{n-1} \kappa_j \quad (n \geq 1),$$

and

$$L_n^{\mathrm{norm}} := \sum_{m=0}^{n-1} \frac{\eta_m}{K_{m+1}}.$$

Then Lemma 5.53 implies

$$\frac{r_n}{K_n} \geq r_0 - L_n^{\mathrm{norm}}.$$

5.9 Countable Restart and normalized summability

Definition 5.56 (Summability of window budgets)**.** *A certified countable Restart run is κ-*normalized summable *if*

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < \infty.$$

It is reserve-safe *if, in addition,*

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < r_0.$$

Ordinary summability $\sum_k \eta_k < \infty$ is not an equivalent condition when K_k contracts.

Definition 5.57 (Summable restart sequence)**.** *A MECE certificate sequence is a* reserve-safe Restart sequence *if the initial reserve is positive, every invoked transition is `certified`, the normalized loss is reserve-safe, every target has a complete passing local record including all invoked representative, eligibility, and Type IV clauses, and no transition or target is undefined.*

Definition 5.58 (Expanded record for Main Definition 6.73: countable Restart specification and hidden-tail prevention)**.** *A countable Restart claim records the complete index enumeration, every transition schema, all exceptional indices, and one tail branch:*

(i) `universally_specified`*: the transition and target clauses are proved for every index, with no finite verifier claim inferred merely from the formula; or*

(ii) `finitely_certified`*: a finite generator or a Chapter 5 finite-to-infinite reduction proves all tail transitions, target clauses, normalized losses, exceptional indices, and the failure boundary.*

A checked finite prefix without one of these branches is not hidden-tail prevention.

Theorem 5.59 (Expanded proof of Main Theorem 6.74: conditional countable Restart principle)**.** *For a reserve-safe Restart sequence, every finite-stage reserve r_n is positive and the ordered family forms an auditable sequential certificate record on the partition cover. More precisely, if*

$$L_\infty^{\text{norm}} := \sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}}, \qquad c := r_0 - L_\infty^{\text{norm}} > 0,$$

then

$$r_n \ge K_n c > 0$$

for every finite n. A uniform lower bound follows only under the additional condition $\inf_n K_n > 0$*.*

Proof. For each finite n, Definition 5.55 gives

$$\frac{r_n}{K_n} \ge r_0 - L_n^{\text{norm}} \ge r_0 - L_\infty^{\text{norm}} = c.$$

Since $K_n > 0$, the pointwise bound follows. The continuation artifacts and complete target records make the record chain reconstructible. Taking infima yields a positive uniform reserve only when $\inf_n K_n > 0$. □

Remark 5.60 (Why ordinary summability is not enough)*.* Even when $\sum_k \eta_k < \infty$, the normalized series can diverge if $K_{k+1} \to 0$. Conversely, pointwise positivity $r_n > 0$ at every finite stage does not imply a uniform positive reserve. The theorem certifies record inheritance, not convergence to a global object.

5.10 Countable overlap atlases

Definition 5.61 (Definable window)**.** *A window is* definable *if it is definable in the declared ambient structure. Definability may support reconstructibility but does not replace local finiteness, simplex records, or coherence.*

Definition 5.62 (Countable definable cover)**.** *A* countable definable overlap cover *is a countable overlap-admitting cover whose windows are definable, whose Čech depth is finite, and whose complete recorded nerve is included in the audit scope. To support a countable atlas theorem it must additionally be audit-locally finite.*

Definition 5.63 (Audit-locally finite cover)**.** *A countable overlap cover is* audit-locally finite *if every bounded subinterval meets only finitely many windows and only finitely many recorded nerve simplices.*

Definition 5.64 (Summable simplex budget)**.** *Let $\varepsilon_\sigma \geq 0$ be the scalarized metric budget of each recorded nonempty simplex. Define the unordered nonnegative sum*

$$\sum_{\sigma \in N_{\mathrm{rec}}} \varepsilon_\sigma := \sup_{S \subseteq N_{\mathrm{rec}},\, S \text{ finite}} \sum_{\sigma \in S} \varepsilon_\sigma .$$

The simplex budget is summable when this quantity is finite. This controls total declared exposure but does not create missing face maps or replace the simplexwise inequalities $\varepsilon_\sigma < \mathrm{gap}_{\tau_\sigma}(\mathfrak{B}_\sigma)$.

Theorem 5.65 (Countable definable-cover gluing)**.** *Let $\{U_n\}_{n \geq 1}$ be a countable, definable, audit-locally finite overlap cover of finite Čech depth. Assume:*

(i) every local window has a complete passing certificate record;

(ii) every nonempty recorded nerve simplex has a complete simplex compatibility record;

(iii) every simplex has positive sharp threshold reserve;

(iv) the scalarized simplex budgets are summable;

(v) all invoked Type IV scopes are defined and correctly typed;

(vi) all face restrictions and artifact references are audit-readable.

Then the family forms an audit-locally finite Čech-coherent certificate atlas on the covered range and has finite declared scalarized simplex exposure.

Proof. Audit-local finiteness makes every bounded audit query finite. Finite Čech depth bounds overlap order. The simplex records and face conditions establish coherence, positive reserves license each thresholded comparison, and the unordered sum records finite total scalarized exposure. Hence the atlas is locally reconstructible and globally indexed as a coherent record system. □

Remark 5.66 (Definability is not decorative, but is not sufficient). Definability constrains the description language. It does not imply audit-local finiteness, summability, positive reserve, or higher coherence. All of those remain explicit theorem hypotheses.

5.11 Budget aggregation across windows and simplices

Definition 5.67 (Global window budget). *For a declared sequence of scalarized local or continuation exposures $\varepsilon_k \geq 0$, define the global window exposure by the unordered or ordered nonnegative sum specified by the run. This quantity is not a quantale element unless the manifest separately records the corresponding $\oplus$-aggregation.*

Definition 5.68 (Global overlap budget). *For a finite or countable recorded nerve, the global overlap exposure is the scalarized simplex sum of Definition 5.64. It includes every declared simplex, not merely pairwise overlaps.*

Definition 5.69 (Total gluing budget). *The* total gluing exposure *is a manifest-declared aggregation of window, continuation, crop, and simplex metric components after preserving their provenance. When represented by a quantale element it is aggregated using $\oplus$; when compared with real clearance it is first scalarized by* $\mathrm{val}_{\mathrm{B}}$.

Proposition 5.70 (Budget auditability under summability). *Finite scalarized window exposure and a summable simplex budget make the declared total metric exposure finite and audit-readable. They do not establish any local or global gate pass without the corresponding profile-scoped strict reserve inequalities and non-scalar obligations.*

Proof. The nonnegative sums are finite by hypothesis, while the provenance-preserving aggregation and scalarization make the value reconstructible. Gate validity remains separate because a finite total may still exceed a local clearance or coexist with a missing artifact. □

Remark 5.71 (Relation to gap_τ). There is no single untyped global inequality $\Sigma\delta_{\mathrm{glue}} < \mathrm{gap}_\tau$ unless the run declares a global profile family, threshold, aggregation law, and scalarization. The canonical requirements are the simplexwise and gatewise strict reserve inequalities actually consumed by the proof. A separately claimed uniform reserve requires an independent positive infimum certificate.

5.12 Compatibility with typed Type IV diagnostics

Definition 5.72 (Overlap Type IV cleanliness). *A local, overlap, simplex, or continuation-target record is* Type IV-clean *when Type IV is in scope and its status is `certified_zero` relative to the declared comparison artifact and terminal-generic data. When Type IV is explicitly outside scope, `not_invoked` is permitted. Status `undefined` is never clean.*

Proposition 5.73 (Overlap cleanliness is required for gluing). *No simplex may participate in a successful certificate atlas with Type IV status `obstructed` or `undefined`. If Type IV is in scope it must be `certified_zero`; otherwise the scope policy must justify `not_invoked`.*

Proof. This is the typed Type IV participation rule of Appendix D, Theorem 4.66, applied to every vertex, simplex, and continuation target. Compatibility cannot repair a rejected or undefined local clause. □

Remark 5.74 (Type IV cleanliness is terminal-generic). On a bounded window, terminal-generic diagnostics count bars cofinal to the right endpoint, even though such bars may be globally finite. Therefore Type IV cleanliness is not accurately described as merely “absence of infinite-bar defect.” It also does not exclude interior noncofinal defects or prove full isomorphism.

5.13 Record-level atlas gluing does not imply Core-DG object descent

Definition 5.75 (Object-level gluing claim). *An* object-level gluing claim *asserts that local filtered complexes, persistence modules, derived objects, sheaves, or categorical objects descend to a global object in a declared category.*

Proposition 5.76 (Certificate gluing does not imply object gluing). *A Čech-coherent certificate atlas does not, by itself, imply an object-level gluing claim.*

Proof. The atlas stores verdicts, profiles, budgets, statuses, face records, and artifact references. Object-level descent additionally requires morphisms or equivalences satisfying the descent axioms of a declared category, stack, or higher stack. Those hypotheses are not consequences of record coherence. □

Remark 5.77 (Where object-level gluing belongs). Object-level descent is outside Appendix E's Core-P and record-globalization authority. In the v21 stable finite lanes it belongs to Main Chapter 7 and Appendix DG: finite upper-Alexandrov sheaf descent, strict bounded-complex descent, or a separately named model-certified lane. Extensions beyond those lanes require a separately admitted Technical Extension or another model-certified theorem. In every lane, mathematical descent and the audit manifest remain independent obligations.

5.14 Typed object-descent handoff to Appendix DG and Chapter 7

Definition 5.78 (Expanded record for Main Definition 6.91: object-level descent interface). *An object-level descent claim requires a declared target category or higher category with restriction functors, actual local objects, actual transition equivalences, strict cocycles or registered higher cells, an effectivity theorem for the declared cover, and an artifact intended to construct the descended object. Metric-reference records and certificate coherence do not supply these data.*

Definition 5.79 (Expanded record for Main Definition 6.92: object-descent eligibility). *An object-descent eligibility record is the canonical nine-field tuple*

$$\mathrm{DescentInput} = (\{E_\alpha\}, \{g_{\alpha\beta}\}, \{c_{\alpha\beta\gamma}\}, \{h_\sigma\}, \mathcal{W}, \mathcal{C}, \mathfrak{M}_{\mathrm{coh}}, \mathsf{T}_{\mathrm{eff}}, \mathcal{A}),$$

with the following independent fields.

(i) E_α *are actual local objects in one declared target category, dg enhancement, model category, stable category, or other named descent environment.*

(ii) $g_{\alpha\beta}$ *are actual overlap equivalences with source, target, restriction direction, variance, and inverse data when used.*

(iii) $c_{\alpha\beta\gamma}$ *are literal cocycle equalities in a strict lane or actual invertible two-cells with declared boundaries in a higher lane.*

(iv) h_σ *are every higher cell required by the selected coherence model; no omitted higher cell receives an identity default.*

(v) $\mathcal{W}$ *is the cover together with its nerve, nonempty intersections, inclusion maps, face maps, and indexing convention.*

(vi) $\mathcal{C}$ *records object-level cover and restriction compatibility. It is not retained-bar coverage, certificate-atlas completeness, pairwise metric agreement, or a global record-zero predicate.*

(vii) $\mathfrak{M}_{\mathrm{coh}}$ *fixes the enhancement, cell types, composition laws, boundary equations, equality notion, and verifier used to interpret* g, c, h*.*

(viii) $\mathsf{T}_{\mathrm{eff}}$ *names the proposed effectivity theorem, its exact source type, target type, supported conclusion strength, and every hypothesis. Naming the theorem does not apply it.*

(ix) $\mathcal{A}$ *contains immutable construction, verification, provenance, replay, and failure-route artifacts for all preceding fields.*

The certificate atlas $\mathfrak{A}$ *remains an independent Appendix E/F/G record and does not populate* $\mathcal{C}$ *by default. Every absent, ill-typed, or mutually inconsistent in-scope field makes the handoff* `undefined`*; a proved false required compatibility clause yields* `reject` *at that clause.*

Definition 5.80 (Active descent-lane selection record)**.** *A descent handoff records exactly one active lane for the current proof-DAG route:*

`strict_alexandrov_sheaf`, `strict_bounded_complex`, `model_certified_derived`.

Several lanes may be mathematically eligible for the same local data, but one run may not mix hypotheses, coherence cells, effectivity conclusions, or uniqueness strengths from different lanes. The active-lane field is a route selection, not an effectivity verdict.

Proposition 5.81 (Certificate-atlas and object-compatibility separation)**.** *A complete certificate atlas does not imply the* $\mathcal{C}$ *field of Definition 5.79. Conversely, a valid object-level cover/restriction compatibility record does not imply detector-atlas completeness, retained-feature coverage, metric exposure summability, or a global record-zero verdict.*

Proof. The two records have different source and target types. A certificate atlas contains predicates, budgets, stages, verdicts, and evidence references; $\mathcal{C}$ concerns actual restriction functors, local objects, and their compatibility with the cover. Neither data type contains the theorem fields required to reconstruct the other. □

Theorem 5.82 (Expanded proof of Main Theorem 6.93: object-descent handoff)**.** *A complete and consistent object-descent eligibility record, together with a well-typed active-lane selection, determines a well-typed* DescentInput *for Appendix DG and Main Chapter 7. Its Appendix E conclusion is only*

`descent_input_ready`.

It does not establish lane applicability beyond type matching, effectivity, descended-object or descended-morphism existence, uniqueness, equivalence strength, joint conservativity, global zero, global isomorphism, or any external Return.

Proof. The nine fields supply the exact inputs expected by the downstream lane matcher and effectivity problem. The selected lane fixes which downstream theorem may consume them. Appendix E verifies neither the effectivity hypotheses nor the theorem application that constructs a global object. Therefore its strongest conclusion is input readiness only. □

Remark 5.83 (Effectivity, conservativity, and global verdicts are separate). Even after Appendix DG applies an effectivity theorem, global zero or global isomorphism requires the additional hypotheses of the selected lane. Outside the canonical finite Alexandrov lane, exact restriction, joint conservativity, local zero-reflecting probes, and compatibility with cones must be supplied separately. Effectivity alone constructs an object; it does not certify that local observations detect its zero object or its isomorphisms.

Proposition 5.84 (Inverse-audit participation)**.** *If a local, overlap, Restart, or atlas claim invokes an inverse-system or apex conclusion, the three fields*

$$\mathrm{InvAudit} = (\mathsf{Sys}, \mathbf{R}^1\varprojlim, \mathsf{Apex})$$

are evaluated independently at the exact required strength. Source-profile overlap, Restart reserve, and direct Type IV do not transport any of these fields without a theorem on the actual inverse systems and apex maps.

Definition 5.85 (v21 failure routing)**.** *Failures are routed without compensation:*

(i) missing or ill-typed evidence gives `undefined`*;*

(ii) a well-typed false mathematical clause gives `reject`*;*

(iii) absent in-scope κ_k is not replaced by 1;

(iv) pairwise pass cannot repair local failure or missing higher coherence;

(v) terminal zero cannot repair inverse-audit or transient failure;

(vi) record coherence cannot repair missing descent effectivity.

5.15 Window, overlap, Restart, and atlas package

Definition 5.86 (Historical v20 conditional closure record)**.** *Within the repaired Core, the following package is available.*

(i) Overlap profiles are evaluated directly on the common window before threshold deletion.

(ii) Metric ledger data are compared with clearance only after bottleneck scalarization.

(iii) A sharp two-sided overlap certificate transfers a pre-threshold bottleneck bound to the post-threshold evaluations.

(iv) Pairwise Overlap Gates do not imply higher Čech coherence.

(v) A finite-depth finite cover yields a coherent certificate atlas only when every simplex record and face condition is supplied.

(vi) Restart is conditional on an explicit κ_k-admissible continuation artifact.

(vii) Finite-horizon reserves satisfy the product-weighted inequality of Lemma 5.53.

(viii) Countable Restart requires normalized reserve-safety; ordinary summability alone is insufficient.

(ix) Pointwise positive reserves do not imply a uniform positive lower bound without control of $\inf_n K_n$.

(x) A countable atlas additionally requires audit-local finiteness, finite Čech depth, simplex coherence, positive simplex reserves, and summable scalarized exposure.

(xi) Undefined artifacts or statuses are never promoted to successful overlap, Restart, or gluing verdicts.

(xii) Record-level gluing does not construct a global mathematical object or a single global B-Gate$^+$ pass.

Remark 5.87 (Closure-record status)*.* The mathematical components cited above retain their individual theorem identities. This record is operational and provenance-only; it creates no new v21 package theorem.

5.16 Explicit exclusions

The following are not asserted in Appendix E.

(i) No unconditional Restart lemma or automatically existing $\kappa > 0$ is asserted.

(ii) No missing continuation artifact is interpreted as $\kappa = 1$, zero loss, or a certified transition.

(iii) No ordinary unweighted summability is sufficient under arbitrary contracting κ_k.

(iv) No pointwise reserve statement is promoted to a uniform positive lower bound without an additional product lower bound.

(v) No pairwise Overlap Gate implies triple or higher coherence.

(vi) No differently thresholded outputs are compared without threshold reconciliation and direct common-threshold re-evaluation.

(vii) No crop-after-collapse profile replaces direct common-window evaluation without a commutation certificate.

(viii) No unscalarized quantale element is compared directly with real threshold clearance.

(ix) No budget, gap, or summability estimate repairs a missing representative, morphism, coherence record, Type IV certificate, claim status, or manifest field.

(x) No finite Čech depth alone implies a finite record set or audit-local finiteness.

(xi) No definability alone implies coherence, summability, or local finiteness.

(xii) No finite prefix is treated as a complete countable audit certificate.

(xiii) No summable atlas is automatically a single global B-Gate$^+$ pass.

(xiv) No certificate atlas is automatically a global filtered object, persistence module, sheaf, or derived object.

(xv) No undefined local, overlap, Restart, representative, or Type IV status is treated as zero or pass.

(xvi) No Search-side map, score, or AI proposal supplies a missing overlap, continuation, or gluing artifact.

(xvii) No global exactness of positive-threshold deletion or global strict collapse functor is assumed.

(xviii) No local detector zero family implies global repaired zero without a global profile interface and retained-bar coverage.

(xix) No pairwise local compatibility constructs the global profile used by a detector-atlas theorem.

(xx) No metric-reference simplex record creates transition morphisms, cocycles, or object descent.

(xxi) No κ-Restart reserve inequality transports a repaired-zero predicate without a separately certified zero route.

(xxii) No terminal Type IV zero implies long-transient absence.

(xxiii) No source-profile overlap certificate transports kernel or cokernel defect profiles.

(xxiv) No boundary-crossing retained bar is ruled out by disjoint local detector zero unless boundary-bridge or another global coverage theorem is supplied.

(xxv) No post hoc reading of a complete global barcode is advertised as nontrivial finite proof compression.

5.17 Provenance and status

The sharp overlap-transfer theorem is inherited from Appendix A, Theorem 1.37. Finite detectors and stage discipline are inherited from Appendix B. Representative and eligible-realization clauses are inherited from Appendices B–C. Typed terminal Type IV and long-transient participation are inherited from Appendix D, especially Theorem 4.66 and Definition 4.83. The non-scalar obligation states are forwarded to Appendix F and their Manifest serialization to Appendix G.

The inherited v19 contribution is the record architecture joining sharp overlap comparison, simplex coherence, conditional κ-Restart, product-weighted reserve bounds, normalized countable summability, and audit-locally finite atlases. The v20 conservative extension adds exact/metric simplex modes, detector-aware global retained-bar coverage, detector-safe Restart, explicit normalized counterexamples, transient-clean atlas participation, and the strict boundary between predicate globalization and object descent.

5.17.0.1 Status labels for Appendix E. *Foundation definitions:* Definitions 5.6, 5.7, 5.9, 5.10, 5.12, 5.13, 5.15, 5.16, 5.17, 5.23, 5.24, 5.26, 5.27, 5.28, 5.33, 5.36, 5.37, 5.41, 5.47, 5.48, 5.49, 5.56, 5.57, 5.61, 5.62, 5.67, 5.68, 5.69, 5.72, 5.75.

Foundation lemmas/propositions/theorems: Propositions 5.35, 5.39, Theorem 5.43, Lemma 5.53, Theorems 5.59, 5.65, Propositions 5.70, 5.73, 5.76, and Theorem 5.86.

Operational cautions: Remarks 5.1, 5.8, 5.11, 5.14, 5.18, 5.25, 5.30, 5.44, 5.54, 5.60, 5.66, 5.74, 5.77, and 5.22.

v20 detector-atlas and coherence extension: Definitions 5.89, 5.90, 5.93, 5.95, 5.96, 5.97, 5.98, 5.110, 5.111, 5.112, 5.122, 5.125, 5.128, 5.133, 5.134; Propositions 5.99, 5.135; Theorems 5.115, 5.131, 5.137; Corollary 5.126; Examples 5.117, 5.120, 5.121.

Explicit exclusions: Subsection 5.16.

5.18 v20 conservative extension overview

Remark 5.88 (Immutable inherited locator discipline). All preceding numbered definitions, propositions, lemmas, and theorems retain their inherited v19 statements and order. The v20 material below is a conservative extension with new labels. It does not silently change the source strength of an inherited Claim Register entry.

5.19 v20 exact and metric-reference coherence modes

Definition 5.89 (Expanded record for Main Definition 6.23: overlap comparison mode)**.** *Every overlap inference declares exactly one canonical comparison mode:*

$$\texttt{exact}, \qquad \texttt{metric\_gap\_certified},$$
$$\texttt{theorem\_controlled\_nonmetric}, \qquad \texttt{not\_compared}.$$

The mode `exact` *consumes an actual equality, isomorphism, or directed morphism together with a proof that this exact relation entails the requested conclusion. The mode* `metric_gap_certified` *consumes a pre-threshold bottleneck bound and the two-sided threshold-gap theorem at the same budget. The mode* `theorem_controlled_nonmetric` *consumes a named theorem, its exact hypotheses, source, target, variance, printed conclusion, and a checked entailment to the requested conclusion. The mode* `not_compared` *is permitted only when the governing claim consumes no overlap inference. No affirmative mode is promoted to another, and exactness does not create a positive robustness radius.*

Definition 5.90 (Expanded record for Main Definition 6.24: exact overlap certificate)**.** *An exact overlap certificate on $(i, U_{\alpha\beta}, \tau_{\alpha\beta})$ contains:*

(i) the directly evaluated common-window pre-threshold and repaired profiles;

(ii) the common threshold and threshold-reconciliation record;

(iii) the requested overlap conclusion $\mathcal{C}^{\mathrm{req}}_{\alpha\beta,i}$;

(iv) exactly one exact relation type:

(a) `identification`: an actual equality or isomorphism;

(b) `directed_morphism`: an actual morphism with declared source, target, and variance;

(v) a theorem or direct proof that the relation entails $\mathcal{C}^{\mathrm{req}}_{\alpha\beta,i}$ at precisely the requested strength;

(vi) immutable source, target, proof, and artifact identifiers.

A directed morphism does not discharge identification, symmetric agreement, zero reflection, or descent compatibility unless the entailment proof states that conclusion. In particular, a zero morphism between arbitrary profiles is not an overlap pass merely because it is actual. No positive threshold-gap reserve follows merely from an exact relation.

Definition 5.91 (Expanded record for Main Definition 6.26: theorem-controlled nonmetric overlap certificate)**.** *A theorem-controlled nonmetric overlap certificate contains a named theorem T, its version and proof artifact, the exact source and target on the common window, every hypothesis and variance field, the printed conclusion $\mathcal{C}_{\mathsf{T}}$, the requested conclusion $\mathcal{C}^{\mathrm{req}}_{\alpha\beta,i}$, and a checked implication*

$$\mathcal{C}_{\mathsf{T}} \Longrightarrow \mathcal{C}^{\mathrm{req}}_{\alpha\beta,i}.$$

Only $\mathcal{C}_{\mathsf{T}}$ and registered weakenings may be consumed. No metric bound, morphism, cocycle, zero reflection, or descent datum is inferred unless it is explicitly part of that theorem.

Theorem 5.92 (Expanded proof of Main Theorem 6.27: sharp typed overlap theorem)**.** *For a well-typed pairwise overlap record, the admissible conclusion is mode-specific:*

(i) exact mode returns the registered exact relation and proved entailments;

(ii) metric-gap-certified mode returns the sharp repaired discrepancy bound, and repaired-zero equivalence only when zero separation is also certified;

(iii) theorem-controlled nonmetric mode returns exactly the named theorem's conclusion;

(iv) not-compared mode returns no overlap conclusion.

No cross-mode promotion is valid without a separate theorem.

Proof. The exact and theorem-controlled branches are exactly their proof artifacts. The metric branch is Appendix A's sharp threshold-gap theorem, together with its separate repaired-zero transport theorem when zero reflection is requested. The last branch is definitional. Since the branches have different source, target, and conclusion types, no stronger common conclusion follows formally. □

Definition 5.93 (Expanded record for Main Definition 6.29: pairwise Overlap Gate)**.** *A v20 pairwise Overlap Gate record has exactly one status*

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{pass}, \qquad \texttt{reject}.$$

Its status is determined as follows.

(i) `not_invoked`*: the governing claim consumes no overlap inference, the comparison mode is* `not_compared`*, and an explicit scope rationale is recorded;*

(ii) `undefined`*: overlap inference is invoked but the comparison mode, threshold reconciliation, proof, provenance, Type IV, transient, crop, or immutable artifact data are missing or ill-typed;*

(iii) `pass`*: an affirmative comparison mode has a complete certificate entailing the requested conclusion in every monitored degree; all consumed representative and realization provenance is compatible; direct Type IV, inverse-audit, and transient fields pass at the required scope; and no required clause is false or undefined;*

(iv) `reject`*: the record is well-typed and at least one required exact relation, metric bound, threshold-gap inequality, named-theorem hypothesis or entailment, direct Type IV field, inverse-audit field, transient field, or other participation condition is evidenced false.*

The mode `not_compared` *never supports* `pass`*.*

Remark 5.94 (Source overlap does not transport defect profiles). An exact or metric comparison between local repaired source profiles does not by itself compare

$$\ker \phi \qquad \text{or} \qquad \operatorname{coker} \phi$$

for tower artifacts attached to those windows. Terminal or transient defect transport requires actual defect-profile comparison data and a separately proved kernel/cokernel stability theorem.

Definition 5.95 (Simplex coherence mode)**.** *Every nonempty nerve simplex used by a coherence claim declares exactly one mode:*

$$\texttt{exact\_cocycle}, \qquad \texttt{metric\_reference}, \qquad \texttt{not\_compared}.$$

The first mode consumes actual isomorphisms and cocycle identities. The second consumes a fixed reference profile and certified metric comparisons to that reference. It supplies bounded record-level discrepancy but no transition morphisms or categorical cocycle.

Definition 5.96 (Exact simplex cocycle record)**.** *Let $\sigma = \{\alpha_0, \dots, \alpha_p\}$ index a nonempty simplex. An exact simplex cocycle record contains actual repaired-profile isomorphisms*

$$g^{\sigma}_{\alpha\beta} : \mathrm{Eval}^{\alpha}_{i,\sigma} \xrightarrow{\sim} \mathrm{Eval}^{\beta}_{i,\sigma}$$

with

$$g^{\sigma}_{\alpha\alpha} = \mathrm{id}, \qquad g^{\sigma}_{\beta\alpha} = (g^{\sigma}_{\alpha\beta})^{-1}, \qquad g^{\sigma}_{\beta\gamma} \circ g^{\sigma}_{\alpha\beta} = g^{\sigma}_{\alpha\gamma}.$$

Restriction to every face agrees with the previously registered face cocycle.

Definition 5.97 (Metric-reference simplex record)**.** *For a nonempty simplex σ, a metric-reference record contains:*

(i) one pre-threshold reference profile $R_{i,\sigma}$ on U_σ;

(ii) one common threshold τ_σ;

(iii) for every $\alpha \in \sigma$, an artifact-supported bound

$$d_B(B^{\alpha}_{i,\sigma}, R_{i,\sigma}) \le \varepsilon_{\alpha,\sigma,i};$$

(iv) the two-sided threshold-gap certificate for $(B^{\alpha}_{i,\sigma}, R_{i,\sigma})$ at that same budget;

(v) reference identity, scalarization, threshold reconciliation, and face-restriction provenance.

No map between two local repaired profiles and no cocycle is inferred.

Definition 5.98 (Metric-reference face compatibility). *Let $\rho \subseteq \sigma$ be nonempty recorded simplices in `metric_reference` mode. Their records are face-compatible only when:*

(i) the threshold on ρ is the restriction of the declared threshold policy on σ;

(ii) there is an actual equality or persistence-profile isomorphism

$$\mathsf{W}_{U_\rho} R_{i,\sigma} \xrightarrow{\sim} R_{i,\rho};$$

(iii) every local-to-reference comparison on σ, after restriction to U_ρ, agrees with the registered comparison on ρ at the declared provenance strength;

(iv) scalarization, gap, reference identity, and artifact hashes commute with the face map.

Without this record, simplexwise metric bounds do not form a coherent atlas. A weaker metric comparison between the two references would require a new three-way threshold-gap and exposure theorem and is not inferred here.

Proposition 5.99 (Pairwise bound through a metric simplex reference). *Under a metric-reference simplex record,*

$$d_B\Big(\mathrm{Eval}^{\alpha}_{i,\sigma}, \mathrm{Eval}^{\beta}_{i,\sigma}\Big) \le \varepsilon_{\alpha,\sigma,i} + \varepsilon_{\beta,\sigma,i}$$

for every $\alpha, \beta \in \sigma$.

Proof. Apply the Appendix A threshold-gap-safe theorem to each local-to-reference comparison and use the triangle inequality through $T^{\mathrm{bar}}_\tau(R_{i,\sigma})$. □

Remark 5.100 (Metric coherence is not an approximate cocycle). A system of distance bounds has no composable transition morphisms. Metric-reference coherence concerns reference identity, threshold, face records, and certified bounds. It is not an exact cocycle modulo an unspecified error.

Definition 5.101 (Expanded record for Main Definition 6.34: simplex coherence mode). *Every nonempty nerve simplex consumed by a coherence claim declares exactly one mode*

`exact_cocycle`, `model_certified_homotopy_coherent`,
`metric_reference`, `theorem_controlled_nonmetric`, `not_compared`.

The modes have distinct data and distinct conclusion strengths.

Definition 5.102 (Expanded record for Main Definition 6.36: model-certified homotopy-coherent simplex cell). *Let actual local objects E_α lie in a declared dg, simplicial, or other higher-categorical model. A model-certified homotopy-coherent simplex-cell record contains:*

(i) the fixed model $\mathfrak{M}_{\mathrm{coh}} = (I, Q, E, V)$, including its information, query, encoding, and verifier semantics;

(ii) actual vertices, edge morphisms, higher cells h_ρ for every required face $\rho \subseteq \sigma$, and their degrees and variance;

(iii) the exact finite list of differential, boundary, degeneracy, and higher-coherence identities required through the declared Čech depth;

(iv) verifier transcripts and immutable encodings for every tested cell;

(v) a theorem that the finite checks imply the named model-relative coherence conclusion.

The record is not an object-descent theorem and is not invariant under a change of model unless a separate comparison theorem is supplied.

Proposition 5.103 (Expanded proof of Main Proposition 6.37: homotopy coherence is model-relative)**.** *A model-certified homotopy-coherent record certifies only the cells, identities, and conclusion represented by its declared $\mathfrak{M}_{\mathrm{coh}}$. It does not imply a strict cocycle, metric bound, model-independent $(\infty, 1)$-categorical coherence, or effective descent without a separate theorem.*

Proof. The verifier receives only the encoded cells and relations specified by $\mathfrak{M}_{\mathrm{coh}}$. Strict equality, metric control, comparison to another model, and descent effectivity are not logical consequences of those finite predicates unless included in the registered theorem. Hence the conclusion is exactly model-relative. ☐

Definition 5.104 (Expanded record for Main Definition 6.38: theorem-controlled nonmetric simplex)**.** *A theorem-controlled nonmetric simplex record contains a named theorem, its version, the exact local inputs and faces, all hypotheses and variance fields, its printed higher-compatibility conclusion, and a checked entailment to the requested simplex conclusion. It creates no transition maps or cells not present in the theorem.*

Theorem 5.105 (Expanded proof of Main Theorem 6.32: no higher coherence from pairwise overlap)**.** *Pairwise exact isomorphisms need not satisfy a triple cocycle; pairwise metric bounds have no composable maps; and pairwise theorem-controlled relations need not possess compatible higher conclusions. Therefore every consumed triple, higher simplex, or homotopy-coherent conclusion requires its own registered cell or theorem.*

Proof. For exact maps, choose three isomorphic objects and pairwise automorphisms whose product around the triangle is nonidentity. Metric bounds are scalar relations and therefore have no composition operation. A theorem whose conclusion is pairwise supplies no higher datum. Thus none of the three pairwise modes entails higher coherence. ☐

5.20 v21 certificate-atlas bundle and finite record soundness

Definition 5.106 (Expanded record for Main Definition 6.42: simplex compatibility record)**.** *For every nonempty simplex σ, a simplex compatibility record contains the direct common-window inputs, common threshold, all face records, one mode from Definition 5.101, the mode-specific proof artifact, requested conclusion and entailment, direct Type IV and inverse-audit participation fields when invoked, transient-defect participation when required, and immutable face-compatible references.*

Definition 5.107 (Expanded record for Main Definition 6.44: certificate atlas)**.** *A certificate atlas is the typed bundle*

$$\mathfrak{A} = (\mathcal{L}, \mathcal{O}, \mathcal{H}, \mathsf{Cov}, \mathcal{R}, \mathcal{D}_{\mathrm{dir}}, \mathcal{D}_{\mathrm{inv}}, \mathcal{D}_{\mathrm{tr}}, \mathcal{M}),$$

where $\mathcal{L}$ is the local-record family, $\mathcal{O}$ the pairwise overlap family, $\mathcal{H}$ the higher-simplex family, Cov *retained- feature coverage when a covered-range predicate is claimed, $\mathcal{R}$ Restart data, the next three fields are direct, inverse, and transient diagnostics, and $\mathcal{M}$ is the manifest and replay bundle.*

Definition 5.108 (Expanded record for Main Definition 6.45: atlas verdict)**.** *The atlas verdict is the noncompensating conjunction of every required field in Definition 5.107. It has status*

$$\texttt{pass}, \qquad \texttt{reject}, \qquad \texttt{undefined}.$$

A favorable metric reserve, local zero, or coherent subfamily cannot compensate for a rejected or undefined required field.

Theorem 5.109 (Expanded proof of Main Theorem 6.48: finite certificate-atlas soundness)**.** *For a finite cover, suppose all required atlas fields are present and every local, overlap, higher-coherence, coverage, Restart, diagnostic, and manifest clause passes at its registered strength. Then the atlas verdict passes. Conversely, an atlas pass implies that no required field is rejected or undefined. The conclusion is record-level only.*

Proof. The verdict is the finite noncompensating conjunction of its typed fields. Thus all passing clauses imply a pass, and any reject or undefined field blocks one. No object-construction rule occurs in the conjunction. □

5.21 Detector-aware local-to-global coverage

Definition 5.110 (Global profile and local restriction interface)**.** *Fix a covered range X, degree i, and one threshold $\tau > 0$. A global profile interface contains a constructible pre-threshold profile*

$$B_{i,X}$$

and, for every atlas window W_α, the exact restriction

$$R_i^\alpha := \mathsf{W}_{W_\alpha} B_{i,X}.$$

The observed local pre-threshold profile B_i^α is connected to R_i^α by exactly one branch:

(i) `exact`*: an actual isomorphism $R_i^\alpha \cong B_i^\alpha$;*

(ii) `metric_zero_transport`*: an artifact-supported bound*

$$d_B(R_i^\alpha, B_i^\alpha) \le \varepsilon_\alpha,$$

the Appendix A two-sided threshold-gap hypotheses for both profiles, and

$$2\varepsilon_\alpha \le \tau.$$

The operational policy may require the strict reserve $2\varepsilon_\alpha < \tau$.

The second branch consumes Theorem 1.47. Pairwise compatibility among local profiles does not construct $B_{i,X}$.

Definition 5.111 (Certified local detector atlas)**.** *Relative to a global profile interface, a certified local detector atlas assigns to every W_α:*

(i) a finite stencil $T_\alpha \subseteq (W_\alpha)_\tau$, or the canonical witness family of a certified birth or right-germ detector;

(ii) an Appendix B detector at source stage `windowed_prethreshold`*;*

(iii) endpoint, critical-set, family, applicability, completeness, and immutable artifact records;

(iv) one interface branch from Definition 5.110;

(v) one mathematical detector value and an independent detector certificate status.

A mathematical zero is consumed globally only when the local certificate is complete at its exact declared strength.

Definition 5.112 (Global retained-bar coverage)**.** *A certified local detector atlas has global retained-bar coverage when every bar I of $B_{i,X}$ retained by T_τ^{bar}, including every genuinely infinite retained bar, has a certified local witness: there exist α and $t \in T_\alpha$ such that*

$$t \in I \cap W_\alpha, \qquad t + \tau \in I \cap W_\alpha.$$

For a birth or right-germ detector, this clause is read using its canonical interval witness. The coverage status is one of

$$\texttt{undefined}, \qquad \texttt{certified}, \qquad \texttt{failed}.$$

It is `certified` *only when the artifact proves the universal quantifier over all globally retained bars. It is* `failed` *when the coverage claim is well-typed and a retained bar without a certified local witness is exhibited. Missing applicability, global-barcode, witness, or provenance data yield* `undefined`*, never vacuous coverage.*

Theorem 5.113 (Expanded proof of Main Theorem 6.53: finite retained-bar coverage)**.** *Assume the total registered local witness set is finite. Global retained-bar coverage holds if and only if every globally retained interval I has at least one registered window and witness t with*

$$t, t + \tau \in I \cap W_\alpha,$$

or the corresponding certified birth/right-germ witness. Under this condition, simultaneous zero of complete local detectors is equivalent to global repaired zero.

Proof. The first statement is the definition specialized to a finite witness registry. If a retained global interval exists, coverage places one of its nonzero τ-span maps inside a complete local detector, forcing a nonzero local value. Conversely, global repaired zero restricts to local repaired zero and therefore to zero complete detector values through each certified interface branch. □

Corollary 5.114 (Expanded proof of Main Corollary 6.54: countable detector-family zero criterion)**.** *For a countable detector family, the same equivalence holds under a complete enumeration, universally quantified local completeness, certified global retained-bar coverage, and hidden-tail prevention. A finite prefix supplies no countable conclusion unless a finite-to-infinite reduction proves the tail.*

Theorem 5.115 (Expanded proof of Main Theorem 6.55: detector-atlas local-to-global zero)**.** *Let a global profile interface, a certified local detector atlas, and a* `certified` *global retained-bar coverage artifact be supplied in the declared constructible interval-decomposable regime. Assume every metric interface branch satisfies its complete repaired-zero transport hypotheses and every local detector is applicable and complete on its declared profile class. Then*

$$T_\tau^{\mathrm{bar}}(B_{i,X}) = 0 \quad \Longleftrightarrow \quad \mathcal{D}_\alpha^{\mathrm{loc}}(B_i^\alpha) = 0 \quad \textit{for every } \alpha.$$

Proof. Suppose first that the global repaired profile is zero. Restriction cannot increase the length of a finite interval, and a globally zero repaired profile contains no infinite interval. Therefore

$$T_\tau^{\mathrm{bar}}(R_i^\alpha) = 0$$

for every α. The exact or metric-safe interface branch gives

$$T_\tau^{\mathrm{bar}}(B_i^\alpha) = 0.$$

Local detector completeness then gives detector zero.

Conversely, suppose every local detector is zero and assume, for contradiction, that a globally retained interval I exists. Coverage supplies α and a witness t with $t, t+\tau \in I \cap W_\alpha$. Hence

$$T_\tau^{\mathrm{bar}}(R_i^\alpha) \neq 0.$$

The exact or metric-safe interface branch preserves and reflects repaired zero, so

$$T_\tau^{\mathrm{bar}}(B_i^\alpha) \neq 0.$$

Completeness of the local detector forces a nonzero detector value, contrary to hypothesis. □

Remark 5.116 (Coverage is the global proof obligation)*.* Coverage may be verified by reading the complete global barcode, but that post hoc route supplies no proof-compression advantage. A nontrivial finite-certificate theorem derives coverage independently from geometry, an endpoint lattice, bounded birth data, a family theorem, boundary-bridge windows, or another registered reduction.

Example 5.117 (Local zero and exact overlap coherence do not imply global zero)*.* Fix $\tau > 0$, set

$$X = [0, 2\tau), \qquad B_{i,X} = \{[0, 2\tau)\},$$

and cover X by

$$W_1 = [0, \tau), \qquad W_2 = \left[\frac{\tau}{2}, \frac{3\tau}{2}\right), \qquad W_3 = [\tau, 2\tau).$$

Every restricted interval has length exactly τ, so every local repaired profile is zero. Every nonempty overlap repaired profile is also zero and admits an exact cocycle. Nevertheless the global interval has length $2\tau > \tau$ and survives. The missing hypothesis is global retained-bar coverage: no listed window contains a full τ-span strictly inside the retained global bar under the hard deletion convention.

Definition 5.118 (Object-level descent interface)**.** *An object-level descent claim additionally declares:*

(i) a category or higher category with restriction functors;

(ii) actual local objects;

(iii) actual transition equivalences;

(iv) an exact cocycle or higher coherence datum;

(v) an effective-descent theorem for the declared cover;

(vi) an artifact constructing the descended object.

Metric-reference records, retained-bar coverage, and a certificate atlas do not substitute for these data.

Remark 5.119 (Predicate globalization is not object descent)*.* Theorem 5.115 proves a predicate about a separately declared global profile. It neither constructs that profile from local data nor descends filtered, persistence, or derived objects.

Example 5.120 (Ordinary summability can coexist with reserve failure)*.* Let

$$r_0 = 1, \qquad \kappa_k = \frac{1}{2}, \qquad \eta_k = 2^{-k-2}.$$

Then

$$\sum_{k\geq 0} \eta_k = \frac{1}{2} < r_0,$$

but

$$K_{k+1} = 2^{-k-1}, \qquad \frac{\eta_k}{K_{k+1}} = \frac{1}{2}.$$

Thus the normalized loss diverges. If equality holds in every Restart recurrence, then

$$\frac{r_n}{K_n} = 1 - \frac{n}{2},$$

so the reserve becomes nonpositive at a finite stage.

Example 5.121 (Pointwise positive reserve need not be uniform)*.* Let

$$\eta_k = 0, \qquad \kappa_k = \frac{1}{2},$$

and take equality in every Restart step. Then

$$r_n = 2^{-n} r_0 > 0$$

for every finite n, but

$$\inf_n r_n = 0.$$

5.22 Detector-aware Restart and boundary-bridge coverage

Definition 5.122 (Detector-aware Restart handoff)**.** *A Restart transition that consumes a repaired-zero predicate records, in addition to reserve transport, exactly one target-zero route:*

(i) a fresh complete detector certificate on the target window whose mathematical value is zero;

(ii) an exact profile isomorphism from a source profile already certified zero to the target profile;

(iii) an Appendix A metric-safe zero-transport record from a source profile already certified zero, including the two-sided threshold gap and post-threshold zero separation;

(iv) a theorem-controlled nonmetric record whose named conclusion contains the required source-zero to target-zero implication with all hypotheses and variance fields verified.

The zero-route status is exactly one of

$$\texttt{not\_invoked}, \qquad \texttt{undefined}, \qquad \texttt{certified}, \qquad \texttt{failed}.$$

It is `certified` only when the source-zero premise, the complete route artifact, and the target-zero conclusion are all verified. It is `failed` when the route is well-typed but a required mathematical premise or conclusion is evidenced false. Missing or ill-typed route data yield `undefined`; an explicit claim that does not consume zero transport uses `not_invoked`. The record identifies source and target stages, threshold, endpoint convention, comparison mode, proof identity, and immutable artifact. A κ_k-admissible inequality alone is not a zero-transport theorem.

Theorem 5.123 (Expanded proof of Main Theorem 6.78: Restart zero transport)**.** *Suppose the predecessor zero predicate is certified and the Restart datum is well-typed. The target zero predicate is certified only through one of the routes in Definition 5.122: a fresh complete target proof, an exact zero-reflecting identification, Appendix A metric-safe zero transport, or a theorem-controlled nonmetric implication. A positive κ-reserve alone is neither necessary nor sufficient for target zero.*

Proof. Each listed route contains an explicit implication to the target predicate. The Restart recurrence is only a real inequality among reserves and contains no logical map between source and target zero predicates. □

Theorem 5.124 (Expanded proof of Main Theorem 6.79: Restart record transport)**.** *Let the predecessor local record pass. If the Restart datum is complete, the continuation artifact is certified, every inherited field satisfies Proposition 5.46, all invalidated or fresh fields are reproved, a target-zero route passes when required, and every remaining target clause passes, then the target local record passes.*

Proof. The continuation theorem authorizes exactly the inherited fields. Fresh proofs supply every invalidated or target-specific field, and the selected zero route supplies target zero. The target verdict is their noncompensating conjunction. □

Definition 5.125 (Boundary-bridge coverage)**.** *For an ordered partition $\{W_k\}$, let $\{V_k\}$ be declared bridge windows meeting both sides of the boundary between W_k and W_{k+1}. The augmented family*

$$\mathcal{W}_{\mathrm{aug}} := \{W_k\}_k \cup \{V_k\}_k$$

has boundary-bridge coverage at τ when every globally retained interval admits a certified τ-span witness in one partition or bridge window. Every bridge window has its own direct evaluation and complete detector certificate.

Corollary 5.126 (Expanded proof of Main Corollary 6.81: detector-safe partition globalization)**.** *Assume a global profile interface for an augmented partition/bridge family, boundary-bridge coverage, and complete zero detector certificates on every partition and bridge window. Then*

$$T_\tau^{\mathrm{bar}}(B_{i,X}) = 0.$$

Proof. The augmented family is a certified local detector atlas with global retained-bar coverage. Apply Theorem 5.115. □

Remark 5.127 (Reserve transport does not detect crossing bars)*.* A positive Restart reserve controls the threshold classification of the profiles named by the continuation artifact. It does not prove that a retained interval crossing a partition boundary is detected inside either disjoint cell. Fresh target detection controls the target predicate; boundary-bridge coverage controls the global predicate.

Definition 5.128 (Metric-reference simplex exposure)**.** *Let N_{met} be the set of recorded simplices in `metric_reference` mode. For $\sigma \in N_{\mathrm{met}}$, let $e_\sigma \geq 0$ be a fixed nonnegative quantity dominating every metric comparison charged to that simplex. Define the unordered sum*

$$\sum_{\sigma \in N_{\mathrm{met}}} e_\sigma := \sup_{\substack{S \subseteq N_{\mathrm{met}} \\ S \text{ finite}}} \sum_{\sigma \in S} e_\sigma.$$

Exact-cocycle simplices are supported by exact proof artifacts and are not assigned fictitious metric exposure.

Definition 5.129 (Expanded record for Main Definition 6.84: countable atlas enumeration and hidden tail)**.** *A countable atlas records a complete enumeration of windows, nonempty nerve simplices, face maps, required local clauses, and a proof that no unindexed tail datum can affect the verdict. Its verification level is exactly one of*

$$\texttt{countably\_specified}, \qquad \texttt{finitely\_certified}.$$

The second level requires a finite generator with a universal schema proof or a Chapter 5 finite-to-infinite reduction with exceptional indices, tail coverage, and a failure boundary. Audit-local finiteness alone does not prove enumeration completeness or finite verifiability.

Theorem 5.130 (Expanded proof of Main Theorem 6.85: countable semantic record coherence)**.** *Let the countable cover be audit-locally finite and of finite Čech depth. If the atlas is completely enumerated, every local record passes, every nonempty simplex has a complete mode-specific record with compatible faces, and every required diagnostic and manifest field passes, then the family is a countably specified Čech-coherent certificate atlas. This conclusion does not require summable metric exposure.*

Proof. Audit-local finiteness makes every bounded audit query finite, while the complete enumeration and finite depth specify every required simplex and face. The mode-specific records establish semantic coherence. Exposure summability is not used in these logical clauses. □

Theorem 5.131 (Expanded proof of Main Theorem 6.87: budgeted countable record-atlas principle)**.** *Let $\{W_n\}_{n\geq 1}$ be a countable audit-locally finite overlap cover of finite Čech depth. Assume:*

- *(i) every local window has a complete certificate record;*
- *(ii) every nonempty nerve simplex has one complete mode-specific record; the permitted v20 modes are `exact_cocycle` and `metric_reference`;*
- *(iii) all exact-cocycle identities and exact face restrictions are proved;*
- *(iv) all metric-reference comparisons satisfy their individual two-sided threshold-gap conditions and every metric-reference face satisfies Definition 5.98;*
- *(v) metric-reference simplex exposure is summable;*
- *(vi) terminal Type IV is correctly `not_invoked` or `certified_zero`;*
- *(vii) when long-transient cleanliness is required, its aggregate Appendix D verdict is `certified_absent`;*
- *(viii) all records and artifact references are audit-readable.*

Then the family is an audit-locally finite Čech-coherent certificate atlas with finite declared metric-reference exposure.

If a global repaired-zero conclusion is additionally invoked, the atlas must also carry:

- *(a) a global profile interface;*
- *(b) complete local detector certificates;*
- *(c) exact or metric-safe local zero transport;*
- *(d) global retained-bar coverage;*

(e) zero local detector values.

Under these additional clauses, the global repaired profile is zero.

Proof. Audit-local finiteness makes every bounded audit query finite, while finite Čech depth bounds overlap order. Exact-cocycle records provide actual coherent morphisms at record level. Metric-reference records provide reference-consistent bounds without claiming a cocycle. The face conditions and local records therefore form a coherent atlas, and the unordered exposure sum is finite by hypothesis. When the detector-globalization clauses are invoked, Theorem 5.115 gives the global repaired-zero predicate. □

Remark 5.132 (Summability is not coherence). Exposure summability does not create missing face maps, cocycles, reference identities, local completeness, or retained-bar coverage. Conversely, a coherent atlas may fail a separately declared exposure policy.

Definition 5.133 (v20 detector-atlas manifest extension). *A proof object invoking local detector globalization additionally records:*

(i) the global profile and covered-range identity;

(ii) every exact local restriction profile;

(iii) every observed local profile;

(iv) the exact or metric-zero-transport interface branch;

(v) every local detector identifier, stage, witness set, endpoint convention, family, completeness source, and value/status pair;

(vi) the global retained-bar coverage theorem and artifact;

(vii) the exact-cocycle or metric-reference mode of every consumed simplex;

(viii) terminal and long-transient participation fields;

(ix) every Restart zero route and boundary-bridge artifact when invoked;

(x) claim, proof, manifest, hash, and provenance references.

Missing coverage or an incomplete local detector yields `undefined`*, not global zero.*

Definition 5.134 (Terminal/transient participation policy). *Each local, overlap, simplex, Restart-target, or atlas record declares separately:*

(i) terminal Type IV scope and status;

(ii) whether long-transient absence is required;

(iii) when required, the fixed $(W, \tau, \tau_{\mathrm{def}}, I_{\mathrm{mon}})$ *and the Appendix D aggregate transient verdict.*

Terminal `certified_zero` *does not imply transient* `certified_absent`*, and neither implies full comparison-map isomorphism.*

Proposition 5.135 (Transient-clean participation rule). *When a gluing claim requires absence of long transient kernel or cokernel defects, no participating record may have transient status* `certified_present` *or* `undefined`*. Every required record must have* `certified_absent` *at the same declared detector component and threshold policy.*

Proof. This is the aggregate verdict discipline of Appendix D, Definition 4.83. Record compatibility cannot repair a present or undefined defect clause. ☐

Remark 5.136 (Source-profile coherence is not defect coherence)*.* Even a complete exact source-profile cocycle does not prove that tower comparison kernels or cokernels glue, agree, or remain absent. Such a conclusion requires actual compatible tower artifacts and defect-profile comparison theorems.

Definition 5.137 (Historical v20 detector-atlas and gluing closure record)**.** *Within the v20 conservative extension:*

(i) the complete inherited v19 overlap, Restart, summability, and record-atlas package remains valid at its original strength;

(ii) pairwise and simplex coherence distinguish exact-cocycle from metric-reference modes;

(iii) metric-reference data do not create morphisms or cocycles;

(iv) local detector zero becomes global repaired zero only through a declared global profile interface, complete local detectors, safe local zero transport, and global retained-bar coverage;

(v) local zero and exact overlap coherence alone do not imply global zero;

(vi) a κ-Restart reserve inequality does not transport a zero predicate without a separate detector or zero-transport route;

(vii) boundary-bridge coverage repairs the blind spot of disjoint partition cells only when each bridge has its own complete detector;

(viii) ordinary summability may coexist with finite-stage reserve failure;

(ix) pointwise positive reserves need not admit a uniform positive lower bound;

(x) terminal Type IV cleanliness and long-transient cleanliness are separate atlas clauses;

(xi) source-profile coherence does not transport kernel/cokernel defect profiles;

(xii) countable atlas coherence, metric exposure, detector globalization, and object-level descent are four distinct conclusions.

Remark 5.138 (Closure-record status)*.* The mathematical components cited above retain their individual theorem identities. This record is operational and provenance-only; it creates no new v21 package theorem.

5.23 v21 manifest extension and replay obligations

Proposition 5.139 (Expanded record for Main Proposition 6.102: window/overlap/Restart manifest extension)**.** *An audit-readable manifest invoking Appendix E records, as applicable:*

(i) the cover type, complete index set, windows, nerve simplices, face relations, Čech depth, and hidden-tail verification level;

(ii) every local pre-threshold and repaired profile, threshold, endpoint, detector, representative, finite-grid, and realization identifier;

(iii) every requested overlap or simplex conclusion, canonical comparison mode, exact relation or theorem identity, metric ledger and scalarization, threshold reconciliation, and entailment proof;

(iv) every strict cocycle, higher cell, coherence model, metric reference, or theorem-controlled nonmetric record and all face restrictions;

(v) the global profile interface, local restriction branches, detector atlas, retained-feature coverage, and boundary/tail witnesses whenever global record zero is claimed;

(vi) every Restart datum, inherited and invalidated field, continuation artifact, κ_k, η_k, r_k, normalized loss, target-zero route, exceptional index, and tail certificate;

(vii) direct Type IV, inverse-audit triple, transient-defect, and full- isomorphism participation fields at their independent statuses;

(viii) the complete canonical nine-field DescentInput

$$(\{E_\alpha\}, \{g_{\alpha\beta}\}, \{c_{\alpha\beta\gamma}\}, \{h_\sigma\}, \mathcal{W}, \mathcal{C}, \mathfrak{M}_{\mathrm{coh}}, \mathsf{T}_{\mathrm{eff}}, \mathcal{A}),$$

the active descent lane, field-by-field statuses, and the proposed downstream artifact whenever descent is handed off; the independent certificate-atlas record is not substituted for $\mathcal{C}$;

(ix) claim, dependency, verifier, hash, provenance, and replay identities.

A manifest serializes evidence; it does not create a missing mathematical proof or increase its conclusion strength.

Remark 5.140 (Canonical manifest location). Proposition 5.139 is an extension schema for Appendix G and the final release manifest. It is not a competing global manifest authority.

5.24 v21 proof-package closure and migration status

Definition 5.141 (Appendix E v21 proof-package closure record). *Appendix E is locally complete at the Foundation layer when:*

(i) all inherited window, overlap, detector-atlas, Restart, normalized summability, and boundary-coverage theorems pass regression;

(ii) every consumed overlap and simplex uses exactly one canonical mode;

(iii) every requested conclusion has an explicit entailment proof;

(iv) local zero, global record zero, and global object zero remain distinct;

(v) finite and countable atlases carry complete enumeration, face, coverage, hidden-tail, diagnostic, and manifest records;

(vi) Restart carries evidence inheritance/invalidation and a target-zero route when zero is consumed;

(vii) object descent is exported only through a complete canonical nine-field DescentInput *and one active-lane selection; the exported status is readiness only, while effectivity, descended-object existence, conservativity, and global verdicts remain downstream;*

(viii) no theorem identity owned by Main Chapter 6 or Appendix DG is duplicated.

This is an operational closure record, not a new mathematical theorem.

Remark 5.142 (G21 contribution ceiling). Appendix E contributes local evidence to G21-2, G21-4, G21-5, G21-8, and G21-9. In particular it can establish a global record-zero predicate and a ready descent input. It cannot by itself pass object-level globalization, effective descent, unified external Return, or release-wide closure.

5.24.0.1 Migration summary.

The v20 record atlas is `inherited`. Four-mode comparison, requested-conclusion typing, model-certified homotopy coherence, countable hidden-tail prevention, inverse-audit participation, and the descent-input schema are `strengthened`. Effective descent and general finite- category reconstruction are `relocated` to Appendix DG and Chapter 7. No v20 mathematical result is rejected.

5.24.0.2 Final nonclaims.

No local or pairwise family constructs a global object. No metric-reference record creates a morphism. No pairwise exact family creates a cocycle. No finite prefix proves a countable tail. No positive Restart reserve proves zero. No direct Type IV verdict determines $\varprojlim^1$ or apex agreement. No record-level atlas, detector-globalization theorem, or DescentInput constitutes an external Return theorem. No certificate atlas populates $\mathcal{C}$ by default. No `descent_input_ready` status applies an effectivity theorem. No effectivity theorem alone supplies joint conservativity, global zero, or global isomorphism. No finite Čech totalization is retyped as object descent without its own comparison and effectivity theorem.

6 Appendix F: Typed Verification, Proof-DAG Semantics, and Governance Kernel

6.1 Scope, canonical ownership, migration, and design boundary

6.1.0.1 Normative status and theorem ownership.

Appendix F is the expanded-proof, counterexample, schema-semantics, proof-DAG, source-quarantine, replay, revision, readiness, and regression owner for the verifier statements canonically owned by Main Chapter 8 of AK-HPDST v21.0.0. Main Chapter 8 retains the canonical theorem identities. This appendix supplies detailed proofs, executable record semantics, adversarial examples, migration rules, and the Chapter 9 handoff contract; it creates no competing Main theorem UID.

The immutable v20 Appendix F remains directly invocable from its frozen artifact at its original strength. The v21 migration is componentwise:

(i) the four atomic statuses, justified scope exclusion, dependency-closed conjunction, non-compensation, countable hidden-tail prevention, immutable repair history, and Appendix G serialization boundary are `inherited`;

(ii) independent statement axes, the canonical four comparison modes, six detector stages, typed proof-DAG nodes and edges, assumption-context propagation, countable well-foundedness, source-statement binding, benchmark quarantine, authority separation, replay hierarchy, impact closure, route-admissible conclusion sets, and `bridge_input_ready` are `strengthened`;

(iii) the former v20 package theorems are retained only as historical closure locators. The canonical v21 package theorem is Main Theorem 8.103;

(iv) Bridge and Return theorem construction is relocated to Main Chapter 9, while final validation remains downstream in Chapter 10.

Appendix F consumes:

(i) Appendices A–E and Main Chapters 1–6 for repaired profiles, comparison modes, detector stages, representatives, eligible realizations, direct/inverse towers, atlases, Restart, and `descent_input_ready`;

(ii) Appendix DG and Main Chapter 7 for finite diagram reconstruction, effective descent, and global zero/isomorphism certificates;

(iii) Main Chapter 8 for canonical verifier ownership;

(iv) Appendix G for serialization only, without transferring status semantics or theorem authority.

The central rule is:

> A v21 verdict is a target-local, dependency-closed, source-bound, artifact-closed conjunction of independently evidenced clauses. No numerical reserve, successful build, graph path, agent consensus, or complete-looking manifest manufactures a missing proof obligation.

Remark 6.1 (Appendix F does not enlarge the mathematical Core). This appendix proves no new persistence, Ext, tower, overlap, descent, or external Return theorem. It governs when such results may be consumed as proof evidence. A status, node, edge, manifest row, hash, or replay record never replaces the mathematical theorem or verified artifact it names.

Remark 6.2 (Normative rather than external-domain content). Most statements below are typing contracts or logical consequences of those contracts. They are mathematically substantive as proof-governance statements, but they do not prove an arithmetic, geometric, analytic, PDE, cryptographic, or other external-domain conclusion.

6.2 Verifier-side datum and independent statement axes

Definition 6.3 (Expanded record of Main Definition 8.3: verifier-side datum)**.** *A verifier-side datum is*

$$\mathfrak{V} = \begin{pmatrix} \mathrm{RunID}, \mathrm{RevisionID}, \mathrm{TheoryVersion}, \mathrm{SchemaID}, \\ \mathrm{ClaimSnapshot}, \mathrm{Scope}, \mathrm{NodeRegistry}, \mathrm{EdgeRegistry}, \\ \mathrm{ArtifactRegistry}, \mathrm{EvidencePolicies}, \\ \mathrm{VerifierProfile}, \mathrm{FailurePolicy} \end{pmatrix}.$$

It is admissible only when every registry is immutable for the run, every identifier resolves uniquely, and every import has an explicit authority, version, scope, and semantic conversion record.

Definition 6.4 (Expanded record of Main Definition 8.4: verifier input admissibility)**.** *A verifier input is exactly one of:*

(i) a theorem or definition with exact statement, hypotheses, proof location, scope, and permitted conclusion strength;

(ii) a typed interface record with source, target, variance, exactness, reflection, information, and artifact fields;

(iii) an immutable artifact with identity, type, provenance, and interpretation policy;

(iv) a metric record with an actual discrepancy theorem and scalarization;

(v) a coherence, descent, reduction, or Return record satisfying its owner chapter's complete input contract;

(vi) an explicit assumption node whose downstream conditionality remains visible.

Prose confidence, visual graph connectivity, successful compilation, a bare citation, or an agent recommendation is not verifier evidence by default.

Definition 6.5 (Expanded record of Main Definition 8.5: independent statement axes)**.** *Every statement record carries the independent tuple*

$$\mathrm{Axis}(c) = \begin{pmatrix} \text{document\_role, evidence\_status, invocation\_role,} \\ \text{mathematical\_verdict, conformance\_status,} \\ \text{workflow\_status, migration\_status} \end{pmatrix}.$$

No component is inferred from another. In particular, `released`*,* `schema_conformant`*,* `replay_passed`*, and human or agent approval do not imply* `proved` *or mathematical* `pass`*.*

Proposition 6.6 (Expanded proof of Main Proposition 8.6: no cross-axis promotion)**.** *A favorable value on one statement axis does not promote another axis without a registered conversion theorem whose hypotheses pass.*

Proof. Each axis is the codomain of a different predicate. A workflow or conformance witness supplies neither a proof term nor a mathematical counterexample. Therefore cross-axis promotion is ill-typed absent an explicit conversion. ☐

Definition 6.7 (Expanded record of Main Definition 8.7: claim-use preflight)**.** *A theorem-level use passes preflight exactly when the claim identity and source statement are fixed; the evidence status is* `proved`*, or* `conditional` *with every condition discharged; the invocation role is theorem evidence; the use does not exceed registered strength; every supersession or migration is explicit; all transitive artifacts resolve; and benchmark or conclusion-derived information has not contaminated upstream construction evidence.*

Proposition 6.8 (Expanded proof of Main Proposition 8.8: preflight blocker)**.** *A claim that fails preflight cannot discharge a theorem, interface, reduction, descent, or Return node.*

Proof. The proposed node would consume an unregistered identity, hypothesis set, role, strength, or independence condition. Hence a necessary dependency remains undischarged and the node cannot pass. ☐

6.3 Immutable obligation atoms and registries

Definition 6.9 (Typed obligation atom)**.** *A* typed obligation atom *is a tuple*

$$\begin{aligned}\mathfrak{o} := (&\text{ObligationID, owner, scope, kind,} \\ &\text{evidence policy, dependencies, failure routes,} \\ &\text{manifest fields}).\end{aligned}$$

Its fields identify:

(i) the immutable clause identity;

(ii) the chapter or appendix owning its mathematical meaning;

(iii) the object, degree, window, threshold, tower, overlap, transition, or atlas scope;

(iv) the class of evidence needed to evaluate it;

(v) the prerequisite obligations;

(vi) every broad Type I–V route that may apply on failure;

(vii) the fields required for manifest serialization and replay.

Definition 6.10 (Obligation kinds). *Each obligation atom has one primary kind:*

(i) mathematical verdict*: for example repaired profile vanishing or eligible Ext vanishing;*

(ii) existence/construction*: for example a representative, comparison morphism, or continuation artifact;*

(iii) compatibility/coherence*: for example profile compatibility, edge realization, face compatibility, or composition coherence;*

(iv) metric-bearing safety*: a clause whose evidence is a declared upper bound consumed by a metric gate;*

(v) operational transition/computation*: for example terminal tameness, scalarization, or a Restart inequality;*

(vi) governance/manifest*: claim status, repair history, artifact identity, scope justification, schema consistency, or replayability.*

The primary kind does not prevent one failed atom from receiving several failure routes.

Definition 6.11 (Declared obligation registry). *For an immutable audit run $\mathfrak{R}$, a* declared obligation registry *is a finite or countable family*

$$\mathsf{Obl}(\mathfrak{R}) = \{\mathfrak{o}_a\}_{a \in A}$$

with immutable identifiers, together with:

(i) the required subset $\mathsf{Req}(\mathfrak{R}) \subseteq \mathsf{Obl}(\mathfrak{R})$;

(ii) an explicit scope-exclusion record for every atom outside $\mathsf{Req}(\mathfrak{R})$;

(iii) the dependency edges and their provenance;

(iv) evidence identifiers and evidence-policy versions;

(v) the registered claim and repair-record versions governing every theorem-level use.

A silent change in any of these data defines a new run or revision.

Definition 6.12 (Evidence satisfaction predicate). *Each obligation atom $\mathfrak{o}$ has a declared predicate*

$$\mathrm{Sat}_{\mathfrak{o}}(E_{\mathfrak{o}}) \in \{\mathrm{true}, \mathrm{false}\}$$

which is evaluable only when the evidence object $E_{\mathfrak{o}}$ is present, typed, version-compatible, and attached to the same immutable run scope. If those typing conditions fail, the atom is not assigned a mathematical truth value by this predicate.

Remark 6.13 (Registry completeness is not proof). A registry lists what must be checked. It does not prove the listed atoms. A complete registry with empty or invalid evidence remains incomplete as a proof object.

6.4 Scalarizable and non-scalarizable obligations

Definition 6.14 (Scalarizable obligation). *An obligation is* scalarizable *only when its evidence policy is a declared metric upper bound that may enter the typed ledger and whose scalarization is proved to dominate the actual bottleneck discrepancy consumed by a specific gate.*

Definition 6.15 (Non-scalarizable obligation). *The following are non-scalarizable:*

(i) representative existence, profile compatibility, monitored-scope correctness, and equivalence strength;

(ii) realization compatibility, amplitude, residual edge extraction, and edge-realization certificates;

(iii) actual profile morphisms, source provenance, terminal tameness, defect profiles, and Type IV typing;

(iv) threshold reconciliation, higher-face coherence, continuation artifacts, and κ-admissibility;

(v) local-pass, target-pass, atlas-completeness, and countable-registry coverage obligations;

(vi) claim status, repair records, artifact identity, manifest completeness, consistency, and replayability.

Proposition 6.16 (No scalar manufacture of non-scalar evidence). *If a required non-scalarizable obligation is absent, ill-typed, inconsistent, or attached to another run, then no ledger value, scalarized tolerance, threshold clearance, or positive reserve makes that obligation pass.*

Proof. A scalar metric record bounds only the declared metric discrepancy named by its evidence policy. It does not construct, type, or verify any of the non-scalar objects in Definition 6.15. ☐

Remark 6.17 (Non-compensation begins at the typing boundary). A large reserve may absorb a larger certified metric discrepancy. It may not absorb missing semantics. This distinction is prior to any aggregate verdict rule.

6.5 Canonical atomic statuses

Definition 6.18 (Atomic audit status). *Every obligation atom in the declared registry has exactly one canonical status*

$$\mathsf{AuditStatus} = \{\mathtt{pass}, \mathtt{reject}, \mathtt{undefined}, \mathtt{not\_invoked}\}.$$

Their meanings are:

(i) `pass`*: the atom is required, well-typed, fully evidenced, and its satisfaction predicate is true;*

(ii) `reject`*: the atom is required, well-typed, fully evidenced, and its satisfaction predicate is false or the declared obstruction is nonzero;*

(iii) `undefined`*: the atom is required but incomplete, ill-typed, inconsistent, blocked by an undischarged dependency, or not computable from the declared evidence;*

(iv) `not_invoked`*: the immutable scope positively declares that the atom is outside the claim.*

Remark 6.19 (Missing is not not-invoked). A missing atom is not automatically outside scope. Without a valid scope-exclusion record it is `undefined`.

Definition 6.20 (Justified scope exclusion)**.** *A scope exclusion is* justified *when it identifies the excluded atom, the governing policy, the reason it is irrelevant to the declared claim, and the immutable run fields on which that exclusion depends. An exclusion inferred only after observing an inconvenient result is invalid.*

Definition 6.21 (Effective aggregation normalization)**.** *For aggregation only, define*

$$\operatorname{Norm}(s,j) := \begin{cases} \texttt{pass}, & s = \texttt{not\_invoked} \text{ and } j \text{ is a valid scope exclusion},\\ \texttt{undefined}, & s = \texttt{not\_invoked} \text{ and no valid exclusion exists},\\ s, & s \in \{\texttt{pass}, \texttt{reject}, \texttt{undefined}\}. \end{cases}$$

The original `not_invoked` *status remains stored; normalization does not erase the exclusion record.*

6.6 Specialized status embeddings from Appendices B–E

Definition 6.22 (Representative lifecycle embedding)**.** *The representative lifecycle statuses of Definition 2.100 map as follows when a representative is required:*

$$\texttt{verified} \mapsto \texttt{pass}, \qquad \texttt{failed} \mapsto \texttt{reject},$$

$$\texttt{constructed}, \texttt{missing} \mapsto \texttt{undefined}.$$

The status `not_required` *maps to* `not_invoked` *only under a valid scope exclusion.*

Definition 6.23 (Ext-clause embedding)**.** *The typed Ext-clause statuses of Definition 3.115 map identically to the canonical statuses with the same names.*

Definition 6.24 (Type IV embedding)**.** *The Type IV statuses of Definition 4.63 map by*

$$\texttt{certified\_zero} \mapsto \texttt{pass}, \qquad \texttt{obstructed} \mapsto \texttt{reject}, \qquad \texttt{undefined} \mapsto \texttt{undefined}.$$

Type IV `not_invoked` *remains canonical* `not_invoked` *only under an explicit no-tower scope decision.*

Definition 6.25 (Restart transition embedding)**.** *The Restart transition statuses of Definition 5.52 embed at the level of the* continuation-law subclause*:*

$$\texttt{certified} \mapsto \texttt{pass}, \qquad \texttt{failed} \mapsto \texttt{reject}, \qquad \texttt{undefined} \mapsto \texttt{undefined}.$$

This map does not assign the complete Restart bundle a passing status.

Definition 6.26 (Complete Restart bundle)**.** *A complete Restart step contains the following required atoms:*

(i) a certified continuation-law subclause;

(ii) a defined positive target reserve;

(iii) a passing target-local certificate record;

(iv) every required endpoint, representative, threshold-reconciliation, scalarization, and comparison artifact;

(v) no required undefined non-reserve obligation.

Its canonical status is the aggregate status of this bundle.

Remark 6.27 (Certified transition is necessary but not sufficient). A certified transition proves only

$$r_{k+1} \geq \kappa_k r_k - \eta_k.$$

The complete Restart step may still reject because the target reserve is nonpositive or the target-local record rejects, and it may remain undefined because another required artifact is missing.

Definition 6.28 (Manifest status embedding)**.** *Manifest statuses embed by*

$$\texttt{complete} \mapsto \texttt{pass}, \qquad \texttt{incomplete} \mapsto \texttt{undefined}, \qquad \texttt{inconsistent} \mapsto \texttt{reject}.$$

A complete manifest supports, but does not itself prove, the mathematical atoms it records.

6.7 Claim-use preflight

Definition 6.29 (Registered claim-use status)**.** *Each invoked claim has exactly one registered status*

$$\mathsf{ClaimStatus} = \{\texttt{proved}, \texttt{conditional}, \texttt{operational}, \texttt{spec}, \texttt{deprecated}, \texttt{rejected}, \texttt{unknown}\}.$$

Their meanings are the canonical Main Chapter 8 meanings: theorem-level permission, conditional permission, procedural use only, target/specification, provenance-only legacy status, explicit mathematical rejection, or absence of authoritative registration.

Definition 6.30 (Claim-use atom status)**.** *For one intended use of a claim:*

(i) `proved` *with matching hypotheses maps to* `pass`*;*

(ii) `conditional` *maps to* `pass` *only when every registered condition is discharged, otherwise to* `undefined`*;*

(iii) `operational` *maps to* `pass` *only for procedural use and to* `reject` *if used as theorem evidence;*

(iv) `spec`, `deprecated`, *or* `rejected` *used as theorem evidence maps to* `reject`*;*

(v) `unknown` *maps to* `undefined`*;*

(vi) a registered status attached to the wrong version, theorem location, or repair state maps to `undefined` *or* `reject` *according to whether the mismatch is unresolved or evidenced inconsistent.*

Definition 6.31 (Audit preflight bundle)**.** *The preflight bundle contains:*

(i) complete run identity and immutable scope;

(ii) admissible claim-use atoms;

(iii) every relevant registered predecessor-to-v21 repair or supersession record;

(iv) required artifacts with matching types, versions, and hashes;

(v) absence of inconsistent artifact resolution;

(vi) the manifest status atom.

Its aggregate status is denoted $\mathsf{Preflight}(\mathfrak{R})$*.*

Proposition 6.32 (Preflight failure blocks Core readiness)**.** *If the preflight bundle is not* `pass`*, the complete Foundation readiness bundle cannot be* `pass`*.*

Proof. The preflight atoms are required dependencies of every theorem-level and manifest-level final verdict. Apply the dependency-closure and aggregate rules below. ☐

6.8 Well-founded dependency registries

Definition 6.33 (Immediate dependency relation)**.** *For obligation atoms* $\mathfrak{p}, \mathfrak{o}$*, write*

$$\mathfrak{p} \prec \mathfrak{o}$$

when $\mathfrak{p}$ *is an immediate prerequisite of* $\mathfrak{o}$*. The transitive dependency closure is denoted*

$$\mathrm{Cl}(\mathfrak{o}) = \{\mathfrak{p} : \mathfrak{p} \prec^* \mathfrak{o}\}.$$

Only dependencies active under the immutable run scope enter the active closure.

Definition 6.34 (Well-founded dependency registry)**.** *A registry is* well-founded *when either:*

(i) it is finite and its dependency graph is acyclic; or

(ii) it is countable and carries a rank function

$$\mathrm{rk} : \mathsf{Obl}(\mathfrak{R}) \longrightarrow \mathbb{N}$$

such that $\mathfrak{p} \prec \mathfrak{o}$ *implies* $\mathrm{rk}(\mathfrak{p}) < \mathrm{rk}(\mathfrak{o})$*.*

Self-dependence and cyclic mutual certification are forbidden.

Definition 6.35 (Dependency-complete pass)**.** *A required atom* $\mathfrak{o}$ *has a* dependency-complete pass *when:*

(i) its own evidence is present, typed, run-consistent, and satisfies its evidence predicate;

(ii) every active required atom in $\mathrm{Cl}(\mathfrak{o})$ *has status* `pass`*;*

(iii) every inactive atom in the declared dependency template has a valid scope exclusion;

(iv) all claim, artifact, and repair-record dependencies agree with the immutable run identity.

Proposition 6.36 (Pass is downward closed)**.** *If* $\mathfrak{o}$ *has a dependency-complete pass, then every active required atom in* $\mathrm{Cl}(\mathfrak{o})$ *has status* `pass`*.*

Proof. This is clause (ii) of Definition 6.35. ☐

Definition 6.37 (Blocked theorem-application atom)**.** *Suppose an atom records the application of a theorem or construction whose active prerequisite is not* `pass`*. The application atom is* blocked *and has canonical status* `undefined`*, even when the failed prerequisite has status* `reject`*. The rejected or undefined prerequisite remains separately recorded.*

Remark 6.38 (Why blocked application is not automatically reject)*.* Failure of a theorem hypothesis blocks that theorem application; it does not necessarily prove the intended conclusion false. The surrounding conjunction or bundle still rejects whenever the rejected prerequisite is itself a required atom.

Proposition 6.39 (Dependency blocker propagation)**.** *If an active prerequisite is* `undefined` *or* `reject`*, then a dependent theorem-application atom cannot pass. It is* `undefined` *as an application atom, while the aggregate bundle preserves any rejected prerequisite.*

Proof. The dependency-complete pass criterion fails. Definition 6.37 types the application atom as `undefined`. Aggregate rejection, when applicable, follows from the rejected prerequisite itself. ☐

6.9 The status aggregation algebra

Definition 6.40 (Effective precedence order). *After applying the normalization of Definition 6.21, use the total order*

$$\texttt{pass} < \texttt{undefined} < \texttt{reject}.$$

For effective statuses s, t, define

$$s \otimes t := \max\{s, t\}$$

in this order.

Proposition 6.41 (Status-algebra laws). *The operation $\otimes$ is commutative, associative, and idempotent, with neutral element* `pass`.

Proof. These are the corresponding properties of the maximum operation on a total order. □

Definition 6.42 (Aggregate audit verdict). *Write*

$$A_{\mathrm{req}}(\mathfrak{R}) := \{a \in A \mid \mathfrak{o}_a \in \mathsf{Req}(\mathfrak{R})\}, \qquad A_{\mathrm{exc}}(\mathfrak{R}) := A \setminus A_{\mathrm{req}}(\mathfrak{R}).$$

Let j_a be the scope-exclusion record, when any, for atom $\mathfrak{o}_a$. The aggregate verdict is

$$\mathrm{Agg}(\mathfrak{R}) := \left(\bigotimes_{a \in A_{\mathrm{req}}(\mathfrak{R})} \mathrm{Norm}\big(s(\mathfrak{o}_a), j_a\big) \right) \otimes \left(\bigotimes_{a \in A_{\mathrm{exc}}(\mathfrak{R})} \mathrm{Norm}\big(s(\mathfrak{o}_a), j_a\big) \right),$$

provided the finite or countable registry satisfies the completeness conditions of this appendix. The two factors make the required/scope-excluded partition explicit; they do not permit an excluded atom to bypass the justified-scope condition of Definition 6.20. Equivalently:

(i) `reject` *if at least one atom in* $\mathsf{Req}(\mathfrak{R})$ *rejects;*

(ii) `undefined` *if none rejects but at least one atom in* $\mathsf{Req}(\mathfrak{R})$ *is undefined, or one purported* `not_invoked` *atom outside* $\mathsf{Req}(\mathfrak{R})$ *lacks valid scope justification;*

(iii) `pass` *if all required atoms pass and each remaining atom carries a justified* `not_invoked` *exclusion.*

Theorem 6.43 (Non-compensation). *No number of passing atoms, zero diagnostics, positive reserves, or favorable scalar values cancels a rejected or required undefined atom in the aggregate verdict.*

Proof. The aggregate is the maximum in the effective precedence order. Passing atoms are neutral, while `undefined` and `reject` remain respectively above `pass`. Proposition 6.16 separately prevents numerical evidence from changing the type of a non-scalar atom. □

Proposition 6.44 (Partition invariance of hierarchical aggregation). *Partition a complete registry into disjoint complete subregistries. Aggregate each subregistry and then aggregate the resulting statuses. The headline verdict equals the direct aggregate of all atoms, provided all rejected, undefined, and justified* `not_invoked` *identifiers remain attached to the intermediate records.*

Proof. Associativity and commutativity of $\otimes$ give equality of headline statuses. Retention of atom identifiers preserves the complete diagnostic detail. □

Remark 6.45 (Headline compression does not erase detail). If one atom rejects and another is undefined, the headline is `reject`, but both atomic facts, dependency blockers, and failure routes remain in the record.

6.10 Failure routing and nonexclusive diagnostics

Definition 6.46 (Failure-routing set)**.** *Every non-pass run records all applicable routes*

$$\mathsf{Routes}(\mathfrak{R}) \subseteq \{\mathrm{I}, \mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{V}\}.$$

Their meanings are:

(i) Type I: a certified nonzero repaired topological predicate in an invoked monitored degree;

(ii) Type II: nonzero categorical obstruction in a complete eligible realization regime;

(iii) Type III: failed or undefined metric, crop, representative, realization, overlap, continuation, Restart, terminal-tameness, or other operational condition;

(iv) Type IV: nonzero terminal-generic tower defect relative to a valid comparison artifact;

(v) Type V: claim-status, scope, manifest, repair-record, artifact-identity, version, replayability, or proposer/verifier governance failure.

The routes are nonexclusive and are not a severity ranking.

Definition 6.47 (Route assignment discipline)**.** *Route assignment obeys the following distinctions:*

(i) a nonzero valid Type IV diagnostic is Type IV, not merely missing evidence;

(ii) an absent Type IV artifact makes the Type IV atom `undefined` *and is routed operationally as Type III and, when omitted or falsely serialized, also as Type V;*

(iii) missing eligibility is not Type II; it is Type III and possibly Type V;

(iv) an inconsistent manifest is Type V and canonical `reject`*;*

(v) an incomplete manifest is canonical `undefined`*, unless another evidenced atom already rejects;*

(vi) reporting `undefined` *as zero or pass creates an additional Type V route.*

Proposition 6.48 (Atomic status and failure route are different types)**.** *Two atoms may share a failure route while having different canonical statuses. In particular, a Type III operational clause may be evidenced false and* `reject`*, or incomplete and* `undefined`*.*

Proof. A route identifies the domain of failure. The atomic status identifies whether the clause was evidenced false or could not be validly evaluated. □

6.11 Bundle readiness without a fifth verdict

Definition 6.49 (Bundle readiness)**.** *For a complete declared obligation bundle $\mathcal{B}$, its* certificate-readiness status *is its aggregate canonical status:*

$$\mathsf{Ready}(\mathcal{B}) := \mathrm{Agg}(\mathcal{B}).$$

The words ready, blocked*, and* rejected *may describe the cases* `pass`*,* `undefined`*, and* `reject`*, but they do not form a competing fifth status system.*

Definition 6.50 (Representative readiness bundle)**.** *The representative readiness bundle is the obligation family of Definition 2.140 with the lifecycle embedding of Definition 6.22 and every active dependency required for the intended use.*

Definition 6.51 (Eligible Ext readiness bundle)**.** *When the Ext clause is invoked, the Ext readiness bundle is the obligation family of Definition 3.140, including the repaired topological input, representative, realization, amplitude, residual-edge, edge-realization, field, theorem-use, and manifest atoms.*

Definition 6.52 (Type IV readiness bundle)**.** *When tower semantics are invoked, the Type IV readiness bundle is the obligation family of Definition 4.145, including actual morphism, source provenance, terminal mode, terminal tameness, defect profiles, interior-defect typing, and the canonical Type IV status.*

Definition 6.53 (Overlap and Restart readiness bundle)**.** *When overlap, atlas, or sequential inheritance is invoked, the corresponding readiness bundle contains the non-scalar obligations of Definition 5.21, all metric reserve atoms, all required simplex records, and every complete Restart bundle of Definition 6.26.*

Definition 6.54 (Foundation readiness bundle)**.** *The complete Foundation readiness bundle contains, as applicable:*

(i) the audit identity, immutable scope, claim-use, repair, and manifest-preflight atoms;

(ii) the gate-scoped metric ledger, scalarization, actual upper-bound theorem, gap, and reserve atoms;

(iii) the representative readiness bundle;

(iv) the eligible Ext readiness bundle;

(v) the Type IV readiness bundle;

(vi) all local, overlap, simplex, atlas, continuation, and complete Restart bundles;

(vii) manifest completeness, consistency, replayability, aggregate verdict, and failure-routing atoms.

Theorem 6.55 (Certificate-readiness criterion)**.** *A Foundation bundle has readiness status* `pass` *exactly when:*

(i) its registry is complete and well-founded;

(ii) preflight passes;

(iii) every required atom has a dependency-complete pass;

(iv) every excluded atom has justified `not_invoked` *status;*

(v) the manifest fragment is complete, consistent, and run-compatible;

(vi) the aggregate verdict is `pass`*.*

Proof. The forward direction follows from the meanings of `pass`, dependency-complete pass, and manifest completeness. The reverse direction follows from Definition 6.42 after all listed conditions are satisfied. ☐

Remark 6.56 (Readiness is scope-relative). A passing readiness bundle certifies only the declared Core audit scope. It does not create a stronger claim, a global object, an object-level gluing theorem, or an external arithmetic, geometric, PDE, cryptographic, or computational conclusion.

6.12 Complete Restart and atlas composition rules

Proposition 6.57 (Complete Restart verdict)**.** *For an invoked Restart step:*

(i) the complete bundle is `pass` *exactly when the transition-law subclause passes, the target reserve is defined and positive, the target-local record passes, and all required non-reserve artifacts pass;*

(ii) it is `reject` *if the certified inequality is false, the defined target reserve is nonpositive, or the target-local record rejects;*

(iii) it is `undefined` *if no required item rejects but a required continuation, reserve, target, representative, comparison, or manifest atom is undefined;*

(iv) it is `not_invoked` *only under an explicit scope exclusion of sequential inheritance.*

Proof. Apply the aggregate verdict to the complete Restart bundle of Definition 6.26. □

Proposition 6.58 (Atlas composition rule)**.** *A finite or countable certificate atlas has readiness status* `pass` *only when:*

(i) every required local vertex record passes;

(ii) every required pairwise and higher-simplex record passes;

(iii) the declared Čech-depth, audit-local-finiteness, threshold reconciliation, and simplex-budget atoms pass;

(iv) every invoked Type IV and Restart bundle has passing status;

(v) the countable registry, when any, has a complete coverage certificate;

(vi) the manifest is complete and consistent.

A rejected required atom rejects the atlas. If none rejects but one required atom is undefined, the atlas is undefined.

Proof. Apply the aggregate and dependency-closure rules to the atlas obligation registry of Appendix E. □

Remark 6.59 (Atlas readiness is not global-object readiness)*.* A passing atlas bundle certifies record-level coherence. It does not construct a single global filtered object, persistence module, derived object, or global B-Gate$^+$ record.

6.13 Countable registries and hidden-tail prevention

Definition 6.60 (Audit-readable countable registry)**.** *A countable obligation registry is* audit-readable *when it has:*

(i) a declared countable index set and immutable enumeration or generator;

(ii) a coverage certificate proving that every obligation in scope occurs in the registry;

(iii) a canonical status, evidence record, and scope decision for every index;

(iv) a replayable dependency generator and well-founded rank;

(v) a universal status certificate sufficient to verify that all required atoms pass or to identify all non-pass atoms;

(vi) no hidden, unindexed, or implementation-dependent tail.

Proposition 6.61 (Finite-prefix checks do not certify countable readiness)**.** *A finite prefix of a countable registry cannot support a pass verdict for the full registry unless a separate coverage proof establishes that the omitted tail contains no required atom or only atoms with valid scope exclusions.*

Proof. Without such a proof, an unchecked tail may contain a rejected, undefined, or unjustifiably excluded atom. The full aggregate verdict is therefore not determined by the finite prefix. □

Theorem 6.62 (Countable aggregate soundness)**.** *For an audit-readable countable registry, the aggregate verdict of Definition 6.42 is sound under the universal status certificate. In particular, a pass verdict requires universal dependency-complete pass over all required indices.*

Proof. The coverage certificate identifies the full registry, the rank proves dependency well-foundedness, and the universal status certificate supplies the required universal quantification. The aggregate cases then follow from Definition 6.42. □

6.14 Repair calculus and immutable verdict history

Definition 6.63 (Repair record)**.** *A repair record contains:*

(i) the predecessor RunID or revision identifier;

(ii) every repaired rejected or undefined atom;

(iii) the changed evidence, dependency, scope, theorem, claim-status, or manifest fields;

(iv) the old and new artifact hashes and schema versions;

(v) the reason the repair is admissible;

(vi) the resulting new atomic statuses, failure routes, and aggregate verdict.

Proposition 6.64 (No retroactive verdict mutation)**.** *Resolving an undefined atom or correcting a rejected atom creates a new run or revision record. It does not rewrite the historical verdict of the immutable earlier run.*

Proof. The obligation registry, evidence identities, and repair state belong to the immutable audit scope. Changing them changes the audited object or its revision identity. □

Remark 6.65 (Repairs need not be monotone in headline status). New evidence may resolve `undefined` to `pass` or `reject`; correction of an invalid rejection may also change the result. Immutability concerns historical records, not monotonicity of future revisions.

6.15 Appendix G serialization handoff

Definition 6.66 (Status-record serialization contract). *For every obligation atom, the Appendix G schema must be able to serialize at least:*

(i) ManifestID, RunID, obligation ID, owner, and immutable scope;

(ii) requiredness, scope-exclusion status, and scope-exclusion evidence;

(iii) obligation kind and evidence-policy version;

(iv) immediate dependencies, closure references, and rank data;

(v) specialized source status and canonical atomic status;

(vi) evidence, theorem, ClaimID, artifact, version, and hash references;

(vii) scalar ledger, scalarization, actual upper-bound, gap, and reserve fields when applicable;

(viii) every Type I–V failure route and every blocker identifier;

(ix) predecessor and repair-record identifiers;

(x) aggregate bundle verdict, registry-coverage status, consistency, and replayability fields.

Proposition 6.67 (Schema conformance preserves semantics). *A chapter-local or appendix-local record is admissible for the canonical manifest only if its serialized fields preserve the obligation identity, status meaning, dependency relation, failure routes, and immutable artifact references fixed by this appendix.*

Proof. Changing one of those meanings would serialize a different audit atom or run under the same identifier, contradicting immutable registry semantics. □

Remark 6.68 (Semantics here, schema there). Appendix F fixes status meanings, dependency closure, aggregation, readiness, and failure routing. Appendix G fixes canonical field names, serialization constraints, conformance, and replayability. Schema completeness alone is not mathematical proof, and a mathematically true unrecorded clause does not produce an auditable pass until its required evidence is serialized consistently.

6.16 Illustrative audit patterns

Example 6.69 (Constructed but unverified representative). A representative candidate exists, but its profile isomorphisms are not verified. Its lifecycle status is `constructed`; its canonical status is `undefined`. A small bottleneck budget and a large threshold gap do not change that status.

Example 6.70 (Undefined Type IV is not zero). An actual comparison morphism or terminal-tameness certificate is absent. The Type IV atom is `undefined`, and no numerical μ, ν pair is asserted. Recording

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

would add a Type V manifest failure.

Example 6.71 (Certified transition with rejected target). A continuation artifact verifies the κ-inequality, so the transition-law subclause passes. The target-local record has a nonzero repaired topological profile and rejects. The complete Restart bundle therefore rejects.

Example 6.72 (Rejected prerequisite and blocked theorem application). Suppose a theorem application requires an eligible realization, but the amplitude certificate is well-typed and false. The amplitude atom rejects. The theorem-application atom is blocked and undefined; the full bundle rejects because it still contains the rejected amplitude atom.

Example 6.73 (Pairwise overlap without higher coherence). Every pairwise Overlap Gate passes, but a required triple-overlap simplex record is missing. The pairwise bundle passes, while the atlas bundle is `undefined`. No numerical overlap reserve manufactures the missing higher-face record.

Example 6.74 (Complete-looking but inconsistent manifest). All expected fields are populated, but two artifact hashes are declared for the same immutable identifier. The manifest is inconsistent and maps to `reject`; the run has a Type V route even if every numerical computation can be replayed separately.

6.17 Retained v20 token sorts and v21 supersession safety

Definition 6.75 (Canonical token-sort registry). *Every serialized token consumed by a v20 audit object has exactly one declared semantic sort. The registry includes at least:*

(i) atomic audit status:

$$\texttt{pass},\quad \texttt{reject},\quad \texttt{undefined},\quad \texttt{not\_invoked};$$

(ii) runtime comparison mode:

$$\texttt{exact},\quad \texttt{metric\_gap\_certified},\quad \texttt{not\_compared};$$

(iii) detector profile stage:

$$\begin{gathered}\texttt{windowed\_prethreshold},\quad \texttt{repaired\_postthreshold},\\ \texttt{kernel\_defect},\quad \texttt{cokernel\_defect},\quad \texttt{diagram\_probe},\quad \texttt{descent\_coherence};\end{gathered}$$

(iv) detector certificate status:

$$\texttt{detector\_certified},\quad \texttt{detector\_failed},\quad \texttt{detector\_pending};$$

(v) detector mathematical value:

$$\texttt{zero},\quad \texttt{nonzero},\quad \texttt{undefined};$$

(vi) terminal Type IV status, full/interior inspection status, aggregate transient status, source semantics, simplex-coherence mode, Restart-transition status, Restart zero-route status, manifest status, claim evidence status, and invocation role.

Equal character strings occurring in different sorts are not identified. Every serialized occurrence records its field name and semantic sort.

Proposition 6.76 (No cross-sort token substitution). *A token of one semantic sort cannot discharge a field of another sort. In particular, a detector value, profile stage, comparison mode, source semantics, simplex mode, manifest status, or Restart status is not an atomic audit verdict. An unresolved cross-sort assignment makes the affected clause* `undefined`*; an evidenced inconsistent or concealed substitution rejects as a Type V governance failure.*

Proof. The fields have different source types and governing predicates. A value of one type supplies neither an inhabitant nor evidence for another. An inconsistent declaration additionally contradicts immutable manifest semantics. $\square$

Definition 6.77 (v20 typed evidence record)**.** *A typed evidence record contains:*

$$\begin{aligned}&(\text{EvidenceID}, \text{ClauseID}, \text{evidence type}, \text{source}, \text{target},\\ &\ \text{theorem/policy}, \text{hypotheses}, \text{token sorts}, \text{transformations},\\ &\ \text{artifacts}, \text{hashes}, \text{versions}, \text{atomic status}).\end{aligned}$$

A bare numerical value, detector value without certificate status, bare isomorphism symbol, comparison mode, or citation without claim-use preflight is not a complete evidence record.

6.18 Retained v20 specialized embeddings

Definition 6.78 (Two-axis detector status embedding)**.** *The Appendix B detector certificate status and mathematical value remain independent axes. When a detector obligation is required, its certificate axis embeds by*

$$\begin{aligned}\texttt{detector\_certified} &\mapsto \texttt{pass},\\ \texttt{detector\_failed} &\mapsto \texttt{reject},\\ \texttt{detector\_pending} &\mapsto \texttt{undefined}.\end{aligned}$$

The mathematical value `zero`*,* `nonzero`*, or* `undefined` *does not itself map to an atomic verdict. For a required zero clause:*

(i) certified detector with value `zero` *passes the detector-based zero atom;*

(ii) certified detector with value `nonzero` *rejects that zero atom and supplies a Type I witness at a topological stage or the applicable defect witness at a defect stage;*

(iii) failed or pending certification cannot be repaired by a favorable value and yields respectively `reject` *or* `undefined`*;*

(iv) detector semantics outside immutable scope map to `not_invoked` *only through a justified exclusion.*

Remark 6.79 (Detector value is consumable only after certification)*.* A sampled value of zero with incomplete applicability, endpoint, family, coverage, stage, or completeness evidence is not a zero proof. A nonzero value from an inapplicable or ill-typed detector is likewise not a typed obstruction witness.

Definition 6.80 (Transient-defect status embedding)**.** *For a required Appendix D aggregate long-transient clause:*

$$\begin{aligned}\texttt{certified\_absent} &\mapsto \texttt{pass},\\ \texttt{certified\_present} &\mapsto \texttt{reject},\\ \texttt{undefined} &\mapsto \texttt{undefined}.\end{aligned}$$

The status `not_invoked` *remains canonical* `not_invoked` *only under a justified scope decision. The detector component* `full_profile` *or* `interior_noncofinal`*, the defect threshold, window, collapse threshold, and monitored degrees are part of the clause identity.*

Definition 6.81 (v20 Overlap Gate status embedding)**.** *The Appendix E pairwise Overlap Gate status embeds identically:*

$$\texttt{pass} \mapsto \texttt{pass}, \qquad \texttt{reject} \mapsto \texttt{reject},$$
$$\texttt{undefined} \mapsto \texttt{undefined}, \quad \texttt{not\_invoked} \mapsto \texttt{not\_invoked}.$$

The runtime comparison mode is stored separately and never substitutes for this atomic status.

Definition 6.82 (Restart zero-route embedding)**.** *When a Restart step consumes a repaired-zero predicate, the target-zero route is a required independent atom with exactly one specialized status:*

$$\texttt{not\_invoked}, \quad \texttt{undefined}, \quad \texttt{certified}, \quad \texttt{failed}.$$

Its canonical embedding is

$$\texttt{certified} \mapsto \texttt{pass}, \quad \texttt{failed} \mapsto \texttt{reject}, \quad \texttt{undefined} \mapsto \texttt{undefined}.$$

The status `not_invoked` *is permitted only when the Restart claim does not consume target zero. A certified κ_k-inequality does not certify this atom.*

6.19 Coverage atoms and the layered obligation architecture

Definition 6.83 (Coverage-obligation taxonomy)**.** *The following are distinct non-scalar obligations:*

(i) registry coverage*: every obligation in a finite or countable audit scope occurs in the registry;*

(ii) retained-bar coverage*: every globally retained bar has a detector witness in the declared local atlas;*

(iii) boundary-bridge coverage*: every globally retained interval crossing a partition boundary has a witness in a declared bridge or another certified coverage window;*

(iv) artifact closure*: every referenced artifact and transitive interpretation dependency is immutable and resolvable;*

(v) release-document coverage*: every mandatory document role and release clause occurs in the advertised release registry.*

A certificate for one coverage type does not discharge another.

Definition 6.84 (Five-layer obligation architecture)**.** *Every v21 readiness graph is organized into the following typed layers:*

(L1) mathematical layer*: repaired profiles, Ext clauses, comparison morphisms, defect profiles, exact equalities, reduction and descent theorems;*

(L2) detector layer*: detector stage, applicability, completeness, witness family, endpoint convention, value, and certification;*

(L3) atlas/transition layer*: overlap, simplex, retained-bar coverage, continuation, Restart, boundary bridge, and local-to-global interfaces;*

(L4) readiness layer*: dependency closure, atomic statuses, non-compensatory aggregation, failure routing, and bundle verdicts;*

(L5) manifest/release layer*: claim use, token sorts, artifact closure, schema conformance, replayability, repair history, and release acceptance.*

Edges may skip layers only when a registered theorem or policy explicitly authorizes the dependency. No higher layer manufactures missing lower-layer evidence.

Proposition 6.85 (Layered non-compensation)**.** *A passing detector, atlas, readiness, manifest, or release-layer atom cannot repair an absent, undefined, or rejected prerequisite in a lower layer. Likewise, a true mathematical clause does not produce an auditable pass until its required upper-layer identity, dependency, and manifest obligations are satisfied.*

Proof. The first assertion follows from dependency-complete pass and Theorem 6.43. The second follows because audit pass is a conjunction of mathematical and governance obligations rather than mathematical truth alone. □

6.20 v21 readiness bundles

Definition 6.86 (Detector readiness bundle)**.** *A detector readiness bundle contains:*

(i) exact source profile identity and authorized detector stage;

(ii) detector mode, information class, endpoint convention, witness set, family, and applicability record;

(iii) soundness and, when a zero conclusion is consumed, completeness;

(iv) detector certificate status and independent mathematical value;

(v) every metric-safe transport dependency when the detector conclusion crosses a comparison or hard deletion;

(vi) artifact identities, theorem locators, claim-use preflight, and manifest fields.

Its readiness status is the aggregate status of these atoms.

Definition 6.87 (Detector-atlas readiness bundle)**.** *A detector-atlas readiness bundle contains:*

(i) every local detector readiness bundle;

(ii) exact local restrictions from a declared global profile;

(iii) the exact, metric-gap-certified, or theorem-controlled-nonmetric local interface branch;

(iv) pairwise and higher-simplex records in their declared strict-cocycle, homotopy-coherent-cell, metric-reference, or theorem-controlled-nonmetric modes;

(v) retained-bar coverage and, when needed, boundary-bridge coverage;

(vi) terminal and long-transient participation atoms;

(vii) complete registry coverage, artifact closure, and manifest fields.

Pairwise local compatibility without the global profile and coverage atoms does not make this bundle pass.

Definition 6.88 (Global repaired-zero readiness bundle)**.** *A global repaired-zero claim has a passing readiness bundle exactly when:*

(i) the global pre-threshold profile and covered range are identified;

(ii) every local restriction and observed local profile is typed;

(iii) every local zero detector is certified and complete;

(iv) every exact, metric-gap-certified, or theorem-controlled-nonmetric zero route passes;

(v) retained-bar coverage passes;

(vi) every consumed overlap/simplex clause passes;

(vii) no required terminal, transient, artifact, claim-use, or manifest atom rejects or remains undefined;

(viii) the detector-atlas local-to-global theorem is admissibly invoked.

A family of local zero values without these dependencies is `undefined`*, not a global zero proof.*

Definition 6.89 (v20 complete Restart bundle)**.** *When a Restart step consumes only reserve inheritance, the inherited complete Restart bundle applies. When it additionally consumes target repaired zero, the v20 complete bundle also requires:*

(i) a passing target-zero route atom;

(ii) a passing target detector or zero-transport subbundle;

(iii) boundary-bridge coverage when the claimed predicate is global rather than target-local;

(iv) stage, threshold, endpoint, source, target, and artifact consistency.

The reserve and zero-route subbundles are independently mandatory.

Definition 6.90 (Finite-to-infinite reduction readiness bundle)**.** *A finite-to-infinite reduction claim contains:*

(i) the actual directed system and transition morphisms;

(ii) the finite witness or stabilization index;

(iii) the genuine colimit or other theorem-authorized source semantics;

(iv) a compatible apex cocone and induced comparison morphism;

(v) apex agreement and every reduction-theorem hypothesis;

(vi) the resulting terminal, transient, and full-isomorphism consequences actually claimed;

(vii) claim-use, artifact-closure, replay, and manifest atoms.

An observed finite plateau, repeated finite barcode, or implementation apex is not a reduction certificate.

Definition 6.91 (Completed object-descent bundle)**.** *A completed object-level descent claim contains:*

(i) the declared category, enhancement, or higher category and its restriction functors;

(ii) the complete canonical nine-field DescentInput *and one active descent lane;*

(iii) *actual local objects, transition equivalences, exact cocycles or higher coherence, and object-level cover/restriction compatibility;*

(iv) *an effective-descent theorem whose exact lane-specific hypotheses are discharged;*

(v) *the constructed descended object or morphism and local comparison artifacts at the declared equivalence strength;*

(vi) *joint conservativity, exact restriction, local zero-reflection, and cone compatibility whenever a global zero or isomorphism conclusion is consumed;*

(vii) *independent certificate-atlas, manifest, artifact-closure, and replay clauses.*

Metric-reference coherence, record-level atlas readiness, `descent_input_ready`*, effectivity alone, or global-record-zero readiness does not substitute for this completed bundle.*

Definition 6.92 (Descent-stage atomic decomposition)**.** *A descent route is normalized into independent atomic clauses for:*

(i) *canonical nine-field input completeness and mutual consistency;*

(ii) *active-lane selection and source-type match;*

(iii) *effectivity-theorem hypothesis discharge;*

(iv) *descended-object or descended-morphism existence;*

(v) *local-to-global comparison and uniqueness/equivalence strength;*

(vi) *exact restriction and joint conservativity when global detection is requested;*

(vii) *global zero or global isomorphism at the exact requested strength.*

Each clause receives its own atomic status. A downstream clause may consume a passing predecessor, but no favorable clause is copied backward or sideways.

Proposition 6.93 (Descent-stage noncompensation)**.** *A passing input-readiness atom does not pass effectivity; passing effectivity does not pass descended-object artifact closure or joint conservativity; and passing local probes does not pass a global zero or isomorphism atom before a global object or morphism and the required conservativity theorem exist.*

Proof. The clauses have different typed predicates and evidence owners. The non-compensating verifier evaluates each predicate from its own necessary sub-DAG, so evidence for one clause cannot manufacture the missing source type or theorem of another. □

Definition 6.94 (v21 Foundation readiness bundle)**.** *The v21 Foundation readiness bundle is the inherited Foundation bundle together with every applicable:*

(i) *token-sort and typed-evidence atom;*

(ii) *detector and detector-atlas readiness bundle;*

(iii) *transient-defect bundle;*

(iv) *global repaired-zero bundle;*

(v) *v20 complete Restart bundle;*

(vi) *finite-to-infinite reduction bundle;*

(vii) *object-descent bundle;*

(viii) *artifact-closure, semantic-replay, schema-migration, and release-acceptance atom.*

Theorem 6.95 (v21 compatibility readiness criterion)**.** *A v21 Foundation bundle has status `pass` if and only if:*

(i) *its immutable registry and token-sort assignments are complete and well-founded;*

(ii) *claim-use preflight passes;*

(iii) *every required specialized status is validly embedded;*

(iv) *every required atom has a dependency-complete pass;*

(v) *every excluded atom has justified `not_invoked` status;*

(vi) *every applicable detector, atlas, transient, Restart, reduction, and descent bundle passes;*

(vii) *artifact closure and semantic replay pass;*

(viii) *the canonical manifest is complete, consistent, and schema-conformant;*

(ix) *the aggregate verdict is `pass`.*

Proof. The forward direction follows from the meaning of a v20 pass and dependency-complete evidence. For the reverse direction, all required atoms normalize to `pass`, every excluded atom has a valid exclusion, and the non-compensatory aggregate is therefore `pass`. ☐

6.21 Retained artifact closure, migration, and replay semantics

Definition 6.96 (Artifact closure)**.** *A manifest has artifact closure when:*

(i) *every referenced artifact has one immutable identifier, type, provenance, version, and content hash;*

(ii) *every transitive dependency needed to interpret a consumed artifact is referenced or belongs to the registered theory release;*

(iii) *no identifier is dangling, ambiguous, mutable under the same hash claim, or resolved differently by two fragments;*

(iv) *no undeclared artifact replacement occurs after run identity is fixed.*

Artifact closure supports replayability but is not mathematical proof.

Definition 6.97 (Semantic schema-migration record)**.** *A migration records source and target schema identifiers, a total map on every consumed field, token-sort preservation, status and scope preservation, explicit defaults only for newly out-of-scope fields, semantic-difference evidence, artifact hashes, and proof that verdict reconstruction at the old supported strength is preserved. A change in theorem strength, evidence, scope, or claim use creates a new run rather than a schema-only migration.*

Proposition 6.98 (No lossy migration of theorem evidence)**.** *A migration that drops a required hypothesis, dependency, token sort, lower-level status, missing-artifact marker, scope boundary, or forbidden stronger reading cannot preserve audit equivalence. The migrated record is incomplete or inconsistent until repaired by a new evidence-backed run.*

Proof. The omitted field participates in clause meaning or verdict reconstruction. Deleting it changes or prevents reconstruction of the consumed claim. □

Definition 6.99 (Computational and semantic replayability). *A run is computationally replayable when artifact closure permits an independent verifier to reconstruct every deterministic profile, detector, ledger, diagnostic, dependency, and serialization computation. It is semantically replayable when the verifier can additionally reconstruct each clause's source type, governing claim, hypotheses, token sort, specialization map, dependency transition, and headline aggregation. Every consumed machine-produced verdict requires semantic replayability for a Core pass.*

Proposition 6.100 (Replay invariance under fixed inputs). *Two semantic replays using the same immutable artifacts, token registry, claim-register snapshot, schema semantics, and verifier version yield the same atomic and headline verdicts. A disagreement evidences an implementation, schema, artifact, or manifest inconsistency and blocks pass.*

Proof. Under fixed inputs, every evidence predicate, dependency transition, specialized embedding, normalization, and aggregation operation is deterministic. □

6.22 v21 extended failure routing and release acceptance

Definition 6.101 (v20 extended failure routing). *The inherited Type I–V routes remain nonexclusive, with the following v20 readings:*

(i) Type I includes a certified complete topological detector with nonzero value for a required repaired-zero clause;

(ii) Type II remains a nonzero eligible categorical obstruction and does not include missing eligibility or a failed detector;

(iii) Type III includes failed or undefined detector stage, applicability, soundness, completeness, canonical four-mode comparison, simplex coherence, retained-bar coverage, transient clause, target-zero route, finite-to-infinite reduction, object descent, or Return theorem;

(iv) Type IV remains a valid semantically authorized tower comparison with nonzero terminal-generic diagnostic;

(v) Type V includes semantic or governance inconsistency: cross-sort substitution, silent stage relabeling, unsupported zero/pass, lossy migration, artifact conflict, silent supersession, unjustified scope exclusion, or proposer output used as verifier evidence.

Mere missing mathematical evidence first yields an undefined Type III clause. Type V is added when the governance record is incomplete, inconsistent, concealed, or semantically false.

Definition 6.102 (Release-acceptance bundle). *A document set advertised as a complete v21 release has a release-acceptance bundle containing:*

(i) mandatory document-role and version coverage;

(ii) canonical Claim Register and supersession closure;

(iii) all mandatory Foundation, interface, manifest, and reproducibility artifacts;

(iv) token-sort, schema, artifact-closure, and semantic-replay clauses;

(v) absence of unresolved mandatory rejected or undefined atoms;

(vi) *the v21 Foundation readiness verdict.*

The release passes exactly when every mandatory clause passes. A mathematically strong chapter does not compensate for an incomplete release registry or inconsistent manifest.

Definition 6.103 (Proposer–verifier re-entry protocol)**.** *A human, Search-side, Platform-side, Companion-side, algorithmic, or AI proposal may influence a Core verdict only after:*

(i) *normalization into a declared Core type;*

(ii) *claim-status and invocation-role classification;*

(iii) *production of required mathematical and artifact evidence;*

(iv) *comparison-mode, ledger, detector-stage, and threshold-gap typing;*

(v) *semantic-difference or transformation-preservation evidence when the proposal was translated, summarized, formalized, or migrated;*

(vi) *manifest registration, token-sort validation, dependency closure, artifact closure, and immutable identity;*

(vii) *execution of every applicable verifier clause.*

Proposal confidence, ranking, or apparent plausibility has no direct status effect.

Remark 6.104 (Retained v20 governance-kernel compatibility record)*.* Within the retained v20 compatibility layer:

(i) the complete inherited v19 obligation, status, dependency, aggregation, readiness, countable-registry, repair, and serialization package remains valid at its original strength;

(ii) token sorts prevent cross-type status substitution;

(iii) detector certificate status and detector mathematical value remain independent axes;

(iv) terminal Type IV, long-transient, Overlap, Restart transition, and Restart zero-route statuses embed without semantic collapse;

(v) registry, retained-bar, boundary-bridge, artifact, and release coverage are distinct obligations;

(vi) the five-layer architecture prevents upper-layer evidence from manufacturing lower-layer proof;

(vii) detector, detector-atlas, global-zero, complete Restart, finite-to-infinite, and object-descent readiness are distinct bundles;

(viii) a v20 pass is a dependency-closed, non-compensatory conjunction of all applicable bundles;

(ix) artifact closure, lossless schema migration, and semantic replay are mandatory for consumed machine-produced verdicts;

(x) v21 extended failure routing preserves Type I–V detail;

(xi) release acceptance is an independent conjunction and is not inferred from mathematical strength alone;

(xii) proposer output affects a verdict only through the typed re-entry protocol.

6.23 v21 specialized embeddings and comparison compatibility

Definition 6.105 (Four-mode comparison claim profile). *For each comparison claim c, declare*

$$\mathrm{Allowed}(c) \subseteq \left\{ \begin{array}{c} \texttt{exact},\ \texttt{metric\_gap\_certified}, \\ \texttt{theorem\_controlled\_nonmetric},\ \texttt{not\_compared} \end{array} \right\}.$$

and for every allowed mode m a downward-closed output-strength set $\mathrm{Out}(c, m)$*. The target relation is* $\mathrm{Strength}_{\mathrm{req}}(c)$*.*

Definition 6.106 (Expanded record of Main Definition 8.21: comparison-mode compatibility). *The comparison clause passes exactly when the runtime mode lies in* $\mathrm{Allowed}(c)$*; source, target, stage, threshold, variance, and artifacts match; all mode-specific hypotheses pass; and some element of* $\mathrm{Out}(c, m)$ *entails* $\mathrm{Strength}_{\mathrm{req}}(c)$*.*

Theorem 6.107 (Expanded proof of Main Theorem 8.22: no comparison-mode substitution). *No canonical comparison mode can be consumed as another without a proved conversion theorem. Exactness supplies no automatic positive-radius robustness; metric closeness supplies no actual morphism or cocycle; a nonmetric theorem supplies only its printed conclusion; and* `not_compared` *supplies no comparison inference.*

Proof. The four modes have different source types and output-strength sets. A target requiring one mode's relation is discharged only when the runtime record inhabits that relation or a passing conversion theorem produces it. □

Definition 6.108 (Inverse-audit embedding). *The inverse-audit triple from Appendix D is embedded componentwise:*

$$\mathsf{Sys} \in \{\texttt{pass}, \texttt{reject}, \texttt{undefined}, \texttt{not\_invoked}\},$$

$$\mathbf{R}^1\varprojlim \in \{\texttt{cleared}, \texttt{obstructed}, \texttt{undefined}, \texttt{not\_invoked}\},$$

$$\mathsf{Apex} \in \{\texttt{agreement}, \texttt{disagreement}, \texttt{undefined}, \texttt{not\_invoked}\}.$$

The last two map to canonical atomic statuses in the evident way, but remain independent clauses. Clearing $\mathbf{R}^1\varprojlim$ *does not pass apex agreement, and conversely.*

Definition 6.109 (Descent-input readiness embedding). *Appendix E's* `descent_input_ready` *is a readiness value for the canonical input*

$$(\{E_\alpha\}, \{g_{\alpha\beta}\}, \{c_{\alpha\beta\gamma}\}, \{h_\sigma\}, \mathcal{W}, \mathcal{C}, \mathfrak{M}_{\mathrm{coh}}, \mathcal{T}_{\mathrm{eff}}, \mathcal{A}),$$

not an object-descent theorem. It maps to atomic `pass` *only for the clause "the Chapter 7 input contract is complete and mutually consistent." The active-lane match, effectivity, descended-object or morphism existence, local comparison maps, uniqueness/equivalence strength, joint conservativity, global zero, and global isomorphism are distinct Chapter 7 nodes governed by Definition 6.92.*

6.24 Typed proof-DAG nodes, edges, and assumption contexts

Definition 6.110 (Expanded record of Main Definition 8.28: proof-DAG). *A proof-DAG is*

$$\mathsf{ProofDAG} = (N, E, s, t, \mathrm{NodeType}, \mathrm{EdgeType})$$

with immutable node and edge registries, a target node, requested strength, and dependency policy. Visual reachability records indexing, not proof.

Definition 6.111 (Canonical proof-bearing node types)**.** *The canonical proof-bearing node types are theorem, interface, artifact, metric, coherence, descent, reduction, and Return. Source-binding, formalization, interpretation, replay, audit, and governance nodes may be added, but their authority is fixed by their exact node type and verifier policy.*

Definition 6.112 (Expanded node semantic record)**.** *Every node n has a record*

$$\mathfrak{n} = \begin{pmatrix} \text{NodeID, type, statement, owner, scope, role,} \\ \text{input\_type, output\_type, required\_strength,} \\ \text{provided\_strength\_set, assumption\_context,} \\ \text{dependencies, verifier\_policy, artifacts, status} \end{pmatrix}.$$

The provided-strength set is downward closed. The statement is the exact typed proposition, not merely a title.

Definition 6.113 (Assumption-context propagation)**.** *Every node carries a finite or admissibly countable set* $\mathrm{Cond}(n)$ *of explicit undischarged assumptions. A derived node takes the union of the contexts of all consumed predecessors, removing an assumption only through a separately passing discharge theorem. Its semantic output is the sequent*

$$\mathrm{Cond}(n) \vdash \mathrm{statement}(n).$$

Proposition 6.114 (Expanded proof of Main Proposition 8.33: no silent assumption erasure)**.** *A conditional passing node cannot discharge an unconditional target by omitting, renaming, or hiding an undischarged assumption.*

Proof. Deleting a premise strengthens the sequent. No weakening rule licenses this direction; a separate theorem discharging the premise is required. ☐

Definition 6.115 (Necessary target sub-DAG)**.** *For target z,* $\mathsf{TargetDAG}(z)$ *is the smallest registered subgraph containing z, every dependency actually consumed by z, every recursively consumed dependency, and all necessary edges among them. Unused advisory or alternative branches are excluded.*

Definition 6.116 (Node discharge certificate)**.** *A node discharge certificate records exact statement, input and output types, required and provided strengths, verifier policy, necessary incoming edges, evidence, artifacts, propagated assumptions, and derived atomic status. A conditional pass remains conditional.*

Proposition 6.117 (Expanded proof of Main Proposition 8.37: node existence is not discharge)**.** *A theorem name, graph node, formal file, proof sketch, or replay record does not discharge its proposition without a valid node discharge certificate.*

Proof. Indexing proves only that a record exists. Discharge is the separate verifier predicate defined by the node's semantic record. ☐

Definition 6.118 (Canonical edge semantics)**.** *Every proof-DAG edge has one primary type: logical, artifact, comparison, bridge, interpretation, formalization, or weakening. Auxiliary source-binding, validation, replay, provenance, and advisory edges carry only their registered authority.*

Definition 6.119 (Expanded edge semantic record)**.** *Every edge e records*

$$\mathfrak{e} = \begin{pmatrix} \text{EdgeID}, s(e), t(e), \text{type, relation, direction,} \\ \text{input\_strength, output\_strength\_set,} \\ \text{assumption\_transport, discharge\_policy,} \\ \text{artifacts, status} \end{pmatrix}.$$

The relation field contains the exact implication, map, equality, comparison, interpretation, or weakening. Assumptions may be preserved, added, or discharged only as explicitly recorded.

Definition 6.120 (Edge compatibility)**.** *An edge is compatible exactly when source output type, target input type, direction, comparison mode, strength, assumption transport, and discharge policy match. A weaker or differently typed edge is misrouted.*

Theorem 6.121 (Expanded proof of Main Theorem 8.41: misrouted edges)**.** *A necessary misrouted edge leaves the target dependency* `undefined` *until a proved conversion retypes it or a compatible passing edge replaces it.*

Proof. The source output does not inhabit the target's required semantic input. The dependency predicate therefore cannot be evaluated at the requested type. □

Definition 6.122 (Weakening edge)**.** *For a declared strength preorder* $(\Sigma, \preceq)$*, an edge from strength* σ *to* σ' *is a weakening edge only when a theorem proves* $\sigma' \preceq \sigma$ *and transports the exact source statement to the exact target statement.*

Proposition 6.123 (No reversal of weakening)**.** *A weakening edge cannot be traversed backward without an independent promotion theorem.*

Proof. A preorder licenses strong-to-weak consumption only; its converse is not part of the data. □

Definition 6.124 (Edge composition policy)**.** *A path may be composed only when adjacent edges are compatible, every intermediate node supplies the exact next input, assumption transport composes, and the result lies in the intersection of all component output-strength sets.*

Proposition 6.125 (Expanded proof of Main Proposition 8.45: reachability is not composition)**.** *A directed path does not by itself prove the path's endpoint implication.*

Proof. A path can contain a reversed weakening, an incompatible type, an undefined node, or an assumption-erasing edge. Only the composition policy excludes all such failures. □

6.25 Finite, countable, and controlled recursive dependency

Definition 6.126 (Ordinary acyclicity and well-founded rank)**.** *Finite ordinary proof dependency is acyclic. A countable necessary sub-DAG additionally carries a rank into a well-founded order such that every necessary edge points from lower to higher rank, or a certified finite-to-infinite reduction replacing explicit tail enumeration.*

Definition 6.127 (Controlled recursive theorem node)**.** *Induction, recursion, coinduction, or fixed-point reasoning is represented by a separately proved theorem node whose internal cyclic semantics is hidden behind an acyclic external interface specifying hypotheses and conclusion.*

Proposition 6.128 (Unlicensed cycle blocks certification)**.** *A directed cycle in ordinary necessary logical dependency blocks target certification unless the cycle is internal to a controlled recursive theorem node.*

Proof. Without a well-founded external rule, each clause relies on itself through the cycle and no primitive evidence base is reached. □

Definition 6.129 (Countable dependency certificate)**.** *A countable target sub-DAG is admissible only with either a checkable well-founded rank plus a uniform verifier schema for all indices, or a passing finite-to-infinite theorem that reduces the infinite closure to a finite verified package. A finite prefix alone is insufficient.*

Theorem 6.130 (Expanded proof of Main Theorem 8.51: determinacy)**.** *Fix immutable registries and deterministic verifier policies. A complete well-founded typed target sub-DAG with determined primitive statuses has a unique recursively computed node-status assignment and headline verdict.*

Proof. Evaluate nodes in well-founded rank order. At each rank all necessary predecessors are already determined, so the node's total verifier policy gives a unique result. Transfinite or countable limit stages are handled by the registered uniform schema or finite-to-infinite certificate. Atomic statuses, assumption contexts, strengths, and failure routes are retained, so aggregation is deterministic without loss of detail. □

6.26 Failure precedence, source binding, and benchmark quarantine

Definition 6.131 (Diagnostic failure precedence)**.** *Repair precedence follows dependency causality: primitive identity, source, independence, typing, and artifact defects precede downstream mathematical obstruction readings. This is a routing order, not the aggregate status order.*

Definition 6.132 (Failure-routing record)**.** *A failure-routing record lists every nonpassing node and edge, atomic status, causal predecessor, Type I–V routes, blocked descendants, and minimal repair frontier. It never suppresses later independent failures merely because an earlier blocker is selected for repair.*

Proposition 6.133 (Typing precedes obstruction)**.** *A raw zero or nonzero value is not interpreted as a mathematical verdict until source type, stage, applicability, authority, and artifact identity pass.*

Proof. Before typing, the value has no registered predicate whose truth or falsity it could witness. □

Definition 6.134 (Source-statement binding)**.** *A source binding is*

$$\mathsf{SourceBind} = \begin{pmatrix} \text{BindingID}, \text{SourceID}, \text{SourceLocator}, \\ \text{ExactStatement}, \text{Hypotheses}, \text{Conclusion}, \\ \text{Version}, \text{Hash}, \text{Authority}, \text{PermittedUse} \end{pmatrix}.$$

It separates storage of a source from proof of a target theorem.

Definition 6.135 (Precursor package and benchmark conclusion)**.** *A recovery precursor package contains only independently authorized source-side hypotheses, constructions, and tools. The benchmark conclusion is the frozen target statement used for terminal comparison. Its proof, truth value, constants, conclusion-derived witnesses, and equivalent oracles are not precursor data.*

Definition 6.136 (Benchmark-quarantine cut)**.** *A quarantine cut partitions a recovery DAG into an upstream construction side and a terminal benchmark side. Only the frozen statement identity, target signature, publicly authorized neutral metadata, and terminal validation comparison may cross in the permitted direction.*

Theorem 6.137 (Expanded proof of Main Theorem 8.61: benchmark leakage)**.** *If prohibited conclusion-side information influences a necessary upstream construction, detector, realization, reduction, or Return witness, the route cannot certify independent recovery.*

Proof. Independent recovery requires that the upstream witness be a function only of the authorized precursor package and declared neutral metadata. Leakage adds target-derived information to that function. The resulting object may still be mathematically interesting, but it no longer witnesses the registered independence predicate, so the affected nodes become `undefined` or `reject` according to whether contamination is unresolved or proved. ☐

Proposition 6.138 (Source storage is not theorem recovery)**.** *Possessing a source statement, source PDF, or exact quotation does not establish an independent recovery proof.*

Proof. Storage supplies identity and provenance. Recovery additionally requires an independent precursor-to-conclusion derivation satisfying the quarantine cut. ☐

6.27 Artifact identity, canonical manifest, and schema migration

Definition 6.139 (Immutable artifact identity)**.** *An artifact identity is*

$$\mathfrak{a} = \begin{pmatrix} \text{ArtifactID, type, format, version,} \\ \text{hash\_algorithm, Hash, size, provenance,} \\ \text{semantic\_locator} \end{pmatrix}.$$

It identifies bytes or a declared canonical representation and names the parser or interpretation theorem giving mathematical meaning.

Proposition 6.140 (Hash identity supports identity, not truth)**.** *Hash agreement proves content identity under the declared canonicalization policy, not mathematical truth, correct interpretation, completeness, or noncircularity.*

Proof. The hash is a function of content. The listed semantic properties are separate predicates requiring theorem or verifier evidence. ☐

Definition 6.141 (Cross-format semantic equivalence)**.** *Two byte-distinct artifacts are theorem-equivalent only under a record containing their parsers or interpretation maps, normalized semantic objects, semantic diff, and a preservation theorem at the exact consumed strength.*

Definition 6.142 (Expanded canonical v21 verifier manifest)**.** *The manifest contains at least run and revision identity, theory and schema versions, immutable target scope and requested strength, complete node and edge registries, statement axes and preflight, every comparison mode and metric ledger, every non-scalar mathematical/interface obligation, source bindings and quarantine, artifact closure and cross-format records, agent and replay records, revision lineage and dirty impact closure, every atomic status, headline verdict, failure route, and repair instruction needed for independent reconstruction.*

Definition 6.143 (Manifest status)**.** *The manifest status is exactly one of*

$$\mathtt{complete}, \qquad \mathtt{incomplete}, \qquad \mathtt{inconsistent}.$$

Its specialization is

$$\mathtt{complete} \mapsto \mathtt{pass}, \quad \mathtt{incomplete} \mapsto \mathtt{undefined}, \quad \mathtt{inconsistent} \mapsto \mathtt{reject}.$$

No manifest status determines mathematical truth by itself.

Definition 6.144 (Semantic-preserving schema migration)**.** *A migration gives a total map on every consumed field and preserves node and edge types, statuses, scope, strengths, missing-artifact markers, source quarantine, failure routes, and verdict reconstruction. A change in theorem strength, evidence, source semantics, or scope creates a new run rather than a schema-only migration.*

Proposition 6.145 (No lossy migration)**.** *Dropping a required hypothesis, edge type, source locator, quarantine boundary, artifact identity, lower-level status, or forbidden stronger reading destroys verifier equivalence.*

Proof. The omitted field changes the target proposition or prevents reconstruction of the earlier verdict. The result is therefore incomplete or inconsistent. ☐

6.28 Agent authority, formalization, and contamination

Definition 6.146 (Agent roles and authority matrix)**.** *The canonical roles are proposer, verifier, auditor, and proof assistant. Each action is recorded under one role. A proposer establishes candidate metadata; a verifier establishes only predicates in its registered profile; an auditor establishes reconstructibility and governance predicates; a proof assistant establishes the exact formal proposition relative to its environment and axioms.*

Definition 6.147 (Proposer-to-verifier conversion certificate)**.** *Conversion requires normalization into a declared type, exact source and target statements, explicit hidden assumptions, statement axes and requested strength, immutable input/output artifacts with semantic diffs, source and quarantine checks, and the complete remaining verifier predicate list.*

Proposition 6.148 (Agent assertion does not establish a Core node)**.** *A human or AI assertion that a theorem, morphism, detector, descent datum, DAG closure, or Return holds does not discharge that node without conversion and verifier evidence.*

Proof. The assertion is a candidate statement about the predicate, not evidence satisfying the predicate's verifier policy. ☐

Definition 6.149 (Formalization and interpretation pair)**.** *A proof-assistant route contains a formalization edge from the intended statement to the exact formal proposition and, when necessary, an interpretation edge from the verified formal theorem back to the intended target. Both edges must pass at the requested strength.*

Proposition 6.150 (Formal verification is not silent interpretation)**.** *A verified formal node does not discharge the intended informal claim when the formalization or interpretation edge is undefined, rejected, or too weak.*

Proof. The proof assistant checks only the formal proposition in its environment. The semantic relation to the intended statement is an independent dependency. ☐

Definition 6.151 (Contamination event)**.** *Contamination includes benchmark leakage, fabricated source data, unsupported citation, detector-stage mutation, hidden target use, artifact drift, unrecorded post-processing, or semantic transformation without preservation evidence.*

Proposition 6.152 (Contamination blocks status inheritance)**.** *A necessary contaminated node or edge cannot inherit its earlier pass. The route is* `undefined` *or* `reject` *until a clean revision is verified.*

Proof. The earlier evidence no longer establishes the required identity, independence, stage, or semantic preservation predicate. ☐

6.29 Replay hierarchy and strict separation

Definition 6.153 (Replay boundary)**.** *A replay boundary fixes input artifacts, code, environment, models, parsers, verifier versions, settings, seeds, external dependencies, and source or network snapshots.*

Definition 6.154 (Replay hierarchy)**.** *The four distinct records are operational replay, artifact replay, semantic replay, and mathematical verification.*

Theorem 6.155 (Expanded proof of Main Theorem 8.84: strict replay hierarchy)**.** *In general,*

$$\textit{operational replay} \not\Rightarrow \textit{artifact replay} \not\Rightarrow \textit{semantic replay} \not\Rightarrow \textit{mathematical verification.}$$

Each promotion requires a new identity, preservation, or proof record.

Proof. A nondeterministic execution may rerun while producing different output. Identical bytes may be interpreted under a different schema or semantic locator. Semantically stable serialization may faithfully encode a false or unproved theorem. Therefore no displayed implication is valid in general. ☐

Definition 6.156 (Semantic replay certificate)**.** *A semantic replay certificate records source and target schemas, statement normalizations, token-sort preservation, node and edge semantic diffs, source-binding and quarantine preservation, status reconstruction, and proof that the target closure has the same mathematical meaning.*

Proposition 6.157 (Replay disagreement blocks silent reuse)**.** *Different semantic nodes, edge directions, statuses, or target verdicts under supposedly identical immutable inputs make the manifest inconsistent until the difference is resolved.*

Proof. The determinacy theorem gives one output for fixed inputs. Disagreement proves that the inputs, environment, or semantics were not identical. ☐

6.30 Revision, impact closure, and immutable repair history

Definition 6.158 (Immutable revision record)**.** *A revision record is*

$$\mathfrak{R}' = \begin{pmatrix} \mathrm{RevisionID}', \mathrm{RevisionID}_{\mathrm{parent}}, \mathsf{SemDiff}, \\ \mathrm{changed\_artifacts}, \mathrm{changed\_nodes}, \\ \mathrm{impact\_closure}, \mathrm{reverification\_plan} \end{pmatrix}.$$

Every proof-relevant change creates a new revision; a changed target scope or evidence basis creates a new run identity when required by the governing policy.

Definition 6.159 (Impact closure)**.** *For changed nodes or edges S, let $V(S)$ contain every changed node and the target of every changed edge. Define*

$$\mathrm{Impact}(S) = V(S) \cup \mathrm{Desc}_{\mathsf{ProofDAG}}(V(S)).$$

All affected descendants are dirty until re-evaluated or protected by a semantic-preservation theorem.

Proposition 6.160 (No automatic status inheritance)**.** *A pass, reject, undefined, replay, audit, or Return status does not automatically transfer across a proof-relevant revision.*

Proof. The status was computed from the predecessor's immutable datum; changed statements, evidence, or dependencies define a different verifier input. ☐

Theorem 6.161 (Expanded proof of Main Theorem 8.92: immutable history)**.** *A valid repair creates a new revision and verdict. Historical artifacts, statuses, failure routes, and replay boundary remain immutable and independently replayable.*

Proof. The historical verdict is a function of the historical verifier datum. Mutating its inputs would destroy identity and replayability; corrected content must therefore be a linked successor record. □

6.31 Typed target closure and route-admissible conclusions

Definition 6.162 (Predecessor closure)**.** $\mathsf{PredDAG}(z)$ *contains every node of* $\mathsf{TargetDAG}(z)$ *except* z *and every necessary edge, including the edges entering* z*. The target is not presumed passed merely because it appears in the projection.*

Definition 6.163 (Main Definition 8.94: derivation readiness)**.** *A target* z *is derivation-ready at context* Γ *and strength* σ *when the target DAG is complete and well-founded; all proper predecessors and necessary edges pass at consumed strengths; the target has a total theorem- backed verifier policy; comparison compatibility, non-scalar obligations, metric no-double-counting, source binding, quarantine, artifact closure, manifest completeness, semantic replay, and revision identity pass; the predecessor context is contained in* Γ *without erasure; and the target output-strength set contains* σ*.*

Theorem 6.164 (Expanded proof of Main Theorem 8.95: typed DAG closure)**.** *If target* z *is derivation-ready at* (Γ, σ)*, recursive evaluation assigns* z *status* `pass` *and licenses the context sequent*

$$\Gamma \vdash \mathrm{statement}(z)$$

at strength σ *and every separately registered weakening. It is unconditional only when* $\Gamma = \varnothing$*. Conversely, any necessary rejected, undefined, misrouted, leaked, stale, out-of-closure, too-weak, or assumption-erasing component blocks consumption at that context and strength.*

Proof. Evaluate the well-founded predecessor graph in rank order. Every predecessor passes and every edge transports the exact type, strength, and context. The total target verifier policy therefore establishes the registered sequent and output strength. In the converse direction, each listed defect falsifies one necessary readiness clause; non-compensation and edge typing prevent a sibling branch from repairing it. □

Definition 6.165 (Route-admissible strength set)**.** *For route* R*, let every required node and edge provide a downward-closed set* $\mathfrak{S}_x$*. Define*

$$\mathfrak{S}_R = \bigcap_{x \in \mathrm{Req}(R)} \mathfrak{S}_x.$$

Several incomparable maximal elements are permitted; a unique scalar ceiling is not presumed.

Definition 6.166 (Route-admissible conclusion record)**.** *Let* $\mathrm{Cond}(R)$ *be the propagated assumption context after registered discharges. The admissible conclusion pairs are*

$$\mathfrak{A}_R = \{(\Gamma, \sigma) : \mathrm{Cond}(R) \subseteq \Gamma, \sigma \in \mathfrak{S}_R\}.$$

An unconditional conclusion requires $\mathrm{Cond}(R) = \varnothing$*.*

Proposition 6.167 (No conclusion beyond route intersection)**.** *A typed-DAG pass does not license a strength outside* $\mathfrak{S}_R$ *or omit an assumption in* $\mathrm{Cond}(R)$*.*

Proof. At least one required route component does not provide the stronger strength or requires the omitted premise, so no compatible composite output exists. □

6.32 Chapter 9 verifier handoff

Definition 6.168 (Expanded record of Main Definition 8.100: unified Bridge handoff)**.** *The verifier handoff is*

$$\mathsf{BridgeDAGInput} = \begin{pmatrix} \text{source\_node}, \text{prebridge\_boundary\_node}, \text{BridgeGoal}, \\ \mathsf{TargetDAG}(\text{prebridge\_boundary}), \text{requested\_context}, \\ \text{requested\_strength}, \text{Cond}(R), \mathfrak{S}_R, \\ \text{node\_registry}, \text{edge\_registry}, \text{source\_bindings}, \\ \mathsf{Quarantine}, \mathsf{ArtifactClosure}, \text{verifier\_verdict}, \\ \text{failure\_routing}, \text{manifest\_identity} \end{pmatrix}.$$

Its readiness token is `bridge_input_ready`*.* `BridgeGoal` *is a Chapter 9 specification, not a node already discharged by Appendix F.*

Theorem 6.169 (Main Theorem 8.101: handoff soundness)**.** *A* `bridge_input_ready` *record gives Chapter 9 a typed, dependency-closed, artifact-closed, source-bound, quarantine-safe pre-Bridge input with exact assumptions and admissible strength set. It neither proves nor presupposes the named Bridge or Return theorem.*

Proof. Typed closure discharges only the pre-Bridge boundary node. The Bridge and Return nodes are owned by Chapter 9 and remain new downstream dependencies. □

Remark 6.170 (No forward semantic dependency)*.* Appendix F may specify the input type expected by Chapter 9, but no verifier theorem here uses a Chapter 9 result as a premise.

6.33 Historical closure locators and v21 package

Remark 6.171 (Historical Appendix F Foundation closure locator)*.* The frozen v19/v20 typed-obligation package remains available at its original strength. In v21 it is a compatibility locator, not a second canonical package theorem. New runs are governed by Main Chapter 8 and the expanded records in this appendix.

Remark 6.172 (Historical v20 governance-kernel closure locator)*.* The frozen v20 governance-kernel package remains directly invocable for v20 runs. It is not silently upgraded to the v21 proof-DAG, source-quarantine, six-stage detector, inverse-audit, or route-strength semantics.

Theorem 6.173 (Expanded proof of Main Theorem 8.103: v21 verification and governance kernel)**.** *Within the v21 architecture:*

(i) *verifier inputs are immutable, typed, and source-bound;*

(ii) *document role, evidence, invocation, mathematical verdict, conformance, workflow, and migration are independent axes;*

(iii) *every required clause has one canonical atomic status and* `not_invoked` *requires positive exclusion;*

(iv) *aggregate verdicts are non-compensating conjunctions;*

(v) *comparison modes are consumed only at compatible output strength;*

(vi) *metric ledgers cannot manufacture non-scalar evidence;*

(vii) *proof-DAG nodes and edges have exact semantics and reachability is not theorem composition;*

(viii) finite dependency is acyclic and countable dependency requires a well-founded rank or finite-to-infinite reduction;

(ix) failure precedence identifies causal blockers without erasing other failures;

(x) Known-Theorem Recovery preserves source binding and benchmark quarantine;

(xi) hashes and manifests support identity and replay, not truth alone;

(xii) proposer, verifier, auditor, and proof-assistant authorities remain separate until conversion and interpretation pass;

(xiii) operational, artifact, semantic, and mathematical replay are distinct;

(xiv) repairs create new immutable revisions;

(xv) only a derivation-ready target DAG licenses consumption at a context and strength in the route-admissible record;

(xvi) Chapter 9 receives only a typed BridgeDAGInput*, not an automatic Bridge theorem.*

Proof. Items (i)–(ii) are the verifier datum and independent-axis calculus. Items (iii)–(iv) are the canonical status algebra and non-compensation. Items (v)–(vi) are comparison compatibility and the scalar/non-scalar boundary. Items (vii)–(viii) are the node, edge, composition, and well-foundedness sections. Items (ix)–(x) are failure precedence and source quarantine. Items (xi)–(xiv) are artifact, agent, replay, and revision semantics. Item (xv) is Theorem 6.164; item (xvi) is Theorem 6.169. □

6.34 Explicit exclusions

Appendix F does not assert:

(i) that a manifest, hash, graph, path, node name, proof-store entry, or successful build proves a theorem by existence;

(ii) that missing data imply `not_invoked`;

(iii) that a majority vote, consensus, confidence score, repeated run, or passing sibling compensates for one necessary reject or undefined;

(iv) that exact comparison is automatically robust, metric comparison creates a morphism, or a nonmetric theorem returns more than its named conclusion;

(v) that a metric ledger or certificate-cost bound creates a representative, morphism, detector completeness, generator completeness, coherence cell, descent theorem, reduction theorem, source authority, or Return theorem;

(vi) that graph reachability is typed theorem composition;

(vii) that ordinary dependency cycles are self-supporting proof;

(viii) that a finite prefix certifies a countable dependency tail;

(ix) that source storage or a benchmark citation establishes independent recovery;

(x) that benchmark truth, proof, constants, or conclusion-derived witnesses may flow upstream through the quarantine cut;

(xi) that byte identity proves cross-format semantic equivalence or truth;

(xii) that lossy schema migration preserves verifier equivalence;

(xiii) that proposer output, AI assertion, human approval, agent consensus, or audit review is theorem evidence without conversion;

(xiv) that a proof-assistant pass establishes the intended claim without formalization and interpretation edges;

(xv) that operational replay implies artifact, semantic, or mathematical verification;

(xvi) that status silently inherits across proof-relevant revisions;

(xvii) that a passing atlas implies object descent, or that `descent_input_ready` implies effectivity;

(xviii) that clearing $\mathbf{R}^1\varprojlim$ implies inverse apex agreement or direct Type IV zero;

(xix) that `bridge_input_ready` proves the Chapter 9 Bridge or Return theorem;

(xx) that a route has a unique scalar conclusion ceiling when its strength preorder has incomparable maxima.

6.35 Provenance and status classification

Appendix F integrates the v21 outputs of Appendices A–E and Appendix DG under the canonical Main Chapter 8 verifier. Its primary contribution is not a new external theorem but an exact proof-object architecture: immutable obligation atoms, four atomic statuses, four comparison modes, six detector stages, typed nodes and edges, assumption propagation, well-founded target closure, source quarantine, semantic replay, immutable revision, and the BridgeDAGInput handoff.

6.35.0.1 Canonical Main identities expanded here. Main Definitions 8.3–8.5, 8.7, 8.9–8.16, 8.19–8.21, 8.23–8.25, 8.27–8.32, 8.34, 8.36, 8.38–8.40, 8.42, 8.44, 8.46–8.48, 8.50, 8.52–8.60, 8.63–8.75, 8.78, 8.80, 8.82–8.83, 8.86, 8.88–8.89, 8.91, 8.93–8.94, 8.97–8.100; Main Propositions and Theorems 8.6, 8.8, 8.17–8.18, 8.22, 8.26, 8.33, 8.37, 8.41, 8.43, 8.45, 8.49, 8.51, 8.54–8.55, 8.61–8.62, 8.65, 8.70, 8.72, 8.76–8.77, 8.79, 8.81, 8.84, 8.87, 8.90, 8.92, 8.95–8.96, 8.99, 8.101, and 8.103.

6.35.0.2 Historical compatibility. All inherited v20 Appendix F labels remain resolvable. Their theorem strength is not silently increased; where a v21 canonical statement differs, the Main Chapter 8 statement and the explicit migration record govern new runs.

7 Appendix G: Canonical Manifest, Proof-DAG Serialization, Artifact Closure, and Replayability

7.1 Scope, dependency position, and canonical boundary

This appendix is the canonical serialization and proof-object layer of the v21.0.0 Foundation architecture. The complete v19/v20 schema below remains an immutable compatibility baseline, but Main Chapter 8 owns the canonical v21 verifier identities and Appendix F owns their expanded status and proof-DAG semantics. Appendix G serializes those semantics without mutation. It consumes the mathematical and record-theoretic disciplines of Appendices A–E, Appendix DG and Main Chapter 7 when available, and the typed verification kernel of Appendix F. Its purpose is not to invent a fifth verdict system, but to serialize without semantic loss the statement axes, obligation atoms, typed nodes and edges, comparison modes, source bindings, quarantine cuts, evidence, statuses, failure routes, coverage certificates, artifact closure, schema migration, replay hierarchy, immutable revision history, Chapter 9 handoff, and release acceptance fixed by Main Chapter 8 and Appendix F.

Chapter 1 supplies the foundational Minimal Manifest requirement. Main Chapter 8 supplies the canonical typed verifier and governance contract. Appendix F fixes its expanded status meanings, proof-DAG semantics, dependency closure, source quarantine, replay hierarchy, and Chapter 9 handoff. The present appendix fixes the canonical logical field schema, conformance rules, identity, immutability, semantic serialization, and replayability requirements. Main Chapter 7 and Appendix DG own effective finite descent and global-object certificates; Chapter 9 owns Bridge and Return construction.

Appendix CR-v20 supplies claim-status governance, Appendix CM-v20 supplies Core micro-theorem references, and Appendix TB-v20 supplies toy-bridge references. Whenever such material is invoked, the manifest records

$$\text{ClaimRef}, \qquad \text{CMRef}, \qquad \text{TBRef},$$

together with the exact permitted use and every discharged hypothesis.

Remark 7.1 (Appendix G is logically machine-readable, not format-prescriptive). This appendix fixes canonical logical fields, field types, conditional requiredness, identity constraints, and replayability conditions. It does not mandate one concrete syntax such as JSON, YAML, CSV, SQL, or a particular proof-store layout. A concrete encoding is admissible only if it preserves the logical schema bijectively enough for independent reconstruction.

Remark 7.2 (Appendix G does not enlarge the mathematical Core). This appendix creates no new persistence, derived, tower, Restart, overlap, or external-domain theorem. It governs admissible records. A complete schema is not a proof, and serialization cannot strengthen the status of a claim, micro-theorem, toy bridge, or proposed external interface.

Remark 7.3 (Relation to Appendix CR-v20). A registered claim is usable only through the claim-use preflight semantics of Appendix F. Registration is not proof. The manifest records the claim identifier, snapshot version, registered status, permitted use, discharged conditions, and non-claim boundary.

Remark 7.4 (Relation to Appendix CM-v20). A Core micro-theorem reference records its identifier, statement version, hypotheses, hypothesis evidence, and exact invocation site. Citation does not discharge hypotheses and does not turn an internal Core lemma into an external-domain theorem.

Remark 7.5 (Relation to Appendix TB-v20). A toy-bridge reference records the toy source category, realization route, repaired profiles, actual comparison artifact, diagnostic conclusion, and explicit toy-scope boundary. No toy theorem is promoted to a problem-specific bridge theorem without a separately proved transport theorem.

7.2 Semantic authority: Appendix F first, Appendix G second

Definition 7.6 (Semantic-preserving serialization). *A serialization of an Appendix F obligation registry is* semantic-preserving *if it preserves:*

(i) every obligation identity and immutable scope;

(ii) requiredness and justified scope exclusion;

(iii) *the dependency relation, closure references, and well-founded rank data;*

(iv) *specialized source status and canonical atomic status;*

(v) *evidence-policy version, artifact identity, and theorem/claim references;*

(vi) *scalar ledger data only for scalarizable obligations;*

(vii) *all Type I–V failure routes and blocker identifiers;*

(viii) *predecessor, revision, and repair-record links;*

(ix) *registry coverage, consistency, replayability, and aggregate verdict.*

Proposition 7.7 (Appendix F semantics are authoritative)**.** *No Appendix G field, default, omission rule, encoding convention, or implementation shortcut may alter the status meanings, dependency closure, non-compensation rule, readiness criterion, or failure routing fixed by Appendix F, in particular Definition 6.66 and Proposition 6.67.*

Proof. An encoding that changes one of those meanings represents a different obligation registry or verdict while reusing the same identifiers. That violates immutable run semantics and schema conformance. ☐

Remark 7.8 (Schema completeness is not proof). A schema may be complete and replayable while recording a rejected run. Conversely, a mathematically true but unserialized clause does not yield an auditable pass. Appendix F fixes the semantics; Appendix G fixes how those semantics are exposed for audit.

7.3 Audit identity, schema identity, and immutable scope

Definition 7.9 (Audit run identity)**.** *An audit run identity is the tuple*

$$\mathfrak{I} = (\mathrm{RunID}, \mathrm{ManifestID}, \mathrm{SchemaID}, \mathrm{Version}_{\mathrm{theory}}, \mathrm{Version}_{\mathrm{claim}}, \mathrm{Version}_{\mathrm{artifact}}, \mathrm{RevisionID}).$$

It identifies the exact theory release, canonical schema release, claim-register snapshot, artifact namespace, and revision under which all fields are interpreted.

Definition 7.10 (Immutable audit scope)**.** *The immutable scope of a run records at minimum:*

(i) *the object or object family under audit;*

(ii) *monitored degrees, windows, thresholds, and terminal modes;*

(iii) *invoked clauses and justified non-invoked clauses;*

(iv) *collapse policy and evaluation order;*

(v) *claim, theorem, micro-theorem, and toy-bridge identifiers used;*

(vi) *immutable identifiers of all source, profile, representative, realization, comparison, diagnostic, continuation, and atlas artifacts.*

Changing any item creates a new run or revision; it does not silently mutate the old verdict.

Definition 7.11 (Schema version boundary)**.** *A schema version boundary is a declared change in field names, field types, requiredness rules, normalization, or conformance semantics. A record from another schema version is admissible only through an explicit, auditable migration map that preserves Appendix F semantics.*

Proposition 7.12 (Identity collision is manifest inconsistency)**.** *If two records use the same* RunID *or* ManifestID *but assign different immutable scope, theory version, schema version, artifact identity, or revision semantics, then the manifest is inconsistent and routes to Type V rejection.*

Proof. One immutable identifier cannot denote two distinct runs under the canonical single-run semantics. The conflict is therefore an internal governance contradiction rather than mere missing data. □

7.4 Proof objects and certificate records

Definition 7.13 (Proof object)**.** *A* proof object *is an abstract mathematical-and-audit entity intended to support a declared claim. It contains or references the mathematical evidence, typed obligation registry, dependencies, ledger records, diagnostics, artifacts, claim-use preflight, and verdict data required by the applicable Core clauses.*

Definition 7.14 (Certified proof object)**.** *A proof object is* certified *for a declared scope only if:*

(i) its Appendix F foundation readiness bundle has aggregate status `pass`*;*

(ii) its manifest is complete, consistent, conformant, and replayable;

(iii) every required obligation in the dependency closure has canonical status `pass`*;*

(iv) every justified `not_invoked` *clause has a valid scope-exclusion record;*

(v) no hidden registry tail or unresolved blocker remains.

Manifest completeness alone does not certify a proof object.

Definition 7.15 (Certificate record)**.** *A* certificate record *is the canonical manifest-level representation of one proof object or one scope-relative proof-object fragment. It serializes evidence and verdict semantics but is not identical to the underlying mathematical object.*

Remark 7.16 (Proof object vs. certificate record)*.* The proof object is the evidenced entity; the certificate record is its audit-readable representation. A record may be complete and record a rejection. A record may also be incomplete even when an unrecorded mathematical argument happens to be true.

Definition 7.17 (Local certificate record)**.** *A* local certificate record *is attached to one declared window or overlap stratum and contains the local obligation subregistry, repaired evaluation data, local ledger, typed diagnostics, local verdicts, dependencies, and artifact references.*

Definition 7.18 (Global certificate record)**.** *A* global certificate record *is a record assembled from local records through a declared record-level Čech atlas, Restart sequence, finite union, or other proved certificate-record composition rule. It retains all local identities, simplex records, transition records, budgets, routes, and dependencies used in the assembly.*

Remark 7.19 (Global record is not object-level gluing)*.* A globally coherent certificate atlas does not construct a global filtered complex, persistence module, derived object, sheaf, or external-domain solution. Object-level descent requires separate mathematical hypotheses.

7.5 Canonical manifest object and field cardinalities

Definition 7.20 (Canonical manifest object)**.** *A canonical manifest is the typed record*

$$\begin{aligned}\mathsf{Manifest} = (&\mathsf{Identity}, \mathsf{Scope}, \mathsf{Objects}, \mathsf{Windows}, \mathsf{Policies}, \mathsf{Registry}, \mathsf{Evidence}, \mathsf{Artifacts},\\ &\mathsf{Ledger}, \mathsf{DomainRecords}, \mathsf{References}, \mathsf{Verdicts}, \mathsf{Routes}, \mathsf{Repairs}, \mathsf{Replay}).\end{aligned}$$

Each component is itself typed and versioned. Conditional components are absent only through a justified scope-exclusion record; silent omission is not permitted.

Definition 7.21 (Canonical cardinality markers)**.** *For schema statements, the markers*

$$[1], \qquad [0..1], \qquad [0..*], \qquad [1..*]$$

mean respectively exactly one, optional singleton, finite or countable family possibly empty, and nonempty finite or countable family. Countable families require a coverage certificate and replayable generator.

Definition 7.22 (Conditional requiredness)**.** *A field is* conditionally required *when an invocation predicate is true. The record stores both the predicate and its evidence. If the predicate is false, the field is absent only together with a justified* `not_invoked` *scope-exclusion record. If the predicate is true and the field is missing, the relevant obligation is* `undefined`*.*

Proposition 7.23 (No silent default theorem)**.** *No omitted representative, comparison morphism, eligibility certificate, continuation artifact, claim status, ledger upper-bound theorem, or coherence record may be reconstructed from a default value.*

Proof. Each listed item is a non-scalarizable proof obligation. A default would manufacture evidence and violate Appendix F non-compensation. □

7.6 Minimal Manifest Axiom

Axiom 7.24 (Minimal Manifest Axiom)**.** *A proof object may support an auditable Core verdict only if its manifest contains, at minimum:*

(i) *audit, manifest, schema, theory, claim-register, artifact-root, and revision identities;*

(ii) *immutable scope, declared objects, windows, thresholds, policies, and monitored degrees;*

(iii) *the full required obligation registry, dependency relation, well-founded rank, and coverage certificate;*

(iv) *every specialized source status and canonical atomic status;*

(v) *evidence-policy versions and immutable evidence/artifact references;*

(vi) *scalar ledger, scalarization, actual metric upper-bound, gap, and reserve records where applicable;*

(vii) *representative, realization, eligibility, Type IV, overlap, Restart, and atlas records when invoked;*

(viii) *claim-use, micro-theorem, and toy-bridge preflight records when invoked;*

(ix) *every blocker and every applicable Type I–V failure route;*

(x) *aggregate bundle verdict, manifest status, readiness status, consistency, and replayability fields;*

(xi) predecessor and repair-record identifiers for revised runs.

Remark 7.25 (Meaning of minimality). Minimality means necessary for the declared Core scope, not small in byte size. Concrete systems may add runtime logs, timestamps, proof-assistant traces, signatures, or redundant indexes, but may not omit a conditionally required field.

Proposition 7.26 (Necessity of manifest data)**.** *If a required field of Axiom 7.24 is absent or not reconstructible, then the manifest is incomplete and the corresponding audit clause is* `undefined`*, unless another evidenced clause already forces aggregate rejection.*

Proof. The missing field prevents reconstruction of at least one required obligation or its evidence. Appendix F maps missing required evidence to `undefined`; the aggregate precedence rule may still report `reject` if a separate clause is evidenced false, while retaining the undefined detail. □

7.7 Obligation registry serialization and dependency closure

Definition 7.27 (Obligation atom record)**.** *For every obligation atom, the manifest stores exactly one primary atom record containing:*

(i) ObligationID*, owner, immutable scope, and obligation kind;*

(ii) requiredness and scope-exclusion state;

(iii) evidence-policy identifier and version;

(iv) immediate dependency identifiers and closure reference;

(v) well-founded rank or finite acyclic-graph certificate;

(vi) specialized source status and canonical atomic status;

(vii) evidence references, blockers, routes, predecessor, and repair links.

Definition 7.28 (Dependency record)**.** *A dependency record is a directed edge*

$$\mathrm{ObligationID}(\mathfrak{p}) \longrightarrow \mathrm{ObligationID}(\mathfrak{o})$$

meaning that $\mathfrak{p}$ *is an immediate prerequisite of* $\mathfrak{o}$*. The manifest stores the edge source, target, dependency kind, evidence for the dependency, and rank check.*

Definition 7.29 (Registry coverage certificate)**.** *A registry coverage certificate proves that the serialized atom family is exactly the declared finite registry, or for a countable registry, that an immutable generator enumerates every required atom without hidden tail. It includes the generator version, index type, rank rule, and universal status certificate.*

Definition 7.30 (Closure reference)**.** *A closure reference identifies a reproducible computation or certificate for the transitive dependency closure of an atom or bundle. It does not replace the immediate dependency edges.*

Proposition 7.31 (Finite prefix cannot certify a countable registry)**.** *A manifest containing only a finite prefix of a countable obligation registry cannot support a pass for the full registry without a valid coverage certificate and universal status certificate.*

Proof. An unchecked tail may contain a required undefined or rejected atom. Finite-prefix success therefore does not determine the countable aggregate. □

7.8 Canonical statuses, classifications, and aggregate verdicts

Definition 7.32 (Canonical status field). *Every obligation atom has exactly one canonical atomic status*

$$\mathsf{Status}(\mathfrak{o}) \in \{\mathit{pass}, \mathit{reject}, \mathit{undefined}, \mathit{not\_invoked}\},$$

with the meaning fixed by Appendix F. The field also stores the specialized source status, normalization rule, evidence reference, timestamp or sequence index if used, and responsible verifier version.

Definition 7.33 (Claim/use classification field). *The claim/use classification is a separate field whose values may include*

$$\mathit{proved}, \ \mathit{conditional}, \ \mathit{operational}, \ \mathit{spec}, \ \mathit{deprecated}, \ \mathit{rejected}, \ \mathit{unknown},$$

or declared layer tags such as Core, Interface, Bridge Candidate, Bridge Program, Toy Bridge, Search Artifact, Companion Template, and Non-Claim. These are not atomic audit statuses.

Definition 7.34 (Aggregate verdict record). *An aggregate verdict record stores:*

(i) the required atom set and justified scope-excluded set;

(ii) the normalized status of every included atom;

(iii) the aggregation rule and Appendix F version;

(iv) the headline aggregate $\mathrm{Agg}(\mathfrak{R})$*;*

(v) all retained lower-level statuses, blockers, and routes.

The headline does not erase an undefined atom merely because another atom rejects.

Definition 7.35 (Manifest status field). *The manifest itself has one of the source statuses*

$$\mathit{complete}, \qquad \mathit{incomplete}, \qquad \mathit{inconsistent}.$$

Under the Appendix F embedding, these map respectively to

$$\mathit{pass}, \qquad \mathit{undefined}, \qquad \mathit{reject}.$$

This status is one obligation atom inside the full readiness bundle, not a replacement for the bundle verdict.

Proposition 7.36 (Status/classification separation). *A strong claim classification cannot compensate for an undefined or rejected audit status, and an audit pass cannot promote a claim beyond its registered classification.*

Proof. The fields have different types and govern different questions. Appendix F non-compensation and claim-use preflight prohibit cross-type substitution. □

7.9 Declared objects and windows

Definition 7.37 (Declared object data). *Declared object data include object identity, ambient type, source provenance, monitored degrees, coefficient field, constructibility or finiteness conditions, and immutable source artifacts. For an interface run, both the external object and its projected Core object are recorded together with the interface version.*

Definition 7.38 (Declared window data). *Declared window data include window identifiers, endpoints, endpoint convention, finite or right-unbounded terminal mode, MECE or overlap-cover role, cover/partition identifier, and all nonempty recorded overlap strata.*

Definition 7.39 (Window identifier). *A window identifier is immutable within one run and links all evaluations, representatives, diagnostics, ledgers, statuses, simplex records, continuation records, and artifacts for that window.*

Remark 7.40 (Window declarations and Appendix E). The MECE/overlap distinction, simplex coherence, Restart, and record-level gluing semantics are inherited from Appendix E. The historical label name of this remark is retained, but the governing main-body chapter is Chapter 6 in v19.

Proposition 7.41 (Window data are audit-critical). *Without declared window and terminal-mode data, the repaired evaluation, threshold clearance, terminal-generic Type IV diagnostic, overlap comparison, and Restart scope are not determined.*

Proof. Each listed construction is window-relative. Omitting the window therefore leaves required mathematical inputs undefined. ☐

7.10 Threshold, collapse, and repaired evaluation records

Definition 7.42 (Threshold policy). *A threshold policy records the threshold value or indexed family, fixed-threshold or sweep mode, common-threshold reconciliation rules, stable-band claims, window dependence, and the threshold-policy version.*

Definition 7.43 (Collapse policy). *A collapse policy records the permitted level of use, including*

$$\texttt{barcode\_level}, \quad \texttt{representative\_verified}, \quad \texttt{tau\_split}, \quad \texttt{exact\_admissible},$$

together with every artifact and hypothesis required by the chosen level. The default is barcode-level hard deletion after window restriction.

Definition 7.44 (Fixed-threshold certificate). *A fixed-threshold record evaluates all declared clauses at one immutable $\tau > 0$, with gate-scoped profile families and clearance records.*

Definition 7.45 (Threshold-sweep certificate). *A threshold-sweep record stores the threshold index set or band, artifact-coherent construction policy, pointwise records, bandwise quantifiers, and any claimed uniform reserve proof. Pointwise positivity is not serialized as a uniform positive infimum without separate evidence.*

Remark 7.46 (Threshold policy and repaired collapse). The threshold policy fixes barcode-level deletion semantics. It does not silently provide global exactness, Serre localization, or a strict filtered collapse functor.

Proposition 7.47 (Threshold and collapse data are audit-critical). *Without threshold, policy version, evaluation order, and collapse status, the repaired profile and the set of licensed downstream operations are not reconstructible.*

Proof. Different thresholds and collapse statuses produce different profiles and different admissible theorem uses. ☐

Definition 7.48 (Evaluation order record). *An evaluation order record declares*

$$\mathbf{P}_i(F) \longmapsto \mathsf{W}_W\mathbf{P}_i(F) \longmapsto T_\tau^{\mathrm{bar}}(\mathsf{W}_W\mathbf{P}_i(F)) = \mathrm{Eval}_{i,W,\tau}(F).$$

It stores the operator versions and every conditional commutation certificate if another route is used.

Definition 7.49 (Repaired evaluation record)**.** *For each monitored triple (i, W, τ), a repaired evaluation record stores the pre-threshold profile, barcode or reconstructive profile artifact, thresholded profile, evaluation status, clearance data, and immutable artifact references.*

Definition 7.50 (Overlap evaluation record)**.** *An overlap evaluation record is computed directly on the common overlap stratum at the reconciled threshold. It includes both local-source identities, the common-window pre-threshold profiles, discrepancy certificate, clearance certificates, and post-threshold comparison.*

Remark 7.51 (No hidden crop-collapse commutation)*.* A restriction of an already-collapsed larger-window profile cannot replace direct common-window evaluation without the window-adapted condition of Definition 1.64 and the conditional criterion of Proposition 1.65, or another explicitly proved exact-admissible route.

Proposition 7.52 (Evaluation records are audit-critical)**.** *A record without reconstructible repaired evaluations cannot determine the topological gate atom.*

Proof. The atom is the typed assertion $\mathrm{Eval}_{1,W,\tau}(F) = 0$ or its evidenced negation. Without the profile and artifacts, neither outcome is auditable. □

7.11 Representative, realization, and eligibility records

Definition 7.53 (Representative policy)**.** *A representative policy records whether a filtered representative is invoked, its lifecycle status, canonical atomic status, monitored degrees, equivalence policy, representative theorem/artifact, and realization compatibility when required.*

Definition 7.54 (Representative record)**.** *A representative record serializes $C_{\tau,W}F$, the relation*

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathrm{Eval}_{i,W,\tau}(F),$$

its proof artifact, construction provenance, equivalence class, status, blockers, and repair links.

Definition 7.55 (Eligibility record)**.** *When the* Ext^1*-clause is invoked, the eligibility record stores the representative, realization functor or procedure, realized object, amplitude certificate, typed residual-edge extractor, edge-realization isomorphism, zero-compatibility, and the canonical clause status.*

Remark 7.56 (No representative, no derived clause)*.* A missing representative or eligibility artifact makes the derived theorem-application atom `undefined`. It is not repaired by topological vanishing, numerical slack, or a claimed Ext value.

Proposition 7.57 (Representative and eligibility records are audit-critical when invoked)**.** *If a run invokes filtered or derived clauses, omission of the applicable representative or eligibility records prevents a dependency-complete pass.*

Proof. Those records are immediate prerequisites in the obligation registry. Appendix F dependency completeness therefore blocks the downstream application. □

7.12 Typed ledger and non-scalar obligations

Definition 7.58 (Ledger policy)**.** *A ledger policy declares an ordered commutative monoid or, when fully justified, commutative quantale; each metric-bearing component; aggregation; bottleneck scalarization; actual upper-bound theorem; gate-scoped pre-threshold profile family; clearance; scalarized budget; and reserve.*

Definition 7.59 (Budget component). *A budget component is a scalarizable metric discrepancy such as algorithmic, discretization, measurement, representative-choice, realization, comparison, crop, continuation, overlap, or Restart discrepancy. Each component stores semantic type, scope, provenance, value, and upper-bound evidence.*

Definition 7.60 (Non-scalar obligation record). *A non-scalar obligation record covers representative existence, eligibility, actual morphisms, terminal tameness, claim status, threshold reconciliation, simplex coherence, continuation and κ-admissibility, target-local pass, manifest identity, and comparable proof obligations. Such a record has a typed status and evidence; it is not a numerical budget component.*

Definition 7.61 (Total budget record). *A total budget record stores the typed metric ledger element, component decomposition, scalarization map, scalarized value, and theorem proving that the value dominates the actual gate discrepancy.*

Definition 7.62 (Safety margin record). *A safety margin record stores the gate-scoped pre-threshold family, each clearance, the minimum gap, scalarized metric budget, reserve, and strict comparison result.*

Remark 7.63 (Strict inequality). The boundary equality between scalarized discrepancy and gap is not a pass. The manifest stores the exact comparison operator and does not round equality into positivity.

Proposition 7.64 (Ledger data are audit-critical). *A numerical gate cannot pass from a ledger value alone; it also requires typed component provenance, scalarization, an actual metric upper-bound theorem, and a positive gate-scoped reserve.*

Proof. A number without a theorem connecting it to the consumed discrepancy is not evidence. The reserve must be computed from the relevant pre-threshold profiles, not from an unrelated global scalar. □

7.13 Type IV diagnostics and comparison artifacts

Definition 7.65 (Diagnostic policy). *A diagnostic policy records monitored degrees, windows, thresholds, terminal modes, source-profile provenance, actual profile-system morphisms, source-to-apex comparison morphisms, terminal-tameness certificates, kernel and cokernel defect profiles, and the canonical μ/ν diagnostics.*

Definition 7.66 (Tower comparison artifact record). *A tower comparison artifact record serializes*

$$\phi_{i,W,\tau} : \mathrm{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \mathrm{ApexProfile}_{i,W,\tau}(F_\infty),$$

including actual source and target objects, morphism identity, construction provenance, terminal-tameness, defect artifacts, and immutable hashes or equivalent integrity identifiers.

Definition 7.67 ($(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0,0)$ record). *A $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) = (0,0)$ record is permitted only when the Type IV specialized status is `certified_zero` and the canonical atomic status is `pass`. The numeric pair alone is insufficient without a valid comparison morphism and terminal-tameness evidence.*

Definition 7.68 ($(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) \neq (0,0)$ record). *A $(\mu_{\mathrm{Collapse}}, \nu_{\mathrm{Collapse}}) \neq (0,0)$ record stores specialized status `obstructed`, atomic status `reject`, the nonzero degreewise diagnostics, defect artifacts, and Type IV failure route.*

Definition 7.69 (Undefined Type IV record). *If Type IV is invoked but the comparison morphism, source provenance, terminal-tameness, or defect artifacts are missing, the record stores specialized status* `undefined` *and atomic status* `undefined`*; it must not serialize* $(0, 0)$ *as a substitute.*

Remark 7.70 (Terminal-generic Type IV meaning). The diagnostic semantics follow Appendix D. On bounded windows, cofinal bars reaching the right endpoint are counted even though they are globally finite. Therefore "infinite-bar only" is not the canonical bounded-window meaning. Zero terminal diagnostics do not imply full comparison isomorphism without the no-interior-defect condition.

Proposition 7.71 (Diagnostic and comparison-artifact data are audit-critical). *A Type IV atom is not well formed without the actual comparison artifact and the terminal-generic diagnostic prerequisites of Appendix D.*

Proof. Kernel, cokernel, and terminal-generic dimensions are defined relative to that typed morphism and regime. Without them, the clause is undefined rather than zero. □

7.14 Overlap, Čech, Restart, and atlas records

Definition 7.72 (Simplex compatibility record). *For every recorded nonempty Čech simplex, the manifest stores the simplex identifier, common overlap window, face identifiers, common-threshold reconciliation, pre-threshold profiles, discrepancy and sharp clearance certificates, face compatibility, typed Type IV status, non-scalar coherence obligations, and artifact links.*

Definition 7.73 (Restart transition record). *A Restart transition record stores source and target windows, source reserve,* κ_k*,* η_k*, the certified recurrence inequality, continuation artifact, transition substatus, target reserve, target-local status, and complete Restart verdict.*

Definition 7.74 (Countable Restart record). *A countable Restart record additionally stores* K_n*, normalized losses, the reserve-safe series certificate, pointwise reserve conclusions, and any separately proved uniform lower bound. Ordinary unnormalized summability is not recorded as sufficient evidence.*

Definition 7.75 (Atlas record). *An atlas record stores the cover identity, finite Čech depth, audit-local-finiteness certificate when countable, every required simplex record, face maps, coherence status, ledger exposure, coverage certificate, and the scope-limited conclusion "certificate atlas".*

Proposition 7.76 (Pairwise records do not imply higher coherence). *A manifest containing all pairwise overlap passes but missing a required higher-simplex record cannot support an atlas pass.*

Proof. Higher coherence is a distinct non-scalar obligation in Appendix E. It cannot be inferred from pairwise data or numerical slack. □

7.15 Gate and bundle verdict records

Definition 7.77 (Gate verdict). *A gate verdict is a typed bundle record, not merely a Boolean. It stores the gate scope, required atom set, justified non-invoked set, atomic statuses, aggregate status, blockers, reserves, failure routes, and dependency closure used to compute the headline verdict.*

Definition 7.78 (B-Gate$^+$ verdict record). *A B-Gate$^+$ verdict record stores distinct atoms for repaired topological vanishing, eligible Ext use when invoked, Type IV status, metric reserve, claim-use preflight, and manifest conformance. The aggregate is computed by Appendix F; no passing atom cancels an undefined or rejected atom.*

Definition 7.79 (Overlap Gate verdict record). *An Overlap Gate verdict record stores the common-window evaluation and sharp two-sided transfer atoms, threshold reconciliation, metric reserve, Type IV status, actual artifacts, and simplex/face obligations.*

Definition 7.80 (Foundation readiness record). *A foundation readiness record serializes the Appendix F readiness bundle and its aggregate status. Readiness is not a fifth atomic status; it is the aggregate status of a declared required bundle.*

Remark 7.81 (Verdicts are not scores). A confidence, ranking, probability, heuristic score, or AI assessment may appear only as proposer provenance. It is never a canonical atomic status or gate verdict.

Proposition 7.82 (Gate verdicts are audit-critical). *A record that stores only a headline verdict without the atomic statuses, dependency closure, blockers, and failure routes is not replayable and cannot support a canonical Core pass.*

Proof. The headline is a compression of typed detail. Without the detail, the verifier cannot reconstruct the aggregation or distinguish rejection from undefinedness. □

7.16 Claim, micro-theorem, and toy-bridge references

Definition 7.83 (Claim-register reference). *A claim-register reference stores* ClaimID*, claim-register snapshot, stable statement pointer, registered status, layer, permitted use, discharged conditions, forbidden use, exact invocation site, and artifacts.*

Definition 7.84 (Micro-theorem reference). *A micro-theorem reference stores* CMID*, statement version, hypotheses, evidence for each hypothesis, exact invocation site, conclusion consumed, and non-claim boundary.*

Definition 7.85 (Toy-bridge reference). *A toy-bridge reference stores* TBID*, toy source category, source object/tower, realization route, repaired profiles, actual comparison artifact, diagnostics, ledger assumptions, exact invocation site, and toy-scope boundary.*

Definition 7.86 (Reference consistency check). *Reference consistency is the claim-use preflight that verifies version identity, registered classification, permitted strength, discharged conditions, and invocation-site compatibility for every reference.*

Proposition 7.87 (Reference data are audit-critical when invoked). *An invoked reference without complete preflight data makes the corresponding theorem-application atom `undefined`; a forbidden or status-inconsistent use makes the claim-use atom `reject`.*

Proof. This is the canonical Appendix F claim-use embedding serialized at the invocation site. □

Remark 7.88 (Reference is not proof). A reference identifies a source and permitted use. It neither proves the source nor discharges its hypotheses.

7.17 Artifacts, integrity, and evidence roles

Definition 7.89 (Artifact)**.** *An artifact is an immutable or versioned evidential object supporting one or more manifest fields, including source data, barcodes, profiles, filtered representatives, realization records, comparison morphisms, defect profiles, ledgers, simplex records, formal proofs, computation traces, or governance snapshots.*

Definition 7.90 (Artifact reference)**.** *An artifact reference stores* ArtifactID*, URI or storage locator, media/type identifier, version, integrity algorithm, digest or equivalent immutable identity, producer, creation provenance, and supported field roles.*

Definition 7.91 (Artifact reference policy)**.** *The artifact reference policy declares which artifacts are required for verification, recomputation, explanatory context, or external transport; which fields they support; and how identity and integrity are checked.*

Definition 7.92 (Evidence role)**.** *Every evidence reference has a declared role such as theorem proof, hypothesis discharge, metric upper bound, artifact existence, identity attestation, deterministic computation, provenance, or explanatory context. Explanatory context cannot satisfy a proof role.*

Remark 7.93 (Artifact references are not themselves theorems). Integrity and provenance establish which artifact was used; they do not establish that the artifact proves the claimed mathematics.

Proposition 7.94 (Artifact references are audit-critical)**.** *A required evidence claim lacking an immutable artifact reference or an equivalent self-contained proof object cannot be independently reconstructed.*

Proof. The verifier cannot identify or inspect the alleged evidence. The relevant obligation remains undefined. ☐

7.18 Repair records and immutable verdict history

Definition 7.95 (Repair record)**.** *A repair record stores predecessor run and manifest identifiers, changed fields, changed evidence, reason, responsible schema/theory versions, affected obligations, re-evaluated dependency closure, and new verdict.*

Definition 7.96 (Revision relation)**.** *The revision relation is a directed acyclic predecessor relation between immutable run records. It is not an in-place overwrite operation.*

Proposition 7.97 (No retroactive verdict mutation)**.** *A repair may change the verdict of a new revision but cannot alter the stored status or evidence semantics of its predecessor.*

Proof. The predecessor is identified by immutable run, manifest, schema, and artifact identities. Changing its contents would destroy audit history and violate Appendix F repair semantics. ☐

7.19 Completeness, consistency, replayability, and audit readability

Definition 7.98 (Audit-readable certificate record)**.** *A certificate record is* audit-readable *if an independent verifier can identify the run, reconstruct the required registry and dependency closure, inspect all required evidence, recompute deterministic records, reproduce atomic statuses and the aggregate verdict, and verify every failure route and repair link.*

Definition 7.99 (Audit reconstruction)**.** *Audit reconstruction is the replay procedure that, under the registered theory and schema versions, recomputes or checks all deterministic profile, ledger, diagnostic, status-normalization, aggregation, and conformance steps used by the run.*

Definition 7.100 (Manifest completeness)**.** *A manifest is complete if every conditionally required field and required atom is present, typed, versioned, and linked to reconstructible evidence, and every omitted conditional field has a valid scope-exclusion record.*

Definition 7.101 (Manifest consistency)**.** *A manifest is consistent if shared identifiers have one meaning, duplicate field assignments agree, dependency ranks are well founded, status normalizations agree with source records, artifact identities match, and no field contradicts immutable scope or version boundaries.*

Definition 7.102 (Replayability)**.** *A manifest is replayable if an independent verifier using only declared immutable artifacts, generators, and registered versions can reconstruct every deterministic verdict computation and identify every non-deterministic or theorem-dependent step with its evidence.*

Proposition 7.103 (Minimal Manifest implies audit readability)**.** *If a manifest satisfies the Minimal Manifest Axiom and is complete, consistent, conformant, and replayable, then it is audit-readable. This conclusion does not imply that its aggregate verdict is pass.*

Proof. The hypotheses provide all typed fields, identities, dependencies, evidence, and replay instructions required for reconstruction. The reconstructed statuses may still contain rejection. □

Proposition 7.104 (Audit readability enables proposer-independent evaluation)**.** *An audit-readable record can be evaluated by a verifier that did not participate in proposing or producing the original candidate, provided the verifier implements the registered theory and schema semantics and has access to the referenced immutable artifacts.*

Proof. Under these hypotheses, the verdict depends on immutable Core inputs, typed evidence, registered semantics, and artifact access, not on proposer identity. □

Remark 7.105 (Relation to Core sovereignty)*.* Replayability supports but does not replace mathematical validity. Search-side confidence, proposal history, or AI ranking is not a verdict field.

7.20 Proposer–verifier separation

Definition 7.106 (Proposer-side data)**.** *Proposer-side data include candidate objects, windows, thresholds, maps, rankings, terrain models, validity scores, bridge programs, generated proof sketches, and AI recommendations.*

Definition 7.107 (Verifier-side data)**.** *Verifier-side data are the typed Core objects, obligation atoms, evidence, artifacts, policies, ledgers, diagnostics, references, statuses, dependencies, and repair records required by the registered audit semantics.*

Proposition 7.108 (Only verifier-side data determine Core verdicts)**.** *Proposer-side data may suggest or locate evidence, but cannot alter a Core verdict until normalized into the required Core types, registered, evidenced, and re-evaluated by the verifier.*

Proof. This is Core sovereignty: proposal authority and verdict authority are distinct. □

Remark 7.109 (Recording proposer data)*.* Optional proposer provenance must be tagged as non-verdict data and may not fill a missing verifier field.

7.21 Abstract schema and concrete execution encodings

Definition 7.110 (Abstract certificate schema)**.** *The canonical logical schema is*

$$\begin{aligned}\mathcal{C} = (&\mathsf{Identity}, \mathsf{Scope}, \mathsf{Object}, \mathsf{Window}, \mathsf{Threshold}, \mathsf{Collapse}, \mathsf{Evaluation},\\ &\mathsf{Registry}, \mathsf{Dependency}, \mathsf{Evidence}, \mathsf{Representative}, \mathsf{Eligibility}, \mathsf{Diagnostic}, \mathsf{Comparison},\\ &\mathsf{OverlapRestartAtlas}, \mathsf{Ledger}, \mathsf{Reference}, \mathsf{Artifact}, \mathsf{AtomicStatus},\\ &\mathsf{AggregateVerdict}, \mathsf{FailureRoutes}, \mathsf{Repair}, \mathsf{Replay}).\end{aligned}$$

Every component is typed, versioned, and conditionally required according to the declared scope.

Remark 7.111 (Schema is abstract but canonical)*.* The tuple fixes logical content and field roles, not punctuation or storage syntax. A JSON object, relational schema, proof-assistant structure, or other encoding may refine it only through a declared conformance map.

Definition 7.112 (Execution schema)**.** *An execution schema is a concrete encoding with a total decoding/conformance procedure into the canonical logical schema, plus validation rules for requiredness, types, identities, versions, and integrity references.*

Remark 7.113 (Appendix G vs. execution companion)*.* Appendix G fixes what a conformant record means. The Execution/Reproducibility Companion may fix concrete field names, file layouts, APIs, signatures, or replay commands, but may not weaken the logical requirements.

Proposition 7.114 (Concrete execution schemas must refine the abstract schema)**.** *A concrete schema is Core-admissible only if decoding preserves identities, types, statuses, dependencies, evidence roles, routes, repairs, and aggregate semantics.*

Proof. Otherwise two non-equivalent audit records could decode to the same canonical record or one canonical field could be lost, violating semantic-preserving serialization. ☐

7.22 Schema conformance and validation profiles

Definition 7.115 (Schema conformance)**.** *A record is schema-conformant if it passes structural type validation, conditional-requiredness validation, identifier uniqueness, version compatibility, dependency well-foundedness, status normalization, artifact integrity, and cross-field consistency checks.*

Definition 7.116 (Validation profile)**.** *A validation profile is a named set of required field families for a declared scope, such as `topological_only`, `eligible_ext`, `typeIV`, `overlap_atlas`, `restart`, or `full_foundation`. Profiles control conditional requiredness only; they do not change field meanings.*

Proposition 7.117 (Canonical manifest conformance)**.** *A chapter-local or appendix-local fragment is admissible only if it embeds into one canonical manifest without changing shared field meanings, identifiers, statuses, version boundaries, or artifact identities.*

Proof. The canonical manifest represents one immutable run. Incompatible fragments produce manifest inconsistency. ☐

Definition 7.118 (Status field, legacy label retained)**.** *The historical phrase "final status field", introduced in v19 and inherited unchanged in v20, is interpreted as the pair*

$$(\mathsf{AtomicStatusRegistry}, \mathsf{AggregateVerdict})$$

together with a separate claim/use classification field. No mixed enumeration of pass/reject with operational, interface, toy, or spec layers is canonical.

Proposition 7.119 (Schema sufficiency for Core audit)**.** *A conformant, complete, consistent, and replayable record is sufficient to reconstruct a Core audit verdict, but not necessarily sufficient to obtain a pass.*

Proof. The record exposes all inputs and rules required for reconstruction. Reconstruction may yield reject or undefined. □

7.23 Incomplete, inconsistent, and failed records

Definition 7.120 (Incomplete certificate record)**.** *A record is incomplete when at least one required field, atom, dependency, evidence reference, coverage certificate, or justified scope exclusion is missing or not reconstructible.*

Definition 7.121 (Failed certificate record)**.** *A record is failed when enough data are present to evidence at least one required atom as false, giving canonical status* `reject`*. A failed record may simultaneously contain other undefined atoms.*

Definition 7.122 (Inconsistent certificate record)**.** *A record is inconsistent when shared identifiers, immutable scope, versions, artifact identities, dependency ranks, source statuses, or duplicate field assignments contradict one another.*

Remark 7.123 (Incomplete vs. failed)*.* Incomplete means audit undefinedness for the affected clause; failed means evidenced rejection. Inconsistency is a Type V rejection. The aggregate headline follows Appendix F precedence while retaining all details.

Proposition 7.124 (Failure is preferable to incompleteness)**.** *For diagnosis and repair, an audit-readable failed record is more informative than an incomplete record because it exposes a reconstructible counterevidence path. This does not make rejection normatively preferable to pass.*

Proof. A failed record identifies the atom, evidence, and route. An incomplete record lacks at least one required reconstructive component. □

7.24 Appendix G manifest package

Theorem 7.125 (Appendix G manifest package)**.** *As a historical v20 compatibility closure, preserving the inherited v19 semantics:*

(i) Appendix F fixes verdict semantics and Appendix G serializes them without mutation;

(ii) immutable run, manifest, schema, theory, artifact, and revision identities delimit one audit meaning;

(iii) the Minimal Manifest includes the full required registry, dependency closure, evidence, artifacts, statuses, routes, repairs, and replay data;

(iv) canonical atomic statuses are exactly `pass`*,* `reject`*,* `undefined`*, and justified* `not_invoked`*;*

(v) claim/use classification is separate from atomic status;

(vi) scalar ledgers contain only metric-bearing discrepancies and cannot encode missing non-scalar evidence;

(vii) Type IV zero requires a valid actual comparison artifact and terminal-generic certificates;

(viii) *overlap, higher coherence, Restart, target-local pass, and atlas records remain distinct obligations;*

(ix) *a complete manifest may record rejection, and manifest completeness never implies theorem truth;*

(x) *countable registries require coverage and universal-status certificates;*

(xi) *repairs create new immutable revisions rather than mutating historical verdicts;*

(xii) *conformance, completeness, consistency, and replayability yield audit readability but not automatic pass;*

(xiii) *proposer data cannot alter verdicts without Core re-entry;*

(xiv) *concrete execution schemas must refine the canonical logical schema.*

Proof. The items collect the definitions and propositions of this appendix together with Appendix F's authoritative status, dependency, aggregation, readiness, and repair semantics. ☐

7.25 v20 token-sort and typed-evidence serialization

Definition 7.126 (Token-sort registry record)**.** *The manifest stores one immutable token-sort registry containing:*

(i) TokenSortID*, registry version, owning theory release, and digest;*

(ii) *every token literal, semantic sort, governing field family, and admissible source types;*

(iii) *the canonical audit-status sort, runtime comparison-mode sort, detector-stage sort, detector-certificate sort, detector-value sort, terminal Type IV sort, transient-verdict sort, source-semantics sort, simplex-mode sort, Restart-transition sort, Restart-zero-route sort, manifest-status sort, claim-status sort, and invocation-role sort;*

(iv) *every permitted embedding into atomic audit status;*

(v) *every prohibited cross-sort conversion.*

Every token occurrence in a consumed record stores both its literal and its TokenSortID*-qualified semantic sort.*

Definition 7.127 (Typed evidence record)**.** *For every evidence-bearing atom, the manifest stores one typed evidence record:*

$$\begin{aligned}&(\,\text{EvidenceID}, \text{ObligationID}, \text{evidence type}, \text{source type}, \text{target type},\\&\ \text{governing theorem/policy}, \text{hypotheses}, \text{transformations}, \text{token sorts},\\&\ \text{artifacts}, \text{versions}, \text{digests}, \text{specialized status}, \text{atomic status}).\end{aligned}$$

The record distinguishes theorem proof, hypothesis discharge, exact isomorphism, metric upper bound, detector certificate, coverage certificate, artifact identity, deterministic computation, provenance, and explanatory context.

Proposition 7.128 (Cross-sort inconsistency is detectable)**.** *A record assigning a token to a field of the wrong semantic sort is schema-inconsistent. If the mismatch is unresolved, the affected obligation is* `undefined`*; if the manifest asserts the mismatched assignment as a valid verdict or conceals the mismatch, the manifest is* `inconsistent` *and routes to Type V rejection.*

Proof. The token registry and field schema determine the unique admissible source sort. An incompatible assignment cannot decode to the canonical Appendix F semantics. ☐

7.26 v20 detector and transient records

Definition 7.129 (Detector policy record)**.** *A detector policy record stores:*

(i) detector identifier, mode, information class, profile stage, family, endpoint convention, and monitored scope;

(ii) exact source profile identity and source artifact;

(iii) witness set, stencil, birth, right-germ, or canonical family data;

(iv) soundness, applicability, completeness, and sharpness references;

(v) metric-safe transport dependencies when the detector conclusion is transferred across a comparison or hard deletion;

(vi) detector-certificate status and detector mathematical value as separate fields;

(vii) canonical atomic status, blockers, routes, and immutable artifacts.

Definition 7.130 (Detector certificate record)**.** *A detector certificate record serializes*

$$(\mathrm{CertificateStatus}, \mathrm{DetectorValue})$$

without identifying the two axes. A required zero atom may pass only when the certificate status is `detector_certified`, the value is `zero`, and every applicability and completeness dependency passes. A sampled zero with pending or failed certification is never serialized as a zero proof.

Definition 7.131 (Transient-defect record)**.** *A transient-defect record stores:*

(i) window, collapse threshold, defect threshold, monitored degrees, and detector component;

(ii) kernel and cokernel defect-profile identities;

(iii) complete detector policies and values for both profiles;

(iv) specialized aggregate status

`not_invoked`, `undefined`, `certified_absent`, `certified_present`;

(v) canonical atomic status, witness bars when present, blockers, theorem locators, and artifacts.

Terminal Type IV status is stored separately.

Definition 7.132 (Terminal/transient participation record)**.** *Every local, overlap, simplex, Restart-target, atlas, and global-predicate record stores:*

(i) terminal Type IV invocation and specialized status;

(ii) whether long-transient absence is required;

(iii) the exact transient clause identity when required;

(iv) independent canonical statuses and artifact references.

Terminal zero is not serialized as transient absence or full isomorphism.

7.27 v20 exact/metric coherence and coverage records

Definition 7.133 (v20 overlap comparison record). *A v20 overlap record stores separately:*

(i) atomic Overlap Gate status;

(ii) runtime comparison mode: `exact`*,* `metric_gap_certified`*, or* `not_compared`*;*

(iii) direct common-window source profiles and reconciled threshold;

(iv) in exact mode, the actual repaired-profile isomorphism and proof;

(v) in metric mode, the pre-threshold bottleneck bound, both gap certificates, scalarization, and post-threshold bound;

(vi) terminal/transient participation and all artifacts.

The comparison mode never substitutes for the atomic status.

Definition 7.134 (v20 simplex-coherence record). *A v20 simplex record declares exactly one mode:*

`exact_cocycle`, `metric_reference`, `not_compared`.

In exact-cocycle mode it stores all transition isomorphisms, identities, inverses, cocycle equations, and face restrictions. In metric-reference mode it stores the reference profile, local-to-reference bounds, two-sided gap certificates, reference face-compatibility isomorphisms, and pairwise bounds. Metric mode does not serialize transition morphisms or a categorical cocycle.

Definition 7.135 (Coverage certificate record). *Every coverage record contains* CoverageID*, coverage type, governing scope, universal quantifier, witness generator or proof, theorem/policy reference, artifacts, status, and dependencies. The coverage type is exactly one of:*

(i) `registry_coverage`*;*

(ii) `retained_bar_coverage`*;*

(iii) `boundary_bridge_coverage`*;*

(iv) `artifact_closure`*;*

(v) `release_document_coverage`*.*

A record of one type cannot discharge another.

Definition 7.136 (Global profile interface record). *A global profile interface record stores:*

(i) global pre-threshold profile, covered range, degree, threshold, and profile artifact;

(ii) every exact local restriction profile;

(iii) every observed local profile;

(iv) for each local profile, either an exact isomorphism branch or a metric-safe zero-transport branch;

(v) every two-sided gap and $2\varepsilon \leq \tau$ certificate used;

(vi) local detector and retained-bar coverage references.

Pairwise local compatibility does not construct this record.

Definition 7.137 (Detector-atlas record)**.** *A detector-atlas record serializes:*

(i) the global profile interface;

(ii) every local detector policy and certificate;

(iii) every consumed overlap and simplex record;

(iv) retained-bar and boundary-bridge coverage records;

(v) terminal/transient participation;

(vi) local and global zero atoms;

(vii) atlas readiness, global-zero readiness, blockers, routes, and artifacts.

Local zero values without complete detector and coverage records do not support a global-zero verdict.

Example 7.138 (Manifest distinction for local zero without global coverage)*.* A record may contain certified zero detector values for every listed local window and exact zero overlap profiles while retaining-bar coverage is `undefined` or `reject`. The local clauses may pass, but the global repaired-zero atom remains `undefined` or rejects according to the evidenced coverage clause. The manifest must retain both facts rather than compressing them to "global zero."

7.28 v20 Restart, reduction, and descent records

Definition 7.139 (Restart zero-route record)**.** *When a Restart claim consumes target repaired zero, the transition record additionally stores:*

(i) zero-route status:

$$\texttt{not\_invoked},\quad \texttt{undefined},\quad \texttt{certified},\quad \texttt{failed};$$

(ii) route kind: fresh complete detector, exact zero transport, or metric-safe zero transport;

(iii) source and target zero atoms;

(iv) profile stages, thresholds, endpoint conventions, comparison mode, theorem/policy, and artifacts;

(v) boundary-bridge coverage when a global predicate is claimed.

The κ_k-reserve substatus and zero-route substatus remain independent.

Definition 7.140 (Finite-to-infinite reduction record)**.** *A finite-to-infinite reduction record stores:*

(i) the actual directed system, transition morphisms, and source semantics;

(ii) stabilization index or finite witness;

(iii) genuine colimit or theorem-authorized source object;

(iv) compatible apex cocone, induced comparison morphism, and apex agreement;

(v) every reduction-theorem hypothesis;

(vi) terminal, transient, or full-isomorphism consequences actually consumed;

(vii) claim-use, artifact closure, replay, and verdict data.

A finite observed plateau or repeated barcode is serialized only as proposer or empirical context unless a reduction theorem is proved.

Definition 7.141 (Object-descent record)**.** *An object-descent record stores, without field contraction:*

(i) category, enhancement, or higher category and restriction functors;

(ii) the canonical nine-field DescentInput

$$(\{E_\alpha\}, \{g_{\alpha\beta}\}, \{c_{\alpha\beta\gamma}\}, \{h_\sigma\}, \mathcal{W}, \mathcal{C}, \mathfrak{M}_{\mathrm{coh}}, \mathcal{T}_{\mathrm{eff}}, \mathcal{A});$$

(iii) the active descent lane and its exact source/target types;

(iv) effectivity-theorem identity, discharged hypotheses, and supported conclusion strength;

(v) constructed descended object or morphism and all local comparison artifacts;

(vi) uniqueness or equivalence strength, recorded independently from existence;

(vii) exact restriction, joint conservativity, local zero-reflection, and cone compatibility when global zero or isomorphism is consumed;

(viii) independent certificate-atlas, readiness, manifest, artifact-closure, and replay clauses.

A metric-reference atlas, certificate atlas, global-record-zero predicate, or `descent_input_ready` *value is not serialized as effective object descent.*

7.29 Artifact closure and semantic replay

Definition 7.142 (Artifact-closure record)**.** *An artifact-closure record contains:*

(i) the artifact-root identifier and complete reachable artifact graph;

(ii) for each node, type, version, producer, provenance, locator, integrity algorithm, digest, and supported evidence roles;

(iii) every transitive interpretation dependency not supplied by the registered theory release;

(iv) uniqueness and non-dangling checks;

(v) ambiguity, collision, and silent-replacement checks;

(vi) closure status and replay instructions.

Artifact closure identifies the complete evidence graph; it does not prove the mathematics stored in that graph.

Definition 7.143 (Semantic replay record)**.** *A semantic replay record contains* ReplayID*, verifier implementation and version, token-sort registry, claim-register snapshot, schema version, theory version, artifact closure, replay environment, and commands or proof-checking procedure sufficient to reconstruct:*

(i) deterministic profiles, detectors, ledgers, diagnostics, and dependency graphs;

(ii) source types, governing claims, theorem hypotheses, specialization maps, and invocation roles;

(iii) specialized-status embeddings and canonical atomic statuses;

(iv) normalization, aggregation, blockers, routes, and bundle verdicts;

(v) every non-deterministic or theorem-dependent step with its evidence.

Computational replay is recorded as a subfield; semantic replay is the stronger Core requirement for consumed machine-produced verdicts.

Proposition 7.144 (Semantic replay is stronger than byte reproduction)**.** *Reproducing files or numerical outputs without reconstructing their source types, theorem hypotheses, token sorts, dependencies, and status embeddings does not establish semantic replayability.*

Proof. Identical bytes may be interpreted under different schemas, theorem versions, or claim statuses. Semantic identity therefore requires the registered interpretation data in addition to file reproduction. □

Definition 7.145 (Replay discrepancy record)**.** *If two replays disagree, the manifest stores both replay identities, immutable inputs, first divergent atom or deterministic step, candidate causes, affected dependency closure, and revised verdict. Until resolved, the affected replayability atom is `undefined`; an evidenced contradiction under identical declared semantics makes the manifest `inconsistent`.*

7.30 Lossless schema migration

Definition 7.146 (Schema migration record)**.** *A migration record contains:*

(i) MigrationID*, source and target* SchemaID*, theory releases, and digests;*

(ii) a total map for every consumed field and token sort;

(iii) requiredness, scope-exclusion, status, dependency, artifact, route, and repair-history preservation;

(iv) explicit defaults only for newly introduced fields proved outside the old claim scope;

(v) semantic-difference evidence;

(vi) source and target replay records;

(vii) a proof that the old supported verdict is reconstructible at exactly its old strength.

A theorem-strength, evidence, scope, or claim-use change is a new run or revision, not a schema-only migration.

Proposition 7.147 (Lossy migration is non-conformant)**.** *A migration dropping a required hypothesis, dependency, token sort, lower-level status, missing-artifact marker, scope boundary, blocker, route, or forbidden stronger reading is non-conformant. The target record is incomplete or inconsistent until repaired as a new evidence-backed revision.*

Proof. The dropped data participate in clause meaning or verdict reconstruction, so the decoding cannot preserve Appendix F semantics. □

7.31 Release manifest and acceptance

Definition 7.148 (Release manifest)**.** *A release manifest is the typed record*

$$\mathfrak{M}_{\mathrm{rel}} = (\mathrm{ReleaseID}, \mathsf{Documents}, \mathsf{Claims}, \mathsf{Schemas}, \mathsf{Artifacts}, \mathsf{Supersession}, \mathsf{Replay}, \mathsf{Acceptance}).$$

For every mandatory document role it stores DocumentID*, title, role, version, digest, schema, dependencies, advertised scope, supersession status, and artifact closure.*

Definition 7.149 (Mandatory release-document roles)**.** *A complete v20 release registry declares the mandatory roles applicable to its advertised scope, including:*

(i) Main;

(ii) Foundation Appendices;

(iii) Claim Register;

(iv) Core micro-theorems and toy bridges when advertised;

(v) problem-interface or search-platform appendices when included;

(vi) canonical manifest and execution/reproducibility artifacts;

(vii) release notes, supersession map, and integrity index.

A role may be excluded only by a release-level scope decision consistent with the advertised product.

Definition 7.150 (Release-acceptance record)**.** *A release-acceptance record stores:*

(i) release-document coverage;

(ii) canonical Claim Register and supersession closure;

(iii) schema conformance and migration closure;

(iv) artifact closure and semantic replay;

(v) every mandatory Foundation readiness verdict;

(vi) all unresolved rejected and undefined mandatory atoms;

(vii) headline acceptance status.

The release passes exactly when every mandatory clause passes. A strong mathematical theorem does not compensate for an incomplete release registry or inconsistent manifest.

Definition 7.151 (Supersession record)**.** *A supersession record identifies predecessor and successor documents, claims, schemas, and artifacts; classifies each relation as replacement, refinement, deprecation, withdrawal, or coexistence; states preserved and changed semantics; and prevents an obsolete claim from silently re-entering an active release.*

Proposition 7.152 (Document completeness and theorem strength are independent)**.** *A release with a proved theorem but missing a mandatory manifest, claim snapshot, replay artifact, or supersession record does not pass release acceptance. Conversely, a complete release manifest may faithfully record a mathematically rejected or incomplete program.*

Proof. Release acceptance and mathematical clause status are distinct required atoms combined by non-compensatory aggregation. □

7.32 v20 canonical schema and validation profiles

Definition 7.153 (v20 canonical manifest object)**.** *The v20 canonical manifest extends Definition 7.20 to*

$$\begin{aligned}\mathsf{Manifest}_{20} = (&\mathsf{Identity}, \mathsf{TokenSorts}, \mathsf{Scope}, \mathsf{Objects}, \mathsf{Windows}, \mathsf{Policies},\\ &\mathsf{Registry}, \mathsf{Dependencies}, \mathsf{TypedEvidence}, \mathsf{Artifacts}, \mathsf{ArtifactClosure},\\ &\mathsf{Ledger}, \mathsf{Evaluations}, \mathsf{Detectors}, \mathsf{Representatives}, \mathsf{Eligibility},\\ &\mathsf{TerminalTransient}, \mathsf{OverlapSimplex}, \mathsf{Coverage}, \mathsf{Restart},\\ &\mathsf{Reduction}, \mathsf{Descent}, \mathsf{References}, \mathsf{Verdicts}, \mathsf{Routes},\\ &\mathsf{Repairs}, \mathsf{Migration}, \mathsf{SemanticReplay}, \mathsf{Release}).\end{aligned}$$

Every field family remains conditionally required under immutable scope.

Definition 7.154 (v20 validation profiles)**.** *In addition to inherited profiles, v20 provides:*

$$\begin{gathered}\texttt{detector},\quad \texttt{detector\_atlas},\quad \texttt{global\_zero},\quad \texttt{transient\_clean},\\ \texttt{restart\_zero},\quad \texttt{finite\_to\_infinite},\quad \texttt{object\_descent},\\ \texttt{semantic\_replay},\quad \texttt{release\_acceptance},\quad \texttt{full\_v20\_foundation}.\end{gathered}$$

A profile selects conditionally required fields; it never changes their meaning or weakens dependencies.

Proposition 7.155 (v20 schema sufficiency for audit reconstruction)**.** *A v20 record that is token-sort valid, conformant, complete, consistent, artifact-closed, semantically replayable, and dependency-complete is sufficient to reconstruct every declared v20 Foundation verdict. It is not sufficient by itself to make those verdicts pass.*

Proof. The hypotheses expose the complete typed evidence and deterministic semantics needed for reconstruction. The reconstructed atoms may still reject or remain undefined. □

Theorem 7.156 (Appendix G v20 manifest and replay package)**.** *As a historical v20 conservative-extension closure:*

(i) the complete inherited v19 manifest package remains valid at its original strength;

(ii) token-sort and typed-evidence records prevent cross-type verdict substitution;

(iii) detector certificate status and detector mathematical value are serialized independently;

(iv) terminal Type IV and long-transient records remain distinct;

(v) exact-cocycle and metric-reference coherence remain distinct;

(vi) registry, retained-bar, boundary-bridge, artifact, and release coverage have separate record types;

(vii) detector-atlas and global-zero records expose every local-to-global dependency;

(viii) Restart reserve and target-zero routes are independently serialized;

(ix) finite-to-infinite reduction and object descent require actual mathematical artifacts;

(x) artifact closure and semantic replay expose the complete interpretation graph of machine-produced verdicts;

(xi) schema migration is admissible only when semantic and verdict reconstruction are lossless at the inherited strength;

(xii) release acceptance is a separate non-compensatory conjunction;

(xiii) the v20 canonical schema is sufficient for audit reconstruction but never creates theorem truth.

Proof. Item (i) is Theorem 7.125. Items (ii)–(iv) are Definitions 7.126–7.132. Items (v)–(vii) follow from Definitions 7.133–7.137. Items (viii)–(ix) are Definitions 7.139–7.141. Item (x) follows from Definitions 7.142 and 7.143. Item (xi) is Definition 7.146 and Proposition 7.147. Item (xii) follows from Definitions 7.148–7.150. Item (xiii) is Definition 7.153 and Proposition 7.155. □

7.33 v21 canonical ownership and supersession boundary

Definition 7.157 (v21 semantic authority boundary)**.** *For every new v21 run, Main Chapter 8 owns the canonical verifier statements, Appendix F owns their expanded status and proof-DAG semantics, and Appendix G owns only semantic-preserving serialization, conformance, replay exposure, and release packaging. When an inherited v19/v20 field or token conflicts with a canonical v21 field, the record contains an explicit supersession or migration edge; no historical field is silently reinterpreted.*

Remark 7.158 (Historical labels are compatibility locators)*.* The labels `thm:G-package` and `thm:G-v20-package`, and the v19/v20 logical fields preceding this subsection, remain valid locators for historical records. They do not form a second v21 verifier kernel. The canonical v21 mathematical-governance package is Main Theorem 8.103, expanded in Appendix F; Appendix G supplies its lossless record realization.

Definition 7.159 (Canonical v21 manifest purpose)**.** *A v21 manifest is a typed reconstruction object for one immutable target run. It answers, without strengthening any claim:*

(i) what exact target, context, and conclusion strength were requested;

(ii) which nodes, edges, obligations, artifacts, and source bindings were necessary;

(iii) which evidence and verifier policies discharged them;

(iv) which assumptions, failures, exclusions, and quarantine boundaries remain;

(v) which verdict is reproducible under the registered semantics.

It is not itself a theorem, Bridge, Return, descended object, or proof of the truth of its payload.

7.34 Canonical v21 manifest object and field families

Definition 7.160 (Expanded serialization of Main Definition 8.68)**.** *The canonical v21 verifier manifest is the typed record*

$$\begin{aligned}\mathsf{Manifest}_{21} = (&\mathsf{Identity}, \mathsf{StatementAxes}, \mathsf{Scope}, \mathsf{TargetRequest},\\ &\mathsf{NodeRegistry}, \mathsf{EdgeRegistry}, \mathsf{TargetProjection}, \mathsf{Obligations},\\ &\mathsf{TypedEvidence}, \mathsf{ComparisonModes}, \mathsf{MetricLedger}, \mathsf{NonScalar},\\ &\mathsf{DomainRecords}, \mathsf{SourceBindings}, \mathsf{Precursor}, \mathsf{Quarantine},\\ &\mathsf{Artifacts}, \mathsf{ArtifactClosure}, \mathsf{FormalizationInterpretation},\\ &\mathsf{AgentAuthority}, \mathsf{Replay}, \mathsf{Revision}, \mathsf{ImpactClosure},\\ &\mathsf{Handoffs}, \mathsf{AtomicStatuses}, \mathsf{AggregateVerdict},\\ &\mathsf{FailureRouting}, \mathsf{Repairs}, \mathsf{Release}).\end{aligned}$$

Every field family is versioned and conditionally required under immutable scope. A conditional family is absent only with a positive scope-exclusion record. The tuple is logical rather than format-prescriptive.

Definition 7.161 (Canonical v21 identity block)**.** *The identity block contains exactly one value for each of*

$$\text{ManifestID},\ \text{RunID},\ \text{RevisionID},\ \text{SchemaID},\ \text{TheoryVersion},\ \text{ClaimSnapshot},$$
$$\text{VerifierProfileVersion},\ \text{TokenSortRegistryVersion},\ \text{ArtifactRootID},\ \text{ReleaseID}.$$

The release identifier is optional for an unreleased run, but its absence is explicit. Any difference in immutable target scope, requested strength, source authority, necessary target sub-DAG, evidence, or artifact root creates a new revision or run.

Definition 7.162 (Target request record)**.** *The target request record contains*

$$\begin{pmatrix} \text{TargetNodeID}, \text{exact\_target\_statement}, \text{requested\_context}, \\ \text{requested\_strength}, \text{target\_owner}, \text{invocation\_role} \end{pmatrix}.$$

The requested context and strength are inputs to verification, not conclusions inferred from the final status. A stronger target statement requires a new target identity or a proved promotion edge.

Definition 7.163 (Independent statement-axis record)**.** *Every consumed statement stores the Main Definition 8.5 tuple*

$$\text{Axis}(c) = \begin{pmatrix} \text{document\_role}, \text{evidence\_status}, \text{invocation\_role}, \text{mathematical\_verdict}, \\ \text{conformance\_status}, \text{workflow\_status}, \text{migration\_status} \end{pmatrix}.$$

The manifest validator treats these as separate semantic sorts. The `relocated` *token, when used, is a disposition modifier attached to a migration or ownership record; it is not itself a canonical migration status.*

Proposition 7.164 (No manifest-axis promotion)**.** *A favorable workflow, conformance, release, replay, build, render, or agent field does not populate a missing theorem-evidence or mathematical-verdict field.*

Proof. The fields decode to different predicates under Definition 7.163. A semantic-preserving serialization cannot introduce a conversion not present in Main Chapter 8 or Appendix F. ☐

7.35 Proof-DAG node, edge, and target-projection serialization

Definition 7.165 (Proof-DAG registry record)**.** *A proof-DAG registry serializes*

$$\mathsf{ProofDAG} = (N, E, s, t, \text{NodeType}, \text{EdgeType})$$

with one immutable target node, one dependency policy, a complete node registry, a complete edge registry, and an exact necessary-target projection. The stored graph may contain advisory or alternative branches, but only the registered necessary target sub-DAG participates in the target verdict.

Definition 7.166 (Expanded node semantic record)**.** *For every consumed node, the primary record is*

$$\mathfrak{n} = \begin{pmatrix} \text{NodeID}, \text{type}, \text{statement}, \text{owner}, \text{scope}, \text{role}, \\ \text{input\_type}, \text{output\_type}, \text{required\_strength}, \text{provided\_strength\_set}, \\ \text{assumption\_context}, \text{dependencies}, \text{verifier\_policy}, \text{artifacts}, \text{status} \end{pmatrix}.$$

The statement is exact, not a title. The provided-strength set is downward closed in the declared preorder. Primitive/derived classification and every assumption identifier are separately stored.

Definition 7.167 (Canonical node-type token sort)**.** *The proof-bearing node-type sort contains*

$$\texttt{theorem},\ \texttt{interface},\ \texttt{artifact},\ \texttt{metric},\ \texttt{coherence},\ \texttt{descent},$$
$$\texttt{reduction},\ \texttt{return}.$$

Source-binding, formalization, interpretation, validation, replay, audit, and governance nodes are auxiliary node types with explicitly limited authority. They do not inherit theorem authority from graph placement.

Definition 7.168 (Node discharge certificate record)**.** *A node discharge record contains the exact target sequent*

$$\mathrm{Cond}(n) \vdash \mathrm{statement}(n),$$

input and output types, required predecessor strengths, incoming necessary edges, supplied evidence, verifier policy, artifacts, propagated assumption context, provided-strength set, and derived atomic status. A passing conditional node remains conditional.

Definition 7.169 (Expanded edge semantic record)**.** *Every necessary edge has the primary record*

$$\mathfrak{e} = \begin{pmatrix} \mathrm{EdgeID}, s(e), t(e), \mathrm{type}, \mathrm{relation}, \mathrm{direction}, \\ \mathrm{input\_strength}, \mathrm{output\_strength\_set}, \mathrm{assumption\_transport}, \\ \mathrm{discharge\_policy}, \mathrm{comparison\_mode}, \mathrm{artifacts}, \mathrm{status} \end{pmatrix}.$$

The comparison-mode field is required only for a comparison edge and is otherwise explicitly not invoked. Assumption transport lists preserved, added, and theorem-discharged assumptions separately.

Definition 7.170 (Canonical edge-type token sort)**.** *The primary proof-DAG edge sort contains*

$$\texttt{logical},\ \texttt{artifact},\ \texttt{comparison},\ \texttt{bridge},\ \texttt{interpretation},\ \texttt{formalization},\ \texttt{weakening}.$$

Auxiliary validation, source-binding, replay, provenance, audit, and advisory edges are separately sorted and have no implicit logical authority.

Definition 7.171 (Necessary target projection record)**.** *For target node z, the target projection stores*

$$\mathrm{TargetDAG}(z), \qquad \mathrm{PredDAG}(z),$$

the completeness certificate proving that every actually consumed dependency is included, and the exclusion reason for every omitted alternative branch. The projection includes every necessary edge, including edges entering the target node.

Definition 7.172 (Well-founded dependency certificate record)**.** *A finite target projection stores an acyclicity certificate and topological order. An infinite or countable projection stores either an independently checkable ordinal rank and uniform verifier schema, or a Chapter 5 finite-to-infinite reduction replacing the infinite closure by a finite certificate and proved tail theorem. A finite prefix is never encoded as the whole countable certificate.*

Proposition 7.173 (Proof-DAG record completeness is not discharge)**.** *A complete node/edge registry, valid graph syntax, or path to the target does not establish a target pass without compatible edge semantics, node discharge, source independence, artifact closure, and target derivation readiness.*

Proof. The graph record serializes candidates and dependencies. The target status is computed only from the verifier predicates attached to those records, as in Main Theorem 8.95. ☐

7.36 Comparison modes, metric ledgers, and non-scalar records

Definition 7.174 (Canonical comparison-mode record). *Every consumed comparison has exactly one runtime mode*

$$\texttt{exact}, \qquad \texttt{metric\_gap\_certified},$$
$$\texttt{theorem\_controlled\_nonmetric}, \qquad \texttt{not\_compared}.$$

The record contains the exact comparison claim, source and target types, $\mathrm{Allowed}(c)$, $\mathrm{Out}(c, m)$, $\mathrm{Strength}_{\mathrm{req}}(c)$*, the artifact-supported relation, and the compatibility verdict. The mode* `not_compared` *can support only a claim that consumes no comparison.*

Definition 7.175 (Canonical non-scalar obligation record). *The non-scalar registry includes every invoked representative, realization, actual morphism, source authority, detector applicability/completeness, generator/cone, coherence, descent, reduction, source binding, quarantine, formalization/interpretation, artifact-closure, and Return obligation. Each has an atomic status and evidence record, never a numerical budget slot.*

Definition 7.176 (Declared metric-ledger record). *A metric-ledger record contains the ordered commutative monoid or justified quantale, every primitive discrepancy with semantic type and gate scope, the aggregation expression, scalarization, actual discrepancy theorem, prethreshold profile family, threshold gap, scalarized budget, reserve, and strict gate comparison.*

Definition 7.177 (No-double-counting ledger index). *Every primitive discrepancy has one charge identifier. The selected target route records where that charge enters the aggregate. A second charge is permitted only through a documented joint-bound theorem replacing the original charges. Aliases, duplicate manifest references, agents, and replay copies do not create independent discrepancies.*

Proposition 7.178 (Numerical records cannot manufacture semantics). *No metric ledger, threshold gap, positive reserve, exposure sum, or score can fill a missing field in Definition 7.175.*

Proof. A metric record proves a bound in its declared codomain. The missing non-scalar field requires an object, theorem, map, compatibility, authority, or governance witness of another type. □

7.37 Canonical token sorts and specialized domain records

Definition 7.179 (v21 token-sort registry). *The token-sort registry contains, at minimum, independent sorts for:*

(i) canonical atomic status $\{\texttt{pass}, \texttt{reject}, \texttt{undefined}, \texttt{not\_invoked}\}$*;*

(ii) the four comparison modes of Definition 7.174;

(iii) detector stage

$$\texttt{windowed\_prethreshold}, \quad \texttt{repaired\_postthreshold},$$
$$\texttt{kernel\_defect}, \quad \texttt{cokernel\_defect},$$
$$\texttt{diagram\_probe}, \quad \texttt{descent\_coherence};$$

(iv) detector certificate status and detector mathematical value;

(v) direct Type IV status, inverse-system validity, $\mathbf{R}^1\varprojlim$ *status, and inverse apex agreement;*

(vi) overlap status, simplex coherence mode, Restart transition and target-zero-route status;

(vii) statement axes, manifest status, replay level, repair relation, handoff status, and release acceptance.

Equal strings in distinct sorts remain distinct values.

Definition 7.180 (Detector record with six-stage identity)**.** *Every detector record stores certificate status and mathematical value as independent fields, together with the exact six-stage token, source profile, information class, family, endpoint convention, witness data, soundness, applicability, completeness, sharpness, transport dependencies, and artifacts. No stage is changed by relabeling an existing certificate.*

Definition 7.181 (Representative and realization record)**.** *A representative record stores workflow lifecycle and canonical atomic status on separate axes, the actual filtered object, monitored profile isomorphisms, equivalence strength, construction and verification artifacts, and blockers. An eligible-realization record additionally stores provenance class, semantic fidelity, action mode, amplitude, residual edge, edge-realization theorem, and the exact Ext clause consumed.*

Definition 7.182 (Direct and inverse tower record families)**.** *A direct tower record stores its profile system, source provenance and semantic authority, declared apex, actual source-to-apex morphism, kernel/cokernel defects, terminal and transient diagnostics, and finite-to-infinite route. An inverse record stores separately*

$$(\mathsf{SystemValidity}, \mathbf{R}^1\varprojlim \mathsf{Status}, \mathsf{ApexAgreement}),$$

together with the Roos complex or alternative theorem, stable-image or ML data, and all apex comparison artifacts. No one field substitutes for another.

Definition 7.183 (Atlas, descent, and global-object records)**.** *The manifest distinguishes and serializes separately:*

(i) certificate-atlas records, including exact, metric, theorem-controlled, or model-certified coherence, detector coverage, countable-tail coverage, costs, and record-level verdicts;

(ii) a `descent_input_ready` record containing all nine canonical fields

$$(\{E_\alpha\}, \{g_{\alpha\beta}\}, \{c_{\alpha\beta\gamma}\}, \{h_\sigma\}, \mathcal{W}, \mathcal{C}, \mathfrak{M}_{\mathrm{coh}}, \mathcal{T}_{\mathrm{eff}}, \mathcal{A}),$$

where $\mathcal{W}$ includes the cover, nerve, intersections and face maps, and $\mathcal{C}$ is object-level cover/restriction compatibility, not certificate-atlas completeness;

(iii) the active descent lane and the lane-matching verdict;

(iv) effectivity-theorem application and hypothesis-discharge records;

(v) descended global-object or global-morphism existence records and the actual output artifacts;

(vi) local comparison maps and uniqueness/equivalence-strength records;

(vii) exact-restriction, joint-conservativity, local-generator, and cone-compatibility records when global detection is requested;

(viii) global zero or global isomorphism certificates at their exact requested strength.

No item populates another by default. In particular, a record-level atlas never populates $\mathcal{C}$, an input-ready record never populates an effectivity result, and effectivity never populates conservativity or a global verdict.

Proposition 7.184 (Lossless descent-stage serialization)**.** *A manifest fragment is semantically lossless for a descent route only when it reconstructs each item of Definition 7.183, its atomic status, owner, assumptions, dependencies, artifacts, and prohibited stronger readings. Omitting or merging $\mathcal{W}$, $\mathcal{C}$, the active lane, effectivity, output existence, conservativity, or global verdict makes the affected route* `undefined`*; it is not repaired by a passing certificate atlas or successful build.*

Proof. Each listed field controls a distinct source type or inference edge in the Chapter 6–Chapter 7 route. A serializer that contracts two fields cannot reconstruct which theorem was applicable or which conclusion strength was proved. The canonical non-compensating status semantics therefore assigns `undefined` to the missing clause rather than inferring it from an unrelated pass. ☐

Definition 7.185 (Return-strength record)**.** *Each node, edge, route, handoff, and Return record stores a downward-closed permitted strength set. For route R, the manifest reconstructs*

$$\mathfrak{S}_R = \bigcap_{x \in \mathrm{Req}(R)} \mathfrak{S}_x$$

and its propagated assumption context. Multiple incomparable maximal strengths are permitted. A single scalar ceiling *field is allowed only when a greatest element is separately proved.*

7.38 Source binding, precursor packages, and benchmark quarantine

Definition 7.186 (Canonical source-statement binding record)**.** *A source-statement binding serializes the Main Definition 8.56 tuple*

$$\begin{pmatrix} \text{ClaimID}, \text{source\_authority}, \text{artifact}, \text{locator}, \text{exact\_statement}, \\ \text{hypotheses}, \text{permitted\_use}, \text{forbidden\_stronger\_reading} \end{pmatrix}.$$

A title, DOI, citation string, PDF, image, or hash without the exact statement and locator is not theorem evidence.

Definition 7.187 (Precursor and benchmark records)**.** *A precursor record enumerates every source-side hypothesis, lemma, construction, normalization, and independently permitted datum available to upstream construction. A separate benchmark record freezes the external target statement, source authority, target strength, and terminal comparison policy. The benchmark is a validation target, not an upstream premise.*

Definition 7.188 (Benchmark-quarantine cut record)**.** *The quarantine record partitions the recovery sub-DAG into construction and terminal-validation sides and lists prohibited crossings: benchmark proof, known truth value, answer table, target constants, exceptional bounds, conclusion-derived witnesses, equivalent oracles, and independent validation artifacts. The only permitted benchmark-to-construction metadata are:*

(i) immutable target identity and exact statement locator;

(ii) terminal-validation comparison strength;

(iii) source-side hypotheses independently present in the precursor package;

(iv) data explicitly authorized as independent target-theorem hypotheses.

The terminal comparison edge is directed from the frozen construction result to the benchmark node, never backward.

Definition 7.189 (Recovery sub-DAG manifest fragment). *A Known-Theorem Recovery fragment has separate source-binding, precursor, realization, Core, reduction, descent, Return, quarantine, and terminal validation nodes. It stores contamination checks and a proof that every construction-side datum is precursor-authorized. Successful recovery has exactly the frozen benchmark strength unless a separate stronger theorem is proved independently.*

Theorem 7.190 (Quarantine serialization soundness). *A recovery manifest can support* `pass` *for independent recovery only if its quarantine cut is complete and no prohibited conclusion-side datum occurs in the construction-side dependency closure. Detected leakage makes the affected independence clause* `reject`*; unresolved provenance makes it* `undefined`*.*

Proof. The record reconstructs exactly whether each upstream datum belongs to the registered precursor package or a permitted crossing. A prohibited consumed datum makes the route conclusion-conditioned, while unresolved provenance prevents evaluation. This is the serialization specialization of Main Theorem 8.61. ☐

7.39 Artifact identity, semantic equivalence, and artifact closure

Definition 7.191 (Expanded immutable artifact identity). *An artifact identity is*

$$\begin{pmatrix} \text{ArtifactID}, \text{media\_type}, \text{canonicalization}, \text{hash\_algorithm}, \text{Hash}, \\ \text{size}, \text{provenance}, \text{semantic\_locator} \end{pmatrix}.$$

It identifies exact bytes or a declared canonical representation and names the parser, interpretation theorem, formal environment, or schema giving those bytes mathematical meaning.

Proposition 7.192 (Artifact identity is not mathematical truth). *Hash agreement and canonical-byte identity do not prove correctness, interpretation, completeness, independence, theorem strength, or target sufficiency.*

Proof. The digest is a function of content under the declared canonicalization. The listed properties are semantic predicates requiring separate evidence. ☐

Definition 7.193 (Cross-format semantic-equivalence record). *A TeX/PDF/JSON/proof-assistant or other cross-format equivalence claim contains source and target artifact identities, parsers or interpretation maps, normalized semantic objects, a semantic diff, and a preservation theorem at the exact consumed strength. Visual similarity or matching titles are not sufficient.*

Definition 7.194 (Expanded artifact-closure record). *Artifact closure contains the complete reachable artifact graph and proves that every referenced and transitive interpretation dependency resolves uniquely, has immutable identity and provenance, matches the consuming node's type and version, and has no undeclared replacement. It stores canonicalization, integrity checks, semantic locators, supported evidence roles, and replay instructions for every node.*

Proposition 7.195 (Dangling or ambiguous artifacts block pass). *A necessary artifact that is dangling, ambiguously resolved, semantically untyped, version-incompatible, or silently replaced leaves the consuming node* `undefined`*. An evidenced identity contradiction makes the manifest* `inconsistent` *and the governance clause* `reject`*.*

Proof. In the first case the verifier cannot reconstruct the required evidence. In the second, two records assign incompatible meaning to one immutable run. ☐

7.40 Agent authority, formalization, and contamination records

Definition 7.196 (Agent action and authority record)**.** *Every human, AI, script, platform, auditor, or proof-assistant action stores one canonical role for that action:*

$$\texttt{proposer}, \quad \texttt{verifier}, \quad \texttt{auditor}, \quad \texttt{proof\_assistant}.$$

The record names the authority profile and exact node/edge predicates the role may establish. One actor may hold several roles across actions, but authority is never inferred from actor identity alone.

Definition 7.197 (Proposer-to-verifier conversion record)**.** *A proposal becomes verifier input only through a conversion record containing normalization into a declared type, exact source and target statements, hidden assumptions, statement axes, requested strength, immutable before/after artifacts, semantic diff, source-authority and quarantine checks, and the complete verifier profile still to be executed.*

Definition 7.198 (Formalization/interpretation pair record)**.** *A proof-assistant route stores two independently evaluated edges: a formalization edge from the intended statement to the exact formal statement, and an interpretation edge from the verified formal theorem to the intended target. The formal environment, imported axioms, parser, theorem identity, and conclusion-strength map are explicit.*

Definition 7.199 (Contamination event record)**.** *Contamination includes benchmark leakage, fabricated or unsupported source data, hidden use of a target conclusion, detector-stage mutation, artifact drift, unrecorded post-processing, or semantic transformation without preservation evidence. The record identifies affected nodes/edges, first contamination point, downstream impact closure, and required clean revision.*

Proposition 7.200 (Consensus and formal build are non-compensating)**.** *Agreement among agents or successful formal compilation does not repair a missing verifier dependency, source binding, formalization/interpretation edge, or artifact-closure clause.*

Proof. Consensus and compilation are distinct statement axes and evidence roles. The required semantic dependency remains absent under the non-compensation theorem. □

7.41 Replay hierarchy, semantic replay, and discrepancy handling

Definition 7.201 (Replay boundary record)**.** *A replay boundary fixes the exact input artifacts, code, formal environments, parsers, schemas, token registry, theorem and claim snapshots, verifier policies, platform versions, nondeterminism controls, and expected output identities for one replay claim.*

Definition 7.202 (Four-level replay record)**.** *The manifest stores four independent replay levels:*

(i) operational replay*: the declared procedure executes;*

(ii) artifact replay*: declared canonical output artifacts are reproduced or semantically matched under a proved policy;*

(iii) semantic replay*: typed statements, dependencies, verifier policies, statuses, and verdict reconstruction agree;*

(iv) mathematical verification*: theorem-dependent predicates are independently checked at the consumed strength.*

No level is inferred from the preceding level.

Definition 7.203 (Semantic replay certificate)**.** *A semantic replay certificate contains the replay boundary, reconstructed node and edge registries, statement axes, source bindings, comparison modes, assumption contexts, token sorts, specialized embeddings, artifacts, normalizations, aggregation, blockers, failure routes, and semantic diffs. It identifies every nondeterministic or theorem-dependent step and the evidence used there.*

Proposition 7.204 (Build and render success have bounded authority)**.** *Compilation, parsing, schema validation, rendering, byte equality, and visual diff success establish only their registered operational, artifact, or conformance predicates. They do not establish theorem truth or semantic identity outside a passing preservation record.*

Proof. Each procedure verifies a specified syntactic or artifact relation. The mathematical interpretation and target theorem require separate semantic predicates. ☐

Definition 7.205 (Replay discrepancy record)**.** *A replay discrepancy record stores both replay identities, common immutable inputs, first divergent artifact/node/edge/status, candidate causes, affected impact closure, and the current atomic status. Unresolved divergence makes replayability* `undefined`*; a contradiction under identical declared semantics makes the manifest* `inconsistent`*.*

7.42 Immutable revisions, impact closure, and repair relations

Definition 7.206 (Expanded immutable revision record)**.** *A revision record contains predecessor identities, changed statement, node, edge, schema, source, artifact, evidence, scope, or verifier fields; semantic diffs; changed hashes; reason; responsible authority; and the new immutable RunID/RevisionID/ManifestID triple.*

Definition 7.207 (Impact-closure record)**.** *For proof-relevant change set S, the record computes*

$$\mathrm{Impact}(S) = V(S) \cup \mathrm{Desc}_{\mathsf{ProofDAG}}(V(S)),$$

where $V(S)$ includes every changed node and every target of a changed edge. Every impacted node is marked dirty until reverified or covered by a passing semantic-preservation theorem.

Definition 7.208 (Repair relation)**.** *A repair relation links a new revision to a historical non-passing revision and records the repaired blockers, changed evidence, re-evaluated impact closure, old and new verdicts, and preserved fields. It is not a mutation or deletion of the predecessor record.*

Theorem 7.209 (Immutable manifest history)**.** *A valid repair or proof-relevant change creates a new immutable revision. Historical statuses, evidence semantics, and artifact identities remain unchanged and addressable. No pass, reject, undefined, replay, or Return status is inherited across revisions unless the required preservation or reverification record passes.*

Proof. Changing proof-relevant data changes the verifier input. The impact closure identifies every downstream record whose former derivation consumed that data. Reusing an old status without preservation would assign one verdict to a new input without verification and would erase historical audit meaning. ☐

7.43 Target closure, Chapter 9 handoff, and release records

Definition 7.210 (Target derivation-readiness manifest). *A target-readiness record stores every Main Definition 8.94 clause: complete well-founded target projection, passing predecessor nodes, passing compatible edges, total target verifier policy, comparison-mode compatibility, non-scalar closure, metric/no-double-counting closure, source binding and quarantine, artifact closure, complete manifest, semantic replay, revision identity, propagated assumption context, and requested output strength. No target pass is entered as an input field.*

Definition 7.211 (Route-admissible conclusion record). *For route R, the manifest stores*

$$\mathfrak{S}_R = \bigcap_{x \in \mathrm{Req}(R)} \mathfrak{S}_x, \qquad \mathfrak{A}_R = \{(\Gamma, \sigma) : \mathrm{Cond}(R) \subseteq \Gamma,\ \sigma \in \mathfrak{S}_R\}.$$

It stores all incomparable maximal strengths and every undischarged assumption. An unconditional conclusion is available only when $\mathrm{Cond}(R) = \varnothing$*.*

Definition 7.212 (Canonical BridgeDAGInput record). *The Chapter 9 handoff serializes*

$$\mathsf{BridgeDAGInput} = \begin{pmatrix} \text{source\_node, prebridge\_boundary\_node, BridgeGoal,} \\ \text{TargetDAG(prebridge\_boundary),} \\ \text{requested\_context, requested\_strength,} \\ \mathrm{Cond}(R), \mathfrak{S}_R, \text{node\_registry,} \\ \text{edge\_registry, source\_bindings,} \\ \text{quarantine\_cut, artifact\_closure, verifier\_verdict,} \\ \text{failure\_routing, manifest\_identity} \end{pmatrix}.$$

Its specialized handoff status is `bridge_input_ready` only when the pre-Bridge boundary is discharged at the requested context and strength and every pre-Chapter-9 verifier/governance obligation passes.

Proposition 7.213 (Handoff is not a Bridge theorem). *A `bridge_input_ready` record is a typed, source-bound, quarantine-safe pre-Bridge input. It does not include or presuppose the Chapter 9 Bridge or Return node named by* BridgeGoal.

Proof. The handoff target projection ends at the pre-Bridge boundary. Chapter 9 adds new necessary realization, composition, rigidity, replacement, Bridge, and Return nodes downstream. ☐

Definition 7.214 (v21 release manifest). *The release manifest is*

$$\mathfrak{M}_{\mathrm{rel},21} = \begin{pmatrix} \text{ReleaseID}, \mathsf{Documents}, \mathsf{Claims}, \mathsf{Schemas}, \mathsf{Artifacts}, \\ \mathsf{OwnershipMigration}, \mathsf{Supersession}, \mathsf{Replay}, \mathsf{Acceptance} \end{pmatrix}.$$

For the advertised scope it records every mandatory document role. These may include the Main, Foundation Appendices A–G, Appendix DG, CM, TB, HT, UB, Companion, the active Claim Register, source-binding registries, execution/reproducibility artifacts, the Migration/Ownership Matrix, release notes, and the integrity index. The Migration/Ownership Matrix may be a release-side governance artifact rather than a mathematical appendix. Its identifier and digest are nevertheless recorded whenever it justifies ownership or locator migration.

Definition 7.215 (Migration and ownership record). *An ownership/migration entry contains predecessor and successor location, canonical owner, disposition, semantic status, proof responsibility, preserved strength, changed fields, cross-references, and implementation state. The semantic migration status is distinct from disposition modifiers such as `relocated`, `split`, or `merged`.*

Definition 7.216 (v21 release-acceptance record)**.** *Release acceptance is a non-compensatory conjunction of mandatory document coverage, canonical ownership, active claim and supersession closure, schema conformance, artifact closure, source-binding availability, semantic replay, Migration/Ownership consistency, and every mandatory Foundation/Bridge/Return readiness verdict advertised by the release. A missing mandatory role is* `undefined`*; an identity or ownership contradiction is* `reject`*.*

7.44 Concrete encodings, validation profiles, and executable checks

Definition 7.217 (v21 execution-schema conformance)**.** *A JSON, YAML, relational, graph, proof-assistant, or other concrete encoding is v21-conformant only when a total decoding map preserves the complete canonical record, including exact node and edge statements, target projection, assumption contexts, permitted strength sets, token sorts, source bindings, quarantine, artifact semantics, lower-level statuses, failure routes, revision history, and forbidden stronger readings.*

Definition 7.218 (v21 validation profiles)**.** *In addition to inherited profiles, v21 supplies at least*

`proof_dag`, `comparison_four_mode`, `inverse_audit`, `known_recovery`,
`descent_input`, `effective_descent`, `semantic_replay`, `revision_impact`,
`bridge_handoff`, `release_acceptance`, `full_v21_foundation`.

Profiles select conditional field families but never change field semantics or weaken dependencies.

Definition 7.219 (Canonical conformance test suite)**.** *A conformant implementation supplies deterministic checks for identifier uniqueness, token-sort validity, requiredness, node/edge type compatibility, target-projection closure, well-foundedness, assumption preservation, comparison-mode compatibility, no-double-counting, artifact closure, quarantine, revision impact, specialized-status embedding, headline aggregation, and release coverage. The suite also contains negative fixtures for every forbidden default or cross-sort substitution.*

Proposition 7.220 (Schema validation reconstructs but does not create pass)**.** *A token-valid, complete, consistent, dependency-complete, artifact-closed, quarantine-safe, and semantically replayable v21 record is sufficient to reconstruct the declared verifier verdict. It is not sufficient by itself to make that verdict* `pass`*.*

Proof. The hypotheses expose the data and deterministic semantics needed for reconstruction. The reconstructed mathematical, interface, or Return nodes may still be rejected or undefined. □

7.45 v21 canonical serialization closure

Theorem 7.221 (Appendix G v21 canonical manifest and replay package)**.** *Within the v21 architecture:*

(i) Main Chapter 8 and Appendix F determine verifier semantics; Appendix G serializes without semantic mutation;

(ii) independent statement axes, canonical statuses, token sorts, and scope exclusions remain distinct;

(iii) proof-DAG nodes, edges, target projections, assumption contexts, and strength sets are serialized at their exact typed meaning;

(iv) four comparison modes, metric ledgers, no-double-counting, and non-scalar obligations are noninterchangeable;

(v) all six detector stages and every direct/inverse, atlas, descent, reduction, and global-object status retain separate field identities;

(vi) source bindings, precursor packages, quarantine cuts, and terminal benchmark direction are reconstructible;

(vii) artifact identity, cross-format semantic equivalence, and artifact closure support identity and interpretation but do not create truth;

(viii) proposer, verifier, auditor, proof-assistant, formalization, and interpretation authorities remain separate;

(ix) operational, artifact, semantic, and mathematical replay are distinct;

(x) repairs create new immutable revisions, and impact closure controls status reuse;

(xi) route conclusions preserve both assumption context and the intersection of permitted strength sets;

(xii) `bridge_input_ready` *serializes only the closed pre-Bridge boundary required by Chapter 9;*

(xiii) release acceptance is independent of mathematical strength and requires document, ownership, claim, schema, artifact, replay, and supersession closure;

(xiv) abstract and concrete schemas reconstruct verifier verdicts but never manufacture theorem truth.

Proof. Items (i)–(iii) follow from Definitions 7.157, 7.160, and 7.165–7.171. Item (iv) is Definitions 7.174–7.177 and Proposition 7.178. Item (v) is Definitions 7.179–7.183. Item (vi) is Definitions 7.186–7.189 and Theorem 7.190. Item (vii) is Definitions 7.191–7.194 and Proposition 7.192. Item (viii) is Definitions 7.196–7.198. Item (ix) is Definitions 7.201–7.203. Item (x) is Theorem 7.209. Item (xi) is Definitions 7.210 and 7.211. Item (xii) is Definition 7.212 and Proposition 7.213. Item (xiii) is Definitions 7.214–7.216. Item (xiv) is Proposition 7.220. ☐

7.46 v21 explicit non-claims

The following stronger readings are explicitly excluded.

(i) No manifest completeness, graph connectivity, hash equality, replay, compilation, rendering, or agent agreement proves a mathematical node.

(ii) No historical v19/v20 token silently overrides a canonical v21 sort.

(iii) No `relocated` disposition is treated as a semantic migration status.

(iv) No metric field manufactures a representative, morphism, source authority, coherence cell, descent theorem, quarantine proof, or Return.

(v) No detector certificate changes source stage by relabeling.

(vi) No direct Type IV status determines an inverse-system or $\mathbf{R}^1\varprojlim$ status.

(vii) No certificate atlas is serialized as $\mathcal{C}$, as `descent_input_ready`, or as effective object descent.

(viii) No input-ready record is serialized as lane applicability, effectivity, descended-object existence, joint conservativity, global zero, or global isomorphism.

(ix) No benchmark statement, target constant, known answer, or validation artifact flows upstream across the quarantine cut.

(x) No source PDF, source binding, or stored known statement is itself a recovered theorem.

(xi) No artifact hash proves semantic equivalence across formats.

(xii) No proof-assistant success proves an intended informal claim without passing formalization and interpretation edges.

(xiii) No operational or artifact replay is serialized as semantic or mathematical replay.

(xiv) No status is inherited across a dirty revision without preservation or reverification.

(xv) No route result exceeds its strength-set intersection or drops a propagated assumption.

(xvi) No `bridge_input_ready` record is a Chapter 9 Bridge or Return theorem.

(xvii) No release manifest compensates for a missing mandatory mathematical, claim, ownership, artifact, or replay record.

7.47 Explicit exclusions

The following are not asserted in Appendix G.

- No concrete JSON, YAML, CSV, database, API, signature, or file layout is mandated.
- No schema field creates mathematical evidence.
- No complete manifest is automatically a passing proof.
- No passing mathematical atom repairs an incomplete or inconsistent manifest atom.
- No numerical budget contains claim-status, representative-existence, morphism-existence, eligibility, coherence, or manifest obligations.
- No omitted field receives a theorem-producing default.
- No `undefined` Type IV record is serialized as $(0, 0)$.
- No `not_invoked` value is accepted without immutable scope-exclusion evidence.
- No claim/use classification is conflated with atomic audit status.
- No headline verdict erases lower-level undefined, reject, blocker, or failure-route detail.
- No finite prefix certifies a countable registry.
- No repair mutates a historical run in place.
- No pairwise overlap family automatically supplies higher Čech coherence.
- No certified transition alone supplies a complete Restart pass.
- No certificate atlas implies object-level descent or a global mathematical object.
- No Search, Platform, Companion, or AI output updates a verdict without Core re-entry.

- No Claim Register entry is proof merely because it is registered.
- No micro-theorem reference discharges its own hypotheses.
- No toy bridge is promoted to an external bridge theorem by serialization.
- No global exactness of positive-threshold collapse or strict filtered lift is assumed.
- No tower comparison artifact is automatic.
- No token literal is interpreted without its semantic sort.
- No detector mathematical value substitutes for detector certification or atomic status.
- No terminal Type IV zero is serialized as long-transient absence or full isomorphism.
- No local detector-zero family is serialized as global zero without a global profile interface and retained-bar coverage.
- No registry coverage substitutes for retained-bar, boundary-bridge, artifact, or release coverage.
- No metric-reference simplex record creates transition morphisms or a categorical cocycle.
- No κ-Restart reserve pass certifies a target-zero route.
- No observed finite plateau is serialized as eventual constancy or finite-to-infinite reduction.
- No certificate atlas or global-zero predicate is serialized as object descent.
- No artifact hash alone proves the mathematical role of the artifact.
- No computational replay alone establishes semantic replay.
- No lossy schema migration preserves an audit verdict at the same claimed strength.
- No mathematically strong document compensates for missing mandatory release records.
- No proposer, Search, Platform, Companion, or AI field has verifier status without typed Core re-entry.

7.48 Provenance and status

This appendix consumes the complete inherited v19 record architecture and the v20 conservative extensions of Appendices A–F. In particular, it serializes detector stages and certificates, terminal/transient separation, exact/metric atlas modes, retained-bar and boundary-bridge coverage, Restart zero routes, finite-to-infinite reduction, object-descent boundaries, artifact closure, semantic replay, and release acceptance.

This appendix consumes:

(i) Chapter 1 for the foundational Minimal Manifest requirement;

(ii) Chapters 2–3 and Appendices A–B for evaluation, threshold, and representative records;

(iii) Chapter 4 and Appendix C for eligible Ext records;

(iv) Chapter 5 and Appendix D for terminal-generic Type IV records;

(v) Chapter 6 and Appendix E for overlap, Čech, Restart, and atlas records;

(vi) Main Chapter 7 and Appendix DG for finite diagram and effective-descent records;

(vii) Main Chapter 8 and Appendix F for canonical typed verifier semantics, proof-DAG closure, aggregation, readiness, non-compensation, source quarantine, replay hierarchy, and repair history;

(viii) Appendix CR-v20, Appendix CM-v20, and Appendix TB-v20 for governed downstream references.

7.48.0.1 Status labels for Appendix G. *Normative schema definitions and axiom:* Definitions 7.6, 7.9, 7.10, 7.13, 7.14, 7.20, 7.27, 7.32, 7.35, 7.110, and Axiom 7.24.

Contract consequences: Propositions 7.7, 7.12, 7.23, 7.31, 7.36, 7.76, 7.97, 7.103, 7.117, 7.119, and Theorem 7.125.

v20 manifest and replay extension: Definitions 7.126, 7.127, 7.129–7.132, 7.133–7.137, 7.139–7.141, 7.142, 7.143, 7.145, 7.146, 7.148–7.151, 7.153, 7.154; Propositions 7.128, 7.144, 7.147, 7.152, 7.155; Theorem 7.156.

Operational cautions and exclusions: Remarks 7.1, 7.2, 7.8, 7.16, 7.19, 7.56, 7.70, 7.81, 7.88, 7.93, 7.105, and Subsection 7.47.

7.48.0.2 Reinforced A–G revision closure. This revision preserves the mathematical strengths of Appendices A–D and repairs the cross-layer schema and ownership boundary for object descent. Appendix E now matches the canonical Main nine-field DescentInput; Appendix F evaluates readiness, lane matching, effectivity, object existence, conservativity, and global verdicts as distinct atoms; and Appendix G serializes those fields without semantic contraction. These changes are a schema correction and governance hardening, not a strengthening of Main Theorems 6.93 or 7.36 and not an external Return theorem.